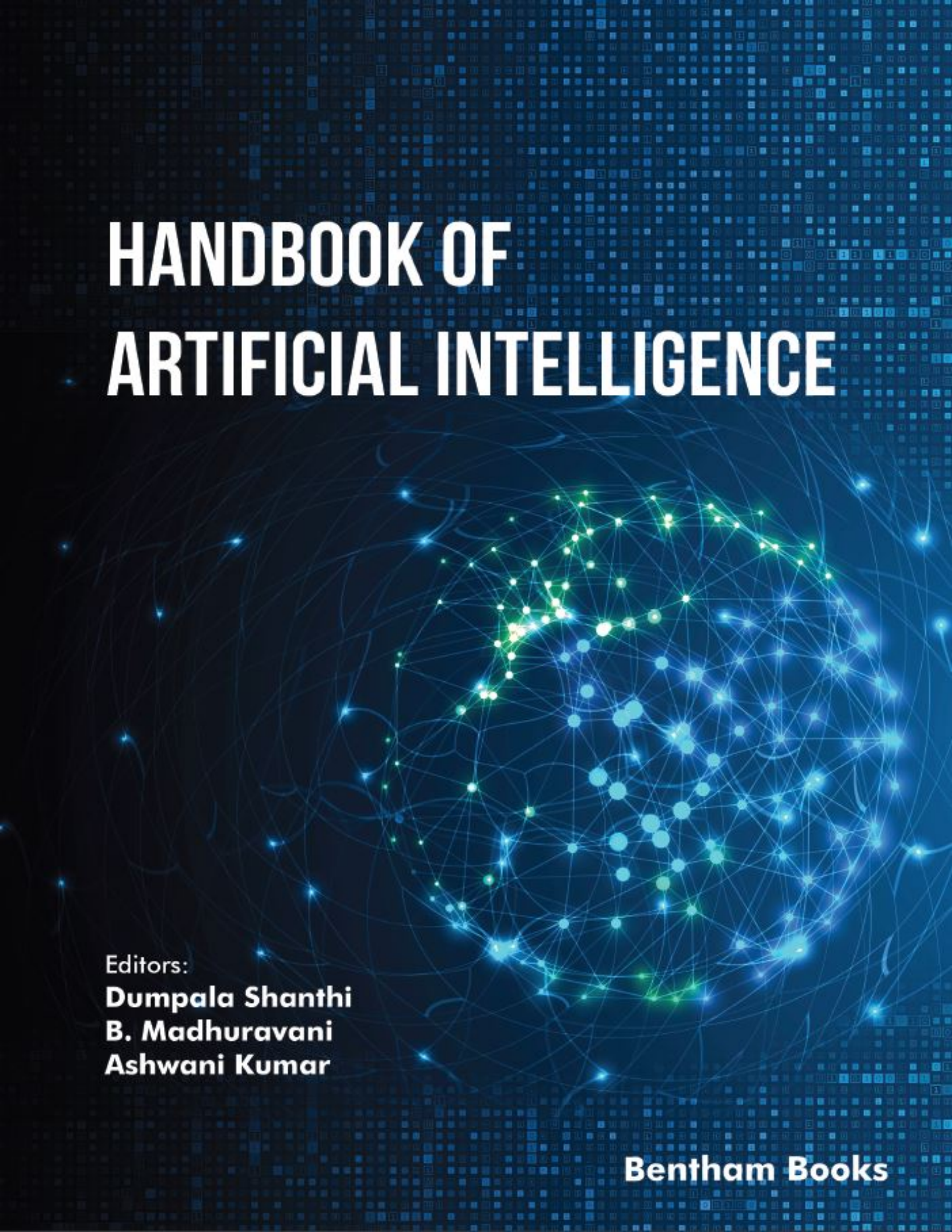


HANDBOOK OF ARTIFICIAL INTELLIGENCE



Editors:
Dumpala Shanthi
B. Madhuravani
Ashwani Kumar

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Handbook of Artificial Intelligence

Edited by

Dumpala Shanthi
Maturi Venkata Subba Rao Engineering College, Nadurgul(v)
Hyderabad, Telangana
India

B. Madhuravani
Marri Laxman Reddy Institute of Engineering and Technology,
Dundigal
Hyderabad, Telangana
India

&

Ashwani Kumar
Sreyas Institute of Engineering and Technology
New Delhi
India

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Singapore 068898

Singapore

Email: subscriptions@benthamscience.net



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PREFACE

Artificial intelligence (AI) is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence. AI is an interdisciplinary science with multiple approaches, but advancements in machine learning and deep Learning are creating a paradigm shift in virtually every tech industry sector. Coming to machine learning, an in-built branch of AI, has changed the shape of the emerging world...!! It's not a mere exaggeration to say that ML has taken living abilities and standards a step forward. Machine learning is an application of artificial intelligence (AI) that allows systems to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it to learn for themselves. The process of Learning begins with observations or data, such as examples, direct experience, or instruction, in order to look for patterns in data and make better decisions in the future based on the examples that we provide. The primary aim is to allow computers to learn automatically without human intervention or assistance and adjust actions accordingly.

This book covers the fundamental concepts and techniques in Artificial Intelligence, Machine Learning and Soft Computing in detail. The main focus is on real-time applications of AI, ML and Soft computing. As AICTE introduced emerging technology courses under UG Program, authors realized that there are no proper

textbooks to teach students with practical orientation. The sequence of the book has been set in an appropriate manner to make it easy for students and teachers to understand concepts. The book provides insights into AI, ML and Soft computing and their applications in core areas like Agriculture, Smart Cities, Environment, *etc.*, which impact society and their vital role in the development of the nation.

Dumpala Shanthi

Maturi Venkata Subba Rao Engineering College, Nadurgul(v)
Hyderabad, Telangana
India

B. Madhuravani

Marri Laxman Reddy Institute of Engineering and Technology,
Dundigal
Hyderabad, Telangana
India

&

Ashwani Kumar

Sreyas Institute of Engineering and Technology
New Delhi
India

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List of Contributors

A. Manjula	Jyothishmathi Institute of Technology and Science, Telangana, India
A. Swapna	Mahatma Gandhi Institute of Technology, India
Adluri Vijaya Lakshmi	B V Raju Institute of Technology, Narsapur, Medak, India
CH. M. Reddy	Marwadi University, Rajkot, Mahender, India
D. Latha	Department of CSE, Adikavi Nannaya University, Rajamanhendravaram, India
D. Abdus Subhahan	B V Raju Institute of Technology, Narsapur, Medak, India
D. Sai Kamesh	Sreyas Institute of Engineering and Technology, India
Dumpala Shanthi	Maturi Venkata Subba Rao Engineering College, Nadurgul(v), Hyderabad, Telangana, India
Divanu Sameera	CSE Department of SSSUTMS-Sehore, MP, India
G. Narsimha	JNTUH College of Engineering, Sulthanpur, Telangana, India
J. Narasimha Murthy	Sreyas Institute of Engineering and Technology, India
Jyothi P. Naga	Department of CSE, GITAM Institute of Technology, GITAM, Visakhapatnam, A.P, India
K. Chandra Sekhar	Department of Information Technology, Maturi Venkata Subba Rao Engineering College, India

K. Sudheer Babu	Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad, India
K. Vedavathi	Mahatma Gandhi Institute of Technology, Hyderabad, Telangana, India
K.V.S.N. Rama Rao	Department of Computer Science & Engineering, KLH, Hyderabad, India
K. Vaishali	Jyothishmathi Institute of Technological Sciences, Karimnagar, Telangana, India
K. Madhuri	Anurag University, Hyderabad, India
M. Sunitha	CVRCE, CSE, Hyderabad, India
Niraj Sharma	Department of CSE, SSSUTMS-Sehore, India
Narmada Kari	Amity University Rajasthan, Jaipur Rajasthan, India
P. Karthik	Department of Information Technology, Maturi Venkata Subba Rao Engineering College, India
Ponnuru Sowjanya	School of Technology GITAM(Deemed to be University), Rudraram, Hyderabad, India
P. Pranitha	Jyothishmathi Institute of Technology and Science, Telangana, India
R.V. Ramana Chary	Department of IT, B V Raju Institute of Technology, Narsapur, Medak Telangana., India
R. Usha Rani	CVRCE, CSE, Hyderabad, India
Rajya Lakshmi	Department of Computer Science and Engineering, JNTUK UCEV, Vizianagaram, A.P., India
Sowmya Gudipati Sri	School of Technology GITAM (Deemed to be University), Rudraram, Hyderabad, India
S.V. Vasantha	CSE Department, Vardhaman College of Engineering, Hyderabad, India

Swathi Gowroju	Sreyas Institute of Engineering and Technology, India
Sanjay Kumar Singh	Amity University Rajasthan, Jaipur Rajasthan, India
S. Suresh	Neerukonda Institute of Technology & Sciences, Sangivalasa, Bheemunipatnam, Vishakapatnam, Andhra Pradesh, India
T. Sajana	Department of Computer Science & Engineering, K LEF, Guntur, India
V. Swathi	Sreyas Institute of Engineering and Technology, India

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Machine Learning Techniques and their Applications: Survey

P. Karthik^{1, *}, K. Chandra Sekhar¹, D. Latha²

¹ Department of Information Technology, Maturi Venkata Subba Rao Engineering College, India

² Department of CSE, Adikavi Nannaya University, Rajamahendravaram, India

Abstract

Machine Learning has become part of our daily life directly or indirectly. Several Machine Learning techniques are being used in several areas to increase the effective usage of computers by human beings. Over the past few decades, the concepts of AI and ML have rapidly increased their importance in various application areas, and a lot of research has taken place and is still in progress. Even then, there is a lot more to explore regarding the applications of AI and ML in our day-to-day lives. The reason to do so is only to improve the existing process of work in various areas and also to give valuable suggestions and scope for further research. In this context, here we tried to discuss basic concepts

related to machine learning and gave a brief overview of various application areas of machine learning.

Keywords: Machine Learning (ML), Artificial Intelligence (AI), Cloud Computing, Big Data Analytics.

* **Corresponding author P. Karthik:** Department of Information Technology, Maturi Venkata Subba Rao Engineering College, India; E-mail: karthik_it@mvsrec.edu.in

1. INTRODUCTION

Artificial Intelligence (AI), the most prominent research area of computer science is basically concerned with the development of machines that can think and work similarly to human beings. AI is an interdisciplinary research area with many sub-branches like Artificial Neural Networks (ANN), Machine Learning (ML), etc. ML, which is said to be a sub-division of AI, is based on the concept of developing machines that can learn from data and formulate patterns from it with very less interference from human beings. Basically ML allows machines to learn from data and improve themselves to solve tasks.

1.1. History of AI & ML

In the year 1950, Alan Turing developed the Turing test. The objective of conducting the Turing test was to check whether a machine can think like a human or not. Even though this was a basic level test, but it later had a huge impact on AI [1, 2].

The Turing test [3, 4] can be defined as a simple question-and-answer game played between a human and a machine. The person who asks questions is also a machine. Now the task of the machine

is to check and analyze the answers given, and analyze whether the machine is human or not. Even though Alan Turing foresaw that by the 21st century, machines will be able to think and act like humans, it didn't happen completely like now. For example, if we consider any kind of Interactive Voice Response System (IVRS) used in mobiles or ATMs *etc.*, we can easily identify it as a machine but not a human. That is how we could clearly differentiate between a machine and a human.

In the year 1952 [[5](#), [6](#)], Arthur Samuel developed the first computer program that was able to learn. The program was the game of checkers, and the IBM computer improved at the game, the more it played, studying which moves made up winning strategies and involving that list of moves inside its program.

In 1957 [[7](#)], the first neural network for computers was designed by Frank Rosenblatt to simulate the thought process of the human brain.

In the year 1960 [[8](#), [13](#)], MIT developed a Natural Language Processing program to act as a therapist. The program was called ELIZA, and it was a successful experiment. But it was still using scripting to do its work. It was a key milestone for the development of a subset of Machine Learning called Natural Language Processing (NLP) which is being widely used even today.

In the year 1967 [[9](#), [11](#), [12](#)], the “nearest neighbor” algorithm was written, which allows computers to start using very basic pattern recognition. Later it was used to form a route map for traveling salesmen, who starts at a random city but ensure they visit all cities in the tour K-Nearest Neighbor or KNN is one of the important and famous Machine Learning algorithms.

In the year 1970 [[10](#), [13](#)], the concept of Backpropagation was developed. Back Propagation is nothing but a set of algorithms used in Deep Learning. They modify deep learning neural networks in such a way that they can correct themselves.

In the year 1979 [\[12\]](#), the students of Stanford University developed a “Stanford cart” that has the ability to navigate the obstacles within a particular room on its own.

In the year 1980 [\[13\]](#), Artificial Neural Network (ANN), which is a multilayered neural network, was developed by Kunihiro Fukushima.

In the year 1981 [\[12\]](#), the concept of Explanation Based Learning (EBL) was introduced by Gerald Dejong. EBL enables a computer system to analyze the given data and formulate a rule that can be followed by discarding data that is not so important. This was a very early stage of developing the concept of a Machine Learning Algorithm.

In the year 1985 [\[12\]](#), Nttalk was invented by Terry Sejnowski and Charles Rosenberg. NETtalk is an automated procedure that was developed to learn how to pronounce the English text written, when text is shown as input and matching phonetic transcriptions for comparison. The purpose of creating NETtalk was to identify the methods of correctly learning to pronounce the text in English.

In the year 1989 [\[13\]](#), the Reinforcement Learning concept was developed. In reinforcement learning, artificial intelligence faces a situation similar to that of a game where it learns to achieve a goal in a potentially complex and uncertain environment. To make the machine to act according to the needs of the programmer, the artificial intelligence gets either a benefit or a fine for the actions it performs. It aims to maximize the total benefit. Here, the programmer will specify game rules but won't give any clue to the machine regarding the rules of the game or the procedure for solving the game. It is up to the machine to decide and formulate how to win the game by following the trial and error method, thereby increasing its own rewards.

In the early 1990's [\[12\]](#), the work on Machine Learning shifted to a “data-driven approach” from a “knowledge-driven approach”. From here, experts have started developing computer programs, and those computers must analyze or gain knowledge from that data.

In 1995 [[13](#), [14](#)], Random Forest Algorithm and Support Vector Machines algorithm were developed. These are considered to be some of the most important Machine Learning algorithms.

In 1997/98 [[13](#), [15](#)], Sepp Hochreiter and Jürgen Schmidhuber introduced LSTM. Long Short-Term Memory (LSTM) [[16](#)] networks are a type of recurrent neural network capable of learning order dependence in sequence prediction problems. This is a required behavior in domains like machine translation, speech recognition, *etc.* A **recurrent neural network (RNN)** [[17](#)] is a class of artificial neural networks where connections between nodes form a directed graph along a temporal sequence. A **feed-forward neural network** [[18](#)] is an ANN where the connections between the nodes are acyclic. In this network, the movement of information is unidirectional, which means it moves in the forward direction from the input nodes, through the hidden nodes and then to the output nodes. The network is acyclic.

In the year 2006 [[12](#)], Geoffrey Hinton used the Deep Learning concept that allows computers to see and differentiate data that is in the form of videos and images.

In the year 2011 [[12](#)], Google Brain was developed, whose deep neural network is capable of identifying and classifying objects similar to that of a cat.

In the year 2012 [[12](#)], Google's X Lab developed an algorithm that can browse videos on YouTube that contain cats in them.

In the year 2014 [[12](#)], Facebook developed an algorithm called "Deep Face" that is capable of identifying human beings in photos uploaded on Facebook just like humans do.

In the year 2015 [[12](#)], Amazon started its own platform for Machine Learning. Stephen Hawking, Elon Musk, and Steve Wozniak have signed an open letter and warned about the danger that can be caused by automatic weapons. They have quoted that these

automatic weapons can select and hit the target without any order or command from human beings.

In the year 2016 [\[12\]](#), Google's artificial intelligence algorithm beat a professional player in the Chinese board game Go, which is considered the world's most complex board game and is many times harder than chess. Deep Mind, which was the algorithm created by Google, won five games out of five in the Go competition.

Even after all these developments in the areas of AI and ML over the past few decades, there are still some experts around the globe who believes that a computer can never think and act like a human. Irrespective of whether their opinion is true or false, the computer's capability to see, identify, verify and analyze is increasing at a quick rate. Also, globalization has become one of the main reasons for the generation of large amounts of data every day in various aspects, so there are a lot of scopes for computers to read, analyze and learn from the data.

1.2. Applications of ML

Some of the real-time application areas of ML are:

1.2.1. Speech Recognition [19]

Machine learning can translate speech into text. Some software applications can convert the live voice or a speech which is recorded into a text file. Ex: Alexa.

1.2.2. Predictive Analytics [19]

Machine learning can classify available data into groups, which are then defined by rules set by analysts. When the classification is completed, the probability of the occurrence of a fault can be calculated by analysts. Predictive analytics is one of the most promising application areas of machine learning. Predictive analytics

is applicable everywhere and for everything, right from the estimation of real estate prices to the development of products.

1.2.3. Product Recommendation [20]

While shopping on e-commerce brands like Amazon, Flipkart, Myntra *etc.*, you get to see recommended items or options like ‘users who bought this product also bought’, ‘users also buy this along with this product’!. Also, while browsing YouTube, we see similar kind of suggestions regarding the videos we browse. The ML methods will learn the patterns from the purchase and the browsing history of users and then make suggestions according to it.

1.2.4. Image Recognition [19]

Image recognition is a well-known and widely mentioned example of machine learning in the real world. It is the concept of identifying any object as an image that is in digital form, depending on the intensity of the pixels. Ex: X-ray scans, Tagging people on social media, Phone unlocking using face *etc.*

1.2.5. Video Surveillance [20]

This is one of the most advanced applications of machine learning and AI. Videos always provide us with an efficient opportunity of extracting useful information from automatic surveillance devices. This is only possible because machines keep a better outlook on objects which are used to compared to human minds. Surveillance footages are the best machine-learning dataset because of its accuracy.

1.2.6. Extraction [19]

Machine learning can extract structured information from unstructured data. Organizations gather together large volumes of

data from customers. ML technique will automate this process of annotation of datasets for predictive analytics tools.

1.2.7. Traffic Alerts [20]

How does Google Maps know that you are on the fastest route despite the traffic is high? When we use Google Maps to find the route to a particular place, we give permission to Google Maps to use the following data of ours like location, the day and time we are using Google Maps *etc.*; this type of information is gathered and saved by the application. Machine Learning algorithms will utilize this kind of information to properly instruct us to travel from our location to the location we want to.

1.2.8. Medical Diagnosis [19]

Machine learning can help with the diagnosis of diseases. Many physicians use chatbots with speech recognition capabilities to recognize patterns in symptoms.

1.2.9. Sentiment Analysis [20]

Based on sentiment analysis, machines are able to analyze sentiments on words. The words which are used in posts are classified into negative, neutral and positive words using ML algorithms. Companies are interacting with customers to use this model to increase efficiency.

1.2.10. Google Translate [20]

Google's GNMT(Google Neural Machine Translation) is working with thousands of languages and dictionaries, to provide the most accurate translation of any sentence or word based on Natural Language Processing. This feature is particularly useful whenever we visit a new place, and we don't know anyone over there.

1.2.11. Virtual Personal Assistants [20]

Siri, Alexa, Google Home *etc.*, are some of the popular examples of virtual personal assistants. As the name indicates, they help users in finding data, immediately after asking for it through our voice. Machine learning helps these personal assistants to organize the information on the basis of past data. Later, this data is used to respond to the instructions given by the users [[21](#), [22](#)].

As said above, ML has become a part of our day-to-day lives and shows its capability of helping human beings.

1.3. Difference Between Traditional Programming Concepts and ML Concepts [[23](#)]

Properly understand the problem, and try to give a solution with a correct program by passing the proper input, which gives the desired output. This is the concept of traditional programming.

1.3.1. Why Must We Learn ML?

ML can automate many tasks on a routine basis for an individual or an organization. It also helps in automating and quickly creating models for data analysis.

1.3.2. Difference Between AI & ML [[23](#)]

AI makes computers think and solve tasks like human beings. ML depends on the input data given by the user, and it makes the machine learn from the external environment like sensors, storage devices *etc.*

1.4. Steps to Learn ML

1. Data Gathering

2. Data Preparation
3. Selecting the Model
4. Training the Model
5. Evaluating the Model
6. Hyperparameter Tuning
7. Predicting

1.4.1. Data Gathering [24]

The accuracy and efficiency of the model depend on the quantity and quality of data that we collect, and will be used for training the model.

1.4.2. Data Preparation [24]

It will prepare the data for training. The term preparing means cleaning the data *i.e.*, discarding duplicate data, identifying and filling the values that are missing, handling the errors in the data, normalizing the data, and performing data (type) conversions wherever necessary. Finally, categorizing the entire data into training sets and evaluation sets.

1.4.3. Selecting the Model [24]

Generally different algorithms are used to solve different tasks. So, depending on the type of task the relevant algorithm has to be chosen.

1.4.4. Training the Model [24]

It is used to analyze the data and formulate the appropriate pattern or to give proper answer to the question given. Every iteration in this process is considered a training step.

1.4.5. Evaluate the Model [24]

The performance of the model is checked using the 'unseen data'. The term 'unseen data' means a subset of the dataset used to analyze the probable future performance of a model. A dataset means a collection of data or the table of a database.

1.4.6. Parameter Tuning [24]

The same machine-learning model may require different constraints, weights or learning rates to generalize different data patterns. These measures are called hyperparameters, which are prepared for the algorithm to solve the given problem in the best possible way [\[25\]](#).

1.4.7. Make Patterns [24]

Based on the test data, we can analyze the possible future performance of a model in the real world.

1.5. Types of ML

Different types of Machine Learning are:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Supervised learning [\[26\]](#) means the ML algorithm is given a dataset that is labeled, and the algorithm is expected to learn from that data by identifying and specifying the relationships between variables present in the dataset given. Improvement of supervised machine

learning algorithms is a continuous process that will continue to find new patterns and relationships as it trains itself on new data sets.

Unsupervised learning [\[26\]](#) means allowing the ML algorithm to learn from data that is not labeled. Here no human efforts are required to make the dataset machine-readable. Algorithms are expected to learn from data that has no labels; these algorithms are generally versatile. Unsupervised learning algorithms can adapt the data by dynamically changing the behavior of the dataset which is provided.

Reinforcement learning [\[26\]](#) is on the basis of human learning strategy; it learns from data and applies the trial-and-error method to every task. Once performing the given task, the algorithm has to interpret whether the outcome is correct or not. If the output is not proper or not as per expectations then the algorithm must reiterate till it gets a correct result.

1.6. Basic ML Methods

A machine learning algorithm can also be referred to as a model. It has a mathematical expression that represents the data from a business point of view. The goal is to figure out how likely things are to happen in the future by moving from facts to insights. For example, an online store might be able to predict quarterly sales with the help of a machine-learning algorithm that has been trained on past data.

Some of the important ML methods are:

Regression is a supervised learning technique that helps to estimate or analyse the numerical value based on previous data. For example, the cost of air conditioners for next summer can be estimated based on their cost this summer and the previous summer.

Classification is another supervised learning technique that explains or forecasts the value of a class. For example, they can help you predict whether an online customer will purchase a

particular product or not. In this kind of case, the answer is either yes or no. Apart from this classification, methods can generate all possible outputs for a prediction. For example, this algorithm can help you check whether a given image has a cat or mouse. In this case, we can get three possible outputs. They are in the image; it can be a 1. Cat 2. Mouse or 3. None of them. One of the classification algorithms is logistic regression. Logistic regression predicts the probability of the occurrence of an event based on the inputs given. For example, based on the entrance exam rank of a student, this algorithm can predict whether that student gets admission to a particular college or not. Since the possible values of probability are either 0 or 1, if the value is 0, the student will not get admission; if the value is 1 or greater than 0.5, the student will get admission.

Clustering is an unsupervised learning technique where a group or cluster is formed for a set of elements with the same characteristics. Here we can use visualisations to check the quality of the cluster. K-means clustering is a clustering algorithm. Briefly, K-means clustering works like this: K is the number of clusters the user chooses to create. The “K” centres are chosen at random from the data. A data point is assigned to the closest of those randomly created centres. Then, the centre of each cluster is calculated again. If centres change very little or don’t change at all, then the process is said to be completed; otherwise, another data point is assigned to the closest of the centres that were created randomly.

Dimensionality Reduction is used to remove the information that is not so important from the data set given. Principal Component Analysis (PCA) is a dimensionality reduction algorithm that is capable of decreasing the dimension of data without losing too much information.

Ensemble Method is a supervised learning technique that combines several predictive models to get higher-quality predictions. If we want to build a computer, we search for the best manufacturer of the respective computer peripherals. Once we assemble them, they will

definitely have better performance than other computers. Ensemble methods work on the same idea. The random-forest algorithm is an ensemble method. This algorithm combines different sets of data into decision trees.

Neural Networks and Deep Learning The aim of neural networks is to get data that is in a non-linear pattern by adding parameter layers to a model. A neural network is said to be a deep neural network if it has many hidden layers inside it. A lot of data is needed by a deep neural network. For the classification of images, audio, video, etc., deep learning techniques are used. Pytorch and Tensorflow are software packages related to deep learning.

Transfer Learning is re-using a part of a previously trained neural network and adapting it to a new but similar kind of task. Only less data is needed to train the neural network.

Natural Language Processing is not an ML technique. It is a technique used to prepare data for machine learning. Researchers at Stanford created NLTK. NLTK stands for Natural Language Toolkit. NLTK is used to filter the data, which comes in different formats like Word, PDF, ppt, etc. To map to a numerical representation, TFM can be used. TFM stands for Term Frequency Matrix. In TFM, each row represents a text document, and each column represents a word. The frequency of each word within a document is specified in TFM. From TFM, TFIDF can be developed. TFIDF is the short form of Term Frequency Inverse Document Frequency. TFIDF holds information about how important each word is in a set of documents.

Word Embeddings captures the context of a word in a document. Word context quantifies the similarity between words, thereby allowing us to do arithmetic in words. We calculate how many words are combined. For this purpose, we use machine learning techniques. But it is only an initial step to apply a machine learning algorithm. For example, consider that we have access to the tweets of several people who use Twitter. We can also assume that we know which of these users purchased a new house. To predict the probability of a new Twitter user purchasing a new house, we can

use a combination of Word2Vec and logistic regression. Word2Vec is a mechanism whose working concept is based on neural networks that map words in a corpus to a numerical vector. We can then use these vectors to find similar words, perform mathematical operations on those words, or even use them to represent text documents.

1.7. ML in Agriculture

Konstantinos G. Liakos [4] has analyzed a new ML that can be useful for agriculture. The author has reviewed around 40 articles published in different journals. Out of all these articles, most of them were related to how ML can be applied in crop management; after analyzing all these articles by the author, it was found that a total of 8 ML models were implemented. Out of these 8, 5 ML models were implemented in crop management. All these models were implemented in various agriculture domains like Animal welfare, livestock production, crop quality, water management, soil management etc. Also, a diagrammatic representation of ML techniques that can be applied to agriculture in the future for each subcategory is given. From the study made by the author it is found that models of ML were applied in several applications for crop management (61%), disease detection (22%), and yield prediction (23%). Also, it is specified that by applying ML to sensor data, farm management gradually evolves in AI systems and moves into an era of knowledge-based agriculture.

1.8. ML in Sentiment Analysis

Jaspreet Singh [6] exploited 4 ML classifiers for sentiment analysis. The author used 3 manually annotated datasets for this purpose. Performance evaluation of the 4 classifiers, i.e., Native Bayes, J-48, BF Tree, and OneR is given in a tabular form. Also, another table that shows the test accuracies of these 4 algorithms for these 3 datasets is given. Freeware WEKA software tool is used for student's classification in the text.

1.9. ML in Stock Prediction

Vatsal H. Shah [8] employed a few ML techniques in stock prediction and explained how stock prices increase and decrease at trading time. This paper focuses on applying SVM, linear regression, Expert Weighting, online learning and prediction using decision stumps, along with the merits and demerits of each method. This paper has introduced the variables and parameters that are useful in pattern recognition in stock prices that can be useful for future stock prediction and also discussed how boosting can be used along with other ML algorithms to improve the accuracy of such prediction systems. This paper briefly provides an overview of the financial prediction system domain. The evaluations of generated results after applying ML algorithms are discussed.

1.10. ML in Disease Prediction

Shahadat Uddin [7] has made an attempt to study the comparative study of various supervised ML algorithms that can be used for disease prediction. Since there will be variation in disease prediction between research data set and clinical data, the author used the same data on which ML techniques were applied for disease prediction and to compare them. One limitation of this work is that no sub-classification of supervised ML algorithms was compared instead, only broad categories were considered for the study. In the study, a table showing the advantages and drawbacks of supervised ML algorithms was given. One most useful information in this paper is a table showing the names, and references of diseases and the corresponding supervised ML algorithms used for prediction is given. Also, out of all the used algorithms, the name of the better algorithm was also mentioned. Above mentioned table provides a summary of 48 articles published that are related to disease prediction using supervised ML techniques.

1.11. ML in Data Mining

Seema Sharma [1] explained ML techniques that are used in the field of data mining. The author discussed applications of some of the techniques like Nearest neighbor, SVM, Bayesian Networks,

Decision trees *etc.*, in data mining. Before that, the basic concepts related to these techniques discussed. According to the author, it is always useful to apply more than one technique rather than applying any single technique. In the paper the author has differentiated among the above-listed 4 algorithms in terms of average accuracy, tolerance noise, speed discrimination and also whether they are generative or discriminative. The author also used the MATLAB classification toolbox in order to show the classification accuracy among the algorithms by using data sets. In the related work section of this paper, the author gave several references about applications of various ML techniques in data mining.

1.12. ML in COVID-19

Yazeed Zoabi [9] discussed the development of prediction models that combine various features to analyze the risk of infection. The Aim of this work is to assist the staff of the medical field worldwide to decide the order of treatment, particularly in case of availability of limited resources of healthcare. An ML approach was established, which was trained on data from 51,831 individuals who were tested. From these people, 4769 have got tested positive for covid-19. The data on which the machine was trained also had data of 47,401 tested individuals of a subsequent week, out of whom 3624 were positive. The model specified in this article gave accurate results using 8 binary features only. They are: Gender, Age greater than or equal to 60 years, who had confirmed contact with the tested individual and appearance of 5 initial symptoms. This work was done based on data that was publicly reported by the Health Ministry of Israel. The model developed detects covid-19 by asking some basic questions, and according to the author, the framework of this model is specifically useful where there are limited resources for testing people.

1.13. ML in Cyber Security

Kamran Shaukat [5] has made a detailed review that specified how quickly the interest in ML and Cyber Security is growing in

academics and as well as in industry. In this paper, an effort was made to bridge the gap between techniques of ML and threats to computer networks by providing a survey of crossovers of these 2 areas. The survey made in this paper depicts the ML techniques that can be used for intrusion detection, malware detection, and spam detection in mobile devices and computer networks. This paper mentioned the applications of ML in the field of cyber security, especially in the last decade. In this paper, a graphical summary of cyber security attacks and existing ML techniques to combat them are represented. In this paper, an overview of present challenges in using ML techniques for cyber security was given. Also, a summary and comparison of: i) ML models in detecting malware over the last ten years, ii) ML models for detecting Spam over the last 10 years iii) ML models over 10 years for intrusion detection. The time complexity, description and limitations of frequently used ML techniques are given in a tabular form. A tabular form that represents the similarities and dissimilarities between deep learning and traditional ML is also given. Overall, this paper covers the following: The evolution of cyber security and ML over the last 10 years, basic cyber security attacks and threats, security data sets that are most commonly used, criteria and metrics to evaluate those data sets, basic concepts related to deep learning and ML, the current state of detecting spam/intrusion/malware both in computer networks and mobile devices and challenges faced by ML to handle cyber security.

1.14. ML in Cloud Computing

In the study done by Umer Ahmed Butt [3], the analysis was made on threats, attacks, and challenges faced by cloud computing. Also, it was discussed how various ML algorithms like SVM, Naïve Bayes, K-means *etc.*, can be explored as solutions for security issues of cloud computing. A review was done on some ML algorithms used for cloud security; also, their advantages and disadvantages were highlighted. The scope of research in this area was also mentioned. Joe Fiala [10] looked at different possible ways to use ML to improve cloud computing. Hence, ML techniques were looked at as a way to manage resources dynamically, with a specific focus on predicting

the size of VM. Also, in this paper, ML techniques were looked at using resources efficiently, specifically focusing on turning off idle machines and workload consolidation. The basic conclusion of this paper was even though multiple ML techniques are available as options to do the required task, choosing a specific ML technique is totally system dependent. Finally, the paper concluded with a brief description of cloud security. Kyriakos N. Agavanakis [11] has explored the abilities and practical aspects of ML. A practical cloud-based solution was built which provides a suitable environment in terms of scalable infrastructure to share knowledge without disclosing the data on their own or risking the IPR of researchers. It is also specified that the integrity of data may be safeguarded by digitally signing the data. The cloud-based approach specified will support the direct usage of third-party models through a reusable set of services which can be integrated with the applications of end users. Ex: websites, mobile apps, social media apps etc.

1.15. ML in Big Data Analytics (BDA)

Kumar Rahul [2] briefly discussed the concepts of ML that are applicable to various applications based on big data. In this paper, the author has mainly focused on how ML is applied in BDA, particularly in areas of healthcare, production and material segment. Also, this paper focuses on open issues in applying ML to industrial applications, which can be considered for the purpose of research. According to the author, retail, travel, healthcare, finance and media are some of the industries where ML can be used extensively. Identifying diseases, helping the discovery of drugs, detecting frauds, recommending suitable products etc., are some of the ML applications in these industries. Martín Noguerol has explained about strengths, weaknesses, and opportunities of ML applications. In the paper, various terms of ML-like NLP, speech recognition, supervised learning, unsupervised learning etc., In this paper, the strength of ML was mentioned in the context of image-based diagnosis, automatic lesion detection etc., Requirement of a lot of data sets for training was specified as a weakness of ML. Athmaja S

has made a survey on compatible ML concepts for BDA. This paper also discusses the prominence of ML applications based on BDA.

1.16. ML in Recommendation System

Being one of the ML techniques the aim of recommender systems is to provide the user with relevant suggestions. The suggestions given for watching videos on YouTube, the news/search suggestions given by Google, the friend suggestions given by Facebook, the follow suggestions given by Instagram, product suggestions we get for purchasing in e-commerce websites etc., are some examples of recommender systems. The suggestions received are basically the prioritized relevance's based on the browsing history of users. For example, if we watch videos about cricket matches on YouTube, then YouTube will start suggesting more videos about cricket matches. Collaborative filtering and Content-based filtering are 2 major classifications of recommender systems. Collaborative filtering is again classified into Model-based and Memory based recommender systems. Methods of collaborative filtering consider the browsing data of the user, and based on that, relevant suggestions are given to the user in the future. This suggestion is purely based on the browsing history of the user only. For example, let's assume that pizza is the most favorite food item of people in a particular city. A memory-based recommender system will focus only on what the user has ordered in the past 1 month without considering what favorite food that person in that city has. A model-based recommender system will always focus a bit more on recommending various types of pizzas to the people of that city because the basic assumption of this model is that many people in that city love to eat pizza. A content-based system in order to make predictions, uses additional information about items or users. For instance, from the above example about pizza, the content-based system might consider the age of the user. If the user's age is 70 years, then the chances of him/her liking pizza are very less, based on the suggestions that will be provided to the user. This is why whenever we register for any website, in most cases, they ask for

details about our age, birth date, gender, nationality *etc.* This data is used by their systems to make predictions in a better way.

1.17. Future Experiments on Real Time Problems Using ML

ML is used directly or indirectly in almost every part of our daily lives in this generation. Apart from the present applications, in the future ML techniques might be extensively used in the fields of education, digital marketing, search engine optimisation, health care, the automobile industry, *etc.*

Teachers can use machine learning techniques to analyse to what extent students have understood the lesson. This can help increase teacher-student interaction in the classroom. Apart from voice recognition, image search, and translation tools, ML allows computers to make sure that they listen and speak the same as human beings. Customer segmentation allows for focus on the appropriate customer at the right time with the right message. ML techniques can be used to do this more effectively. ML techniques can be used to detect diseases or any other underlying health issues more efficiently so that they can be treated faster. ML can be used in the automobile industry to create safer self-driving cars. Tesla's self-driving car, built using ML, is the best in the industry.

CONCLUSION

This chapter gives an overview of the brief history, types, and techniques of machine learning and some of its real-time applications. We discussed the applications of ML in various areas like data mining, big data analytics, cloud computing, sentiment analysis, *etc.*, and also tried to give information regarding how ML can be used in dealing with COVID-19 and the development of agriculture. Finally, we concluded the paper by giving some basic information about the scope of future experiments to solve real-time problems using ML.

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Applications of Machine Learning

K. Sudheer Babu^{1, *}, CH. M. Reddy², A. Swapna³, D. Abdus Subhahan⁴

¹ Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad, India

² Marwadi University, Rajkot, Mahender, India

³ Mahatma Gandhi Institute of Technology, India

⁴ B V Raju Institute of Technology, Narsapur, Medak, India

Abstract

In this chapter, we briefly discuss various real-time applications of machine learning algorithms. Machine Learning Algorithms explain the following topics: Introduction to ML algorithms, Supervised Learning, Classification, Regression (**Linear Regression, Logistic Regression, Decision Tree, Naive Bayes, Support Vector Machine, Random Forest, AdaBoost, Gradient-Boosting Trees**), and Unsupervised Learning (**K-Means Clustering, Gaussian Mixture Model, Hierarchical Clustering, Recommender Systems, PCA/T-SNE**). Application of Machine Learning explains various real-time applications like augmentation, automation, finance, government, healthcare, marketing, traffic alerts, image recognition, video surveillance, sentiment analysis,

product recommendation, online support using chatbots, Google translate, online video streaming applications, virtual professional assistants, machine learning usage in social media, stock market signals using machine learning, auto-driven cars, and real-time dynamic pricing.

Keywords: Machine-Learning, Supervised learning, Un-Supervised learning, Naive bayes, Support vector machine, Random Forest, AdaBoost, Gradient-boosting.

* **Corresponding author K. Sudheer Babu:** Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad, India; E-mail: sudheerkuntumalla@gmail.com

1. INTRODUCTION

1.1. Machine Learning

Machine-learning is a set of computer algorithms that are able to learn from experience and improve themselves without any necessary direct programming.

One part of AI is machine learning, which employs mathematical methods to combine data and predict a result that may be used to make decisions [1, 2].

This new idea is based on the concept that a computer could learn from a dataset (for example) in order to come up with proper results on its own. Machine learning is closely related to both data mining and Bayesian predictive modelling. The computer takes information as input and uses an algorithm to come up with answers [3].

Providing a recommendation is a common machine-learning problem- for example, entire proposal of films or series depending on the client's history of information for those with a Netflix account. Unsupervised learning is being used by tech businesses to improve the customer experience by personalising recommendations [4].

1.2. Machine Learning vs. Traditional Programming

Traditional programming (Fig. 1) is very different from machine learning. In conventional programming, a programmer writes all of the instructions with the help of an expert in the field for which the software is being made. Every instruction is based on something logical, and the computer will follow the logical assertion, and make something happen. As the system gets more intricate, more rules need to be written. It can quickly become impossible to keep up [5, 6].

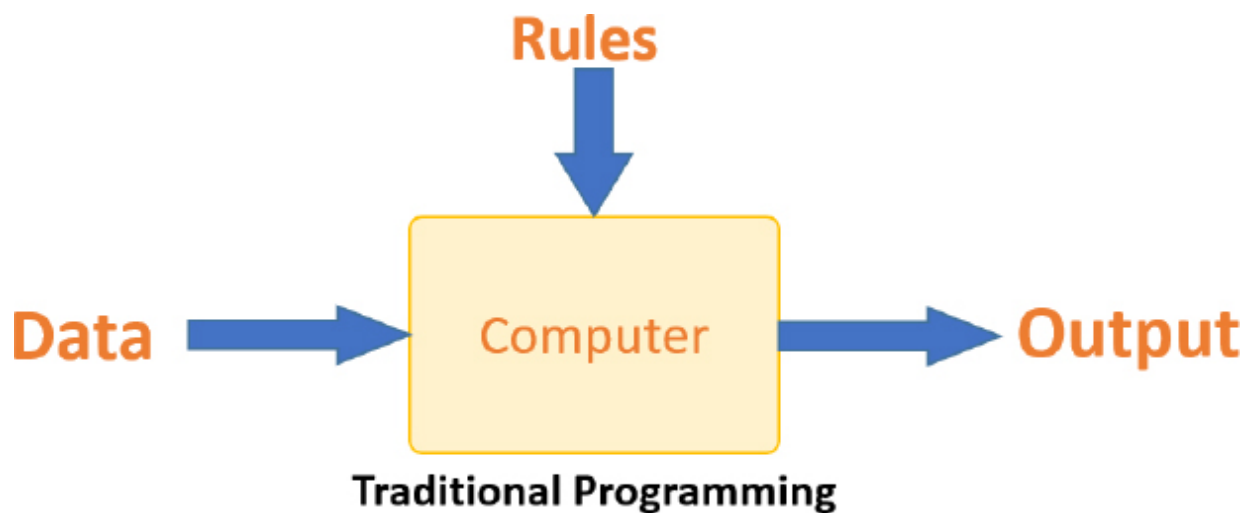


Fig. (1)

Traditional Programming.

This problem is meant to be solved *via* machine learning. The machine deduces the relationship between the input/output information and creates the instructions. Each time there is fresh data, the programmers do not need to design new rules. To improve efficacy over time, the algorithm adjusts in reaction to novel information and involvements [7, 8].

1.2.1. Machine Learning Work

We will now study how Machine Learning (ML) works:

The “brain” of the machine learning system is where all learning takes place. Learning occurs in exactly the same way on the computer as it does in people. Experience is the single most important teacher for humans. When we have more knowledge at hand, it is much simpler to make intelligent predictions. By analogy, the chances of succeeding in an unknown circumstance are lower than the chances of succeeding in a known situation. A similar instructional method is used for machines. In order to generate an accurate forecast, the machine investigates a specific example. When we provide the machine with a scenario that is comparable to what might occur, it is able to make accurate predictions [9]. On the other hand, just like a person, if the computer is given a scenario that it has never encountered before, it will struggle to accurately forecast what will occur in the future.

As per the (Fig. 2) learning and inference are the two main goals of machine learning. First and foremost, the machine acquires by perceiving designs. The information helped in making this finding. One of the most important tasks for a data researcher is to painstakingly select the information to provide to the computer [10, 11]. A feature vector is a collection of attributes used to solve an issue. A feature vector can be considered a subset of information that will be utilised in solving the issues.

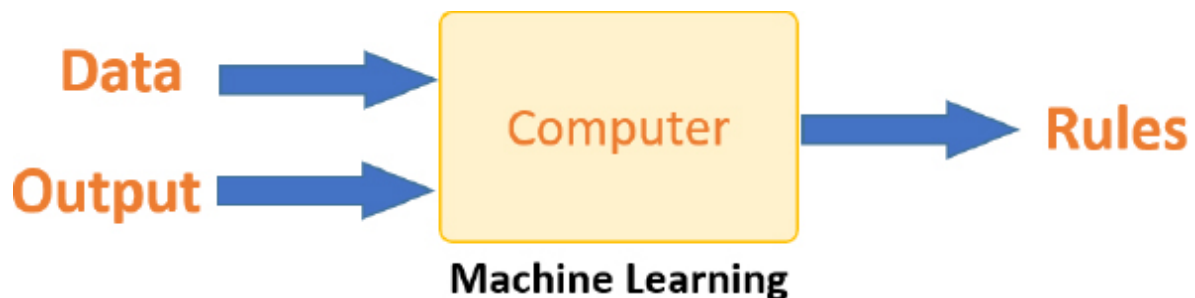


Fig. (2))
Machine Learning.

The computer employs a few sophisticated algorithms to improve realism and turn this finding into a model. As a result, the step in learning is utilized to define and summarise the information in order to create a model.

Consider an example; shown in Fig. (3) the device which attempts to comprehend the connection between a person's pay and his or her possibility of dining at a posh restaurant. It turns out that the system detects a link between pay and dining at a very good quality establishment: It is referred to as model Inferring [[12](#), [13](#)].

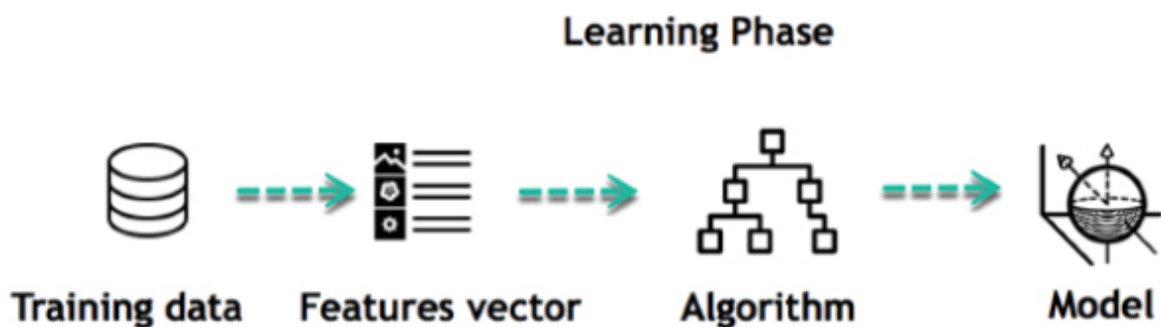


Fig. (3))

Learning Phase.

After the model has been developed and tested with never-before-seen data to evaluate its effectiveness [[58](#)], shown in Fig. (4) the new data is turned into a features vector, which is then passed via the model, and a forecast is made. This is every part of the fascinating world of ML. There is nothing other compelling reason to retrain the model or otherwise alter the rules. You can make inferences on new data using the model you previously trained [[14](#), [15](#)].

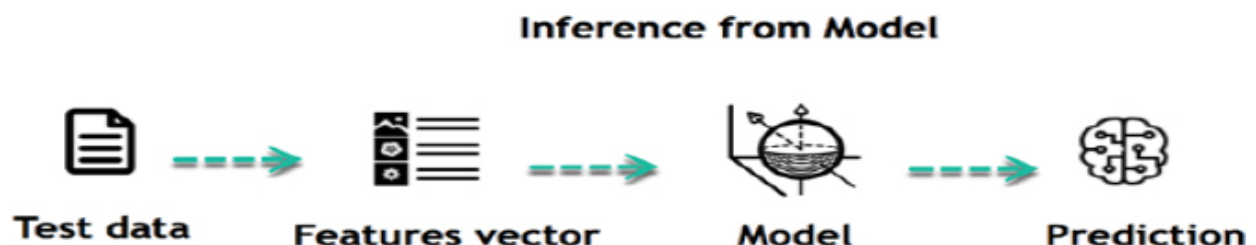


Fig. (4))

Inference from Model.

Machine Learning programmes have a clear life cycle that can be stated as follows:

1. Define an inquiry
2. Data Collection
3. Data Visualization
4. Algorithms Training
5. Algorithms Testing
6. Feedbacks Collecting
7. Algorithms Refining
8. Loop 4-7 until the outcomes are fulfilling
9. Make a forecast by using the model

When the algorithms have mastered generating the correct conclusions, it applies what it has learned to new data sets.

2. MACHINE LEARNING ALGORITHMS

2.1. Introduction

We will explore where ML algorithms are utilised in this ML lesson for beginners:

There were two classifications of ML, one was supervised, and the other was unsupervised learning Fig. (5). There were various algorithms accessible.

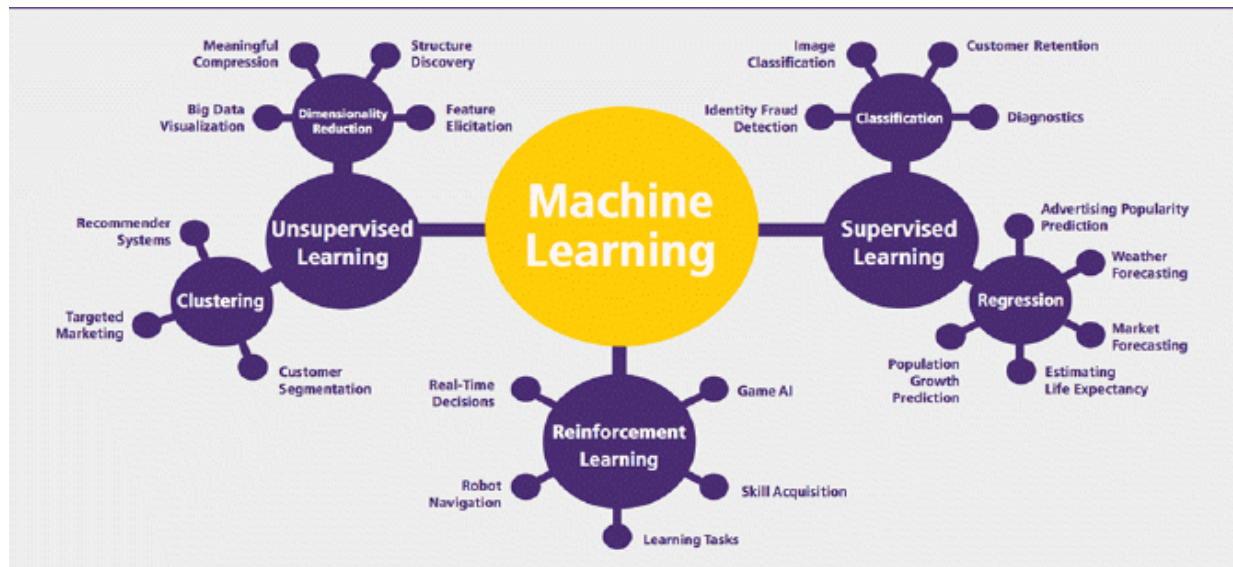


Fig. (5))
ML Algorithms.

2.2. Supervised Learning

The algorithms learn the link between given inputs and outputs by utilising training data and human feedback. For example, a practitioner can forecast canned sales using marketing expenses and climate forecasts as information [16, 17].

When output data is known, supervised learning can be used. New data will be predicted by the programme.

Supervised learning is divided into two categories:

- Classification task
- Regression task

2.2.1. Classification

Expect you need to classify the gender of a client for an ad. From your customer database, you'll begin collecting information such as height, weight, occupation, *salary, shopping basket, and so on*. You know every one of your customers' gender; it must be masculine/feminine [18]. The classifier's goal will be to allocate a

likelihood of being masculine/feminine (*i.e.*, the mark) in view of the data (*i.e.*, highlights you have gathered). You can use additional data to create a forecast once the model has figured out how to detect masculine/feminine [19]. For example, you've just received fresh information from an unknown consumer and want to know whether he or she is the gender of masculine/feminine. If the classifier guesses 70%, masculine implies that the algorithm is 70% certain that this consumer is a man and 30% certain that he is a woman [20, 21].

There might be at least two classes on the label. Although the example of ML above has two classes, a classifier that necessities to foresee an article will have thousands of them (*e.g.*, glass, desk, shoes, *etc.*, Individual article addresses a class).

2.2.2. Regression

The task is called a regression when the output is a continuous value. A monetary investigator, for example, might be required to estimate the worth of a stock dependent on a variety of factors such as fair play, prior standard performance, and macro-economic record. The algorithm will be prepared to gauge stock prices with the very least amount of inaccuracy possible in Table 1.

Table 1 **Regression Algorithm.**

Name of the Algorithm	Explanation	Types
Linear regression	Discovers a mechanism to link every element to the yield so that forthcoming qualities can be predicted.	Regression
Logistic regression	The application of linear regression for arrangement undertakings has been extended. Instead of continuous, the output variable 3 being double (for example, merely white or dark) (for example, an endless series of possible colours)	Classification

Name of the Algorithm	Explanation	Types
Decision tree	At decision nodes, an exceptionally interpretable grouping or regression model separates information highlight standards into divisions (for example, if a component is colour, every individual potential colour turns into another branch) till an ultimate choice output is formed.	Regression Classification
Naive Bayes	The Bayesian technique is a classification approach that uses the Bayesian-theorem as its foundation. This replaces an event's prior knowledge with the independent likelihood of individual attributes that can influence it.	Regression Classification
Support vector machine	The Support Vector Machine (SVM) is generally utilized for classification. The classes can be divided in a well-organized way by a hyperplane which was found by SVM. It works best when combined with a nonlinear solver.	Regression (uncommon) Classification
Random forest	The method depends on a choice tree, which extraordinarily further develops precision. The random forest creates various modest choice trees and with the use of the 'majority vote' method, it determines which label to return. The final forecast for the classification occupation will be unique with maximum votes, while the last forecast for the regression challenge would be the normal estimation of the multitude of trees.	Regression Classification
AdaBoost	An order or regression technique that evaluates an assortment of models	Regression Classification

Name of the Algorithm	Explanation	Types
	relying upon their accuracy in predicting the result prior to settling on a choice.	
Gradient-boosting trees	Gradient-boosting trees are the state of the classification and regression technique. It centres on the preceding trees' errors and endeavours to fix them.	Regression Classification

2.3. Unsupervised Learning

Without using any specific output variable, a method in unsupervised learning investigates the input data (*e.g.*, to recognize patterns, client's demographic information has been explored).

This is the algorithm to employ to do the classifications in the cases of lack of knowledge in categorising the data and the need of an algorithm to detect patterns.

We'll learn how to pick ML algorithms in these fundamentals of ML Table 2.

Table 2 **Unsupervised learning Algorithm.**

Algorithm	Explanation	Types
K-means clustering	Places information into groups (k), every one of which contains information with a similar arrangement of qualities (as controlled by the model, not ahead of time by people). This is the algorithm to employ to do the classifications in the cases of lack of knowledge in categorising the data and the need of algorithm to detect patterns.	Clustering
Gaussian mixture model	A generalisation of k-means grouping that considers greater group size and	Clustering

Algorithm	Explanation	Types
	shape freedom (groups)	
Hierarchical clustering	Structures an order framework by dividing bunches along a hierarchical tree. For clients who have a Cluster loyalty card, it can be utilized.	Clustering
Recommender system	Assist in defining the information needed to make a suggestion.	Clustering
PCA/T-SNE	Normally used to lessen the information's dimensionality. The methods decline the measure of attributes to three or four highest variance vectors.	Dimension Reduction

There is a wide variety of ML algorithms to choose from. The algorithms are selected with consideration given to the end goal see Fig. (6).

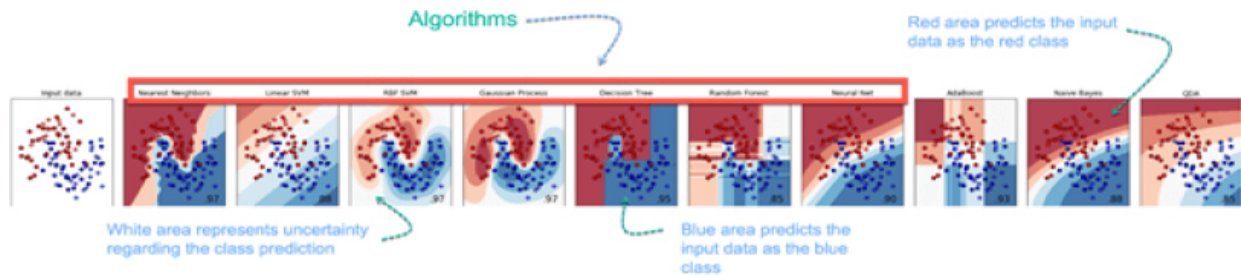


Fig. (6)
Selection of machine learning Algorithm.

The goal of the ML example underneath is to anticipate the sort of flower from 3 different variations. To make the forecasts, the extent and thickness of the petal are used; the illustration shows the outcomes of ten distinct algorithms. The dataset is depicted on the upper leftward of the image [22, 23]. The information is divided into 3 groups: light blue, dark blue and red. Also, there are a few classifications. Consider an example, in the subsequent picture, everything at the upper leftward has a place with the red group; in the centre, there is a mix of ambiguity and bright blue; and at the base,

everything has a place with the dark group [24, 25]. Different designs portray various algorithms and how they attempt to classify information.

3. APPLICATION OF MACHINE LEARNING

3.1. Augmentation

Machine learning is a type of artificial intelligence that helps people with their everyday work, whether they are doing it on their own or for a business, without giving them full control over the results check Fig. (7). It is used for an assortment of purposes, including remote helpers, data analysis, and programming solutions [26, 27]. The primary goal is to eliminate human bias-related errors.

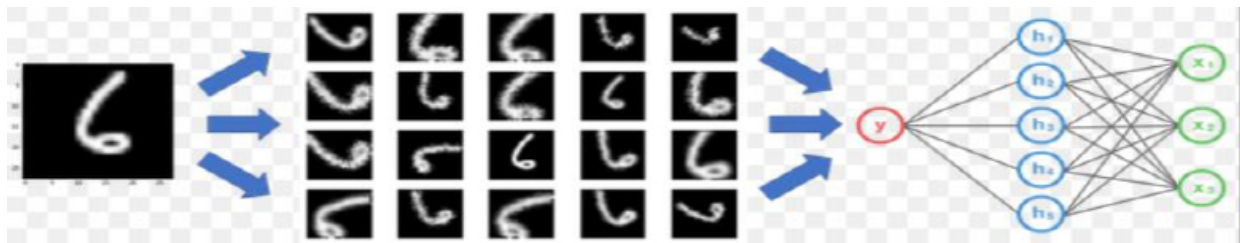


Fig. (7))
Augmentation.

3.2. Automation

As per the Fig. (8) machine learning is a type of artificial intelligence that can perform independently in any field without the need for human interaction. In manufacturing factories, for example, robots conduct critical process stages.

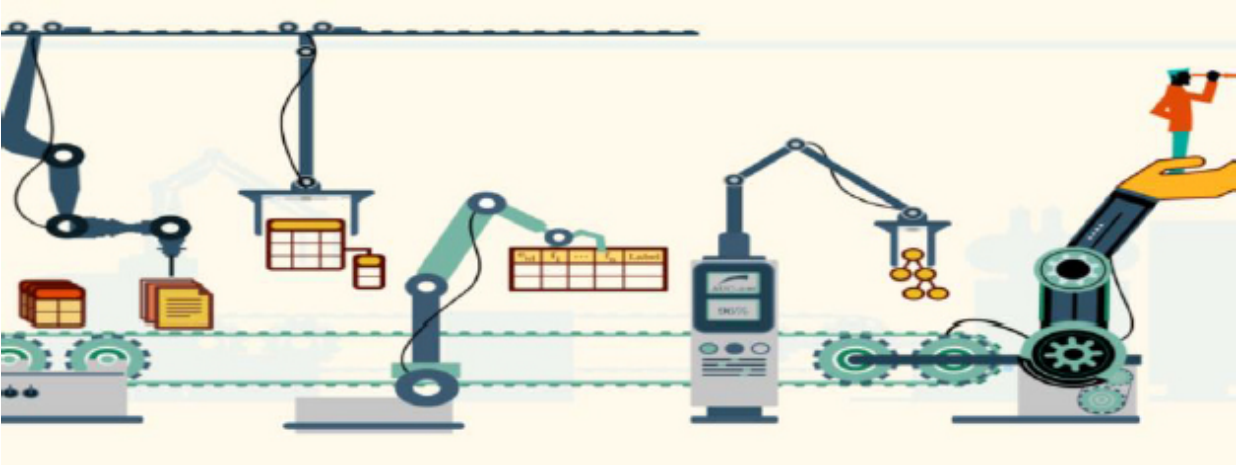


Fig. (8)
Automation.

3.3. Finance Industry

In the banking industry, machine learning is gaining traction. Banks are mostly utilising machine learning to detect trends in data, but it is also being used to prevent fraud.

3.4. Government Organization

The government uses ML to regulate conveniences and public well-being. Consider, for instance, China's enormous facial acknowledgment system. To prevent jaywalking, the government employs artificial intelligence.

3.5. Healthcare Industry 2

When it comes to using machine learning for picture identification, the healthcare industry was one of the first to adopt the technology.

3.6. Marketing

AI is commonly employed in advertising because of the abundance of data available for analysis as shown in Fig. (9). Before big data [\[28-30\]](#), researchers relied on sophisticated mathematical tools like

Bayesian analysis to estimate a client's worth. As data continues to expand, marketing departments are looking to AI to help them better connect with customers and run more effective campaigns see Figs. (10 & 11).



Fig. (9)
Finance Industry.



Fig. (10))

Government organization.



Fig. (11))

Healthcare industry.

3.7. Traffic Snarl

How does Google Maps recognize you are on the quickest path even though there's a lot of traffic?

As a result, it is based on several criteria, including how many individuals are currently utilizing Google Guides (Maps), historical recorded information for that course, and constant algorithms [31]. At the point when you practice Google Guides (Maps), the application authorization to use information, for example, your location, will be given to you.

- What is your average travel speed?
- Responses to questions such as, "Does the course actually have a stream of traffic?"
- The date, time and some special events

The application captures and stores all of this information. AI and ML algorithms use this data to draw the correct conclusions and provide you with accurate information as shown in Figs. **(12 & 13)**.

Fig. (14). Google Maps' upgraded functionality also allows us to perceive the distance away the following transport is from a given stopover and uniform gauge transport deferrals. Moreover, the devices are savvy to the point that the devices can illuminate people how bustling the transport or train is so that people can call to jump aboard [32, 33].



Fig. (12))
Marketing.



Fig. (13))
Google Maps.



Fig. (14))
Image Recognition.

3.8. Image Recognition

It is quite simple for people to identify any given image. Imagine an automobile, for example see the Fig. (15). I am confident that you can examine the vehicle's image and, unexpectedly, it is shading. However, for a computer, images are nothing more than a collection of mathematical numbers, which is why image processing algorithms are used to look for patterns in complex images (recordings, graphics, or still images) [34, 35]. Algorithms allow computers to

perceive patterns, and ML algorithms can recognise any form of image [36, 37].



Fig. (15))

Video reconnaissance.

So, how does facial recognition work with machine learning?

Is it possible to unlock a phone just by glancing at it? If you answered yes, you are utilising machine learning. Your phone's high-resolution camera recognises 80 nodal points on a human face and uses machine learning to calculate the variable of a person's face and unlock the phone [38-40].

Unravelling phones is currently one of the most prevalent applications of ML. We, the Gen-Z generation, currently experience a daily reality in which we can make speedy bank payments using face recognition. Other applications of picture recognition include:

- Drone
- Engineering
- Vehicles which can be self-driven
- Military reconnaissance
- Activity of forest

3.9. Video Reconnaissance

This is quite possibly the most exceptional AI and ML application. When compared to any other source, videos provide an enhanced potential nearby obtaining important data from robotized reconnaissance gadgets. This is only feasible because of the greater perspective of machines in articles contrasted with humanoid personalities see the model in Fig. (16).

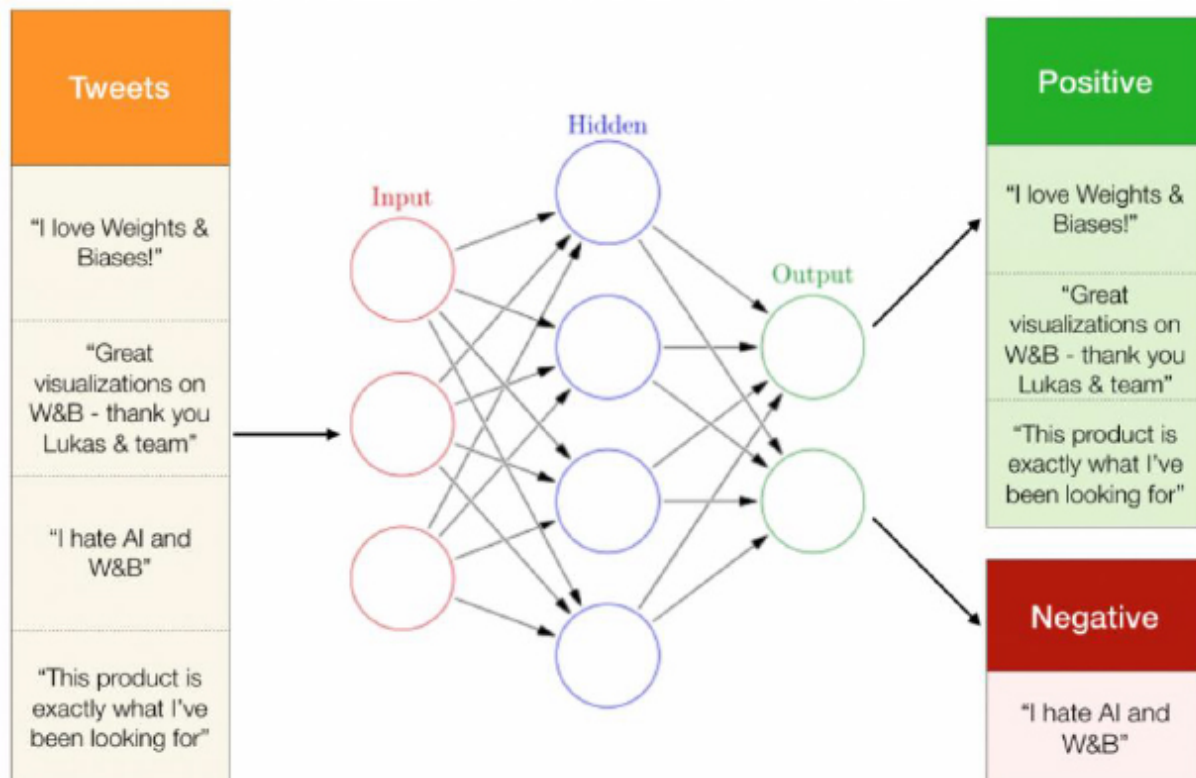


Fig. (16))
Sentiment Analysis.

Video reconnaissance is utilised on behalf of a variety of applications, including:

- Robbery counteraction of Copper
- To detect a strange occasion
- Safekeeping of capability
- To monitor the operation
- Area to park
- Traffic checking

Surveillance video is the most accurate dataset in machine learning, yet it is tough to collect. Consequently, object detector training is essential so that these devices can easily identify focal points in ordinary images.

3.10. Analysis of Sentiment

Sentiment analysis and high-quality ML applications include a grouping of sentiment, mining of opinion, and emotional analysis. Machinery prepares itself to analyse notions dependent on words using this technique. They are able to tell whether the words are being spoken in a favourable, negative, or neutral tone [41]. They can likewise characterize the significance of those words.

Data-miners utilize NLP to automatically extract and conclude an opinion by evaluating both supervised and unsupervised data using supervised and unsupervised machine learning methods in Fig. (17). This model is used by companies who deal with customers to further develop client experience dependent on feedback [42-44].

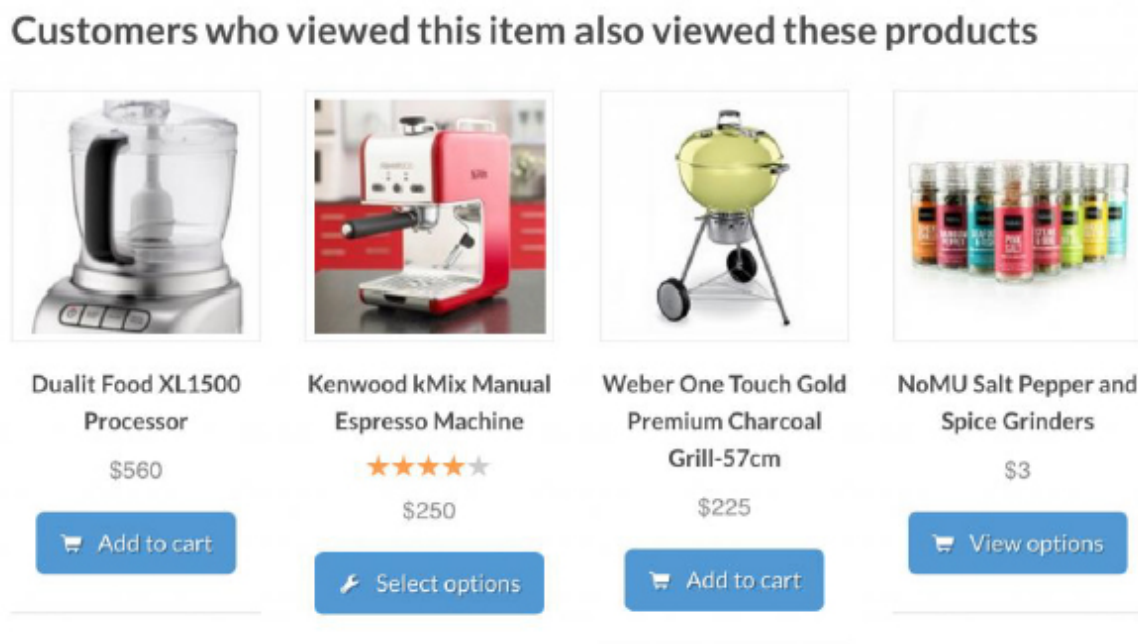


Fig. (17))
Product suggestions.

The Machine Learning model example given under reads the users' tweets and divides them into pros and cons concepts. This provides the company with a fast snapshot of whatever their customers are speaking about them, allowing them to respond quickly to any unfavourable feedback.

Music applications are another example of machine learning. Ganna.com and Jiosaavn, for example, recommend music based on client assessments by evaluating the historical backdrop of tunes played, preferred playlists and, surprisingly, the hour of music postings [\[45\]](#).

Sentiment analysis has many advantages:

- Viable brand and web-based media observing
- Improved client service
- Serious investigation
- Well-tracked staff input and client-contributed content (user-generated content like audits)

3.11. Recommendation of a Product

You can view suggested things or alternatives like 'customers who purchased this item additionally purchased' or 'customers likewise purchased this along with this item' while buying on e-Commerce sites like Amazon and Flipkart! refer Fig. (18).

3.13. Google Translate

Venturing out to a different location seems constantly exciting, yet the solitary question is figuring out how to communicate in the local language see the Fig. (19). To address this issue, Google has released an app that allows for quick interpretation of any language.



Fig. (19))

Google Translate.

Google employs 'Google Neural Machine Translation,' which is capable of absorbing numerous ling os, words, and vocabularies and converting any of the sentences into the target speech.

3.14. Online Video Streaming Applications

Video streaming was pioneered by online video programmes such as Netflix and Amazon Prime. Almost every television station now has a video content online streaming application that targets audiences based upon their particular preferences [49, 50].

Due to a method known as machine learning, many apps, particularly Zen X, have attracted a large audience. The apps collect information about the user's activity and provide video recommendations based on that information [51].

What sort of information do these applications keep track of?

- Content watching day and time
- Type of content you like to observe
- Break, Wind back or Quick Forward actions
- Design of browsing
- Earlier, essentially watching the movie/show, the trailers you watch

Businesses can engage their audience by providing great streaming experiences based on these trends, machine learning frameworks, and applications.

3.15. Virtual Professional Assistants

Based on ML, VPA is one of the greatest well-known applications of machine learning. Machines are becoming smarter in adopting human habits as the use of smart gadgets increases in Fig. (20).

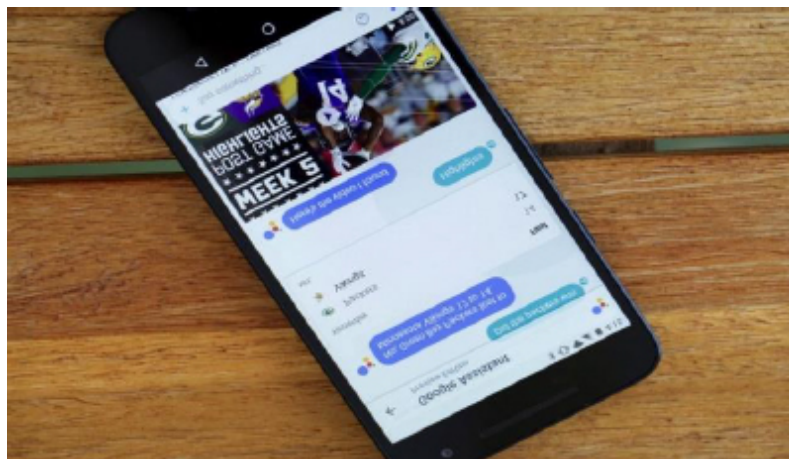


Fig. (20)

Virtual Professional Assistants.

Have you told Google Assistant that you want to be awake at 6.00 a.m.? Have you asked Siri to provide you with directions to your favourite eatery?

Indeed, you are on the correct track!

Machine learning may be seen in all of these situations. Machine learning algorithms are at the heart of everything from turning on

smart appliances to ordering an Uber on demand.

3.16. Usage of ML in Social Media

At the point when you will tag somebody on Facebook see in Fig. (21). Facebook provides you a suggestion before you even mention the name of the person in the photograph and 99.99 percent of the time, it is correct. Facebook has no way of knowing the individual person's name you are going to tag in a picture [\[52\]](#).



Fig. (21))
Social Media.

Indeed, you are right; it is doable, so machine learning is to be appreciated. It is only a minor aspect of ML for huge data. Based on the data acquired, AI and machine learning provide precise insights. Marketers can gain insight into the following topics by using machine learning: How to increase the email promoting's open rate?

- Which missions have performed fine and why?
- How does the customer feel after purchasing a product?
- What types of material do consumers like to peruse or watch according to their age, gender, and geographic area?
- When will be the optimal moment to run a movement or publish a post?

Such vital data guess the trades to increase their volume of sales *via* online broadcasting [\[53, 54\]](#).

3.17. Stock Market Signals Utilising ML

Certainly, you declaim that correctly. Applications of ML are also often used to obtain indications that aid in the making of reasonable stock market trading decisions.

From Fig. (22) stock market price prediction used to be difficult; meanwhile, the introduction of fast-changing ML algorithms to monetary business sectors, dealers would now be able to create more consistent judgments.



Fig. (22))
Stock Market.

ML outlines are then being developed headed for discovering the sentimentality of communal slashes and analysing specialized markers to provide stock traders with meaningful results.

3.18. Cars that are Auto-Driven

We have all gotten that cars that are self-driven remain the way to the future in the auto business. Yes, based on Machine-Learning, Artificial-Intelligence and Deep-Learning, the concept of self-driving is also mentioned. The following are some of the most commonly used ML tools in self driving cars:

- Scale-invariant feature transform (SIFT)
- Texton Boost

- You only look once (YOLO)
- AdaBoost

3.19. Dynamic Pricing- Real-Time

Once purchasing a trip for a night before New Year or getting vehicle of Uber during ultimate office periods for sample check Fig. (23), we can notice a significant difference in between typical cost and pricing for that particular event. When demand is high, corporations usually charge surge prices [55, 56].

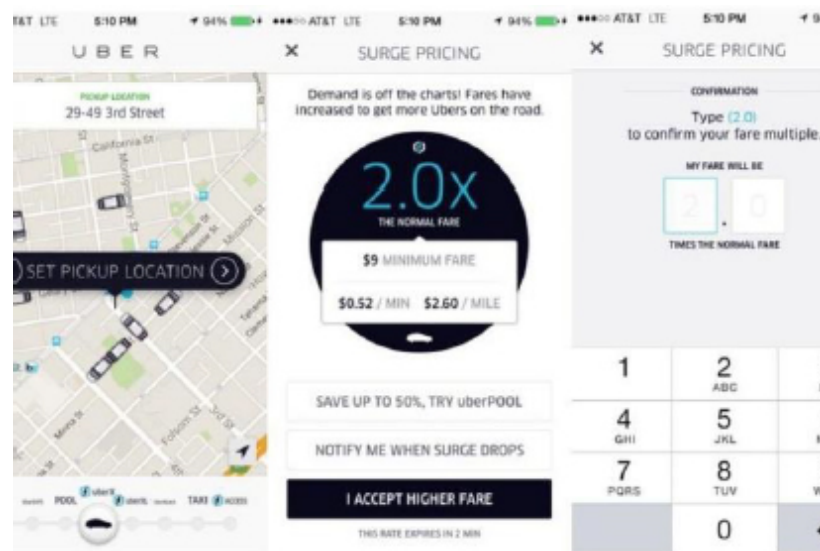


Fig. (23))
Surge Pricing.

The question is, how do businesses come up with such dynamic pricing?

The techniques of Machine-Learning, Artificial-Intelligence and Data-Analysis are the answers. This cutting-edge skill answers dual business questions:

- How might clients respond to price increases?
- Advising businesses on the best pricing strategy to avoid losing clients.

Factors to consider while optimising vibrant valuing include:

- Rivalry
- Meteorological conditions
- Demand
- Event
- Indigenous problems (might be any)

Algorithms of Artificial-Intelligence and Machine-Learning aid with solutions such as best pricing, discounted prices, and promotional prices, as well as calculating the rising prices [[57-59](#)].

CONCLUSION

Various real-time accomplishments of ML methods are briefly addressed in this chapter. Introduction to Machine Learning, Machine Learning vs. Traditional Programming, and Machine Learning Work discussed. Introduction to Machine Learning Algorithms, Supervised learning, Classification, Regression (Linear regression, Logistic regression, Decision tree, Naive Bayes, Support vector machine, Random Forest, AdaBoost, Gradient-boosting trees), Unsupervised learning (K-means clustering, Gaussian mixture model, Hierarchical clustering, Reciprocal clustering) used in various real-time applications of machine learning, such as augmentation, automation, finance, government, healthcare, marketing, traffic alerts, image recognition, video surveillance, sentiment analysis, product recommendation, online support using chatbots, Google Translate, online video streaming applications, and virtual professionals.

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Prediction using Machine Learning

Adluri Vijaya Lakshmi^{1, *}, Sowmya Gudipati Sri², Ponnuru Sowjanya², K. Vedavathi³

¹ B V Raju Institute of Technology, Narsapur, Medak, Telangana, India

² School of Technology GITAM(Deemed to be University), Rudraram, Hyderabad, India

³ Mahatma Gandhi Institute of Technology, Hyderabad, Telangana, India

Abstract

This chapter begins with a concise introduction to machine learning and the classification of machine learning systems (supervised learning, unsupervised learning, and reinforcement learning). ‘Breast Cancer Prediction Using ML Techniques’ is the topic of Chapter 2. This chapter describes various breast cancer prediction algorithms, including convolutional neural networks (CNN), support vector machines, Naive Bayesian classification, and weighted Naive Bayesian classification. Prediction of Heart Disease Using Machine Learning Techniques is the topic of Chapter 3. This chapter describes the numerous heart disease prediction algorithms, including Support Vector Machines (SVM), Logistic Regression, KNN, Random Forest Classifier, and

Deep Neural Networks. Prediction of IPL Data Using Machine Learning Techniques is the topic of Chapter 4. The following algorithms are covered in this chapter: decision trees, naive bayes, K-Nearest Neighbour Random Forest, data mining techniques, fuzzy clustering logic, support vector machines, reinforcement learning algorithms, data analytics approaches and Bayesian prediction techniques. Chapter Five: Software Error Prediction by means of machine learning- The AR model and the Known Power Model (POWM), as well as artificial neural networks (ANNs), particle swarm optimisation (PSO), decision trees (DT), Nave Bayes (NB), and linear classifiers, are among the approaches (K-nearest neighbours, Nave Bayes, C-4.5, and decision trees) Prediction of Rainfall Using Machine Learning Techniques, Chapter 6: The following are discussed: LASSO (Least Absolute Shrinkage and Selection Operator) Regression, ANN (Artificial Neural Network), Support Vector Machine, Multi-Layer Perception, Decision Tree, Adaptive Neuro-Fuzzy Inference System, Wavelet Neural Network, Ensemble Prediction Systems, ARIMA model, PCA and KMeans algorithms, Recurrent Neural Network (RNN), statistical KNN classifier, and neural SOM Weather Prediction Using Machine Learning Techniques that includes Bayesian Networks, Linear Regression, Logistic Regression, KNN Decision Tree, Random Forest, K-Means, and Apriori's Algorithm, as well as Linear Regression, Polynomial Regression, Random Forest Regression, Artificial Neural Networks, and Recurrent Neural Networks.

Keywords: Reinforcement learning, Convolutional Neural Networks, Weighted Nave Bayes Classifier, Fuzzy clustering logic, Linear Classifiers k-Nearest Neighbours.

* **Corresponding author Adluri Vijaya Lakshmi:** B V Raju Institute of Technology, Narsapur, Medak, Telangana. India; E-mail: vijaya.a@bvrit.ac.in

1. INTRODUCTION TO MACHINE LEARNING

ML is a rapidly evolving technology that automatically allows computers to learn from previous data. Machine learning employs a variety of algorithms to create mathematical models and make predictions based on past data or knowledge. It can be used to identify pictures, recognize speech, filter emails, auto-tag in Facebook, recommend systems and many more activities [1, 2].

This machine learning course introduces ML and the many ML techniques available, including supervised, unsupervised and reinforcement learning. Regression and classification models, clustering approaches, hidden Markov models, and other sequential models will all be covered [3].

According to (Fig. 1), a Machine Learning system learns from previous data, constructs prediction models, and predicts the result whenever fresh data is received. The amount of data helps to construct a better model that predicts the output more precisely, hence the accuracy of anticipated output depends on the amount of data [4].

If we have a complex situation for which we need to make predictions, rather than writing code for it, we may just input the data to generic algorithms, and the machine will develop the logic based on the data and forecast the outcome. Machine learning has shifted our perspective on the issue [5, 6]. The following block diagram depicts how the ML algorithms work.

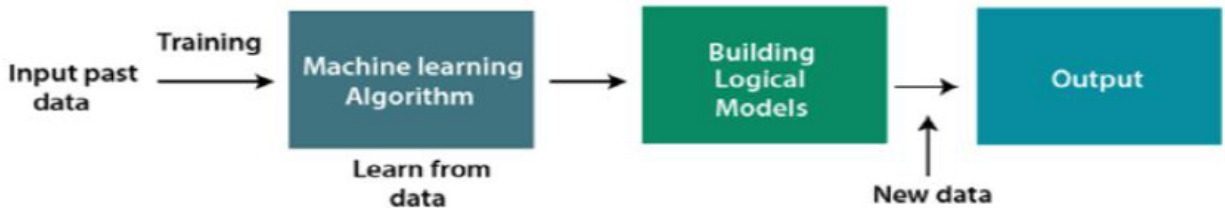


Fig. (1)
ML prediction model.

2. CLASSIFICATION OF MACHINE-LEARNING

ML can be categorized into three types on a general level mentioned in Fig. (2):

1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning



Fig. (2)
Machine-Learning Classification.

2.1. Supervised Learning

It can be defined as a form of ML method in which the ML system sample labelled data is provided to train it on, and it then predicts the

result. To interpret a set of data and to acquire details about each data, the system constructs a model using labelled data in Fig. (3). After the training and processing are complete, the model can be tested by supplying a sample dataset to see if it accurately predicts the output [7, 8].

The goal of supervised learning is to plot the input into output data. It depends upon the supervision, as well as it is similar to learning by a student under the guidance of a teacher. Spam filtering is one example of supervised learning.

2.2. Unsupervised Learning

It is a kind of learning in which a machine learns without any human intervention. The machine is taught given a collection of data that hasn't been labelled, classified, or categorised, and the algorithm is expected to act on it without supervision in Fig. (4). The objective of unsupervised Learning is to reorganise input data into new features or a collection of objects with similar patterns [9, 10].

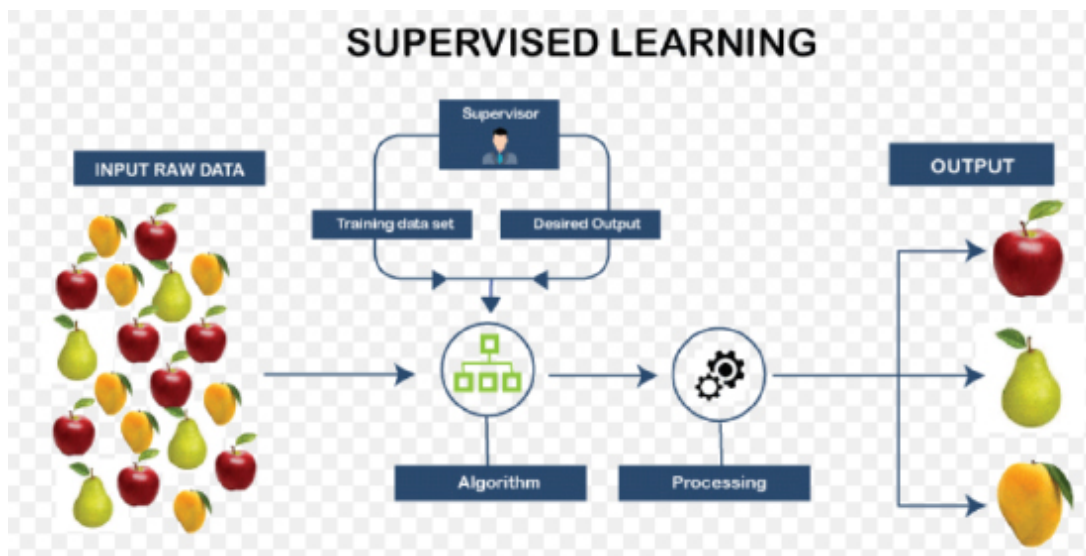


Fig. (3))
Supervised Learning.

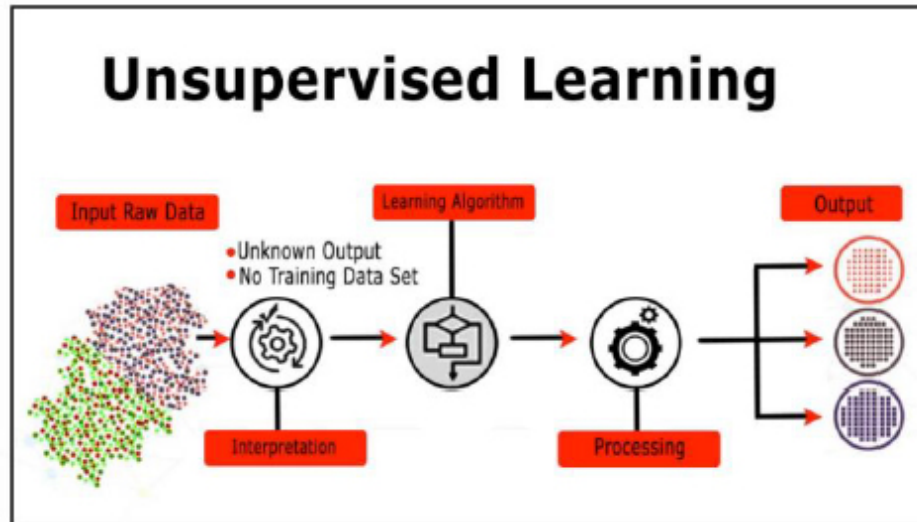


Fig. (4)
Unsupervised Learning.

2.3. Reinforcement Learning

Reinforcement learning is a feedback-based strategy of knowledge by which a learning agent is rewarded for correct actions and punished for incorrect ones. With this feedback, the agent learns automatically and improves its performance. The agent interacts with and investigates the environment in reinforcement learning. An agent's purpose is to earn the greatest reward points, so it enhances its performance [[11](#), [12](#)].

Reinforcement learning is demonstrated by the robotic dog, which automatically learns how to move his arms in Fig. ([5](#)).

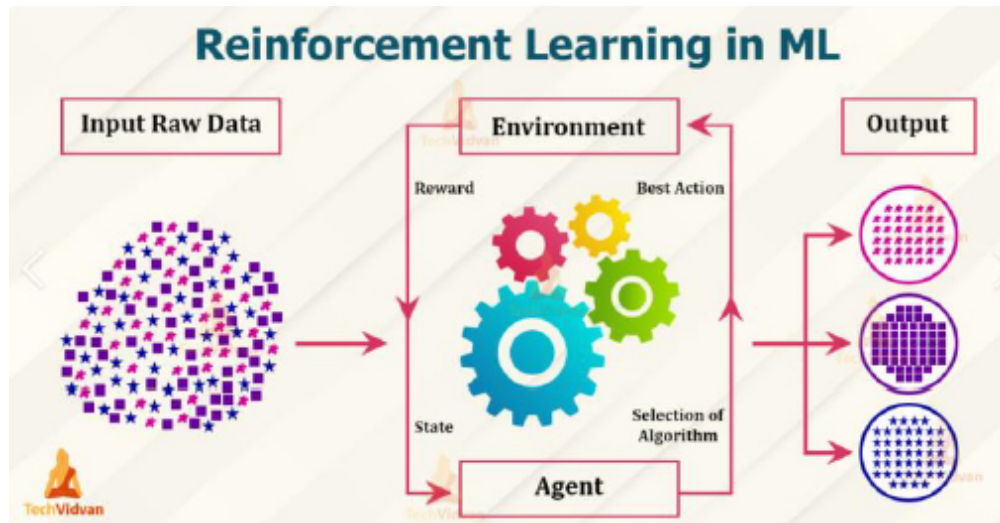


Fig. (5)
Reinforcement Learning.

3. BREAST CANCER PREDICTION USING ML TECHNIQUES

3.1. Introduction

CANCER is a disease that affects both men and women. The death rate from cancer has been significantly lowered in recent years, especially because of improved treatments. Although sophisticated cancer treatments are being developed, early disease identification is one of the most difficult aspects of this profession [13, 14]. Several types of cancer are present, as shown in Fig. (6), the effected body component. Breast cancer is a type of cancer that only affects women. It is usually one of the most preventable kinds of disease. Breast cancer is the leading cause of death in females between the ages of twenty and fifty-nine and the most occurring types of cancer overall. A diagnostic approach that uses X-rays to analyse the human breast is widely utilised as a device for diagnosis and screening. This diagnostic procedure aims to detect breast cancer early, mainly by detecting distinctive micro calcifications/multitudes. Mammography is a merely operative as well as practical method for detecting breast cancer, especially in the case of small tumours. Benevolent injuries are modifications in the conventional structure of the breast tissues

which aren't linked to the development of cancer [15, 16]. Deep learning architectures are used to classify breast cancer using histopathology images in this work. Convolutional Neural Networks (CNN) constitutes the strength to classify applications with images among the current methods. The Convolutional layer's goal is to acquire a map feature. For low-level feature detection, we usually start with a small number of filters. The further we delve into the Convolutional Neural Networks, the more mesh we employ to identify advanced possibilities [17, 18]. Component recognition is accomplished by 'scanning' the input using a filter of a specific size and doing lattice calculations to generate an element map.

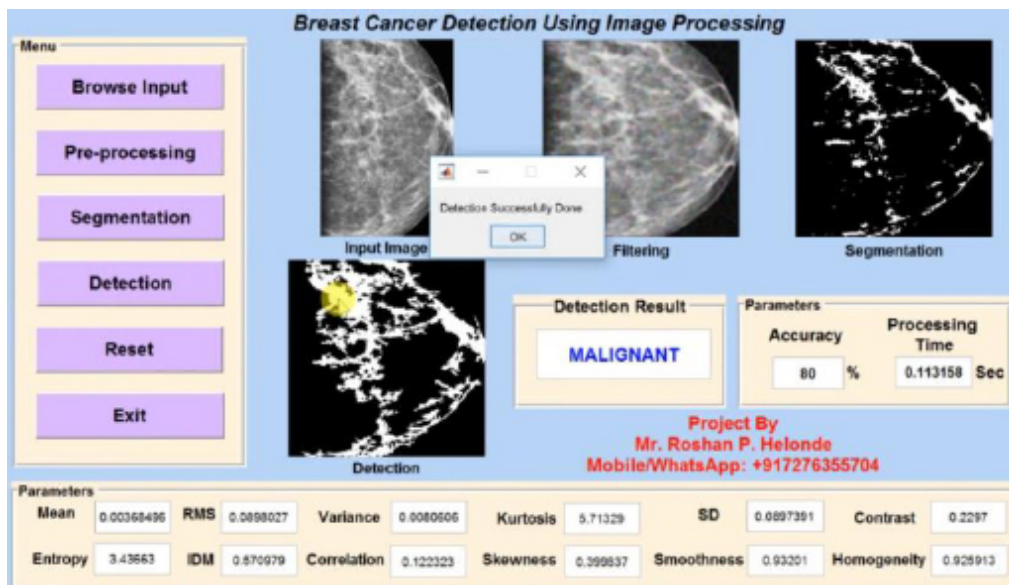


Fig. (6)

Breast Cancer Prediction Using ML and image processing techniques.

3.2. Related Works

There are a few workings that are present for the classification of breast cancer. Likewise, most of the researches use abundant information for their investigations [19, 20]. Yet, inside the current work, we are utilizing just 50% of the dataset to

lessen the calculation by applied case and acquired accuracy around 87%. A total of four hundred images has been utilized to classify the

problems of four classes.

The following task was completed with the help of a pre-equipped Inception Resnet V2 that attained 76 percent of accuracy. For feature extraction, CNN is employed, while support vector machineries are used for grouping [21, 22]. They have the capability to accomplish an accuracy of roughly 77.8%. CAD System for Breast Microcalcification Classification and SSNE Performance Evaluation Breast-cancer is classified as benevolent or spiteful which in turn depends upon cytological pictures of tiny needle biopsies using computer-assisted procedures. From microscopic biopsy images, Alexander Brook used a machine to diagnose breast cancer. With the use of multi-class support vector machines on non-exclusive choices derived from image-level set data, they have the capability to get a great rate of recognition [23, 24]. The neural networks, which are deep convolutional, are used to diagnose skin cancer.

Wavelet and Symmetric Stochastic Neighbour Embedding Based Computer Aided Analysis were used to analyse breast cancer. Image mining for Retinal Defects Detection by Machine Support vector machine was used to classify brain images using various wavelets. DWT, SNE, and SVM were used to analyse the CAD System performance for Breast Mass Classification [25, 26].

Diana Dumitru published an article titled Prediction of recurrent events in breast cancer using the Nave-Bayesian Classification. The Nave Bayes Classifier was used to classify patients in the Wisconsin Prognostic Breast Cancer (WPBC) dataset in this work. The identification accuracy of the testing was 74.24 percent.

Breast cancer is one of the utmost general cancers among all women over the globe. At regular intervals of 13 mins, breast cancer was diagnosed in a woman, leading to death. These findings have provoked the researchers to continue their research into a method for detecting breast cancer in womankind, particularly grown-up females who are in complex danger [27, 28].

Shweta Kharya and Sunita Soni published the paper Weighted Nave Bayes Classifier: A Predictive Model for Breast Cancer Detection in 2016. This research looked at how to combine a classic nave bayes classifier with a unique technique to assigning weights to its features to create a probabilistic classifier for breast cancer diagnosis that experts may utilise in making decisions shown in Fig. (7). For cancer alignment treatment, the software can approximate human diagnostic competence [29, 30].

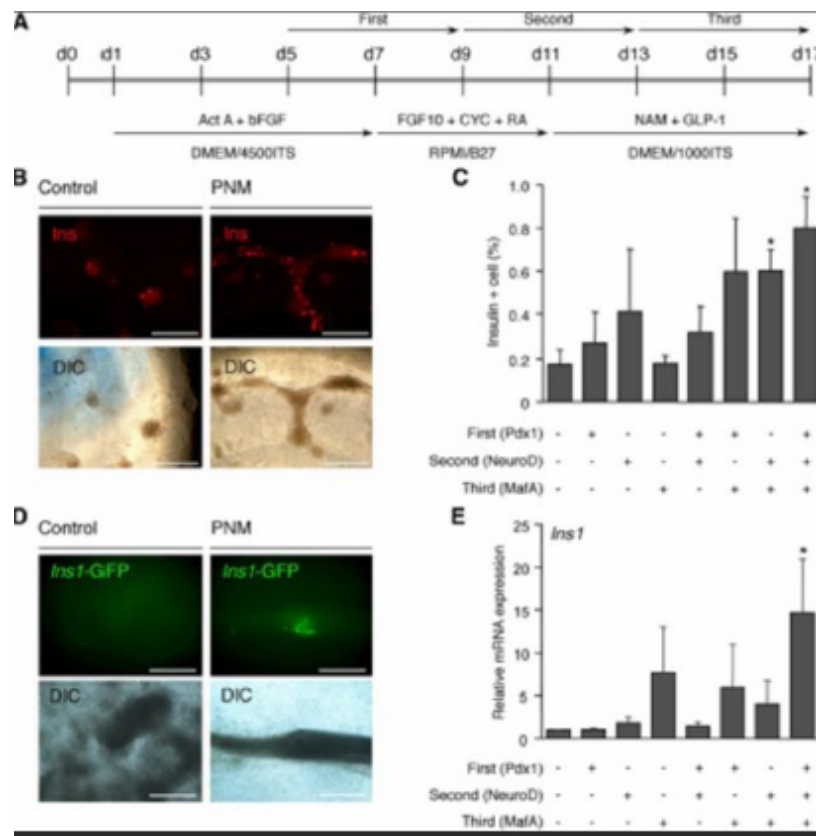


Fig. (7))
Breast Cancer Prediction using ML Techniques.

4. HEART DISEASE PREDICTION USING MACHINE LEARNING TECHNIQUES

4.1. Introduction

In the modern world, heart disease prediction has become a prominent notion that is influencing society's health. The fundamental idea is to use the Random Forest method to determine the age group and heart rate in Fig. (8). This research explains how a system calculates the heart rate and condition based on the user's inputs, such as blood pressure and other factors. When compared to other algorithms, RFA implementation provides a better experience and more accurate results. This aids in the early detection of sickness and is utilized in different ways, including providing input to determine the heart rate based on their health status [31].

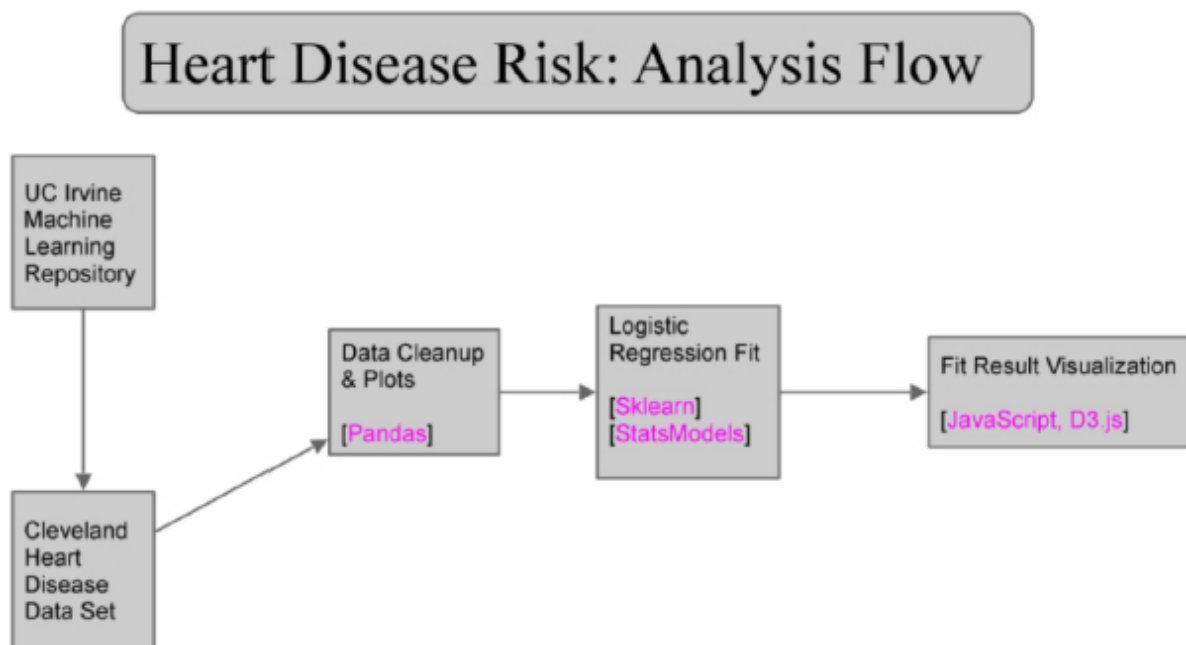


Fig. (8)

Heart Disease Risk Prediction Using Machine Learning Techniques.

4.2. Existing System

The patient provides the input details for this system. The cardiac illness is then analysed utilizing algorithms of machine learning based on the user inputs. The generated findings are now compared to those of current models in the same domain and found to be superior. Patterns are discovered using DT, NN, SVM, and Naive Bayes using data from heart disease patients obtained at the UCI laboratory. With these algorithms, the production and accuracy of the outcomes are

compared. In comparison to other current methods, the suggested hybrid method produces results of 87 percent for F-measure [32].

The peaceful environment of this work was inspired by a large body of work on the analysis of Cardio-vascular heart disease using ML algorithm. This article includes a quick review of the literature. Various algorithms, like Logistic Regression, KNN, RFC, and others, have been used to make an effective cardiovascular disease prediction. Each algorithm has its own strength in terms of registering the set targets, as shown in the results. Using the prior and new ML model and deep learning models, the model containing IHDPS was able to determine the decision boundary. It simplified the most basic and significant factors/knowledge, such as family history, which is linked to any cardiac condition. However, the accuracy of such an IHDPS model was considerably less than that of a new emerging model for diagnosing coronary heart problems employing ANN and other machine-learning as well as deep-learning methods. McPherson et al. identified the dangerous aspects of CHD or Athero-sclerosis using an internal implementation method along with Neural Network techniques and was only able to accurately identify whether the patient was suffering from the condition or not [33].

R. Subramanian introduced the use of neural networks to diagnose and predict heart disease, blood pressure, and other characteristics. A profound Neural-Network had been built using certain disease features to produce an output supported by the output perceptron, which includes just about 120 concealed layers, the very elementary and pertinent strategy to ensure accurateness in consequence of having coronary illness on condition that the model is used in the Test Data-set in Fig. (9). The use of an administered network for heart disease diagnostics has been recommended once the model deployed into the doctors system then the doctor started using it. The model works well and predicts the result with good accuracy, though he tested the model with different data [35].

5. Predicting IPL results using ML Techniques

5.1. Introduction

T20 cricket was first introduced in England in 2003. It became highly popular due to its shorter format. T20 was introduced to India due to the love of high-octane action. In 2008, the BCCI launched the Indian Premier League (IPL), a 20-over cricket event. Every year, the BCCI hosts the IPL T20 cricket event. It is critical to employ analytical tools in various facets of cricket, including result prediction. Due to the game's popularity and the large sums of money at stake, the algorithm that best predicts the outcome of cricket is in high demand. As a result, analysing IPL results becomes increasingly crucial. In order to guess the match result, algorithms of machine-learning have been used, which is a major part of cricket. Models which predict the triumphant group can be made utilising historical player performance data and other related data [36]. This model may be built with machine learning methods like Decision Tree, Naive Bayes, and K-Nearest Neighbor, and the outcomes can measure up using Assessment Measures like accuracy, precision, review, compassion, and rate of errors in Fig. (10).

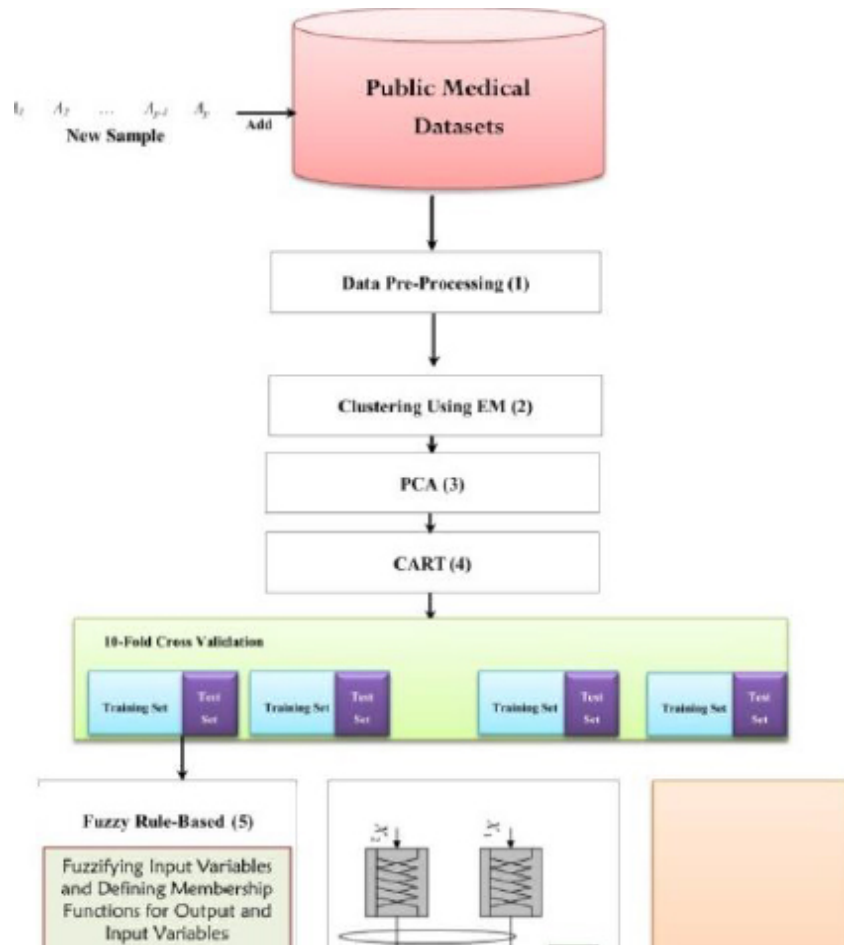


Fig. (9))

Heart Disease Prediction Using Machine Learning Techniques.

	Team	Group	Previous appearances	Previous titles	Previous finals	Previous semifinals	Current rank
0	Mumbai Indians	A	12	4	5	8	1
1	Chennai Super Kings	A	10	3	8	10	2
2	Delhi Capitals	A	12	0	0	4	7
3	Kings XI Punjab	A	12	0	1	2	6
4	Royal Challengers Bangalore	A	12	0	3	5	8

Fig. (10))

Prediction of IPL Data Using Machine-Learning Method.

5.2. Related Work

The study on the Data-Mining of Cricket data-set provides an overview of the various data mining strategies applied to the IPL

dataset, like Decision Tree, Naive Bayes, KNN, and Random Forest, to build a model for predicting match results. The Wrapper and Ranker approach was used to identify the best qualities, and then the classification was completed. WEKA assisted in the completion of this project [37, 38].

According to Gupta *et al.*, management must always select the best team in order to achieve the best results. Instead of using the time-consuming traditional method, the study proposes an ideal solution for selecting the best team utilizing the techniques of data mining. When we set a time for a certain championship, we must choose the best squad, which makes the team's chances of being champion much easier [39].

The authors suggest a fuzzy clustering logic in their paper. With the Data Mining Technique – Clustering, the outcomes of the IPL batting statistics were classified into several clusters, resulting in efficient and accurate findings. MATLAB was used to complete this project. Clustering is used to identify batting statistics from the IPL that contain ambiguous data into acceptable clusters [40].

Raza Ul Mustafa *et.al* offered an examination that looked into the attainability of utilizing Twitter data to predict match results. The goal of this study is to see how effective machine learning techniques are when applied to data acquired from social-media organizations and other true events occurrences that have been expected. Support Vector Machine, Naive Bayes Classifier, and Linear Regression were employed in their research. The SVM method is still valid [41].

The creation of the model that guesses the score for the team which is chasing and will appraise the score of the second innings of the match is described in the Live Cricket Score and Winning Prediction work. Linear Regression, Naive Bayes Classifier, and Reinforce Learning Algorithms are used in the suggested work. The result of the coin toss, the team's ranking, and the home team's advantage were all taken into account.

Sankar Narayanan proposes the notion of developing a prediction system that uses historical data to forecast the outcome of upcoming matches. They employed Linear Regression, Nearest Neighbouring, and Clustering techniques to provide the outcome mathematically and demonstrate algorithm's presentation in forecasting the model's results.

Parag Shah suggested a model predict the match result after each ball in his work on forecasting the result of a live match. Duckworth & Lewis used the par score idea, which calculates the likelihood and makes it clear who will get the victory in the match [[42](#), [43](#)].

Kaluarachchi employed AI techniques, notably Bayes classifiers in ML, for categorizing elements, such as home-game benefit, the effect of day or night on tossing, and first batting, all of which have an impact on the match's outcome. The end product of this effort is a software tool named CricAI, which was offered as a software utility. The programme calculates the chances of captivating depending upon input criteria such as home-game benefit, which was provided during the game's start. When teams are playing cricket, the CricAI can be employed in real-world applications. It is being utilized in altering key parameters in order to improve one's chances of winning in the actual world [[44](#), [45](#)].

Data Analytics approaches from the machine learning area are used in the models. Predicting the effect of a rain-affected contest can be tricky. Batting, bowling, fielding, selecting team, predicting results, and revision of target are all critical in a rain-affected match [[46](#)]. The numerical model constructed dependent on the visions of previous matches' outcomes can solve the match prediction. The SVM is being utilized in generating the Predictor models. Deep Mayo Predictor [[47](#)] is used in this study.

The Bayesian Prediction Technique and parameter-based filtering make a dream team with software built for Statistical Analysis and Data Mining based on Jayshree Hajgude's work. The current IPL players' information is stored in the database. This work aids in the

extraction of necessary data for the use of a prediction algorithm to gain each player's statistical analysis.

Using the ODI dataset, a prescient model has been created to anticipate cricket scores as well as the performance of the players. Performed SVM and Nave Bayes are examples of supervised approaches. To classify accurately, clustering algorithms such as KNN and MLP are used.

ML approaches are utilized in predicting the winner of IPL matches. Every year, there is high anticipation for who will get a victory for the renowned title in the Indian Premier League shown in Fig. (11). In the Indian Premier League (IPL), the results of games can change in a matter of seconds, making it difficult to predict who will come out on top. SVM, Logistic Regression, Nave Bayes, Decision Tree, and KNN are examples of machine learning algorithms.

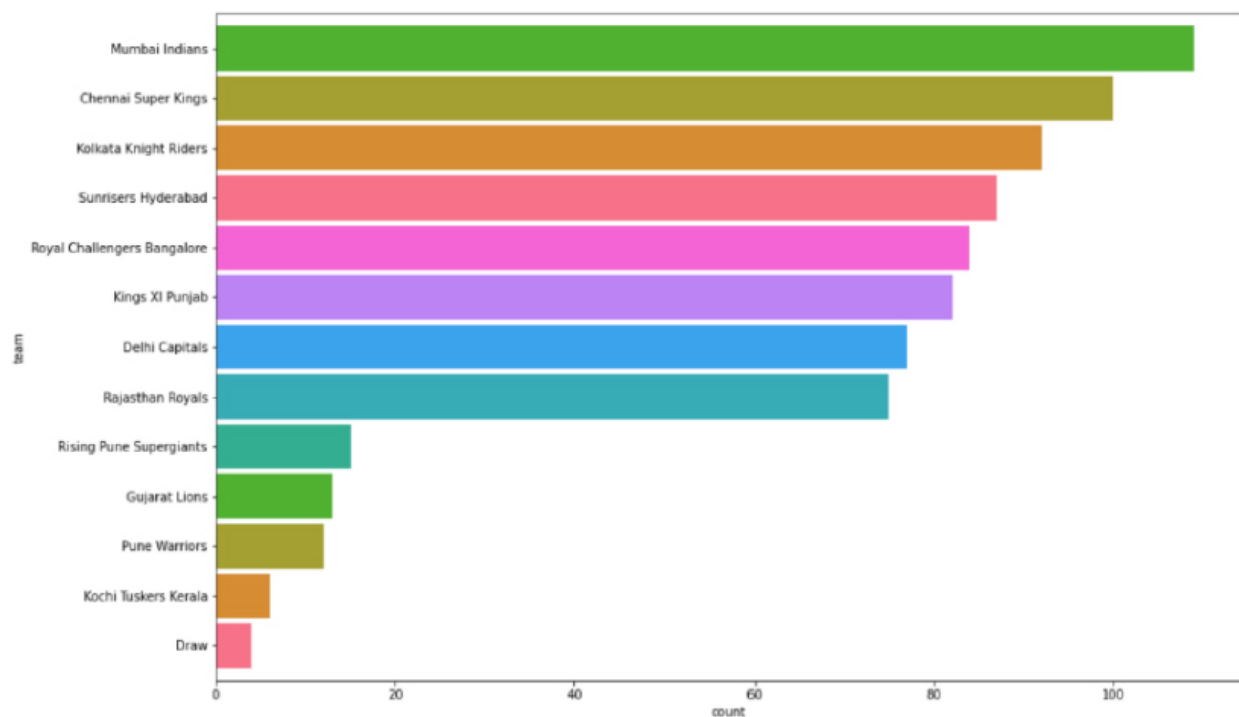


Fig. (11))

Prediction of Indian Premier League Data Using ML Strategies.

6. PREDICTION OF SOFTWARE BUG UTILISING ML TECHNIQUE

6.1. Introduction

Software flaws have a significant impact on software reliability, quality, and maintenance costs. It takes a lot of effort to create bug-free software, even if the software is used correctly because hidden problems are common. Furthermore, establishing a software bug prediction model that can detect problematic segments in the earlier stages of development will be a significant issue in software-engineering [\[48\]](#).

Predicting software bugs has been an important part of developing the software. This is on the grounds that foreseeing problematic segments preceding programme release increases overall software performance and user happiness. Furthermore, anticipating software bugs early improves programme adaption to various settings and maximises resource efficiency.

To address the problem of Software Bug Prediction (SBP), various strategies have been presented. Machine-Learning approaches are the utmost recognized. According to Fig. [\(12\)](#), SBP makes substantial use of approaches of machine learning to forecast defective modules dependent on historic failure information, key measurements, and different programming computing techniques.

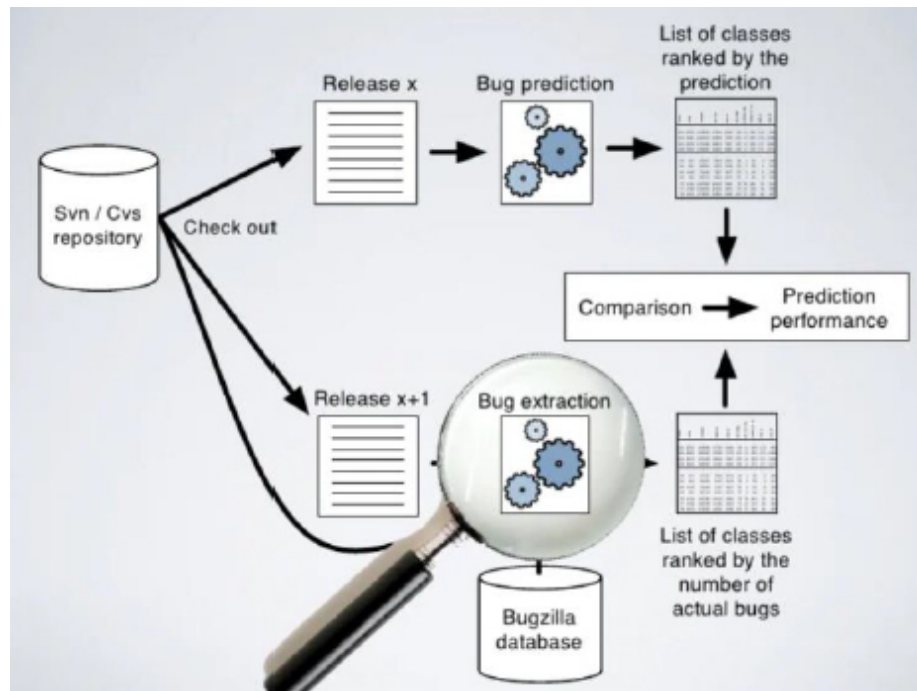


Fig. (12))

Software Bug Prediction using Machine Learning Approach.

6.2. Related Work

There have been numerous examinations on the utilization of techniques of machine learning to forecast software bugs. For example, an undeviating Auto-Regression technique was presented to forecast problematic modules. Based on previous information of software-gathered flaws, the investigation forecasts future software problems. The Auto-Regression model and the Known Power Model (POWM) were also examined and compared using the Root Mean Square Error (RMSE) method. Furthermore, the investigation evaluated three sets of data, and the outcomes were positive.

This research looked into the applicability of several approaches of machine learning for expectations of flaws. Sharma and Chandra supplemented their research with the latest investigations on each ML technique as well as current developments in software bug prediction utilising ML. This research is able to serve as a foundation or stepping-stone for forthcoming work in prediction of software bugs [\[48\]](#).

R. Malhotra introduced an exhaustive study of software bug prediction strategies utilising Machine-Learning. This article incorporated an audit of all investigations published somewhere in the range of 1991 and 2013, an analysis of the techniques such as Machine Learning for predicting software bug models and an assessment of their presentation, a comparison of Machine-Learning and statistical strategies, a comparison of various strategies of ML as well as a summary of ML techniques' strengths and weaknesses.

The publication established a standard to enable a typical and valuable correlation of various bug-prediction algorithms. This examination included a full correlation of renowned algorithms to predict bugs, as well as a new methodology that was introduced and evaluated by constructing a decent correlation with existing ways utilizing the standard provided.

A typical object-oriented Software Bug-Prediction System was created by D. L. Gupta and K. Saxena (SBPS). The study combines defect datasets of comparable categories that may be found at the Promise Software Engineering Repository. The model proposed was tested utilising the presentation extent in the study (accurateness). At long last, the examination's findings revealed that the suggested model's average accuracy is 76.27 percent [50].

Rosli introduced a flaw in inclination forecast applications based on hereditary algorithms. The uses get its standards from an open source software project, such as values based on object-concerned with and tally measurements. The evolutionary algorithm takes data from the application by means of input in developing instructions that are then used to classify software components as blemished or non-faulty. At last, use the hereditary algorithms applet to visualise results.

A lot of object-based criteria were evaluated by using a mix of machine learning techniques (like decision trees and neural networks) and mathematical methods (like logical regression and linear regression), linear regression, and correlation. The study's findings revealed that the Coupling between Object measurements stands for the utmost accuracy in predicting class defects, followed

by Line of Code, while the Depth of the Inheritance Tree and Number of Children are unreliable measurements.

Artificial Neural Networks, Particle Swarm Optimization, Decision Tree, Naive Bayes, and Linear Classifiers are five popular ML strategies used in detecting software issues, according to Singh and Chug (LC). The analysis established that Artificial Neural Networks have the least mistake level, trailed by Decision Tree, however, the direct classifier outperforms different algorithms in terms of deformity forecast exactness, and the greatest mainstream strategies utilized for predicting software defects are: BL, Decision Tree, SVM, Artificial Neural Networks, RBL, and EA; and that the most general measurements utilized in the study predicting in software defects are: Line Of Code (LOC) metrics, object orig.

Furthermore, the research covered several algorithms of machine learning and demonstrated the capability of machine learning in software deformity forecast. The research aided the engineer in utilising valuable software measurements and appropriate data-mining techniques to improve software quality. Response for class, Line of code, and Lack of Coding Quality were shown to be the best measurements for predicting imperfections in a study published.

The most prominent data-mining techniques (k-Nearest Neighbours, Naive Bayes, C-4.5, and Decision Trees) were presented by Bavisi. The study looked at four algorithms and compared them, as well as explained their benefits and drawbacks. The study's findings revealed that a variety of factors influence the accurateness of techniques separately, including the idea of the issue, the data-set employed, and the matrix performance.

The studies looked at the association of object-oriented measurements and a class's flaw inclination. LOC, CBO, RFC, and WMC are active in scheming faults, and the AUC is an effective measure that may be utilised to forecast broken segments at the earlier stages of software advancement and in increasing the accurateness of Machine-Learning approaches, according to Malhotra and Singh.

The supervised ML algorithm attempt to construct surmising capacity by establishing links and conditions amongst the known inputs as well as outputs of the labelled training data, so with the goal that output values for fresh input data can be predicted using the derived inferring function. The following is a synopsis of the selected supervised ML algorithm.

Nave Bayes (NB) is a fast and basic possible classifier centered on the Bayes theorem and the supposition of feature independence [41]. NB is a group of calculations dependent on a typical rule, which accepts that the existence or non-existence of a particular component of the class isn't identified with the presence or nonattendance of some other provisions.

Artificial Neural Networks (ANNs): Artificial Neural Networks are modelled after biological neural networks. Neural networks are a type of non-linear classifier that is able to describe intricate input-output interactions [42]. A neural network is made up of a group of dispensation units called neurons which act in tandem in generating output. Every neuron figures its yield utilizing the non-linear capacity of the amount of all neuron inputs, and each connection amongst neurons can transfer a signal to different neurons.

Decision Tree (DT): In data mining, DT is a typical learning method [43]. The term Decision Tree denotes a hierarchical and prescient model that employs object observations as divisions to arrive at the target value of the objects in the leaf. The decision tree is a tree possessing decision-nodes that possess several branches and leaf nodes that indicate the decision.

7. PREDICTION OF RAINFALL USING MACHINE LEARNING TECHNIQUES

7.1. Introduction

Rainfall forecasting is critical because heavy and irregular rainfall can have a variety of consequences, including crop and farm destruction

and property damage. A better forecasting model is required for early warnings that can reduce risks to life and property while also better managing agricultural farms. This forecast primarily benefits farmers, additionally the efficient use of water resources. Predicting rainfall is a difficult task, and the findings must be correct. Many hardware devices are available for predicting rainfall based on weather parameters such as temperature, humidity, and pressure. As traditional approaches are inefficient, we can achieve accurate results by employing machine learning algorithms. We can simply achieve that by analysing previous rainfall data and forecasting rainfall for future seasons. We can use a variety of approaches, like classification and regression, to meet our needs, and we can also calculate the difference between the actual and predicted values, additionally the accuracy. Different methodologies provide varying degrees of accuracy, it is critical to select the appropriate algorithm and model it according to the requirements.

Rainfall is now widely regarded among the most significant determinants of most of the world's major events. Agriculture is the key variable in determining India's economy, and agriculture is completely reliant on rainfall. Apart from agriculture, rainfall is also more significant in coastal locations around the world, where understanding rainfall is critical to protecting people from floods and excessive rainfall. To set up a rainwater harvester in some of the drought-stricken areas, accurate rainfall forecasting is required.

The goal of this venture is to use machine learning and neural networks to predict rainfall. In this project, we are conducting a comparative analysis of ML and neural network methodologies, and then selecting the best approach for rainfall prediction. The pre-processing phase begins with the representation of the input dataset in graph form. Histograms, bar graphs, and other graphs are examples. LASSO (Least absolute shrinkage and selection operator) regression and the ANN (Artificial Neural Network) method are utilised in machine learning approaches. Following the calculations, the accuracy and mistakes of the LASSO and ANN are compared, and a result is reached. In this case, the rainfall prediction is made using the approach that produces the most accurate results. Rainfall

prediction is carried out using a dataset including rainfall-related data shown in Fig. (13).

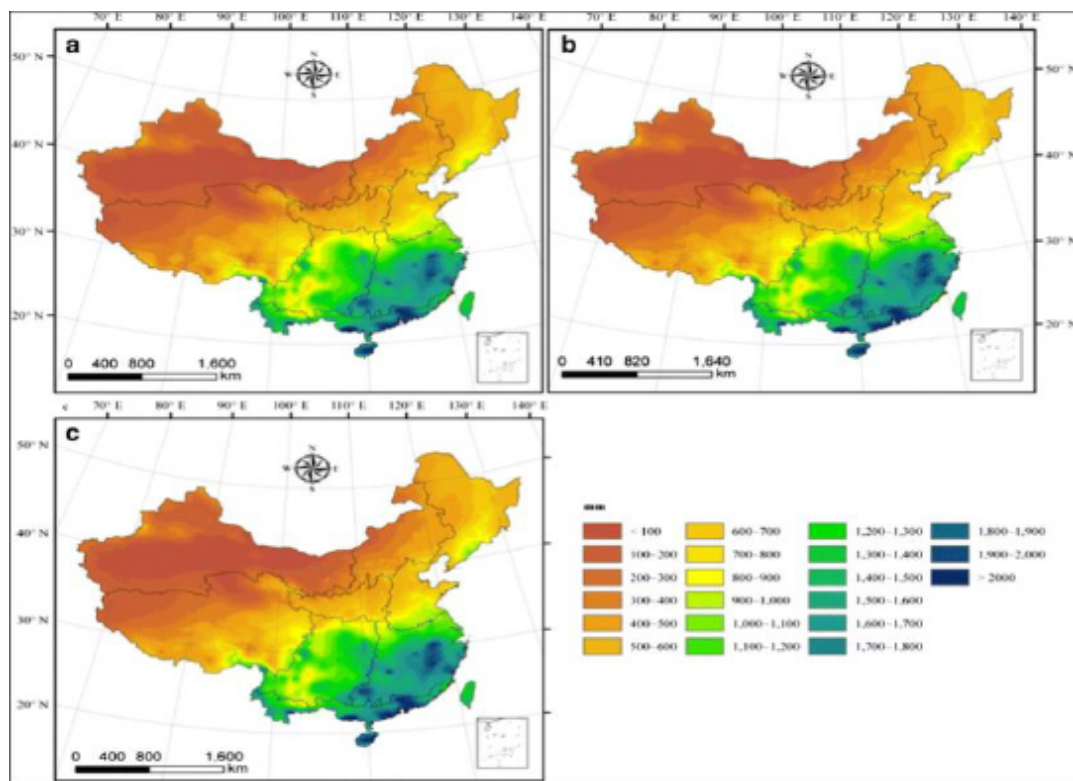


Fig. (13))

Prediction of Rainfall Using Machine Learning Techniques.

7.2. Related Work

Literature review on flood prediction using machine learning methods, a body of water (Switzerland); floods are among the most devastating natural disasters, yet they are difficult to predict. The research on advanced flood prediction models contributed to policy recommendations, risk reduction, a reduction in human life lost in floods, and a reduction in property damage caused by floods caused by heavy rainfall.

Machine learning approaches have been the finest in improvement of prediction systems over the last two decades, providing greater performance and extremely reasonable solutions to emulate the complex mathematical expressions of physical processes of floods.

The popularity of ML has skyrocketed among hydrologists due to its numerous advantages and possibilities. Researchers are attempting to create more effective and accurate prediction models by incorporating new machine-learning approaches and hybridising existing ones.

This paper's aim is to determine the state of the art of ML models in predicting flood and to raise awareness of the most appropriate models. The literature where models of ML were established as a standard through a qualitative examination of accuracy, robustness, speed, and effectiveness, is explored in depth in this work in order to provide a comprehensive overview of the many ML algorithms employed in the area. The performance comparison of models of machine learning allows for a more in-depth understanding of numerous ML techniques within the context of an evaluation and discussion. As a result, the most effective and accurate approaches for short- and long-term floods are presented in this work. The flood prediction methods are also being looked into for quality improvement. Hybridization, algorithm ensemble, data decomposition, and optimization model are among the methodologies for improving ML methods that have been published.

This survey can help hydrologists and climate scientists choose the best machine-learning method for their projects based on their demands. ANN is for Artificial Neural Network, SVM refers to Support Vector Machine, MLP stands for Multi-Layered Perception, DT refers to Decision Tree, ANFIS refers to Adaptive NeuroFuzzy Inference System, WNN stands for Wavelet Neural Network, and EPS stands for Ensemble Prediction Systems. When compared to a single algorithm, the hybrid approach enhances flood prediction accuracy. It takes a lot of time and resources.

Weather predictions and strategies are examined. A concise synopsis of Daily weather forecasting is being utilized for numerous purposes in different types of industries, including agriculture, transportation, and so on. The weather accuracy is shown in forecast reports, which is highly important. A review is undertaken in this study to examine a higher method of prediction that is used for various types of

prediction. Among these, the ARIMA model with time series predicts with no mistakes and higher output accuracy. Box and Jenkins proposed a fairly successful version of the ARIMA model, the seasonal ARIMA (SARIMA), for seasonal statistic prediction.

The ARIMA model's popularity is growing, owing to its capacity to easily represent a wider range of statistics. Still, because of the late Bx-Jenkins approach for some optimal model, an ARIMA model is built, which is eco-friendly in nature and provides more productivity and consistency. It explored further into training data and developed a slew of related approaches that are extremely useful in weather forecasting. This study examines a variety of approaches, with a focus on the ARIMA MODEL methodology for daily meteorology.

This technique uses a variety of criteria to forecast daily weather, including rainfall, humidity, temperature, cloud status, and the weather of the day. The major resolution of this research work is to compare present meteorological models and choose the best model to support their ability to anticipate. The ARIMA model is employed. It aids in the forecasting of future weather conditions based on data. The accuracy is less than 60%.

Open access data mining study of various rainfall estimation & prediction techniques for the use of water resources effectively and optimal planning of water supplies and availability, it is critical to make accurate rainfall predictions. Various strategies and models are created for this goal in various studies employing data mining techniques to estimate rainfall. There are a variety of strategies that might be utilized in predicting rainfall and estimating precipitation with accuracy and precision.

The applications of techniques of data mining to predict rainfall and its solutions may be important in predicting efficient rainfall, which will assist farmers in making crop decisions and increase the agriculture sector's growth. This study compares the outcomes of various rainfalls forecast and estimating systems to the actual rainfall value. PCA and K-Means algorithms are employed in this paper. It is

straightforward and requires little time and effort. The accuracy is less than 65%.

A Review of ML Techniques for Predicting Rainfall Heavy rainfall causes numerous calamities, hence forecasting rainfall is very crucial. Rainfall forecasting aids individuals in taking preventative precautions for their crops and properties, and forecasting should be more precise and efficient. There are two sorts of rainfall forecasts: short-term forecasts and long-term forecasts. In terms of forecasting, mostly short-term forecasting may provide us with accurate results.

The key task ahead of us is to develop a model for long-term rainfall forecasting. As it is intimately related to the economy and human lives, heavy precipitation prediction could be a serious problem for weather forecasting departments. Natural catastrophes, such as floods and droughts, are a problem that people face every year in the overall world. Predicting rainfall with more accuracy is critical for countries such as India, whose major economy is entirely reliant on agriculture.

Because of the vibrant nature of the atmosphere, techniques of applied mathematics have failed to produce a reasonable and accurate precipitation statement. The regression technique may be used in the prediction of precipitation utilising machine learning techniques. The goal of this scheme is to deliver non-experts with easy access to the approaches and techniques used in the field of precipitation prediction, as well as a comparison of the numerous ML algorithms. SVM and Logistic Regression are employed in this strategy. It has an accuracy of up to 89 percent. We can only utilise it for prediction and not for forecasting.

A comprehensive analysis of the published research on the topic of forecasting rainfall by utilising the methods of data mining, predicting rainfall is among the most difficult aspects of climate anticipating. Exact and appropriate rain-fall forecasting can be extremely useful in preparing for ongoing building missions, transport accomplishments, agronomic jobs, aviation procedures, and flood situations, among other things. By finding concealed outlines from accessible aspects of

previous meteorological data, data mining algorithms can efficiently predict rainfall.

This study adds to the body of knowledge by giving a comprehensive examination and assessment of the most recent data mining techniques for rainfall prediction. This study looked at publications that were published between 2013 and 2017 and were found in reputable internet search libraries. This study will aid academics in analysing recent rain-fall forecasting work by means of a focus on data-mining approaches, as well as providing a starting point for future instructions and judgements. We discovered the difficulty in predicting rainfall. There is no implementation in real-time.

ML Techniques for Rainfall Prediction Rainfall forecasting is an important component of the country's economy, and it can assist people in avoiding catastrophic natural disasters. Agriculture is dependent on rainfall, and agriculture is the principal occupation of several states, therefore, some parts in India are economically dependent on it. This prediction aids in the identification of agricultural patterns in addition to the efficient management of crop water resources. Seasonal rainfall prediction is frequently done using linear and non-linear models. CART, SVM and Genetic Algorithms are a few of the algorithms utilised for rainfall prediction.

In this paper, we utilised methods of classification like Artificial Neural Networks (ANN), Support vector machines (SVM), and Logistic Regression to perform qualitative and accurate analysis. The hydrological department of Rajasthan provided the data for this classification application. Overall, we examine which algorithms are the most effective at qualitatively predicting rainfall. Support vector machines (SVMs), Artificial Neural Networks (ANNs), and logistic regression approaches are employed. The accuracy of ANN was found to be 87 percent. It takes more time to train the model because it necessitates abundant data.

An overview of deep-learning and unsupervised-learning for time series modelling is presented in this study, which includes a discussion of current advances in deep learning and unsupervised

feature learning for time-series challenges. Whereas these techniques have exposed potential in the modelling of data statistics, for example, computer vision, their application to time series data is acquiring traction.

This article delivers an overview of some specific challenges that are available in time series information, as well as an audit of workings that have spread-over time series data to unsupervised feature learning algorithm or committed for changes in feature learning algorithm to represent the difficulties that are available in time series data. The method of Recurrent Neural Network (RNN) is applied. Time series modelling has a high level of accuracy. There has been a necessity to concentrate less on the pre-processing pipeline for a precise time series issue and further on learning improved feature demonstrations for a broadly useful organized information strategy, paying little heed to the application.

In this paper, rainfall rate prediction based on weather radar measurements, the clouds that back scatter the electromagnetic radiations are measured using weather radars. Clouds produce more rain because more electromagnetic radiations are backscattered by the clouds, which leads to larger rain droplets and hence produces more rain. The goal of this article is to use weather radar to estimate rainfall rates rather than rain gauges, which are being utilized in measuring rainfall on the ground. During an experiment over the Italian Swiss Alps in June and August 1997, precise data was acquired from weather radar and adjacent rain gauges at the same time.

For the grouping job, the radar data was used as an input, and the rain-gauges measurements as output, and the numerical KNN classifier and neural SOM were used. With an average inaccuracy rate of 23%, the rate of rainfall on the ground was forecast using radar reflections. The findings of this study suggest that rainfall rate prediction using weather radar measurements is possible [64]. The KNN algorithm is utilised. The system outperformed the usual power-law relationship with a reasonable success rate. Time series data are not supported.

Artificial Neural Networks for Forecasting the State of the Art: The popularity of ANNs for predicting has sparked a surge in research during the last decade. While ANNs hold a lot of promise, they also come with a lot of unknowns. The effect of ANNs major elements on forecasting effectiveness is still a mystery to researchers. This paper gives a current state-of-the-art overview of ANN forecasting applications. Our goal is to present (1) a summary of published research in this field, (2) insights on ANN modelling challenges, and (3) recommendations for future study. The acronym ANN is used. Overall, ANNs do admirably when it comes to forecasting [45] shown in Fig. (14). There may be a limit to how much ANNs can learn and predict from data.

International Journal of Forecasting, Forecasting the NN5 Time Series using Hybrid Models. In this present work, we projected a basic method for foreseeing time series using data mining techniques, as well as establishing a baseline for future directions and comparisons. We've discovered the problems in predicting rainfall. There is no implementation in real time of the cyclical nature of the seasons.

In the pre-processing stage, the missed standards of the time series are assessed and introduced. By adopting the prejudiced mean of predictions created by means of time-domain models that had been approved on missed parts of the time-series, we integrate different forecasting methodologies [48]. A neural network group, a group of closest direction models and a model constitutes even-day series make up the hybrid model. The NN percent time series competition data set was used to test this method. Nearest neighbour models are employed in a simple seven-day cycle model the accuracy rate is over 70%.



Fig. (14))
Prediction of Rainfall Accumulation.

8. WEATHER PREDICTION USING MACHINE LEARNING TECHNIQUES

8.1. Machine Learning

Machine learning is relatively impervious to disturbances and does not require any additional physical factors to anticipate. As a result, machine learning offers a greater chance in the improvement of weather forecasting. Weather forecasting was a difficult nut to crack before the advancement of technology. Weather forecasters rely on satellites and a less accurate data model of atmospheric conditions. Since the previous 40 years, the utilisation of the Internet of Things has dramatically improved the accuracy and predictability of weather prediction and analysis. Scientists can now anticipate weather with high accuracy and predictability thanks to data science and artificial intelligence advances.

8.2. Use of Algorithms

There are various methods of predicting climate using Regression and various types of Functional Regression, in which datasets are used to carry out the counts and investigations 34% of the information is used to train the computations, and 14% of the information is used to test the calculations. For example, if we want to use these Machine Learning calculations to forecast the weather in Austin, Texas, we'll use 6 years of data to produce the calculations and 2 years of data as a test dataset [\[48\]](#).

In contrast to Weather Forecasting using Machine Learning Algorithms, which is based primarily on re-enactment based on Physics and Differential Equations, Artificial Intelligence is also used to forecast the weather: this includes models such as Neural Networks and Probabilistic Bayesian Networks, as well as Vector Machines. Among these models, the Neural Network is extensively used since it is effective at capturing more circumstances from historical weather reports and forecasts for the future.

In any event, certain machine learning computations and Artificial Intelligence Models, such as using Bayesian Networks and machine learning calculations in parallel, are computationally expensive. To sum up, ML and AI have fundamentally altered the way we estimate weather with great precision and predictability. Furthermore, in the coming years, further development will be made employing these advancements to correctly forecast the weather in order to avoid disasters such as typhoons, tornadoes, and thunderstorms.

ML has the subsequent foremost algorithms:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

This is a fixed prediction for supervised learning. These predictors are variables that are not reliant on each other. This learning algorithm's goal is to make predictions based on this set of independent variables. The result variable is the variable for which the prediction is made. This is an example of a dependent variable. A function that

enables the allocation of our inputs to the desired outputs is constructed using the independent variables set. The computer is constantly trained to achieve a specific level of precision in our training data. Logistic regression, Linear regression, random forest, KNN decision tree, and other supervised learning techniques are examples [\[49\]](#).

Uncontrolled learning: There is no specific aim or outcome that is able to be approximated or predicted in this method. It divides people into distinct groups, which can then be segmented for specific interventions. K-Means and Apriori's algorithm are two examples of unsupervised learning. Reinforcement learning: when using this technique to train the machine, some decisions were made. It functions in such a way that the machine is exposed to conditions in any setting. The trial-and-error method is used to train the machine on a constant basis. The machine learns from prior experience by capturing the greatest possible knowledge in order to make precise business judgments. The Markov decision process is an example of learning through reinforcement.

The task of predicting the atmosphere at a future time and in a specific place is known as weather prediction. This was done in the beginning using physical calculations that treated the atmosphere as a fluid. The environment's current condition is examined, and the future state is anticipated by numerically solving those equations; however, we can't predict the weather for more than ten days, which can be improved with the help of science and technology.

Linear Regression, Polynomial Regression, Random Forest Regression, Artificial Neural Network, and Recurrent Neural Network are examples of machine learning calculations. These models are created based on the most up-to-date information available in the given area. For example, the contributions to these types of models are offered on the occasion that expecting temperature change mean air weight, and mean wetness for about 2 days continuously. In light of this, the minimum and maximum temperatures for the next 7 days will be met.

CONCLUSION

This chapter begins with a brief explanation of machine learning, followed by a classification of machine learning (Supervised Learning, Unsupervised Learning, and Reinforcement Learning). Breast Cancer Prediction Using Machine Learning Techniques (Chapter 2) explains the various breast cancer prediction algorithms such as Convolutional Neural Networks (CNN), Support Vector Machine, Nave Bayesian Classification, and Weighted Nave Bayes Classifier. Heart Disease Prediction Using Machine Learning Techniques is the third chapter, and it explains the numerous heart disease prediction algorithms, such as Support Vector Machines SVM, Logistic Regression, KNN, Random Forest Classifier, and Deep Neural Network. The following algorithms are discussed in Chapter 4: Decision Tree, Naive Bayes, and K-Nearest Neighbour Random Forest, Data Mining Techniques, fuzzy clustering logic, Support Vector Machine, Reinforce Learning Algorithm, Data Analytics Approaches, and Bayesian Prediction Technique. The Known Power Model (POWM), Artificial Neural Networks (ANNs), Particle Swarm Optimization (PSO), Decision Tree (DT), Nave Bayes (NB), and Linear Classifiers k-Nearest Neighbours, Nave Bayes, C-4.5, and Decision trees are explored in Chapter 5. Rainfall Prediction (Chapter 6) LASSO (Least absolute shrinkage and selection operator) regression, ANN (Artificial Neural Network), Support Vector Machine, Multi-Layered Perception, Decision Tree, Adaptive Neuro-Fuzzy Inference System, Wavelet Neural Network, Ensemble Prediction Systems, ARIMA model, PCA and K-Means algorithms, Recurrent Neural Network (RNN), statistical KNN classifier, and Recurrent Neural Network (RNN) Bayesian Networks, Linear Regression, Logistic Regression, KNN Decision Tree, Random Forest, K-Means and Apriori's Algorithm.

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Machine Learning Algorithms for Health Care Data Analytics Handling Imbalanced Datasets

T. Sajana^{1, *}, K.V.S.N. Rama Rao²

¹ Department of Computer Science & Engineering, K LEF, Guntur, India

² Department of Computer Science & Engineering, KLH, Hyderabad, India

Abstract

In Machine Learning, classification is considered a supervised learning technique to predict class samples based on labeled data. Classification techniques have been applied to various domains such as intrusion detection, credit card fraud detection, *etc.* However, classification techniques on all these domains have been applied to balanced datasets. Balanced datasets are those which contain equal proportion of majority and minority examples. However, in real-time, obtaining balanced datasets is difficult because majority of the datasets tend to be imbalanced. Developing a model for classifying imbalanced datasets is a challenge, particularly in the medical domain. Accurate identification of a disease-affected patient within time is critical as any misclassification leads to severe consequences. However the imbalanced nature of most of the real-time datasets presents a challenge for most of the conventional machine learning algorithms. For the past few years, researchers have developed models using Conventional machine learning algorithms (linear and nonlinear) are stating unsatisfactory performance in classifying imbalanced datasets. To address this problem of skewed datasets several statistics techniques & robust machine Learning techniques have been developed by the researchers. The discussion on handling imbalanced datasets in the healthcare domain using machine learning techniques is a primary focus of this chapter.

Keywords: Healthcare, Skewed distribution, Machine learning techniques, Imbalanced datasets.

* **Corresponding author T. Sajana:** Department of Computer Science & Engineering, K LEF, Guntur, India;
E-mail: sajana.cse@kluniversity.in

1. INTRODUCTION

Now-a-days, people are affected by various kinds of diseases and diagnosing them has become a real challenge. Especially in the case of vector-borne diseases like malaria and dengue, the disease should be diagnosed immediately; otherwise, it may even lead to the death of the patient [1].

Hence, in diagnosing these types of diseases, some level of expertise is required. Pathologists are specialized professionals who analyze medical data samples and diagnose the disease. Pathologists need to assess data thoroughly before concluding the patient's status. In several cases, it was observed that there are variations among pathologists' conclusions, *i.e.*, one pathologist concludes that the patient has a disease, and other pathologists may predict that the patient is not affected by the disease [2]. There was a conflict among decisions on the same data. These variations are due to the following reasons:

- Human vision/ bias causing errors
- Lack of expertise
- Manual analysis complications.

As a result, an automated system is necessary to minimise the drawbacks brought about by such manual diagnosis errors [2, 3]. The Director-General of the World Health Organisation (WHO), Dr. Tedros Adhanom Ghebreyesus, indicated that the usage of computer-based models in the medical arena is producing accurate findings in analysing the patient situation [4]. The training of automated systems is quite good, which results in better outcomes. These better results are helpful for complex data processing, which in turn assists clinicians [5].

2. Machine Learning– An Intelligent Automated System

To analyze complex data with computer-aided models, an expert artificial system is required. Machine learning is an application of Artificial Intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed [6]. Machine learning focuses on the development of computer programs that can access and learn from data. But the primary goal is to allow computers to learn automatically without human assistance. The learning process

basically starts with data collection or observations [5]. Examples include learning to diagnose patients in the medical field, learning to drive an autonomous car, learning to identify new astronomical structures, and so on. Methods or algorithms for learning by machines are often categorized into supervised learning, unsupervised learning, semi-supervised learning and Reinforcement learning. The classification of disease-affected patients is the primary focus of this work, which uses supervised learning. Supervised learning is also known as classification, in which predictions can be done based on the labeled examples in the dataset [7].

3. TYPES OF DATASETS-by nature

By nature, datasets can be typically categorized into two types: balanced and imbalanced. Balanced Datasets are those whose class samples ratio is equally proportionate to each other, whereas, in imbalanced datasets, one class sample ratio dominates the other class sample ratio *i.e.*, the class samples ratio are not proportionate to each other [8].

3.1. Balanced Datasets

Balanced datasets are those whose class samples ratio is equally proportionate to each other. A well-balanced dataset is suitable for the classifiers to perform classification efficiently.

Example: Consider the aircraft dataset as shown in the Table 1.

Table. 1 **Aircraft classification dataset.**

Wing type	Engines	Nose	Intake	Fuselage	Class
Mid	Low	Flat	Nose	Cigar	Foreign
Mid	Low	Flat	Nose	Sleek	Foreign
Low	Low	Snub	Nose	Sleek	Foreign
High	Medium	Point	Body	Thick	Domestic
High	low	point	Body	Thick	Domestic
Low	Medium	Point	Body	Sleek	Domestic

From Table 1, the aircraft dataset has of 6 tuples, which are defined with attributes-wing type, engines, nose, intake and fuselage. The class label for the dataset is defined with two values – foreign and domestic. From the dataset, 3 tuples belong to one class called “foreign” and the remaining 3 tuples are belongs to the “domestic” class. The ratio of Examples of both classes is equally proportionate to each other. Hence, the dataset is called a balanced dataset.

3.2. Imbalanced Datasets

Datasets in which one class sample ratio dominates the other class sample ratio is called imbalanced dataset [9].

Example: Let us consider the Website Phishing dataset, which consists of 10 parameters as shown below in Table 2.

Table 2 shows 9 independent attributes and one dependent attribute, *i.e.*, class label – “result”. Out of the given 10 rows of the dataset, 9 rows belong to the result class ‘0’ called suspicious, and only one row belongs to result class ‘1’ called legitimate class. Hence, the ratio of tuples of both classes is not equally proportionate to each other, *i.e.*, the result class ‘0’ tuples dominate the tuples belonging to result class ‘1’. Such a kind of dataset is called an imbalanced dataset.

Table 2 **Phishing dataset.**

Server Form Handler	Pop up Window	SSL Final State	Request URL	URL Anchor	Web_traffic	URL Length	Age_of Domain	Having ip Address	Result
1	-1	1	1	1	1	1	1	0	0
-1	-1	-1	-1	-1	0	1	1	1	1
1	-1	0	0	-1	0	-1	1	0	0
1	0	1	-1	-1	0	1	1	0	0
0	-1	1	1	1	0	1	-1	0	0
0	0	1	1	-1	0	1	1	0	0
1	0	1	1	1	0	1	-1	0	0
-1	-1	1	1	1	0	1	1	0	0
1	1	1	1	1	0	1	-1	0	0
-1	-1	0	0	-1	0	-1	1	0	0

A well-balanced dataset is suitable for the classifiers to perform classification efficiently [10]. On the other hand, imbalanced datasets suffer from several issues due to the unequal distribution of samples in a dataset, which is discussed in the next section.

4. ISSUES WITH IMBALANCED DATASETS

The imbalanced nature of the datasets is prone to face the following issues.

4.1. Class Imbalance Problem

Because of the highly skewed distribution nature of samples in imbalanced datasets, minority-class samples are dominated by majority-class samples. Such an imbalance proportionate ratio between the classes is called a class imbalance problem [11].

4.2. Classifiers Learning On Imbalanced Datasets

With imbalanced datasets training, the classifiers are a challenge. Classifiers are well-trained with majority class samples only but not with minority class samples. Hence, classifiers tend to produce wrong results by biasing towards the majority class sample and ignoring the minority class sample [12].

4.3. Taxonomy of Various Techniques on Imbalanced Datasets

In this chapter, historical reviews on modeling imbalanced datasets that can take part by various researchers are presented. The survey explores the research direction on the class imbalance problem and various researchers' contributions made by classification of Imbalanced datasets in different applications is explained in further sections [13, 14]. The taxonomy of the survey is shown in Fig. (1).

Consider the research direction to learn with imbalanced datasets as suggested by various authors as described below:

From (Fig. 1), initially, imbalanced datasets are examined with conventional machine learning and data mining algorithms. Due to the problems with these algorithms, later research has been direction of handling imbalanced datasets at data and algorithmic levels [15].

5. Application of Conventional Data Mining & Machine Learning Techniques for Handling Class Imbalance Problem

This section presents research contributions made by various authors on imbalanced dataset handling methods using data-mining and machine-learning algorithms. It proves that due to the biasing problem, only classifiers are not producing satisfactory results [16, 17], as stated below.

N.Poolswa Detal [18] has experimented on the LIFELAB dataset, which was a clinically collected imbalanced dataset. Authors found that almost all conventional data-mining & machine learning algorithms have a strong bias towards the majority class and are subject to error rates indicating poor recall.

Yazan F. Roumani *et al.* [15] collected ICU–intensive care unit dataset described with various parameters, namely–age, time of visiting hours, gender, discharge status, complications and at least two visits to ICU and conducted a comparison study between performance of various classifiers namely Classification and regression tree (CART), C5, logistic regression, discriminant analysis and Support Vector Machine (SVM) algorithms to accurate identification of patient status. It was observed that logistic regression shows the best performance.

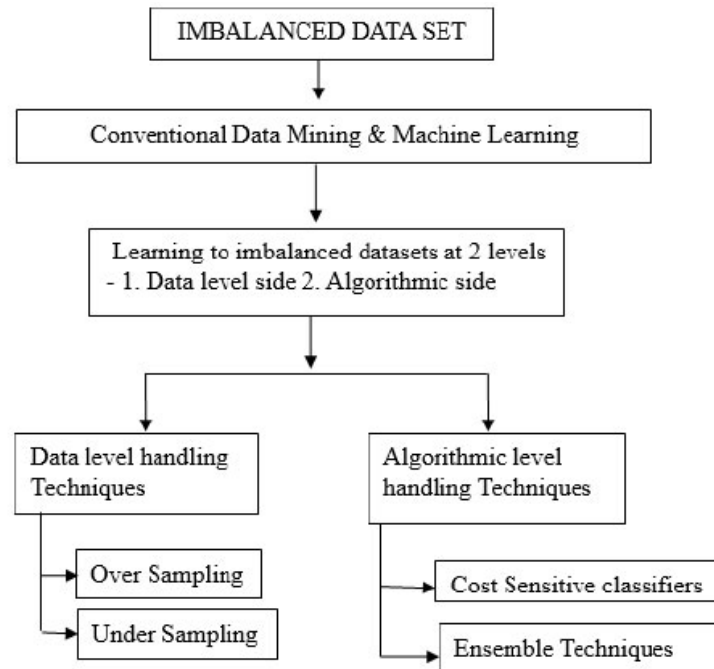


Fig. (1))

Taxonomy of literature survey on imbalanced datasets.

Jia Pengfei [20] also reiterated that traditional classifiers always show poor performance in high-density imbalanced datasets. The authors proposed a new sampling algorithm, namely distinct-border line sampling, for handling high-density imbalanced datasets. Experiments are conducted on the sat image, user, segment, transfusion, breast and Australian datasets that are collected from The University of California, Irvine (UCI) repository and proven that distinct borderline sampling algorithm shows good results in accuracy.

But, the main drawback of traditional data mining and machine learning algorithms (linear and nonlinear) are stating unsatisfactory performance in classifying imbalanced datasets [22].

6. Application of Data Level Methods for Handling Class imbalance problem

To overcome the biasing problem of conventional machine-learning & data-mining algorithms in order to balance the classifiers accuracy levels on both majority and minority classes, researchers applied data-level methods to handle skewed datasets [23, 24].

Typically, there are two data-level approaches for handling imbalanced data sets, namely oversampling and under-sampling [25, 26]. The over-sampling approach is defined as a method of balancing the minority class examples with majority class

examples [27] whereas in the under-sampling method majority class examples are reduced to balance with minority class examples [28].

6.1. Undersampling

Few researchers applied under sampling methods for treating the imbalanced datasets as stated below:

M. Mostafizur Rahman *et al.* [10] explained that the class imbalance problem in the medical domain. Applied cluster based under sampling method and over sampling methods on cardiovascular datasets. Datasets are collected from Hull and Dundee clinical sites. Conducted comparison between cluster-based under-sampling with other over-sampling techniques like Synthetic Minority Over Sampling Technique (SMOTE) and concluded that the proposed cluster-based under-sampling method is best for generating training sets during the classification process. It was suggested that the proposed method overcomes the class imbalance problem in various domains including medicine.

Show-Jane Yen *et al.* [29] developed a cluster based under sampling approach to improve the classifiers' accuracy levels in the classification of imbalanced datasets especially in the case of minority class examples. They conducted a comparison study between proposed algorithms with other under-sampling techniques on synthetic datasets and examined the dis-ordering of sample distribution on imbalanced datasets.

Yuchun Tang *et al.* [30] developed a granular support vector machine–repetitive under-sampling (GSVM-RU) model based on the “granular computing-based learning” framework. Some examples of this are learning how to diagnose patients in the medical field, drive a self-driving car, recognise new structures in the sky, and so on.

Chris Seiffert *et al.* [31] proposed a Random Under Sampling Boost-RUS Boost algorithm for solving the class imbalance problem. They conducted experimental studies on 15 datasets collected from UCI repository, Nortel networks and National Aeronautics and space administration metric data program websites. They suggested that the RUS Boost algorithm is faster and good when compared with SMOTE Boost algorithm.

Whereas Wing W.Y. Ng *et al.* [32] designed an approach of diversified sensitivity under-sampling technique (DSUS), which overcame the drawbacks of a random under-sampling approach. They examined the 14 imbalanced datasets classification results with Radial Basis Function Network with a diversified sensitivity-based under-sampling approach in which most class samples are clustered together. Using stochastic sensitivity measures under-sampling has been done to balance the class distribution.

Marcelo Beckmann *et al.* [33] suggested an under-sampling approach using k-Nearest Neighbor to handle imbalanced datasets. They collected 33 imbalanced datasets from

the UCI repository and conducted a comparative study between C4.5, SMOTE with C4.5, EditedNearest Neighbor Rule (ENN) with C4.5, Neighbor Cleaning rule (NCL), Random Under sampling with C4.5, K-Nearest Neighbor (KNN)-Undersampling with C4.5 and KNN-Under sampling with 1-NN classifiers. It has been demonstrated that KNN-under sampling with 1-NN produces satisfactory results when applied to classification reports.

Xiaoheng Deng *et al.* [34] designed Automatic Clustering and Under Sampling(ACUS) method, which reduces the loss of data and decreases the complexity of the under-sampling technique on the classification of imbalanced datasets. An experimental study was conducted on 22 KEEL datasets with multi-labeled values and evaluated the performance of various clustering methodologies along with the proposed ACUS. The presented algorithms have demonstrated successful results in terms of accuracy, level-high simplicity, and stability.

The main drawback of under-sampling technique is data losing [35-37]. Instead of data losing, *i.e.*, applying the under-sampling technique, research has taken into the direction of over-sampling the minority class examples [36-38].

6.2. Oversampling

Without losing the data and without changing the statistical properties of the imbalanced datasets, over-sampling techniques are invented. The synthetic minority over-sampling technique (SMOTE) is a popular method that is used for over-sampling [39, 40]. Few researchers has applied SMOTE technique and few authors are used SMOTE with other algorithms to handle imbalanced datasets, as described below:

Enislay Ramentol *et al.* [41] investigated the performance of the over-sampling method on rough set theory. Benchmark datasets are collected from the KEEL database and then applied SMOTE, later examine the results with C4. Proven that decision tree can handle rough set theory.

V.Garcia *et al.* [42] experimented with 17 highly imbalanced real datasets by applying both under-sampling and over-sampling techniques. Various classifiers, namely Multi-Layer Perceptron-MLP, Support Vector Machine (SVM), Naïve Bayesian classifier (NBC), Decision tree -J48, Radial Basis Function (RBF), 1-Nearest Neighbor(1-NN), 7-Nearest Neighbor (7-NN), 13- Nearest Neighbor (13-NN) are applied by using weka toolkit with default parameter settings. Classifiers performance is evaluated with metrics like True positive rate, True negative rate, index of balanced accuracy and Geometric mean. Proven that over-sampling techniques are better than under-sampling techniques. Based on Multi-Dimensional Scaling (MDS) with respect to the geometric mean, the authors are suggested that SMOTE and Gabriel-graph-based SMOTE (gg-SMOTE) show equal performance, but with respect to the Index of Balanced Accuracy (IBA) metric SMOTE shows good performance among all the comparison techniques.

Rok Blagus *et al.* [43] investigated the SMOTE technique, conducted theoretical and empirical studies on Sotiriou, Pittman and Miller datasets and concluded that SMOTE efficiently handles low dimensional data when compared with a high dimensional dataset.

Gregory Ditzler *et al.* [44] designed a new concept drift learning algorithm for handling imbalanced datasets. They improved the Non-Stationary Environment algorithm (NSE algorithm) with the replacement of SMOTE using a sub-ensemble method to balance the class distribution.

Jia Pengfei *et al.* [20] proposed a borderline over-sampling method (borderline SMOTE) for the classification of imbalanced datasets with high density. A comparison study of the random oversampling technique, the SMOTE technique, and the Borderline-SMOTE technique was carried out. It was proved that Border-

line-SMOTE had shown the best accuracy performance on the classification of imbalanced datasets.

Li Ma *et al.* [45] designed a step-wise methodology to handle imbalanced datasets. Applied Clustering Using Representatives (CURE) algorithm with SMOTE to overcome the drawbacks of various sampling strategies like safe level-SMOTE, Borderline-SMOTE, C-SMOTE and k-means - SMOTE. Learning on 5 imbalanced datasets has done with feature selection of the Random Forest algorithm and measured the classifier performance with various metrics like accuracy, sensitivity, specificity and precision.

Jiaojiao Li *et al.* [46] proposed a model for the classification of hyper-spectral images in remote sensing applications. The methodology defines a step-wise procedure in which the selection of majority class samples can be done by applying Orthogonal Complement Subspace Projection (OCSP) and applied oversampling on the instances of the minority class. They conducted a comparative study between SMOTE and OCSP algorithms and among them that OCSP model is best to handle the classification of hyperspectral images.

Amritanshu Agrawal *et al.* [47] suggested SMOTUNED - an auto-tuning version of SMOTE for finding software defects. The proposed algorithm is an improved version of SMOTE, designed to balance the class distributions and increase the sample ratio. Experimental results are conducted on different datasets collected from the SEACRAFT repository, where all datasets belong to open-source JAVA object-oriented systems.

Rashmi Dubey *et al.* [48] conducted a study on Alzheimer's disease. Applied data oversampling and under-sampling techniques were to balance the class distribution, and then a comparative study has done with the classifiers Random Forest and support vector machine based on majority Voting concluded that SVM majority voting showed the best performance.

The authors concluded that over-sampling methods are best for handling imbalanced datasets when compared with the performance of under-sampling techniques [49, 50].

However, these over-sampling and under-sampling techniques have certain drawbacks. Loss of data is a major issue in the case of the under-sampling technique whereas over-sampling techniques are more inclined towards overfitting.

7. Application of Algorithmic level methods for Handling Class imbalance problem

It was clearly observed that the skewed distribution nature of imbalanced datasets is not changed either by over-sampling or under-sampling. Hence, research on solving the class imbalance problem took another direction, *i.e.*, concentrating on algorithmic-level methods rather than data-level methods [51-53]. Researchers proposed different algorithmic level methods such as Cost sensitive classifiers [54-58] and ensembles [59, 60].

7.1. Cost-Sensitive Classifiers

Gang-Song Xiao *et al.* [61] developed a graph model-based cost-sensitive classifier called Graph Classification Based on Cost Sensitivity (GCBC) algorithm. They conducted a comparative study between the computational complexity of GCBC with Graph Boost Algorithm (gboost) on various graph datasets which are collected from carcinogenic potency database, and proved that GCBC algorithm is efficient in handling graph-based imbalanced datasets.

Salma Jamal *et al.* [16] developed a predictive model of anti-malarial molecules inhibiting apicoplast formation for imbalanced malaria disease dataset using Cost sensitive Naive Bayesian, Cost sensitive Random Forest and Meta Cost J48 algorithms. Found that Cost sensitive Random Forest algorithm is the best predictive model to classify imbalanced malaria datasets.

The main drawback of cost-sensitive classifiers is these classifiers suffer from high misclassification cost rates during classification [62, 63]. Hence there is a need to improve the accuracy levels of classifiers in the classification of imbalanced datasets [64].

7.2. Ensemble Techniques

To improve the classification accuracy levels on imbalanced datasets, ensemble techniques have been introduced. Ensembles are a group of classifiers that work together [60].

Voting is a basic ensemble technique in which predictions are made based on grouping the decisions from all the classifiers with majority voted value. B. Romera-

Paredes *et al.* [65] applied the Voting technique to accelerometric and gyroscopic data. Authors experimentally demonstrated that groups of classifiers (ensemble) are performed better when compared to single classifiers. However, voting techniques failed to classify the imbalanced datasets as the base classifiers are traditional data-mining and machine learning algorithms.

Later, research was driven in the direction of training the classifiers on imbalanced datasets. For this, boosting and bagging ensemble techniques are found to be appropriate to train the classifiers [12].

Taghi M. Khoshgoftaar *et al.* [59] examined the performance of boosting and bagging ensemble technique's by conducting experiments on 4 different imbalanced datasets collected from the UCI repository. They conducted a comparison between ensemble techniques with the noise values and concluded that the bagging technique is the best ensemble.

Mikel Galar *et al.* [60] reviewed ensemble techniques on bagging, boosting and hybrid approaches on 44 datasets collected from the KEEL dataset repository to solve the class imbalance problem of non-ensemble algorithms like C4.5, SMOTE with C4.5, Classics ensembles, Cost-sensitive Boosting Ensembles, Boosting-based Ensembles, and Bagging-based Ensembles. Hybrid ensembles concluded that Bagging techniques show good results in handling the class imbalance problem on different datasets.

Yanmin Sun *et al.* [66] suggested that various application domains suffering from class imbalance problems are stated. Class imbalance learning problems of various standard classifiers like decision tree-C4.5, Back Propagation Neural Network, Naïve Bayesian classifier, Support Vector Machine, Association classifier and K-NN are described. They started the evaluation metrics F-measure and G-mean for measuring the performance of the classifier derived from the confusion matrix. They are suggested solutions to handle imbalanced datasets at the data level, algorithmic level, cost-sensitive and ensemble learning techniques, which are a little typical. Also, they define the advantages and disadvantages of learners and design a strategy for handling multi-class imbalanced datasets.

Phan Thanh Noi *et al.* [67] conducted a comparative study between the Random Forest, k-Nearest Neighbor classifiers and support vector machine classifiers' performance on Sentinel-2 multispectral images. Applied bootstrap sample technique to learn the imbalanced images and then analyzed classifiers' performance using performance metrics like accuracy etc. Experimentation was done on the performance of the classifier using datasets of spectral pictures that were both balanced and imbalanced.

Shuo Wang *et al.* [68] developed a Dynamic AdaBoost. NC algorithm for detection of software defects. They collected 11 different types of software defect prediction datasets which are available from the public PROMISE repository, conducted a comparison study between the developed model and RUS, Balanced Version of

Random Under Sampling (RUS-bal), AdaBoost, Negative Correlation (AdaBoost.NC), SMOTE Boost, SMOTE, Threshold-moving, Naïve Bayesian and Random Forest algorithms. Classifiers performance is measured with balance, G-mean and Accuracy metrics and proven that proposed dynamic AdaBoost NC shows good classification results.

Ewan Cameron *et al.* [69] investigated forecasting study on Plasmodium Falciparum parasite influence using the ensemble method Bayesian statistical procedure by combining functional regression-based model emulation with Markov Chain Monte Carlo sampling technique suggested that the prevalence of P. Falciparum parasite is more.

Daniel Ruiz *et al.* [70] investigated the records of temperature, rainfall etc. and suggested an ensemble malaria model for assessment of the long-term impact of climate conditions.

Tilman Gneiting *et al.* [71] suggested that ensemble methods are best for forecasting the weather conditions which cause to finding infectious diseases like malaria. Experiments are conducted on nearly 100 different computer runs, each defined with initial weather forecasting conditions. Apart from these boosting and bagging techniques, few researchers used a hybrid approach to classifiers [72-81].

Yale Song *et al.* [82] proposed a distribution-sensitive prior learning algorithm for balancing the class distribution of imbalanced gesture datasets which are available from Microsoft Research Cambridge-12 (MSRC-12) and Naval Air Training and Operating Procedures Standardization (NATOPS) aircraft handling signals conducted a comparative study on a proposed algorithm with under and oversampling techniques. Experimental results prove that the proposed model solves the imbalance problem of gesture datasets and is also superior to Hidden Conditional Random Field (HCRF) algorithm.

L'aszl'o A. Jeni *et al.* [83] conducted experiments on major databases Cohn-Kanade Extended Facial Expression (CK+) database, RU-FACS and UNBC-McMaster pain archive to recognize the facial action units applied simulated classifiers and binary SVM's for recognition of facial action units. Several performance metrics like F1-score, Kappa, Alpha, Accuracy and Area under the Precision-Recall curve (AUC-PR) are applied on composed databases concluded that Receiver Operating Characteristic Curve (ROC Curve) was not affected by imbalanced whereas precision-recall curves are affected by skew.

Yilin Yan *et al.* [84] developed a deep learning model using a Convolutional neural network with the integration of bootstrapping technique for the classification of multimedia datasets. The training dataset is divided into majority and minority class samples generated new sample by adding a bootstrap sample of the minority class to train the neural network, showing the best results on the highly imbalanced Text REtrieval Conference Video Retrieval Evaluation (TRECVID) dataset.

Shoujin Wang *et al.* [85] proposed deep neural network for the classification of imbalanced datasets. The proposed neural network was designed with Mean False Error (MFE) lossless function and its improved version Mean Squared False Error (MSFE), to overcome the bias-variance of samples. The performance of the proposed deep learning model is measured with Accuracy and F-measure metrics and validated on collected 8 imbalanced datasets, out of which 3 are image datasets, and 5 are document datasets that are gathered from Newsgroup dataset.

Chunkai Zhang *et al.* [86] developed a new classification model based on minimizing a loss function. They examined the results of collecting 8 data sets and proved that the proposed method is best for handling the imbalanced datasets.

Salman H. Khan *et al.* [87] developed a model based on Cost-Sensitive Learning (CoSen) for the selection of features of deep neural networks. Examine the performance of CoSen – Convolutional Neural Network (CoSen – CNN) methodology on various datasets with a comparative study with Hierarchical – KNN, Hierarchical – Bayes, Flat –KNN and Baseline CNN.

Kevin A Mc Carthy *et al.* [88] proposed Epidemic logical MODeling software (EMOD) - parameter ensemble model for estimating the pre-erythrocyte vaccine. The model helps to find the causes to reduce malaria disease impact by approximately 10-40%.

Dave A Macleod *et al.* [89] demonstrated a dynamic malaria model for forecasting the severity and timing of seasonal conditions. They said that seasonal conditions affect the prevalence of vector-borne diseases.

Li Peng *et al.* [90] collected shuttle, abalone, yeast and churn datasets from the UCI repository and proposed AdaBoost – Support vector machine classifier shows good results when compared with Support vector machine classification performance on imbalanced datasets.

Cyril Caminade *et al.* [91] suggested that climate change variations can cause vector-borne diseases like malaria that have a highly infectious effect on the global population.

Raquel M. Goncalves *et al.* [92] observed that the Plasmodium vivax parasite has more effect on human beings when compared with the effect of the Plasmodium

falciparum parasite by conducting Kruskal-Wallis test (chi-square test) between cytokine levels of both parasites.

Thomas Smith *et al.* [93] proposed an ensemble model which consists of 14 individual stochastic simulation models to predict the impact of the malaria vaccine on different age people.

Bruno B Andrade *et al.* [94] conducted clinical experiments and suggested that the severest age of malaria disease causes the reduction of inflammatory cytokines,

which is a high-level imbalance class problem in the medical domain.

Victoria Lopez *et al.* [95] experimented with various imbalanced learning algorithms, namely data preprocessing, cost-sensitive and ensemble learning for classification of imbalanced datasets. Collected 66 imbalanced datasets from KEEL repository, C4.5 decision tree, Support vector machine and k-Nearest Neighbor classifiers are used as a set of base classifiers for all the experiments. Applied SMOTE, SMOTE with ENN, Border-SMOTE, Safe Level-SMOTE, Adaptive Synthetic Sampling(ADASYN), SPIDER2 and DensityBased-SMOTE techniques as data preprocessing techniques. Of these re-sampling techniques, SMOTE and SMOTE with ENN show the best results. With respect to Cost-sensitive learning algorithms, the authors applied Cost Sensitive – Weighted, Metacost, and Cost Sensitive-Classifiers. Accuracy levels are tested on these classifiers; Metacost- C4.5 is also shown as best among all. Compare the performance of cost-sensitive classifiers with the Shaffer test to detect the significant differences between the results. Again Metacost- J48 stood best. For the same base classifiers, authors have conducted experiments with ensemble techniques using AdaBoost Multi Class (AdaBoost.M1), Ada Boost with costs outside the exponent (AdaC2), Random under Sampling Boost – RUSBoost, SMOTE Bagging (SBAG) and Easy Ensemble. SBAG shows the best performance with base classifiers. After applying the Shaffer test on ensemble techniques Boosting techniques are shown outstanding performance.

Aida Ali *et al.* [17] suggested that SMOTE with an improved SVM and k-NN algorithm are used for data-level approaches and suggested the usage of one-class learning, cost-sensitive learning, ensemble methods – AdaBoost, bagging and Random Forest techniques, hybrid approaches to solve the class imbalance problem analyzed the advantages and disadvantages of each methodology.

Guo Haixiang *et al.* [18] listed the published papers on imbalanced datasets in top journals. Along with the word cloud of imbalanced datasets, the authors state the basic strategies behind the algorithms, *i.e.*, over-sampling is described with cluster-based, distance-based, evolutionary-based and other novel methods. Under-sampling algorithm is defined with cluster-based, distance-based, and evolutionary-based methods, whereas hybrid sampling has defined with both under-sampling and over-sampling algorithms. A summary of feature selection methods, cost-sensitive methods, iterative-based, parallel based and based classifier choice ensemble methods are presented in their survey. Also, it tabulated various strategies for handling the multiclass imbalanced algorithms and stated the various performance evaluation metrics, and applications of imbalanced datasets in various domains are defined.

Bartosz Krawczyk [12] presented various challenges and problems of many real-world applications like text mining, cancer detection, behavior analysis, sentiment analysis, etc., suffering from handling different imbalanced datasets.

Shaza M. Abd El-Rahman *et al.* [19] conducted a survey on class imbalance problems in machine learning and data mining fields. And also suggested various solutions like

sampling methods, cost-sensitive based methods, ensemble technique's and recognition-based methods for handling the imbalanced datasets.

Based on the survey, researchers concluded that in AdaBoost, the error of each classifier is carry forwarded to the next classifier during its training phase, which leads to more computations, whereas the Random Forest ensemble technique uses a replacement strategy for the selection of tuples in the process of classifier training [96, 97] hence, there is scope for improving AdaBoost and Random Forest classifiers in terms of tuples selection and attribute selection.

CONCLUSION

Handling imbalanced datasets is the barrier that needs to be overcome to address this issue; initially, traditional machine learning and data mining methods are applied to handle uneven datasets. However, classifiers gain a greater amount of experience when trained with a sample from the majority class, while they gain less experience when trained with a sample from a minority class. Hence, because of the biasing problem, the classifiers produce errors during classification. At the next level, imbalanced datasets are handled from the side of the dataset. In this context, under sampling and over sampling methods were designed. These methods suffer from data loss and over-fitting issues. To resolve this, research has been shifted to another direction, *i.e.*, learning on imbalanced datasets at the algorithmic level instead of the data level. Data-level, also cost-sensitive methods were implemented. Experimental results indicated that the misclassification rate in algorithmic-level methods is high. Hence, neither cost-sensitive methods nor data-level methods performance are not superior in the case of imbalanced datasets. In order to fix the issue, a group of classifiers, *i.e.*, ensemble techniques, have been proposed to classify imbalanced datasets. However, ensemble technique such as voting uses a set of base classifiers. As these base classifiers are conventional ML algorithms, they were unable to effectively classify minority class samples. Hence, we can conclude that data-level methods are necessary in order to learn the skewed datasets. Boosting and bagging are the techniques that support these cost-sensitive classifiers, and voting ensemble techniques have not been found to work effectively on imbalanced datasets. Therefore, research was directed towards training classifiers.

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AI for Crop Improvement

S.V. Vasantha^{1, *}

¹ CSE Department, Vardhaman College of Engineering, Hyderabad, India

Abstract

The introduction of high-performance genomic technologies into plant science has resulted in the generation of huge volumes of genomic information. Moreover, for biologists to deal with such complex, voluminous data and infer some significant findings in order to improve crop quality and quantity has presented a big challenge to them. The advent of Artificial Intelligence (AI), Machine learning (ML) and Deep Learning (DL), facilitated automated tools for more efficient and better analysis of the data. Another crucial process that needs to be automated in field farming is the timely and precise diagnosis of crop diseases which plays a vital role in the prevention of productivity loss and reduced quantity of agricultural products. ML provides a solution to solve these problems by automatic field crop inspection. Recently, DL techniques have been widely applied for processing images to obtain enhanced accuracy. This chapter describes the need of AI in Agri-Genomics; it also includes various contemporary AI solutions for the Crop Improvement process and presents the proposed AI-based Crop Improvement Model (AI-CIM).

Keywords: Plants, Genomics, Crop improvement, Artificial intelligence, Machine learning, Deep learning.

* **Corresponding author S.V. Vasantha:** CSE Department, Vardhaman College of Engineering, Hyderabad, India;
E-mail: s.v.vasantha@gmail.com

1. INTRODUCTION

Engineering the crop for the benefit of humankind, is as aged as the agricultural science itself. Around decamillennium ago, our ancestors made a shift from foraging or hunting to the cultivation of crops for their survival. Due to this shift, the continuous program on the improvement of crops that serves as a staple food has begun.

To meet up the demands of an ever-increasing population, dynamically changing environmental factors, and scarce resources for agriculture, crop improvement has become inevitable for

humankind [1]. Therefore, the goal of plant science is to boost the productivity and quality of the crop and, at the same time, to maintain

the ecological system for sustainable agriculture [2]. Majorly, two methods are being practiced for improving the crops, according to Lawrence, a plant biologist at Harvard University. The first method is Selection, which is based on the plant's genetic sequence variation. Early farmers selected plants with beneficial traits, such as one that bore large-sized fruits or one that was easy to harvest. This selection of plants or seeds is used for subsequent year's crop cultivation. The traditional method of natural selection is being strengthened by artificial selection.

The second method is breeding, which is employed to develop new varieties of plants, usually called cultivars, that have desired traits by crossing the selected parent plants. It involves the production of progeny that possibly have traits of both the parent plants after selecting each parent with a preferred trait. But this process was a miss or hit for early farmers due to the lack of awareness of genetic propagation of traits and even being unable to predict the outcome traits of a specific cross until 1900. Mendel, a renowned biologist, has thrown light on plant inheritance and deduced that every single trait, such as colour, is taken from the two parent genes, one from each parent [3].

Plant Breeders select plants with desired traits or characteristics, for example, plants with greater yield and resistance to major diseases or better-adapting capability to varied environments, based on performance statistics, plant pedigree and genetic information. Traditional breeding techniques may take around ten years to produce a new variety of crops from the same kind of species [1]. With the advent of Genomics and Genetic Engineering along with AI techniques in agriculture, crops are drastically improved, and several different cultivars are being produced in various species.

2. GENOMICS FOR AGRICULTURE

Application of Genomics knowledge to the field of agriculture is coined as Agri-Genomics as depicted in functions and respondig. (Fig. 1), which is one of new emerging fields in plant science with an objective to increase the productivity as well as sustainability of the crops and livestock.

Involvement of Plant Genomics in agriculture has a significant role in the improvement of crops over the conventional breeding process [4]. Genomics technologies have contributed significantly to modern agriculture by forming the basis for a better interpretation of how a plant functions and respond to the prevailing environment and attaining the desired objectives in breeding programs to improve the quality, disease resistance and productivity of the crops. Genetic markers related to key traits have enabled knowledge-based breeding, which has reduced the traditionally time-consuming process of an extensive selection of plants from the field [5].

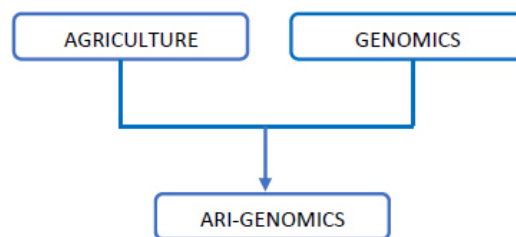


Fig. (1))
Agriculture and Genomics into Agri-Genomics.

Usage of marker-assisted recurrent selection technique for maize crop, a 3% yield raise per cycle under the water stress circumstances was reported [6]. For legumes, authors [7] reported a 26.5kg /ha/year genetic gain by examining the historic data over 80 years of soybean farming.

Genome editing technologies, such as CRISPR/Cas, permit the intended mutation of nearly any plant genome sequence producing a novel variant with expected traits of interest [8]. Genomic selection (GS) presents the opportunity to boost staple crop production in reduced time with intended traits. The main task of GS in Genomics deals with the prediction of Breeding-Values Estimated from the Genome, *i.e.*, GEBVs for each sample available with its genotype (Genes) data [9, 10].

3. AI FOR AGRICULTURE

Modern-day agriculture employs the usage of Argi-Genomics along with Artificial Intelligence (AI) techniques to automate the processing of huge and complex genomic data generated for the betterment of the crop yield, which is depicted in Fig. (2).

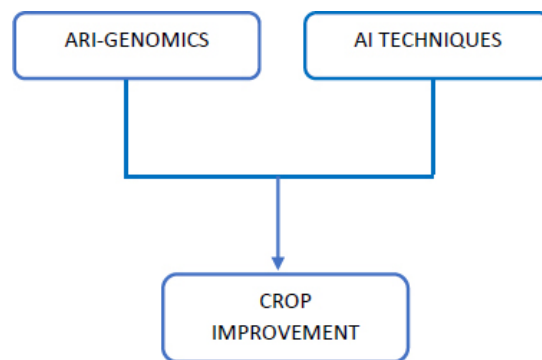


Fig. (2))
Modern Crop Improvement Process.

The availability of high-performance genomic tools and technologies has build-up with massive volumes of genomic information. Consequently, biologists are confronted with how to efficiently analyse this huge data. The advent of AI in Agri-Genomics provides Machine Learning (ML) solutions for enhanced and more effective analysis of this data [9].

ML methods are applied for GS to predict breeding values accurately in a reduced time. The AI model is usually trained from training populations with genotype and corresponding phenotype values to obtain GEBV of an individual of the testing population, which includes only genotypic data [10, 11].

Statistical methods of ML are trained to find out useful patterns in very large and complex datasets. These methods are applied for analyzing the interactions of plant–pathogens in diversified application areas, such as monitoring of plants for infections/diseases, GS-based prediction for disease resistance and pathogen effectors, *etc* [12].

Some of the specific supervised learning model applications include the prediction of regulatory/non-regulatory sections in the maize genome, prediction of mRNA expression intensity, rice plant sequence tagging, stress phenol typing, prediction of polyadenylation spot in *Arabidopsis thaliana* and prediction of nutrient deficiency in tomato. Several ML models are designed and used for the recognition/classification of plant diseases however, subsequent to the advancements in ML *i.e.*, Deep Learning (DL), they have exhibited greater potential in terms

of improved accuracy. DL methods such as fine-tuned CNN models are trained to predict the disease with which the plant is infected or the classification of different kinds of diseases the crop is infected with. DL is applied over different areas that involve large datasets for its analysis to resolve time-consuming and complicated biological problems, such as proteomics, genomics, metabolomics, transcriptomics, and systems biology [13-16].

4. AI TECHNIQUES FOR CROP IMPROVEMENT

This section details the current research work done for crop improvement aided with various sophisticated AI techniques, and also it presents details of some of the current and out-performing ML/DL solutions designed by various contemporary researchers in this field.

An ML approach is designed for prescriptive breeding in [17]; the authors discovered the ability to combine high dimensional phenotype data with an ML method to facilitate tools for plant breeders to perform in-season prediction of seed yield and prescriptive cultivar growth for desired agro-management process, for example, seeding density. Phenotypic data fields such as chlorophyll content, light interception, canopy temperature, and hyperspectral reflectance are collected from a set of sensors during different stages of growth, and SY is found from the machine harvest. Random forest algorithm (RF) with Recursive Feature Elimination (RFE) is employed to train the Seed Yield prediction models based on the phenotypic variables. The observed balanced accuracy ranges from 72% to 88%.

Ensemble ML models were proposed to improve the phenological discovery accuracy of rice plants in the paper [18]. Three kinds of ensemble ML models, first is hard or majority voting, second is soft or weighted majority voting, and last is model stacking, are proposed to detect the main phenological phases in rice plant breeding experiments with UAV- RGB imagery and five ML algorithms as base models, involving SVM, RF, LR, KNN and GNB were examined for the detection of phenological characteristics. The prediction results for phenology discovery is demonstrated in the proposed ensemble ML models that the proposed ensemble ML models exhibited an overall accuracy increase of 1% to 5% and, coming to F1 - Score, a mean increase of 0.2% to 7% when compared to GNB, the best base classifiers over the validation dataset. Amongst the proposed three ensemble ML models, soft-voting has shown the best performance, next to it hard-voting and followed by model stacking.

Authors [19] have designed a Statistical and ML model for detecting the Fusarium Head-Blight(FHB), a pathogen that affects the wheat crop by a major yield loss. They have revealed the association between resistance for FHB along with the biochemical and morphological traits in 20 wheat samples by applying statistical tools. Also, cluster analysis is done for morphological traits to detect analogous genotypes under infected and controlled conditions. Moreover, the SVM classification method is proposed to identify the diseased/infected plants using morphological traits.

DL is applied for the phenotyping of plant images in [20]. They have demonstrated approaches for plant phenotyping with an accuracy of more than 97% for shoot and root features recognition and localization. DL-based trait detection is proposed to detect quantitative trait loci (QTL) in root datasets. They have observed that the majority of manually recognized QTLs are also found using their designed CNN approach.

A mobile application is designed to diagnose the different kinds of plant diseases using DL in tomato plants by the authors in [21]. They have trained and implemented five different CNN models on Plant Village data, such as VGG16, ResNet v2, VGG19, Inception, ResNet50 and Xception, for tomato plant leaf images. They analyzed around 18,160 leaf images distributed

across ten group labels. Among all the models designed the ResNet50 has been demonstrated to be a highly accurate tool for prediction.

In the paper [22], the authors designed a DL-based model to detect pests and diseases in plants. They have implemented and evaluated nine different powerful approaches of DNNs for disease discovery in plants. They have employed pre-trained models for the extraction of essential features and, subsequently, to fine-tune the constructed model. The extracted features are then classified using the SVM model, ELM model and KNN model. The results demonstrate that deep feature extraction models and SVM or ELM classification models show enhanced results over transfer learning. Fc 6 layers of the Alex Net model, VGG16 model, and VGG19 model showed improved accuracy scores over other layers.

The authors in [23] have applied the ANN and Multi Linear Regression (MLR) models for predicting the Oil Content of Sesame (OCS). The six traits, including no. of capsules for each plant (NCP), plant height (PH), no. of days starting from flowering stage to its maturity (NDFM), seed weight, seed yield and length of the capsule, were considered as input features with sigmoid axon as transfer function and two hidden layers are chosen for constructing the ANN final model. Evaluation results has shown that the ANN accuracy for OCS prediction($R^2=0.86$, RMSE=0.56, and MAE=0.43) is higher than the MLR($R^2=0.67$, RMSE=0.74, and MAE = 0.55). The evaluation results have shown the ANN model as the most promising tool for OCS prediction with very good performance. According to sensitivity tests, NDFM follows NCP and PH, the main influential features in OCS prediction in the two models.

Plant diseases can lead to a major reduction in the quantity and quality of agricultural products if not diagnosed timely. To resolve this issue authors in [24] presented a DL-based method for automating the process of banana leaf disease classification. In particular, they have employed the LeNetModel as CNN for disease classification of banana leaf images. The results exhibited the approach's efficiency under challenging circumstances, such as complex backgrounds, illumination, varying resolution, pose, size, and orientation of images.

Capsicum plants are often exposed to various fungal/ virus/ bacterial diseases. Paper [25] has discussed various fungal/ bacterial diseases in capsicum plants. These disease symptoms can be differentiated by inspecting different parts of the plant, such as leaves, stem, and fruit. The presented algorithm automatically classifies capsicum plants as normal or diseased ones and recognizes the infected disease. The infected area is extracted using a k-means clustering algorithm; GLCM features are extracted. These features are used by SVM to classify different bacterial/fungal diseases of capsicum. Several different types of classifiers, such as KNN, Tree, SVM and Linear Discriminant, were used for training and testing purposes; among these classifiers, SVM and KNN exhibited superior results. This system, when tested over a small set of around 62 capsicum and its leaf images, it has classified with 100% accuracy into healthy/diseased type.

Scientific as well as commercial plant breeders need to do phenotyping of a big number of populations. Usually, phenotyping is done manually on the field, which is a more time-consuming and costly task. This task can be automated with the help of DL algorithms. A DL-based estimation method for Seed-per-pod is designed for plant breeding [26]. They have come up with a method based on a computer vision technique, that is used to estimate the count of seeds per soybean pod, which is not an easy task that necessitates the involvement of human experts. Their approach is built on the extraction of necessary features then, followed by an SVM classifier. They have also implemented a simple CNN model and achieved 50.4% and 86.2% accuracies, respectively, demonstrating the DL potential over ML in computer vision applications.

In the paper [27], the physical characteristics (phenotypes) of an individual were predicted from its genotype as well as environmental knowledge. They have considered and examined three problems: the first one is simple yeast, and the other two are rice and wheat. They have implemented ML methods, including elastic-net, support vector machines (SVM), gradient boosting machines (GBM), Random Forest (RF), and ridge and lasso regression. They also implemented a 2-step sequential linear regression method and genomic BLUP. They have discovered that standard ML methods performed more superior than conventional statistical methods of genetics. For yeast problems, GBM has performed better, next lasso regression and then genetic statistical methods. Coming to wheat and rice analysis, SVM and BLUP performed the best. RF has shown its robustness for the data with missing values and noise. ML approaches designed [28] reveal the genomic sequence sections related to brown-rust resistance in sugarcane.

The authors [29] have considered the 5 main yield traits in soybean, including the count of nodes/plant (NP), count of non-reproductive nodes/plant (NRNP), count of reproductive nodes/plant (RNP), count of pods/plant (PP), and the ratio of the count of pods to count of nodes/plant (P/N). They have implemented ML algorithms like RF, MLP and Radial Basis Function (RBF), independently and collectively using an ensemble approach based on bagging-strategy (E-B). The RBF was selected as a meta-classifier for the E-B model as it exhibited the highest value of R^2 , i.e., 0.81, with the lowest MAE and RMSE of 148.61 and 185.31 kg ha⁻¹, respectively. Employing the E-B algorithm, they have noticed a rise in prediction accuracy R^2 , MAE, and RMSE by 0.1, 0.24 kg ha⁻¹ and 0.96 kg ha⁻¹, respectively. The results have shown an improved understanding of the association between the yield of soybean and its components, which can be further useful in parental line selection and developing promising cultivars. Table 1 includes some of the most recent and out-performing solutions based on AI methods for the Crop Improvement process.

Table 1 **Some of the recent best-performing AI solutions for Crop Improvement.**

Plant Species	Findings	AI Technique/Method	Performance	Method
Maize	Prediction of yield, nitrate loss(N Loss)	Meta models:XGBoost and Random forests(RF)	XGBoost for yield prediction-RRMSE: 13.4% RF for N loss-R ² : 0.78	ML-Meta models(2019) [30]
Rice	Plant Height(PH)PhenotypePrediction:	RF Classifier	Mean Accuracy: 92%	ML-BVP(2021) [31]
Apple, Cherry, Corn, Grape, Orange, Peach, Pepper, Potato, Soybean, Tomato,	Leaf disease identification/classification	EfficientNetModel	Accuracy of B5 Model:99.91% Accuracy of B4 Model:99.97%	EfficientNetbased DL technique(2021) [32]

Plant Species	Findings	AI Technique/Method	Performance	Method
Strawberry, Raspberry				
Rice	Diseases detected:Hispa, Leaf-Blast(LB) and Brown-Spot(BS)	Faster_RCNN	Avg_Accuacy: 98.7%	DL-based faster R-CNN framework (2021) [33]
Tomato	Identification of Fusariumoxysporum(FO) Fungus	Naive Bayes method	Accuracy: 96%	Flawless FO identification using ML(2020) [34]
Rice	Diseases classified: Brown Spot(BS), Bacterial Leaf Blight(BLB) and Leaf Smut(LS)	AlexNet neural network	Accuracy: 99%	Deep transfer learning by AlexNet(2020) [35]

5. AI-BASED CROP IMPROVEMENT MODEL (AI-CIM)

5.1. Automation of Modern Crop Improvement Process

To meet the increasing population's basic requirement, the productivity of agriculture needs to be improved drastically through enhanced breeding programs. The present breeding programs are insufficient to withstand uncertain climatic conditions.

Multimomics form large-scale high-dimensional heterogeneous data with multilevel complex structures, such challenging datasets are processed using AI techniques for predicting significant patterns or novel interactions or associations, uncovering important biological findings. Phenomics and genomics, along with AI knowledge utilized together, can speed up the process of producing a progeny that is climate resilient, improved yield, Enhanced tolerance to various kinds of abiotic and biotic stresses, simultaneously delivering higher genetics gains in the agricultural fields in a reduced amount of time.

Existing manual field inspections for plant disease discovery is a tedious task that can be automated using AI to diagnose the plant's diseases early and analyse its severity level; accordingly, farmers can promptly plan for recovery actions. Hence there is a need for a well-designed AI model to assist farmers in overcoming challenges that are confronted during the breeding process and early disease diagnosis of crop fields.

5.2. AI Model for Enhanced Crop Breeding

The two important phases of a crop are, first is the selection of seeds for breeding the plants with desired traits and next is to monitor the health of the crop till its final stage, *i.e.*, harvesting. The proposed model uses artificial intelligence methods to improve the crop breeding program so as to overcome the time-consuming traditional breeding process and also to early diagnose the plant diseases before major yield loss occurs. This model has two main components; they are a) Automated Selective Breeding System and b) Automated Plant Health Monitoring System.

5.2.1. Automated Selective Breeding System

Usually, farmers collect plants for breeding with specific desired traits for the next generation of farming based on manual field selection. This process can be automated with the help of an image sensor-based field selection system [36] that captures images of plants from the field, analyses their health condition, and picks some top healthy plants with expected traits. The next process is to study the germplasm of selected or randomly gathered samples for the expected phenotypic values. Machine Learning algorithms can be applied to predict the phenotype values from the corresponding genotype of the samples [31, 37].

Next biotic and abiotic stress analysis is performed over the samples to pick up the samples that have stress resistance to major harmful diseases and withstand environmental effects. This process employs automated tools for finding the stress resistance-related Transcription Factors (TFs) in the germplasm of the sample. A Plant TF tool called iTAK [38] can be used to identify the genome-wide plant TFs for different kinds of stress resistance for various plant species.

Genotypes of the selected sample can be modified to reflect the desired set of traits and/or with stress resistance characteristics by performing the alteration to the sample's germplasm based on the recommendations yielded from the AI-based prediction system for estimating the best sub-population or category-specific phenotypic value for the corresponding genotype [31]. The steps involved in the proposed automated breeding program are depicted in Fig. (3).

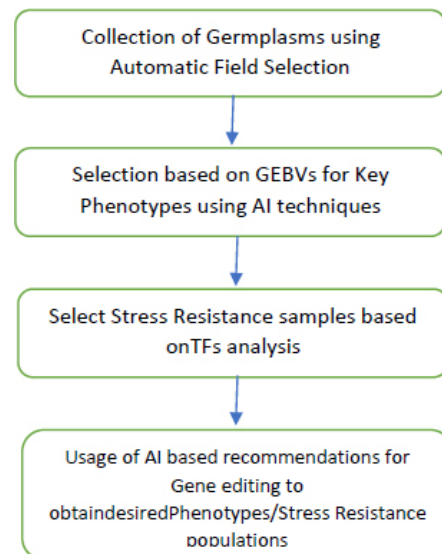


Fig. (3))
Steps of Automated Selective Breeding System.

5.2.2. Automated Plant Health Monitoring System

Timely diagnosis of plant health can be performed using an automated monitoring system which is depicted in Fig. (4). Crop health is continuously monitored by capturing and processing images of plants periodically, along with other sensing information [39]. For accurately predicting the health of the crop, a high-resolution imaging system is employed for periodic capturing of the plant images [40].

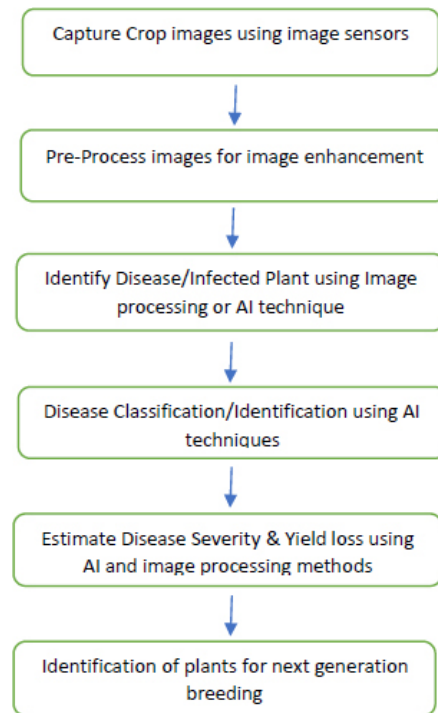


Fig. (4))
Steps of Automated Plant Health Monitoring System.

Collected plant images undergo a pre-processing phase, which includes resizing the image to a smaller scale, removal of noise using filters and conversion to HSV format for further image processing process [41]. Next, Machine Learning or image segmentation methods can be applied to identify disease-infected plants from the field at the early stage so that major yield loss will not occur [42].

Once it is identified that plants are unhealthy or infected with some diseases, then Machine/Deep Learning mechanisms can be applied to accurately diagnose the disease(s) that are infected by the plant; accordingly, the farmer can plan for remedial program that explains usage of appropriate fertilizers, pesticides etc., to recover and result in a better crop yield. The type(s) of disease the plant is infected with is identified, or disease classification is performed from the analysis of lesion, spot, colour and texture of plant by applying various popular deep-learning or transfer learning methods [13-15, 43].

Accurate and timely estimation of disease infection range or level and yield loss/gain using deep learning and/or image processing methods can help to protect the crop from major yield loss. CNN variant models have shown their potential in this area [44, 45]. Superior plant samples can be collected from a healthy plant pool.

CONCLUSION

The proposed AI-CIM automates the two main phases of the crop, *i.e.*, one is breeding, and the other is monitoring its health condition till the harvesting stage; thereby, the crop is improved in terms of quality and quantity. Also, superior cultivars can be produced, which can be used for the next-year crop. Genomic technologies are employed to automate the process of plant breeding, which creates voluminous data for analysis. For handling such data and accurately predicting valuable outcomes, such as identifying the population with the highest yield, greater grain

volume, a greater number of seeds, etc., along with pest/disease resistance, various ML/DL techniques can be applied. RF has shown the most promising results, around 92% overall accuracy, which can be employed in this phase. Coming to automation of plant health monitoring processes, various computer vision and DL techniques have shown their potential in the accurate and timely prediction of various diseases. Variants of CNN models like the faster_RCNN can be employed in this phase, as it has shown the most promising results up to 98.7% mean accuracy.

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Real-Time Object Detection and Localization for Autonomous Driving

Swathi Gowroju^{1, *}, V. Swathi¹, J. Narasimha Murthy¹, D. Sai Kamesh¹

¹ Sreyas Institute of Engineering and Technology, Hyderabad, Telangana 500068, India

Abstract

The term “object detection” refers to a technology that enables humans to recognise specific types of things present in visual media. One of the important applications of the technique is autonomous driving cars. In the application, the activity is to detect the various objects present in the single image frame. Examples of objects belonging to multiple classes are trucks, bikes, persons, cars, dogs, and cats. For this task, we use object localization and classification as we have to locate multiple objects in the image. Various techniques available in the market based on Deep Learning use inbuilt architectures such as VGG-16 and InceptionV3. Using these techniques to solve the problem is a reasonable solution but the response time from these architectures may not be feasible as the autonomous vehicles have to react in less than 0.02 milliseconds in order to avoid collisions of all sorts. So using YOLO, we simply predict the classes and the bounded coordinates of the object in a single run of the model and detect multiple objects from the image rather than focusing only on the interested regions of the image as formerly employed by

various models. YOLO is fast and accurate with the help of Convolution Neural Networks and is less likely to produce localization errors.

Keywords: Object localization, Autonomous driving, Convolution neural networks.

* **Corresponding author Swathi Gowroju:** Sreyas Institute of Engineering and Technology, Hyderabad, Telangana 500068, India; E-mail: swathigowroju@sreyas.ac.in

1. INTRODUCTION

We know that autonomous driving or self-driving cars is a topic that has been in the news lately. The integral part of the self-driving application is object detection using a camera that captures nearly 40 to 80 frames per second. Once the image is captured, meaning a frame is under process, the image should be fed to the mainframe for detection and classification, which means object localization should be done. We call the process PERCEPTION, which is simultaneously detection, prediction, localization, mapping *etc.*

There may be many challenges because the control system should receive the detection results quickly, say in milliseconds, and the process should be repeated multiple times in 5 seconds. Hence, the car adjusts its acceleration and lane system. If too many parameters are present, the system becomes complex, and the time taken to classify increases. To undo this mistake, CNN comes into the picture.

2. LITERATURE SURVEY

Gene Lewis explains the use of SimpleNet architecture [1], which is a near-real-time model. The architecture is capable of producing results at a processing rate of 11 frames per second with an mAP accuracy of 12.83% on the validation set. OverFit [2, 3] is a CNN model which performs object

detection, recognition and localization. It is the most successful model, which is eight layers deep and depends on the overlapping bounding box scheme, where it iteratively merges them to the high-probability predictions. VGG16 [3] is an attempt to work better than OverFit with a 16-layer and 19-layer architecture with functionality similar to OverFit Table 1.

Table 1 **Literature Survey.**

Author	Methodology Used	Accuracy (%)	Dataset Used	Classes Used	Time taken for Prediction
S. Gidaris <i>et al.</i> [4]	MR_CNN_MORE_DATA	93.4	Own dataset with 5000 images	20	0.05 sec
R. Girshick <i>et al.</i> [5]	R-CNN VGG BB	91.4	Own dataset of images	20	0.053 sec
J. Dong <i>et al.</i> [6]	BabyLearning	91.7	Own datasets	25	-
S. Gidaris <i>et al.</i> [7]	MR CNN S CNN	90.4	Own dataset	30	-
S. Ren [8]	NoC	83.3	Own dataset	50	0.043 sec
R. B. Girshick <i>et al.</i> [9]	R-CNN BB	73.3	Own dataset	40	-
B. Hariharan <i>et al.</i> [10]	SDS	70.7	Own dataset	20	-

Vehicle detection and its statistical information are considered significant for tracking vehicles on highway and identifying them through an aerial view. With many of the cameras on the highways and on the roads, monitoring and keeping surveillance is highly needed in obtaining information through traffic video footage, etc. When viewing the vehicle at a certain angle, it is mandatory to install the camera to cover greater

distances. But it is a risky dataset for the researchers as the object becomes smaller when the distance between the camera and the vehicle increases. If the illumination is not proper on the images, the detection rate will fall down [\[11\]](#).

Currently vision-based vehicle detection is divided into two types, namely, traditional-based machine vision methods and deep learning-based machine vision methods. Traditional methods generally use the motion of the vehicle to separate it from the static background and fixed background. Such as background subtraction, change in the frame difference, and optical flow method. This method is best suited for moving vehicles. The optical field represents the direction of the pixel and also the momentum of the pixel.

In the video frame difference method, the variance of two or three consecutive frames is calculated from the pixel intensities to check the background. The foreground is separated by using the thresholding method. This method works well for stopped or parked vehicles. Each frame image is compared with the background and suppressing noise, we can also detect the moving vehicle. When each frame is compared to the background model, we can segment the moving object.

Scale In-variant Feature Transform(SIFT) and SURF methods are also used for detecting moving vehicles on the road. Using the 3D ridges, the correlation curves were detected to classify the vehicles.

In spite of the scarcity of availability of areal images through satellite, it is always a challenging task to find the vehicles from the images. Due to the low resolution and complexity of surrounding images, there are numerous algorithms that were proposed for detection, specifically on a real image. These models can be categorized into two types. One is object modelling based algorithms. Another is Object appearance-based algorithms. The algorithms such as R-CNN, YOLO models basically depend upon the quality of the input image. On the other hand, object appearance-based models such as MAD and Histogram Oriented Gradient (HOG) feature extraction models work with statistical information of the predicted object. HOG method, clubbed with SVM(Support Vector Machine), is used to identify the location of the vehicle.

CNN (Convolutional Neural Networks) are state-of-art methods nowadays due to the use of deep convolution neural networks; the CNNs have

achieved success in every field of computing and vision processing. The ability to learn in CNN has shown an amazing success rate in object detection, too. They can perform different tasks, such as segmentation and classification, using various algorithms. Based on the regression bounded box method, various algorithms were designed. These methods can be divided into two categories. The One-stage method does not generate a bounded box, and a two-stage method generates the candidate box and bounded box and then classifies the object. In one stage method, the algorithm directly converts the problem into a regression problem to detect the object.

In the two-stage method, R-CNN (Region-CNN) is used to select the region of the image. In this method, the input image size must be fixed for n-inputs. And also, the depth of CNN depends upon the number of CNNs we are using to construct the R-CNN. If the depth is more, model learning time increases and requires huge storage and processing power. The model uses fully connected layers with an ROI pooling layer that learns the weights from the convolutional layers. It is an end-to end learning layer network.

On the other hand, SPPNET (Spatial Pyramid Pooling) is a fixed-size CNN network that uses all Fully Connected (FC) layers. Each convolution layer takes the image as input and gives the feature map with respect to the image. The SPPNET gives the ratio of subsampling at every feature map. As the FC layer takes fixed length input, we perform a pooling layer at every FC layer. In this way, we can usually map the object of any type in the input image to the feature map. As the object coordinates get projected in the feature map, one can select the only region to the SPP layer to extract the features.

Mask R-CNN is the extension of R-CNN that is aimed at separating different objects in an image or video. It takes the input as an image or video and gives the bounding boxes or images and their classes as well as masks. It carries the job in 2 stages. Firstly, it generates the region proposals of objects in the input image. Secondly, it finds the class of each object based upon the first stage. The model is built through two paths FPN style network. In upsampling path, the model can be of any Convolution Network. In the down-sampling path, it generates a pyramid map similar to upsampling path.

SSD (Single Shot Box Detector) has reached the record in performance when comes to object detection. It is built on VGG-16 architecture with no fully connected layer. VGG-16 is used as it shows the best performance with high-quality images. It contains a set of auxiliary convolutional layers, enabling the network to extract the features at each subsequent layer. This network majorly is a combination of confidence loss and location loss. Confidence loss measures the cross-entropy value for the network. And location loss is the L2-Norm of the predicted bounding box [[12](#)].

YOLO (You Only Look Once) is also one of the object detection systems, which processes the input image at 30FPS on the COCO test dataset. The biggest advantage of YOLO is its speed. At nearly 40 frames per second, it processes the images. The detection is stated as regression problem. It performs the detections on feature maps. It divides the image into $n \times n$ grids, and for each grid, the model predicts the bounding boxes. The network is formed with fewer CNN layers; hence the error is easy to optimize [[13](#), [14](#)].

The YOLO has been modified with VGG style CNN and named YOLO2, which is also known as DarkNet. The object score has been calculated for each $n \times n$ grid, and going to estimate the object prediction based on the score. YOLO2 extracts the raw output to the bounding box. For the low-quality boxes YOLO2 filters the boxes before predictions. Hence it suits best for areal images as the view of images may be low quality.

YOLO3 is introduced next with pre-trained weights for object detection. Vehicle detection from the video data can be processed with minimal effort using YOLO3. The model first implements the surface extraction and performs the segmentation. It uses Gaussian mixture modelling to detect the background of the image. Once the road surface is extracted, the resultant vehicle information is segmented using multi-object tracking. This algorithm works well in the replacement of SURF and SIFT techniques. According to the vehicle tracking trajectory, the vehicle direction is also predicted. On the image at half position, an intersection line is placed. If the vehicle is crossing the line, the count has been incremented. This count is later used to find the traffic flow from the input video.

Multivariate Alteration Detection (MAD) is developed based on canonical correlations analysis. It is used to detect the variance in the images in a

short duration. It is one of the best-suited techniques to detect the moving vehicle. It is also implemented to detect vehicle density on the road. In the traffic monitoring system [15] using motorways A95 and A96 testing environments, canny edge detection is performed to find the steepness of the edge. On the predicted image k-means algorithm is used to detect the vehicle mode. The color models [2] are used to identify the shadowed area and remove interference while detecting the vehicle from the image.

Proposed enhancement filter that enhances the image and improves the contrast of the image in real images. The system used isotropic wavelet transformation. Each areal image is enhanced with a unimodal and threshold method. For each region in the threshold image, the symmetric pairs were defined, and the symmetry score was calculated; the score was used to detect the vehicle presence using the maxima of Gaussian blurred Hough space [16].

3. PROPOSED METHOD

3.1. Proposed Architecture

There are various techniques available for object detection, as mentioned earlier, which are based on vehicle proximity, Markov random field, HOG descriptors, and heat signature maps. We use digital image processing techniques available in OpenCV to acquire the image from the frames of the video and then use Convolution Neural Networks to localize the objects in the image. We have trained the network with 6000 images of 80 classes and validated with 1000 images. The proposed algorithm is shown in Fig. (1).

Proposed Algorithm:

- Step: 1. Input live Video or Image.
- Step: 2. Convert the video to frames.
- Step: 3. Resize the image into 416 x 416.
- Step: 4. Convert the image into array and feed the array into the YOLO model.
- Step: 5. Use the optimal learning rate of 0.001 and batch size 4.
- Step: 6. The network goes through a series of convolution layers and pooling layers and reduces the trainable parameters using batch normalization.
- Step: 7. Add an extra fully connected layer at the end of the model and use sigmoid activation function in the series of layers. Use logistic regression to classify the objects in the image.
- Step: 8. Compile the model with “adam” optimizer and set loss to “categorical cross entropy”
- Step: 9. Now use the “fit_generator()” to train the model with 100 epochs.
- Step: 10. Predict the class of objects and draw a rectangular box over the localized object.

Fig. (1))

Algorithm of Proposed Method.

Image acquisition is creating a digital image from the physical scene. That is, we capture the frames from the video, which captures the driving environment, which may contain multiple vehicles moving at various speeds [17].

The data is actually captured through frames using the libraries available in Open CV. Then the image is then resized into the size 416 x 416 and converted into any array. Post pre-processing the image array is then fed to the YOLO algorithm, which contains a series of dense layers of convolutions and max pooling layers. Multiple 1x1 and 3x3 filters are applied to the image so that different scales of the objects can be detected, *i.e.*, we can categorize them into large, medium, and small-sized objects with different densities, rather than using multiple heavy component layers for object recognition. Localization, the algorithm uses the simple convolution layers with multiple filters to process the image. 1x1 and 3x3 feature maps are applied on the image at different scales, that is, 1x1 and 3x3 filters for large-size objects of range [512,1024], medium objects of the range [384,512] and small objects of the range [128, 384]. In order to further reduce the features of the image, we use the batch regularization technique so that response time can be increased

and accuracy can be improved, preventing the model from overfitting. At the end, we replace the fully connected layers with a simple average pooling layer which decreases the number of parameters.

4. IMPLEMENTATION

As described in Fig. (2), the proposed system uses convolution neural networks to localize the image. We also use the concept of the bounding boxes, anchor boxes and non-max suppression along with the CNN to accurately predict the object and its co-ordinates [12, 13].

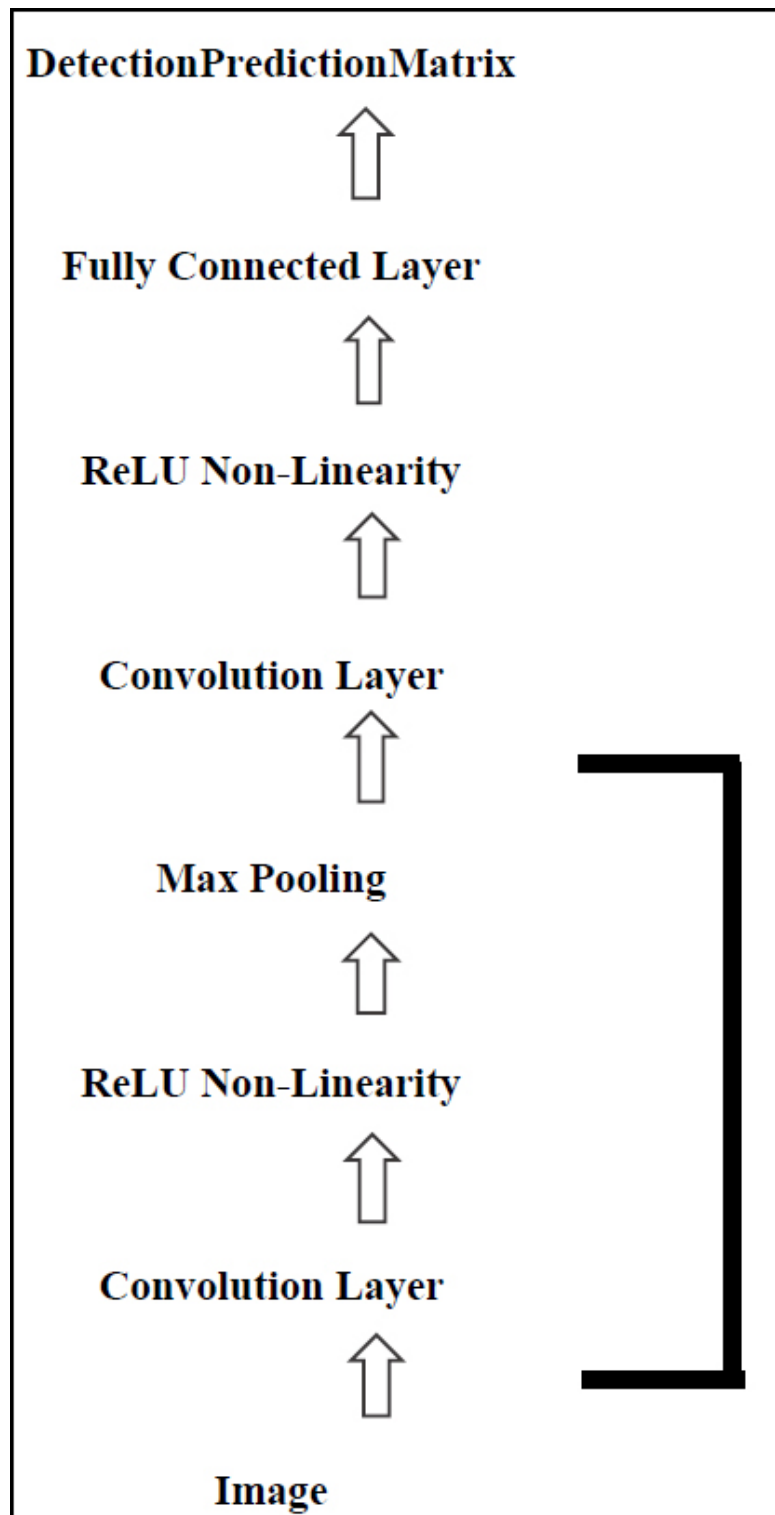


Fig. (2))
Architecture.

4.1. Bounding Boxes

Initially, the image is divided into a GxG grid to make processing easier. In each grid, the objects are located, scanning all the G*G grids so that even the lowest detail in the image is read. Each grid stores its information as a tuple of 5 values, which indicates whether an object is present in the grid or not, and the x and y coordinators the midpoint of the localized object and the width and height of the object in the grid with respect to the grid. After the whole image is scanned for objects, multiple bounding boxes are drawn around the object indicating class confidence. Fig. 3 shows the image is divided into GxG, i.e., a 3x3 grid.

From the proposed architecture in Fig. (3), considering grid 6, we can conclude the following values represented in Fig. (4).



Fig. (3))
Example image with 3x3 grids [14].

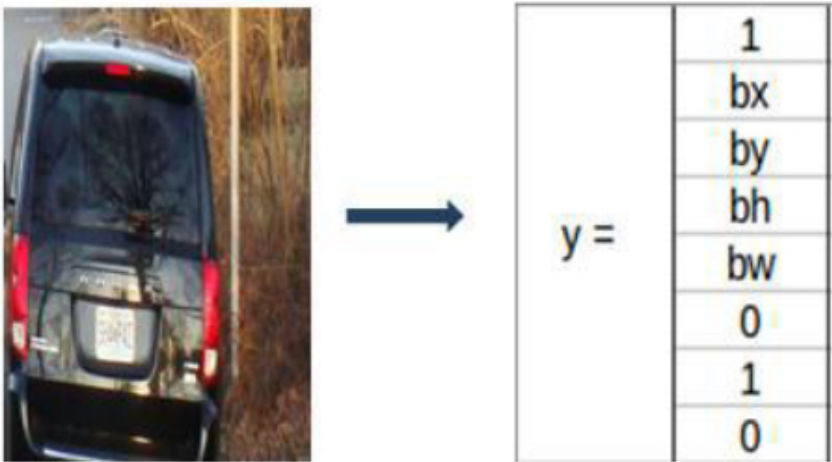


Fig. (4))
Bounding box and Class values of grid 6 [14].

4.2. Anchor Boxes

The GxG grids may contain multiple objects in a single grid. To deal with this problem, we use the concept of the anchor boxes, representing each object with different anchor boxes, and maintain the multiple tuple information about each grid in case more than one object is present. It contains data similar to the bounding box tuples, but in the case of multiple objects, the tuples are also increased, indicating the coordinates of each object.

The vertical box in Fig. (5) represents the human, and the horizontal box represents the car.

4.3. Non-max Suppression

Once the multiple bounding boxes are represented on the image, the accurate bounding box is calculated using the intersection over union (IoU) and non-max suppression. All the bounding boxes less than the required threshold value are eliminated, and only boxes with a threshold value of 0.5 are kept. Then, the intersections over the union are calculated as the ratio of the area of the intersection of bounding boxes to the area of the union of bounding boxes. Further, the probability boxes with a threshold of 0.5 are suppressed until only a single box representing the object is present in the image.

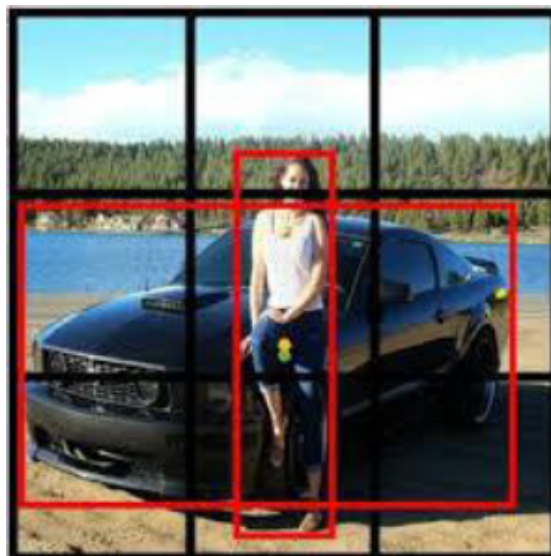


Fig. (5))

An example image for anchor box [\[14\]](#).

5. ReLU activation

Rectified Linear unit activation function can be described as follows:

$$\text{ReLU}(x) = \max\{0, x\}$$

6. Loss Function

Two vectors are represented as V_1 and V_2 . Here $\text{vec}(X)$ denotes the reshaping of the matrix operation. $\mathbf{M}$ denotes the prediction matrix, and $\mathbf{N}$ denotes the ground truth detection matrix.

$$V_1(\mathbf{M} - \mathbf{N}) = \sum |\text{vec}(\mathbf{M} - \mathbf{N})_t|$$

$$V_2(\mathbf{M} - \mathbf{N}) = \sum \text{vec}(\mathbf{M} - \mathbf{N})_t^2$$

7. TRAINING PARAMETERS

In the implementation, we considered 4 parameters under training the model. They are accuracy, value loss, value accuracy and loss. Loss is calculated using the Log-loss function [11]. The training parameters are shown in Table 2. The parameter values are considered at every 10th epoch and tabularized. The model is trained with a batch size of 4 and 100 epochs.

Table 2 **Training Parameter values of the proposed at every 10 consecutive epochs.**

EPOCHS	TIME	LOSS	ACC	VAL_LOSS	VAL_ACC
1	296s 819ms	1.9388	0.4469	1.6624	0.5400
10	276s 819ms	0.9865	0.5634	1.3392	0.6095
20	255s 817ms	0.7810	0.6339	0.9267	0.6414
30	255s 824ms	0.5268	0.6548	0.5599	0.6743
40	266s 815ms	0.2275	0.6906	0.4244	0.7238
50	277s 819ms	0.1710	0.7281	0.3535	0.7348
60	266s 820ms	0.1694	0.7495	0.2528	0.7590
70	265s 824ms	0.0903	0.7976	0.1111	0.7795
80	240s 830ms	0.0798	0.8473	0.0875	0.8371

EPOCHS	TIME	LOSS	ACC	VAL_LOSS	VAL_ACC
90	230s 818ms	0.0563	0.8977	0.0279	0.8514
100	244s 827ms	0.0318	0.9436	0.0062	0.9033

8. RESULTS

The proposed system uses 80 classes. The sample dataset is collected from the DarkNet database and contains 6000 images in the training set and 1000 images in the test set under 80 classes. The average time taken to run the trained model under each epoch is 200sec. To predict the output class, the proposed system took 0.68 seconds with an accuracy of 95%. (Figs. [6-9](#)) show the input and output prediction.

Results for random images collected from the internet.

Result for random images captures randomly from various locations.

Result for random images captured on the road.

Result for random images captured in the traffic.

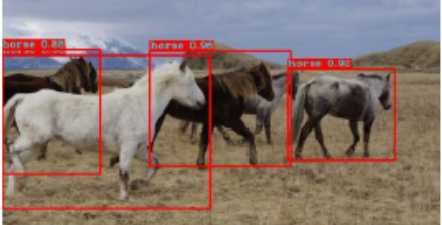
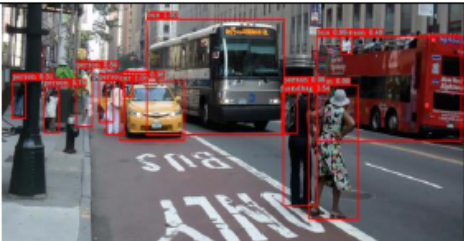
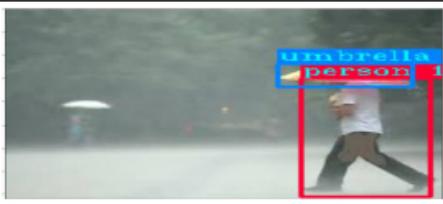
Input Image	Output of Prediction
	
	
	
	
	
	

Fig. (6))
Results for random images collected from the internet.

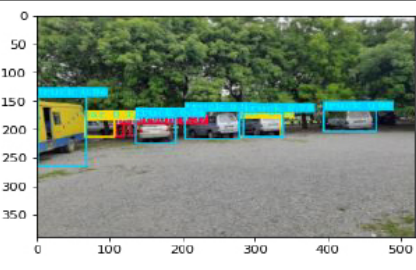
Input Image	Output of Prediction
	
	
	
	
	
	

Fig. (7))
Result for random images captured randomly from various locations.

Input Image	Output of Prediction
	
	
	
	
	
	

Fig. (8))
Result for random images captured on the road.

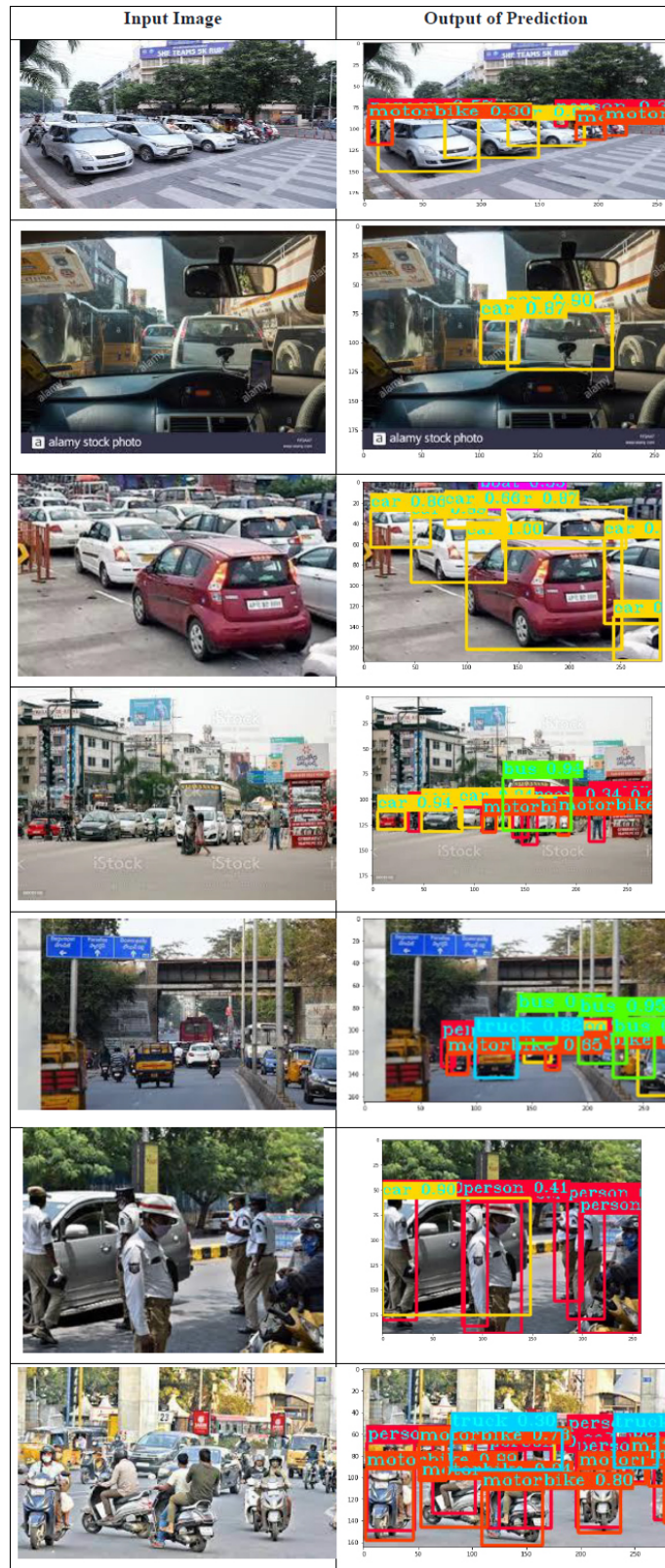


Fig. (9))
Result for random images captured in the traffic.

CONCLUSION

Object detection for autonomous driving using YOLO is the best approach as it is easy and has better accuracy. It has a faster response time of 0.06 seconds compared to other models. The best accuracy is achieved using the convolution architecture of 50 layers and multiple usages of 1x1 and 3x3 filters similar to that of the Inception model in transfer learning. The uniqueness of the YOLO algorithm is Multiple object classification in a single run. YOLO adapts well to new domains; it generalizes positively for new and unseen images, making it ideal for applications that rely on fast, robust object detection.

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All Images are my original results.

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Machine Learning Techniques in Image Segmentation

Narmada Kari^{1, *}, Sanjay Kumar Singh¹, Dumpala Shanthi²

¹ Amity University Rajasthan, Jaipur, Rajasthan, India

² Maturi Venkata Subba Rao Engineering College, Hyderabad, India

Abstract

Image is an important medium to express information easily. This paper deals with the content of image segmentation with machine learning. Segmentation is the process of extracting the information required from the image. Machine learning is the process that helps to classify to obtain good results. A number of algorithms are designed for the segmentation process. The algorithms are selected based on the application. Quality segmentation can be applied if the algorithm is fixed at the application level. Standalone methods can be used for real-time applications. Schematic segmentation is one of the best techniques used for segmenting images. Machine learning combines basic techniques to produce good results. The algorithms vary for different input images like MRI, CT Scans, Colour images, etc. Algorithms like k-mean clustering are mostly used in processing. Many problems occur in segmentation which can be removed by Bayesian architectures. The usage of

machine learning improves accuracy and efficiency. Labeling, training and testing are some of the methods used in segmentation through machine learning.

Keywords: Image segmentation, Feature classification, Machine Learning types.

* **Corresponding author Narmada Kari:** Amity University Rajasthan, Jaipur, Rajasthan, India;
E-mail: narmadakari@gmail.com

1. INTRODUCTION

Information is an important means of communication; the information is of various types, of which images are one of the mediums.

Image is a two-dimensional medium. Over the entire image, the information exists only at some part of the image [1].

Therefore, instead of processing the entire image, the informative part of the image can be separated and processed. This separation process is called image segmentation. This is an important process in the image processing field [2].

Image segmentation is the primary step in all applications. Various segmentation techniques are developed, but there will always be a gap in the research for the segmentation process. Up to now, there is no suitable technique for all the input images. Relative measures are considered for computing the values. There is a difference between computer-generated results and human-generated results. The human eye cannot see the sharpness in the segmented process. The system also easily finds minute errors, so the simulation is important to segment the image [3]. Machine learning

helps to increase the accuracy rate to the point where the exact information extracted is visualized. Images are basically of three types:

- Black and white images (Binary images)
- Greyscale images
- Color Images

The researcher chooses any one type for their segmentation. Generally, medical images are greyscale images. Operating greyscale and color images are difficult compared to binary images. The segmentation method is compared to wide applications instead of narrow applications. Good performance segmentation is not an easy task to generate [4].

Exploiting the process of image segmentation is important. In most of the applications, it is easily viewed whether the segmentation is flexible or inferior to the process. Segmentation is calculated by maximum human-aided techniques, which the researchers mostly use. Human-aided is perfect in performance but takes more time to compute and is tedious. The combination of segmentation with machine learning is always a great process for researchers. Standalone segmentation is also equally good for accuracy. Segmentation is estimated with the characteristics of the image. Reference images are not needed for any type of segmentation.

In real-time, the image is captured, and automatic segmentation can be done on the image with the help of machine learning. If the system is very big, then the segmentation is used in pre-processing of an image. Pre-processing deals with

- Amplification
- De-noising
- Filtering
- Smoothing
- Sharpening
- Segmentation

The pre-processing improves the quality of an image and increases to obtain a good quality of segmentation. The basic segmentation of an image is shown in Fig. (1).



Fig. (1)
Image segmentation.

Generally, many techniques are performed by the simulated algorithm for segmentation in the system. Around 20 different techniques are performed with segmentation, and all the results are stored in the system [5]. Many number of segmentation techniques are available; some of the important techniques are given below.

- Thresholding
- Histogram-based bundling
- K means clustering
- Region growing
- Watersheds
- Active contours
- Conditional Markov random fields
- Graph cuts
- Sparsity
- Schematic
- Instance

The main objective of this paper is to determine the required area through the segmentation process with the help of advanced segmentation algorithms. The required area can be visible or invisible, so detection of this object is not an easy task through segmentation. Particular area segmentation and general classification of the image vary. General image classification classifies the known parameters, but in area segmentation, unknown parameters are classified with known ones also. One of the interesting applications of image segmentation is image forgery detection. Generally, the signature is stored in the database [6].

Each part of the image is segmented, and features are extracted for the signature image. If any question arises, the algorithm tests the features and checks for matching. Segmentation is very useful in forgery detection. The other important application is baggage detection. The image of the baggage is taken, and similarly, the segmentation is done, and features are stored in the dataset. Later, if any doubt arises, they will match the new image with the old stored dataset, which easily verifies the baggage. Schematic is the main technique used by many researchers all over the world. Schematic can be applied to any type of image whether it is a color or grayscale image [7].

The results are different, and if a simple image is considered, the results are shown in Fig. (2). Naturally, the segmentation results are used. As many of the researchers prefer mostly the two techniques for the segmentation of images which are stated as:

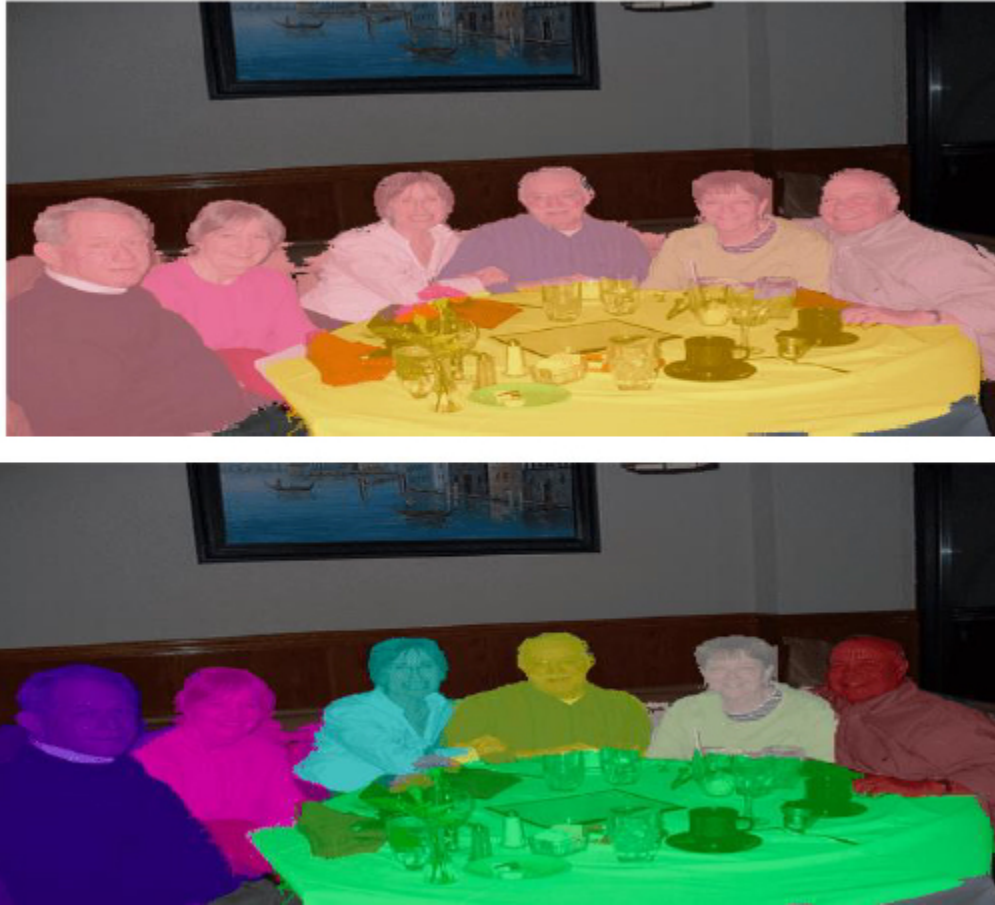


Fig. (2))

(a) Schematic segmentation **(b)** Instance segmentation.

- Schematic segmentation
- Instance segmentations

Both these techniques are good in their respective areas and applications. Now the comparison algorithm compares the system results, and then the best result is chosen for these firefly algorithms, or standalone algorithms are used. Many high-level multinational companies and software users the same way at a time, applying few techniques and identifying the best from the results. This process is done since no researcher can say which method is suitable for the input image, so all the techniques are applied, and the final result is selected. Moreover, for the same application, different segmentation techniques may be suitable. Moreover, for the same input at a different time, different techniques may be applicable; this is

because, all the time, the same training and testing process cannot be done [8].

Standalone techniques have good performance in comparing and selecting the best technique for the input image. Individually for every image, this technique can be applied, or it can be applied for a group of images also. A certain threshold is maintained with the parameters, and the best results are extracted. There is a technique called meta-evaluation technique consists of base design mixed with machine learning; with the help of weights, the segmentation and comparison are done. For every base estimator, one decision tree is generated.

All the training data is labeled. All depended on the parameters or characteristics where the suitable technique is founded. There are millions of images in the database. Even the cost of MRI, scan, etc., is heavy; there are several medical images in the database. This database number of algorithms is designed to detect diseases in medical images. It has been the practice to develop advanced segmentation algorithms widely used by hospitals [9].

Identifying the data resources are became easy now been easy to get the dataset now; many agencies collect the data from hospitals or diagnostic centers and sell it to the researchers on a cost basis. This segmenting process is accelerated because many researchers are easily analyzing the data and processing it. Generally, segemntation process has some steps which are stated below.

- Calculating the intensity
- Normalization of image
- Registration to a common template
- Segmentation of specific substructures
- Statistical analysis.

This process is generic and can be applied to any type of image. Automatic segmentation also uses the same process but simulates with volumetric measures or shape statistics. Fig. (3) shows the segmentation done on a medical image MRI, which is a brain image.

Most experts will do the segmentation manually, where they will define an ROI (Region of Interest) in sequential sections of each area. Before the computer and segmentation were developed, the experts used to directly select the ROI to extract the segmented image. But it was time taking process, especially for the minute extraction. A small example was a researcher want to detect a tumor in the MRI, and it took manually two hours to segment the tumor. The tumor is scanned 3 times, so 3 scans x 2 hours gives 6 hours per scan. Now the area of the tumor is 1000 pixels, so the total time taken would be $1000 \times 6 = 6000$ hours. So for a researcher to detect a simple tumor from the 3 scans, he needs to work for 6000 hours which is quite a big task. So now the doctors are going for automatic segmentation algorithms.

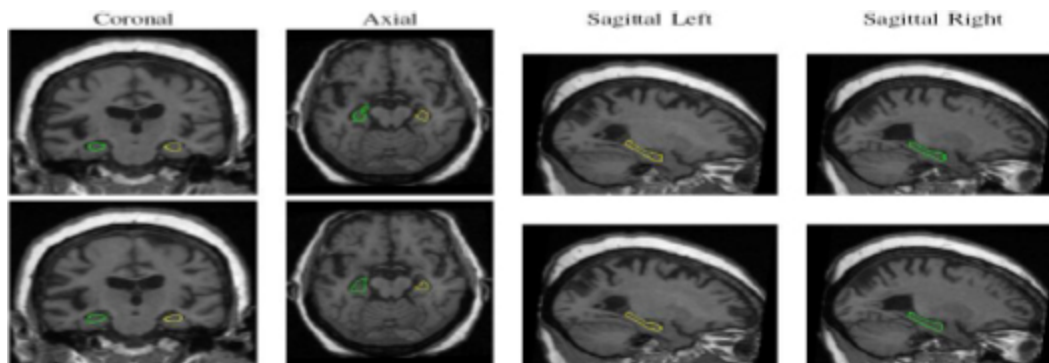


Fig. (3))
Segmentation of Brain MRI.

2. LITERATURE REVIEW

Segmentation algorithms can be categorized into two types. Many researchers produced thousands of segmentation algorithms, but a maximum of these algorithms come under these two types, as stated below:

- Low-intensity algorithms
- High-intensity algorithms

Low-intensity algorithms are also named appearance-based models, which are used to extract the minimum shapes. And high intensity is

used for maximum-order shapes [[10](#)].

Collins *et al.* explained how brain MRI images are segmented. These researchers worked on 3D images. Labels are shifted and compared to generate accuracy. Christensen *et al.* discussed the coupling of nonlinear image registration. These researchers used elastics and fluid shifting to produce optimal results.

Ashburner *et al.* derived extra statistics and maps from the deformation field. The labels are then applied by morphometric statistics, increasing the accuracy. Toga *et al.* designed a new approach used in the opposite path. The name was given as spatial normalization. This was also an effective technology in the segmentation process. Klein *et al.* proposed segmentation with the cross-correlation program. Later, the researchers used mutual information. These parameters are extracted and compared, which has given effective results. Miller *et al.* designed a large deformation method, which is a huge process they updated it with LDDMM, which is a metric mapping technique. They focused on metric pattern theory [[11](#)].

Christensen *et al.* updated their research and worked on the vector value flow field. They used partial differential equations in their design. It also uses continuum mechanics. Chou *et al.* worked on multiple registrations where the image was segmented using voting. This algorithm also considers some statistical techniques [[12](#)].

Lepore *et al.* also worked on spatial normalization and gave better results as compared to previous algorithms [[13](#)]. Kochunov *et al.* used only single registrations and reduced errors. The deformation is also smooth and sharp [[14](#)].

Davatzikos *et al.* discussed how the boundaries are extracted if it's in snake curves. The moving curves are very difficult to operate. They used a predefined algorithm to obtain the accuracy.

Cohen *et al.* proposed an excellent technique that is used now. They designed an algorithm for energy minimization. This algorithm mainly

operated for 3Db images with spherical topology [[15](#)].

Han *et al.* worked on gradient vector flow. They used the concept of convolution in estimating the regions and successfully used the procedure. Pitiot *et al.* researched genetic programming. They used evolutionary algorithms for their research. They worked on a complementary segmentation approach which has given the best results. Wells *et al.* proposed an important technique called Bayesian tissue classification. They also designed an EM algorithm [[16](#)].

Shattuck *et al.* updated the EM algorithm and ensured that the advanced EM was used for all the images. Pitiot *et al.* used ANN (Artificial neural network) to classify the image and generate accuracy.

Golland *et al.* used PCA (Principal Component Analysis) to classify the image and generate accuracy. Fischl *et al.* mixed images and shaped data into a software suite. The name of the software is FREESURFER. This software is based on a Markov random field. This software is free and used by many researchers for processing their images [[17](#)].

Mangin *et al.* simulated the Bayesian prior distribution algorithm. This classifies the image into a different category. This technique is an old technique, but the researchers replaced the single layer with multiple layers and used it for medical images only, which has given moderate results. Mangin *et al.* again updated the research 3D morphological operations. A morphological process like skeletonization is used to generate the exact boundaries which have given better results [[18](#)].

3. METHODOLOGY

A method is part of the research design. This process is the heart of the research. Methods can be different, but the methodology is the same for all segmentation processes. The thought is a very simple

extraction of information from an image but requires 7 steps in which advanced algorithms are used to segment the image in every step. The main aim of the design is labeling. Each pixel is assigned a label. The methodology is simple to understand. The steps are given as:

- Collection of Data
- Pre-processing the images
- Training options
- Defining the label IDs
- Features extraction
- Features reduction/selection
- Feature Classification (Training and Testing)

3.1. Collection of Data

Data collection is the stage where the inputs are collected. If a researcher wants to work on MRI images, he/she will collect the input images and store them in their disk for further processing. The more data is collected, the more will be the accuracy. Any images can be collected and stored in image file format [[19](#), [20](#)].

3.2. Pre-processing of Images

This step is the first step done on the system. The collected images are manipulated and, changed according to our desired format or shape. Generally, pre-processing uses techniques to De-noise an image, smooth an image, sharpen an image, and filter an image. After all this is done, the image is ready for further processing [[21](#)].

3.3. Training Options

The image is grouped into various classes for example in Fig. ([4](#)). There are many elements like a pole, sky, trees, buses, white cars, grey cars, orange cars, humans, cyclists, etc., 14 elements are considered [[22](#)].

Each element is grouped into a classification, and labeling is done. Training is done with the label, where machine learning is used, which is further discussed in Step 7.



Fig. (3))
Training an Image.

3.4. Define Label IDs

Every group formed is defined by a label ID. Label ID is defined with RGB values (each 0-255 range) since it's a color image. If the image is gray-scaled, only 0-255 values are used; for the black and white image, only 0 or 1 are used. For example the building in Fig. 4 is defined as (128x128x128). Similarly, in one group or element tree, where ID is given as (192x190x000) like this, for every 14 elements, the IDs are generated. This is used in the training and testing process. Values should approximately match the color of the image. Own values cannot be chosen [\[23-28\]](#).

3.5. Feature Extraction

The groups are now further processed to extract the parameters in the elements. Many feature extraction techniques are developed, like DWT (Discrete Wavelet Transform), DCT (Discrete Cosine Transform), machine learning, deep learning, vector analysis, Bayesian technology, etc. These are the most commonly used technologies for extracting the features [\[29-33\]](#)

3.6. Feature Reduction/Selection

All the extracted parameters are stored. All the extracted features are useless, so unwanted features are reduced and removed. LDA (Linear Discernment Analysis) is the most popular technique for feature reduction [[34-36](#)].

3.7. Feature Classification

Feature classification is to classify the features or parameters. Machine learning is a process used to extract and classify image parameters. In which machine learning has an important role compared to others. Many researchers worked with this machine learning and produced accurate results [[37](#)].

3.8. Machine Learning

Machine learning evolved in the 1950s; at that moment, computing had just started comparatively, it's just a fraction as compared now. Artificial intelligence uses the values predicted by a machine. Generally, statistical researchers similarly predict the values. So ML is evolved from statistics and its techniques. Statistics techniques are regression, decision trees, classification, random process, and clustering. These are the most commonly used statistical methods for finite data [[38](#), [39](#)]. But the data nowadays is huge, so statistical techniques are not enough for evaluation; there comes AI, which is an advanced version of the statistical method. There are various types of machine learning techniques as given. All these types can be used for training and testing the data. The evolution is also from top to bottom, as given below [[40-42](#)].

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Deep Learning
- Deep Reinforcement Learning

The process of all machine learning evolved one after the other. First, supervised learning is evolved where process like house pricing prediction, population prediction, *etc.*, is done using statistical analysis. Later, unsupervised learning evolved where there was no need for supervision of the computer. It automatically analyses the data and produces the results. Next, researchers designed reinforcement learning which rewarded the machine. This is also used in many local applications. But these techniques were used for a limited amount of input data and non-homogenous data. The next method is deep learning, which is the process where the artificial brain is attached to a network and behaves as a human thinks. This brought drastic changes in the technology era. Since the computer can think like a human and the effectiveness is too perfect, and the speed of operations is very high.

Therefore, many researchers used deep learning for a few decades and then came deep reinforcement learning, where the system was rewarded and incentive was given, which helped it to increase speed and accuracy. At last, the computing speed was very high, and a general human can not operate the same data for years. Many unsolved problems are easily solved.

3.8.1. Supervised Learning

Supervised learning is an example of training a child to write. First, we will show how to write, and next, we will fold his hand and make him write, and lastly, we will see his writing. The same way the training is done. Total supervision is given while this process. The regression process is used, which gives the solved examples for the training process. $x_1, x_2, x_3, \dots$ are the inputs and $y_1, y_2, y_3, \dots$ are outputs.

A perfect relation is founded between x and y ; enough data points would be available for training the data. With these data points, y is predicted from x . we know the real value of x , but we will check whether the prediction is perfect or not. So like this, verification is done such that at least 80 to 90% accuracy is generated. If the

accuracy is generated, then any data predictions are done by testing. This procedure is called regression. Next is classification. The classification can also be done by supervised learning; for example, I have 100 students in my class, I divide the 100 students into three classes based on their heights, and so three groups are formed of different heights. Now I train my data into three groups. A new student enters my class now. I need to see that student in the same height group. Now the new height is taken and tested with the trained data. That will give u accuracy on which group he belongs to. Similarly, I have different labels trained in the images. The segmentation uses the testing principle, and work can be done. If the data inputs are a few thousand, it's enough to apply supervised learning.

3.8.2. Unsupervised Learning

The unsupervised is the method where humans don't interfere with the result of the machine. Unsupervised learning requires huge data inputs, which are around a few million. That much data is easily available on social media or any other company. A huge amount of data is required since the researcher will give the input x and ask which group it belongs. He may ask what you can tell about x .

He asks to find ten groups closer to the input x , so for this type of testing, huge data is required. This learning doesn't take any actions from the human brain so it is under no supervision. Fig. (5) shows how an image is classified into different classes and a boundary is extracted.

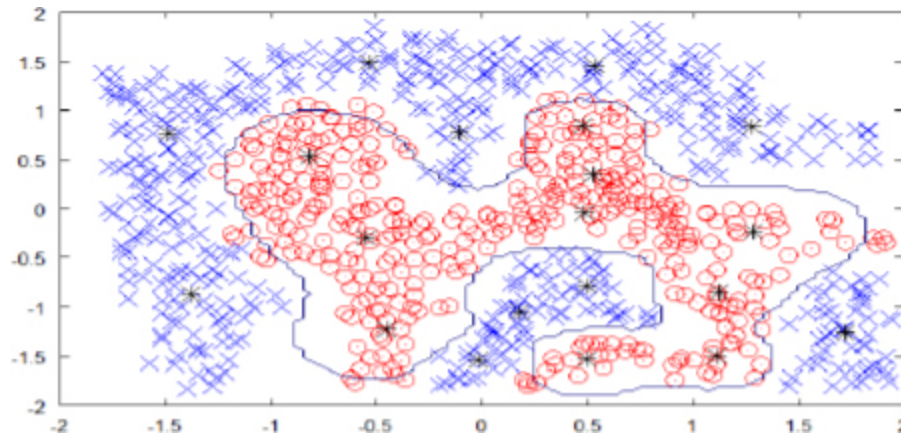


Fig. (4))

Unsupervised learning.

3.8.3. Reinforcement Learning

This can be explained with a simple example, we have a dog, and we throw the ball, and the dog will bring the ball to us, it is supervised learning. And if we don't throw the ball, and a dog sees some other ball, it will bring it to us, it is unsupervised learning.

Now reinforcement is awarding the dog. If a dog brings the ball correctly every time, we award it with a biscuit; Next time, it will do correctly because it knows it will get the biscuit. Similarly, the machine will play a game some algorithm predicted, and it is tested as right it is saved as the right location, and if it gives an error or predicts wrong, it is saved in the wrong location. So system easily differentiates what is wrong and what is right. This is reinforcement learning which is applicable on any data or big data. The process is shown in Fig. (5).

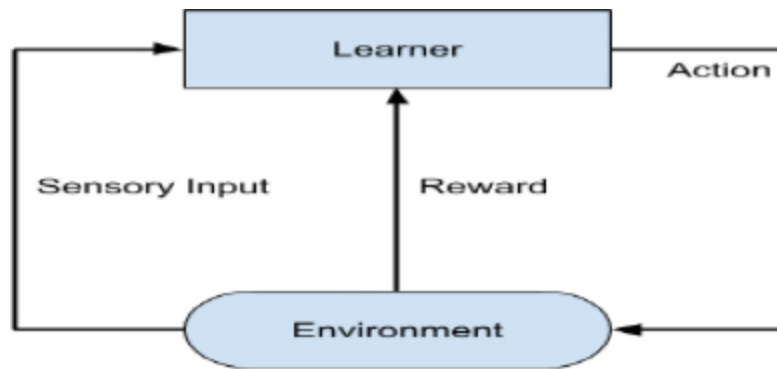


Fig. (5))
Reinforcement learning.

3.8.4. Deep Learning

Deep architecture depends on ANN (artificial neural network) or CNN (convolutional neural network). There are many other networks like DBN (deep belief network), RNN (recurrent neural network), *etc.* These architectures are very successful in huge data. This is used in various applications like speech recognition, NLP, bioinformatics, drug design, medical images, games, and deep learning uses only homogenous data.

3.8.5. Deep Reinforcement Learning

DRL is the mixing of deep learning and reinforcement learning. Both of these advantages are combined and give excellent results. Example Q learning from reinforcement learning is combined with the CNN model, generating an excellent and effective model. This DRL is used in robots, finance, healthcare, games, *etc.* Active research is going on this DRL, which is more applicable in all the applications.

CONCLUSION

This paper is on image segmentation using machine learning. This paper first introduces images and various types of images. The next explained about the types of segmentation done, and the suitable

technique for certain inputs have been discussed. The application of images and advantages of various images are covered. The preprocessing steps are clearly explained and taught about when the smoothening and sharpening are done. Various types of labeling are covered with examples. After the labeling is made, then the feature extraction techniques are seen. What suitable feature extraction techniques are overviewed? Later the reduction or selection of features is explained. The last section is classification, explaining how machine learning is used for classification. Various types of machine learning are discussed in brief. Learning like supervised, unsupervised, and deep reinforcement are discussed in depth. This paper helps beginners to understand the basic concepts, and how the segmentation is done using machine learning would be clear.

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Optimal Page Ranking Technique for Webpage Personalization Using Semantic Classifier

P. Pranitha^{1, *}, A. Manjula¹, G. Narsimha², K. Vaishali³

¹ Jyothishmathi Institute of Technology and Science, Telangana, India

² JNTUH College of Engineering, Sulthanpur, Telangana, India

³ Jyothishmathi Institute of Technological Sciences, Karimnagar, Telangana, India

Abstract

Personalized webpage ranking is one of the key components in search engines. Moreover, most of the existing search engines focus only on answering user queries, although personalization will be more and more important as the amount of information available on the Web increases. Even though various re-ranking algorithms are developed, providing prompt responses to the user query results in a major challenge in web page personalization. Therefore, an efficient and effective ranking algorithm named the Oppositional Grass Bee optimization algorithm is developed to re-rank the web

documents in the webpage personalization system. The proposed algorithm is designed by integrating the Oppositional Grass Hopper (OGHO) and Artificial Bee Colony optimization (ABC) algorithms. The concept of fictional computing and the foraging behavior realize the re-ranking process more effectively in the web environment. However, the semantic features extracted from the web pages make the process more effective and achieve optimal global solutions through the fitness measure. The proposed OGBEE Ranking algorithm effectively captures and analyzes the ranking scores of different search engines in order to generate the re-ranked score result.

Keywords: Page ranking, Artificial bee colony optimization, Web mining, Grass hopper optimization, Feature Extraction, Web Personalization, Factor based Features, Rank-based features, Hyperlink-based ranking, Hyperlink induced topic search, ENN-based Opposition Behavior Learning.

* **Corresponding author P. Pranitha:** Jyothishmathi Institute of Technology and Science, Telangana, India;
E-mail: ppranitha7@gmail.com

1. INTRODUCTION

A search engine is a piece of software that enables users to enter a query and transmit it along with other relevant websites relevant to the information they are looking for. There are countless online papers on the WWW (*i.e.*, on every page

of the website), some of which may be found on various parts of the site with a multitude of information, many sophisticated steps that need to be done by the search engine in order to acquire information from the Internet [1].

Several search engines follow the following sequence of operations: searching, indexing, sorting, and data ranking. Although search engines are helpful for obtaining useful material on the internet, they are important tools for discovering the latest information. All queries, processing and ranking, are performed through the use of search and indexing techniques without accessing the source code of web pages. An important approach to boost search rankings is by automating web pages based on the subject of a user's interest [2]. Bespoke experiences are created by leveraging acquired information, such as the user's navigational activities and navigational patterns, to alter the overall design of a site. It is a method for websites that includes data on how people use the site as well as structural data to rank and provide suggested web page links to visitors. Web Personalization permits the modification or priority delivery of content depending on individual users' explicit or implied preferences. Details supplied either by the user or the implicit/derived behavior of the user and preferences, such as links clicked, or pages seen, are the priority or page ranking for a certain content or web page [3].

One of the most successful strategies when it comes to looking for query phrases is rating the query word. Web pages may be categorized into those that are navigation-oriented and those that include top-notch content [4]. The ranking algorithms in use now were designed within the context of the words and phrases that existed at the time. Page-Rank is the most well-known approach for ranking websites. Although previously mentioned, the amount of connections connecting websites is key to the Page-Rank algorithm [5]. The Page-Rank algorithm assigns thousands of web-based websites to a page ranking score on the World Wide Web [8]. The Hyperlink Threaded Search (HITS) algorithm and Page-Rank algorithm are excellent web-ranking and crawling techniques [6]. The

number of hyperlinks pointing to an inbound link or web page is taken into consideration by the hyperlink-based ranking algorithm, which is used early in the search process [7].

In contrast to the link-based ranking strategy, which is used to discover websites related to the specific page query, the content-based ranking approach aims to return the web pages that are relevant to the query. The queries the user enters are paired with keywords found in online logs and web documents that the user maintains, thereby enabling these algorithms to work effectively (fairy tale) [8, 9].

This project aimed to build a website customization re-ranking technique that used the OGBEE algorithm. Pre-processing involves the web page's design, scripting, and optimizing the webpage for search engines, followed by feature extraction and re-ranking. Next, the question is sent to Google and Bing search engines, which are responsible for returning relevant web pages for the user. Before processing, all the pages are delivered to the number of pages received. Then, noise is eliminated from the pages before they are supplied to the machine [10]. Also, the HTML file is processed to become a text file beforehand. In the features extraction stage, essential properties of online documents are extracted and made available. Finally, the approach that was proposed is utilized to re-rank the competitors. The major contribution of this research is explained as follows:

- **Proposed Ranking Algorithm for Personalization:** For re-ranked websites in search engines, an efficient and optimum web page ranking web is created. The ranking of the web pages allows the re-ranking of measurements. The re-rank measure analyses the distance from the query to the online sites and is the lowest possible answer [11].

2. LITERATURE SURVEY

A number of different existing page ranking approaches are tested to discover whether they provide benefits and pitfalls, with the hope of coming up with a new methodology. Vijaya P and Chander S [1] developed a new method to re-rank the websites, and this algorithm produced a new rating for the sites called the "lion rank." This technique located and obtained the documents that were needed. While the results were favorable, the performance of this approach was subpar. Makkar A and Kumar N [2] created a user behaviour rating system using the learning automata approach. This method reduced computing costs and energy consumption while improving page rank scores for websites. This method did not work for detecting spam pages. Desarkar M.S *et al.* [3] created the unsupervised rank algorithm in order to minimize the number of poor rankers. This product not only surpassed the current market standard, but also was capable of meeting the unique meta-search criteria. It was created by Philip L.H. and Xu H [4] to maximize the achievable expectations of websites. As the rankings increased exponentially, and cognitive classification levels were established, the classification system analyzed the degree of cognitive categorization. Different cognitive rankings and origins did not make them a good group. The multi-layer perceptron (MLPNN) neural network created by Banaei H and Honarvar A.R [5] incorporates machine learning to evaluate this website. With increased precision, performance improved. The neural network's settings were more challenging to set. To determine page rank, You K *et al.* [6] created a Markovian Kaczmarz method. When dealing with spamming nodes, this method worked well. In a short time, it produced many linkages, but the dynamics of opinion were not integrated into social networks. A dual model based on the algorithm of the rank page was developed by G. Masterton and E.J. Olsson [7]. This method offered a precise assessment of websites but did not enable users to explore larger web charts as regards the link model. This technique was used by Plansangket S and Gan J.Q [8] who created the GCrank page ranking system. This method substantially improved ranking performance by better fulfilling user needs; however, re-ranking large-sized web documents was ineffective.

2.1. Challenges

In this section, the many difficulties connected with using page rank as a ranking strategy for research articles are examined.

- It is difficult to search and get important information from the website document list, since books and catalogues are maintained on the web pages [9].
- In the study [1], the meta-search engine is built on the lion algorithm used to rank the web pages again. Additionally, a big challenge with this approach is classifying and integrating search results from several search engines to design an aggregate ranking algorithm.
- One of the primary issues of page ranking is obtaining and using supervised data for the aggregation technique, which rewards increases in page rank and penalizes drops in rank to enhance ranking results [12].
- In the study [13], the SEO technique is used to choose the Keyword and to establish the backlink. The selection of the SEO keyword is a challenge because of the indexing and optimization of the Websites by keywords. The most adequate and relevant keyword selection in online traffic is thus a tough job.
- The replacement-associated strategy is used to re-rank the web page to provide successful results, however, the time the user spends on a certain page is frequently a major problem when it comes to measuring the importance of the page [14].

2.2. Motivation of Research

Developing an effective search engine may be challenging since multiple query languages must be available for different engines, and unrevealed ranking methods are also used by engines. Furthermore, most prominent bandwidth meta-search engines must pay for ongoing high-volume access [15] and negotiate with main engines. A reclassification of search results by adjusting them to user interests to get the most important documents, is helpful and is a well-known issue in the field of information collection [16]. A more difficult issue

is web meta-searching than ranking grouping. The ranks may be loud, incomplete or disjointed, individually. Thus, the only criteria influencing the ranking should not be them. Additional processing is needed to filter the results and to liberate an undesired, deceitful and unjustly rated web page from the end result of the Meta search engine.

2.3. Proposed Methodology

With the rapid expansion of network resources, web users are confronted with difficult problems in obtaining useful information from the web. The majority of ranking algorithms focused primarily on relevant websites and not on entertaining web pages. The research uses an optimal and efficient ranking algorithm to build a dynamic re-ranking procedure for websites in the Web environment [17]. Alternatively, the text is processed to eliminate stop words and facilitate word segmentation by utilizing the stop word removal and stemming methods. Additionally, the pre-processing method of HTML to word conversion is done. The feature must be extracted once the documents have been pre-processed. Based on text analysis, factor analysis, rank analysis, and statistical analysis, results for these textual qualities are obtained from website documents. After extracting the features, the suggested method is used to perform the re-ranking measure procedure during the re-ranking phase [18]. The suggested method for web page re-ranking is shown in Fig. (1).

2.4. Generation of Web Pages

To get the appropriate content, the input user query will be Q passed on to search engines, Google α and Bing β . Let us take into account the user request sent to Google, Bing and Google. The search engines retrieve the web documents according to the query so that the web pages found are shown as R_1 . The web page containing the information interested in the user is filtered and sent into the pre-processing phase of the obtained documents [19, 20]. The censored

web pages are nevertheless seen as R_2 being sent on to the preprocessing step.

3. PRE-PROCESSING

This phase accepts the web pages as input, utilizing the stop word removal and stemming procedure as an advanced processing mechanism [21-23]. In addition, the documents are changed into the Word file in the pre-processing stage from the HTML file. In a system rating the web, pre-processing is an important stage. The preprocessing step lowers the indexing size of documents and makes the retrieval procedure more effective and efficient.

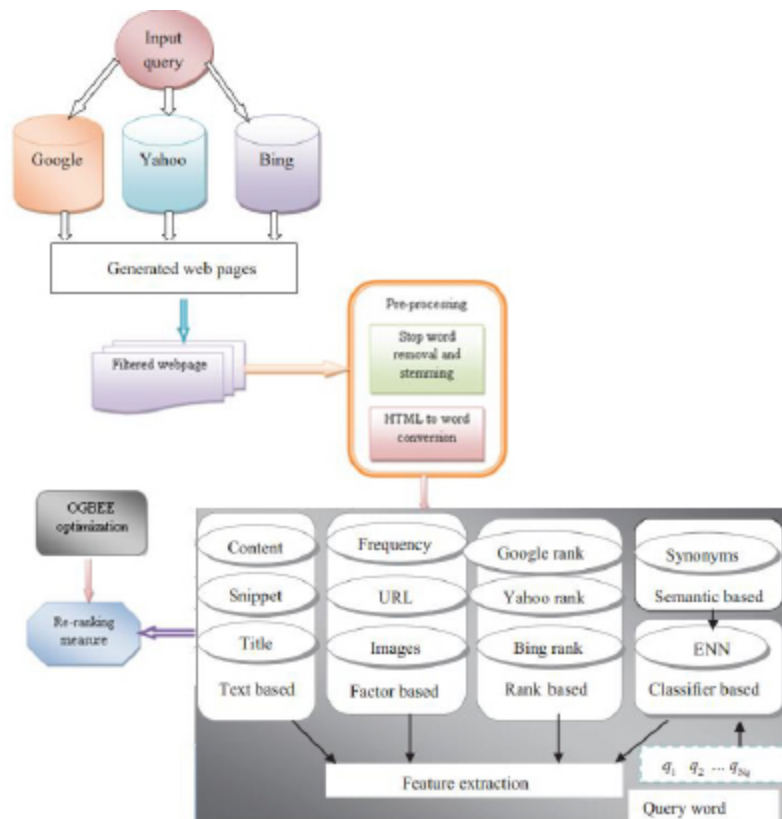


Fig. (1)

Schematic diagram of the proposed ranking algorithm.

Stop word removal: The non-informative words are termed stop words, like conjunctions, prepositions and articles. The elimination of stop words is called the removal of all common words such as “and”

“are,” “this,” “all,” etc. This reduces the data dimension in the pre-processing step [24].

Stemming: It combines various versions of words with the common representation known as the root or stem word. The terms, such as “presented,” “presenting,” and “presentation,” may be simplified to provide a common representation. The word stem here may be obtained by removing the information about the inflection. The stemming procedure does not, however, alter the data in the web picture to decrease the functionality growth effectively.

HTML to word conversion: The contents in the HTML formats are here transformed into a word file, to increase the efficiency and effectiveness of feature extraction.

Finally, the pre-processed document is represented by P , the feature extraction module to extract from the web documents necessary and appropriate functions.

3.1. Feature Extraction and Web Page Ranking

P characteristics are successfully retrieved and used to execute the re-ranking operation. It has successfully extracted at the feature extraction step in order to conduct the re-ranking procedure. Text-based, rank-based, factor-based, statistical features and correlation-based features are among the features retrieved from the pre-processed document [25].

a) Text-based Features: Text-based characteristics, such as snippets, titles, and content of internet sites, easily extract features and are described as follows:

$$M_1^{snippet} = \frac{1}{X} \left[\sum_{j=1}^Y g_j \right] \quad (1)$$

$$M_2^{snippet} = \left[\frac{1}{X \times Y} \sum_{j=1}^Y g_{jb} \right] \quad (2)$$

Where, Y is the total number of query, g_{jb} denotes the frequency of joint queries included within the snippet, g_j denotes the frequency of query words contained within the snippet, and X denotes the total number of snippet words.

$$M_1^{Title} = \left[\frac{1}{A \times Y} \sum_{j=1}^Y g_j \right] \quad (3)$$

Here, A specifies the number of title words.

$$M_2^{Title} = \frac{1}{A \times Y} \sum_{j=1}^Y g_{jb} \quad (4)$$

So, since the web pages' text-based characteristics were retrieved, these features were expressed in text form,

$$g_1 = \frac{1}{4} [M_1^{snippet} + M_2^{snippet} + M_1^{Title} + M_2^{Title}] \quad (5)$$

b) Factor-based Features: Factor-based characteristics are those which are determined by looking at the number of online pages with photos or URLs, as well as the web page frequencies. While mathematical, general, numeric, and marketing phrases are extracted from the factors, this does not mean they are not the same factors. As a result, the factor-based characteristics retrieved by the re-ranking method are represented as follows:

$$g_2 = \frac{1}{4} [z_1 + z_2 + z_3 + z_4] \quad (6)$$

$$z_1 = \frac{1}{2} \left[\frac{C^{snippet}}{X} + \frac{C^{Title}}{X} \right] \quad (7)$$

$$z_2 = \frac{1}{2} \left[\frac{B^{snippet}}{X} + \frac{B^{Title}}{X} \right] \quad (8)$$

$$z_3 = \frac{1}{2} \left[\frac{D^{snippet}}{X} + \frac{D^{Title}}{X} \right] \quad (9)$$

$$z_4 = \frac{1}{2} \left[\left(1 - \frac{E^{snippet}}{X} \right) + \left(1 - \frac{E^{Title}}{X} \right) \right] \quad (10)$$

Where, B is numerical words, E is spam words and C stands for non-dictionary words.

c) Rank-based Features: This feature is based on the search rank of Google and Bing, among other search engines.

$$t_1 = [(Y - J_1) + (Y - J_2)] \times \frac{1}{2} \quad (11)$$

Where, J is the Google rank, J_2 represents the Bing rank, and Y represents the web pages relevant information.

$$t_2 = 1 - \frac{1}{K_X} \{ [\lambda(J_1, J_2)] \} \quad (12)$$

$$g_3 = \frac{1}{2} [t_1 + t_2] \quad (13)$$

3.2. ENN-based Semantic Features

To measure the influence of semantics, search engine rankings are collected using semantic data. Semantic markup can be characterized by the number of synonyms that are feasible in representing the user's request. Following that, the synonyms created in the laboratory are compared to the terms used by the search engines. A likely synonym produced based on the user query input is presented below.

$$Sn = \{Q_a, 0 \leq a \leq h\} \quad (14)$$

Where ash is the number of potential synonyms depending on the user query, and Q_a represents the a -th synonym of the input query.

The neighbors of the input query and neighbors of the neighbors of the input query are used to locate the relevant text of the user query. The computation of the distance between the semantic-based vector and the top keywords produced from search engine websites may be expressed as follows:

$$S_b(S_v, W_{is}) = \frac{\alpha.Y_b + \beta.YY_b}{\alpha + \beta} \quad (15)$$

Where α and β are weighted constants, Y_b is the neighbor to the original input query, and the neighbors of the first query sample are called YY_b , which is connected to the class b^{th} . Y_b and YY_b inquiry results are then measured using the nearest extended scoring value. The availability of Y_b and YY_b data is the determination of the score values of each document.

The sorting value is the highest for the input query after computing the score from the sites. Finally, the documents are collected from the database and presented to the users according to the sorting order. The ranking score is computed using the trial results of the semantically retrieved characteristics based on the ENN classification, which are stored in the upward order and indicated as:

$$g_4 = g(S(S_v, Dwi)) \quad (16)$$

Where, S_v represents the extracted semantic-based feature results and Dwi denotes the database based on the user query.

In order to measure how similar a given input query is to a certain function library, the matching method is implemented. Similarity measurements are used to discover values that are comparable because of how frequently and how similar the papers around the value are. Using this analysis, we found out which keywords appear in queries on the website and how similar pages are in regard to these keywords. This logarithmic option is used to constrain the data range. It is expressed as follows using the input query as a guide.

$$e_Q^{w_i} = \log \left(1 + \sum_{x=1}^{n_x} f_{q_x}^{e_i} \times idf_{q_x} \times B_{e_i} \right) \quad (17)$$

Where idf_{q_x} indicates the inverse document frequency of the query term q_x , Q signifies the query's similarity, and w_i denotes a website. The number of keywords identified based on the input query is denoted by n_x , and $f_{q_x}^{e_i}$ is the frequency with which the query keyword q_x appears on the site e_i . The inverse document frequency is computed by dividing the total number of documents in the database by the number of documents that include the query keyword q_x .

Finally, the final rank of the web pages is represented as,

$$S(q_r) = \mu_1 \cdot g_1(q_r) + \mu_2 \cdot g_2(q_r) + \mu_3 \cdot g_3(q_r) + \mu_4 \cdot g_4(q_r) \quad (18)$$

Where, " $S(q_r)$ represents the final rank, and μ_1 , μ_2 , μ_3 , and μ_4 are the optimization parameters."

4. RE-RANKING OF WEB PAGES

The re-ranking of the pages is done according to the range in which they are found. Using the antagonistic grass bee optimization algorithm as shown in the following equations, the web pages are efficiently re-ordered once characteristics are found:

$$Fit = \frac{1}{F} \left[\sum_{i=1}^F |R_{meta}^i - R_{relevant}^i| \right] \quad (19)$$

The suggested algorithm, which is based on the hybridization of GHO and artificial bee colony algorithm, is used to re-rank the WebPages (ABC). First, the suggested method discusses the GHO algorithm's basis and ABC. At that point, the suggested Hybrid grass bee algorithm-based feature selection is explained in more detail [\[26\]](#).

4.1. Grass Hopper Optimization (GHO)

The following steps are included in the GHO algorithm:

1. Assess every grasshopper's fitness.
2. Find each grasshopper's social interaction.
3. Assess the strength of gravity for each grasshopper.
4. Calculate the entire grasshopper's wind advection.
5. Use social interaction, gravity and wind advection to update the grasshopper position.

These processes are continued until to meet the stopping condition.

4.1.1. Social Interaction Calculation

The social interaction (s_a) is calculated using equation (20).

$$s_a = \sum_{\substack{b=1 \\ b \neq a}}^n st(d_{ab}) \hat{d}_{ab} \quad (20)$$

$$e_{ab} = |G_b - G_a| \quad (21)$$

$$\hat{e}_{ab} = \frac{G_b G_a}{e_{ab}} \quad (22)$$

4.1.2. Solution Updating

$$F_a = s_a + g_a + w_a \quad (23)$$

In the below equation, the values of s, g and w are substituted in (23).

$$F_a = \sum_{\substack{b=1 \\ b \neq a}}^n st(|F_b - F_a|) \frac{F_b - F_a}{d_{ab}} - G \hat{e}_G + C \hat{e}_W \quad (24)$$

Number of grasshoppers as indicates F_a . Using the conditions described above, grasshoppers reach a safe location quickly, and the swarm does not congregate at a predetermined spot. To

overcome this problem, a modified version of this condition is shown below.

$$F_a = C \left(\sum_{\substack{b=1 \\ b \neq a}}^n C \frac{U_P - L_P}{2} \sin(|F_b - F_a|) \frac{F_b - F_a}{d_{ab}} \right) + T \quad (25)$$

$$C = C_{\max} - t \frac{C_{\max} - C_{\min}}{t_{\max}} \quad (26)$$

4.2. Artificial Bee Colony Algorithm (ABC)

ABC is a technique of optimization based on bug rummaging (honey bees). The ABC algorithm includes the following steps:

1. Assessing the fitness of each food source
2. Perform the operation of bee for each source of food
3. Carry out a bee operation of the observer for each food source
4. Conduct scout bee activity to identify new sources of food

4.2.1. Employed Bee Operation

Produce a new solution by equation (29),

$$G = G_{ab} + \gamma(G_{ab} - G_{cb}) \quad (27)$$

“Where, G_{ab} is the employed bee and G_{ab}^{New} is a new solution. G_{cb} is the neighbor bee of G_{ab} in employed bee population and γ is a random number”.

4.2.2. Probability Calculation

Assign for each employed bee a selection probability using the following equation,

$$P_a = \frac{fitness_a}{\sum_{n=1}^{SN} fitness_n} \quad (28)$$

Where fitness a is the fitness value of ath employed bee;

4.2.3. Onlooker Bee Operation

The food source will be chosen according to the probability during the onlooker bee stage. The observer bee stage selects the finest food supply acquired by the employee bee stage.

The discretionarily delivered population is calculated as follows:

$$F_{ab} = F_{ab\min} + rand[0,1](F_{ab\max} - F_{ab\min}) \quad (29)$$

Where, $F_{ab\max}$ and $F_{ab\min}$ is upper and lower boundaries for F_{ab} ;

4.3. Oppositional Grass Bee Optimization (OGBEE)

The objective behind this investigation is to choose characteristics that are reliant on the OGBEE algorithm in a viable manner. In this case, a hybrid optimization algorithm is created by combining the opposing grasshopper optimization with the artificial bee colony algorithm. A disadvantage of the OGHO technique is that it is simple to slip into the other two sets of optimum and moderate union speeds. However, the arrangement will see an increase in computing costs, which means it will slow down as well. By mixing this, OGHO and ABC will balance each other out. In order to utilize the ABC optimization algorithm's scout bee phase, the OGHO solution has been changed.

4.3.1. Opposition Behavior Learning (OBL)

Our main addition to MAFS above traditional AFS is Oppositional Behavior Learning, which results in reverse population. It will

improve task scheduling accuracy. Opposition-based learning (OBL) is a technique that computes the opposing solution for each solution in the population before selecting the best answer based on the value of its objective function f . In general, this technique is utilized to improve the convergence of MH algorithms in order to discover the global solution. Tizhoosh developed an Oppositional based learning method to improve the computing speed and rate of convergence of different optimization algorithms. It is impossible to predict the value optimum solution for a randomly generated population number, which is why the time needed to find the optimal solution, is long. Along with population numbers, opposing numbers are introduced in OBL. Due to the inclusion of these opposing numbers, it is discovered that OBL has the capacity to find an optimum solution in the shortest amount of time. The suggested HFSBEE-based feature selection method is described below, and the flowchart for HGBEE is shown in Fig. (2).

4.3.2. Fitness Calculation

When the solution is initialized, every solution's fitness estimate is evaluated and saved for the future. The condition characterizes the fitness capability.

$$Fit = \frac{1}{F} \left[\sum_{i=1}^F |R_{meta}^i - R_{relevant}^i| \right] \quad (30)$$

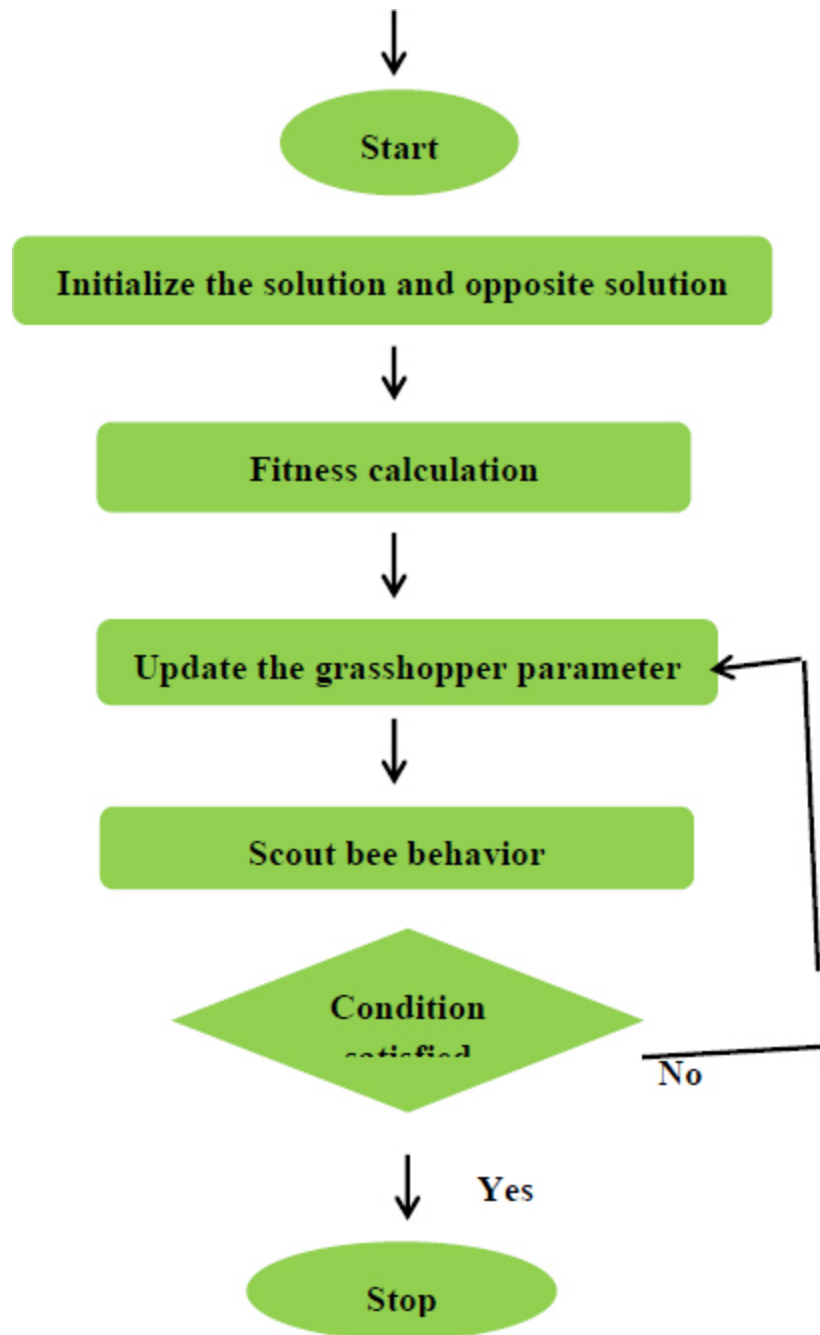


Fig. (2))
Flowchart for OGBEE.

The actualized approach computes the fitness work for each solution based on the given criterion. The solution with the highest fitness capacity picked up is deemed the best. Then, unused arrangements are updated in order to identify the optimum option.

4.3.3. Updating using Grasshopper Optimization

The scientific model used to simulate grasshopper swarming behavior is shown in condition.

$$F_a = C \left(\sum_{\substack{b=1 \\ b \neq a}}^n C \frac{U_p - L_w}{2} st(|F_b - F_a|) \frac{F_b - F_a}{d_{ab}} \right) + T \quad (31)$$

When it comes to finding the optimal answer, the range of acceptable solutions (referred to as “T”) is shrinking. C is a coefficient that decreases this range.

4.3.4. Scout Bee Operation

The fitness estimations of the refreshed solution can't be improved after a specific number of cycles, which means that the refreshed solution is withdrawn, and that solution takes on the role of scout. Once the scout has been selected, he is relocated based on the condition.

$$F_{ab} = F_{ab \min} + rand[0,1](F_{ab \max} - F_{ab \min}) \quad (32)$$

Where, $F_{ab \max}$ and $F_{ab \min}$ is upper and lower boundaries for F_{ab} ;

4.3.5. Termination Criteria

If the suggested technique performs more cycles, the method suspends execution and selects the best fitness available at the time. This solution was selected since it has the most features set. When a method for the OGBEE algorithm achieves the highest fitness, the chosen feature is sent into the multimodal fusion process.

5. RESULTS AND DISCUSSION

The suggested algorithm is implemented in the JAVA tool, and its performance is compared to current approaches.

5.1. Evaluation Metrics

The performance and the measures such as accuracy, recall and F-point are used to assess the suggested re-ranking measurement on the web pages.

5.1.1. Precision

For precision, it is common to say that it is the ratio of the total number of positive observations to the total number of observations that provide a positive result.

$$P = \frac{T}{T+V} \quad (33)$$

Where, P denotes the precision, T represents the true positive rate, and V represents the false positives.

5.1.2. Recall

Recall is the ratio of expected to actual observations. As a consequence, recall is denoted by the following equation.

$$R = \frac{T}{T+U} \quad (34)$$

Where, Y denotes the recall, and T represents the false negatives.

5.1.3. F-measure

The F-score is defined as the average weight of recall and accuracy, which results in false negatives and false positives; this makes the F-score an invalid measure to use.

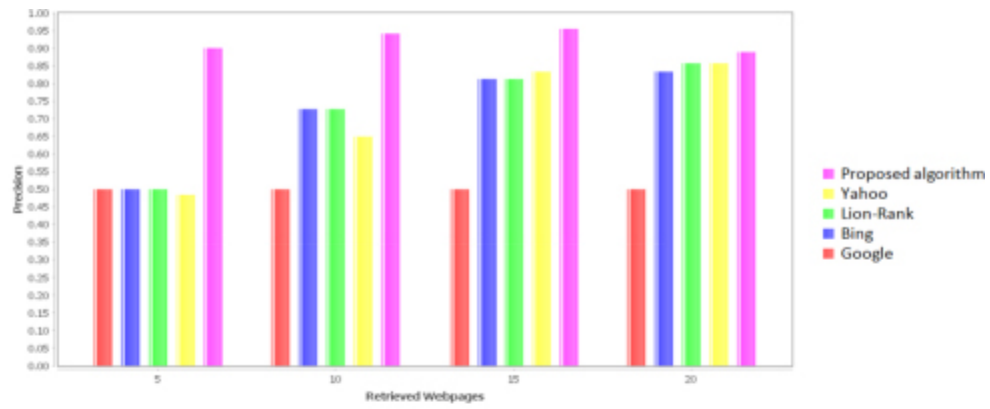
$$F = \frac{2 \times (P \times R)}{(P + R)} \quad (35)$$

The performance of the proposed Re-Ranking algorithm is validated by comparing it to ranking techniques such as Google, Bing, Yahoo, and Lion-Rank [1]. This section explains the comparative analysis of the proposed algorithm using two different queries, namely “information extraction management” and “Web mining”.

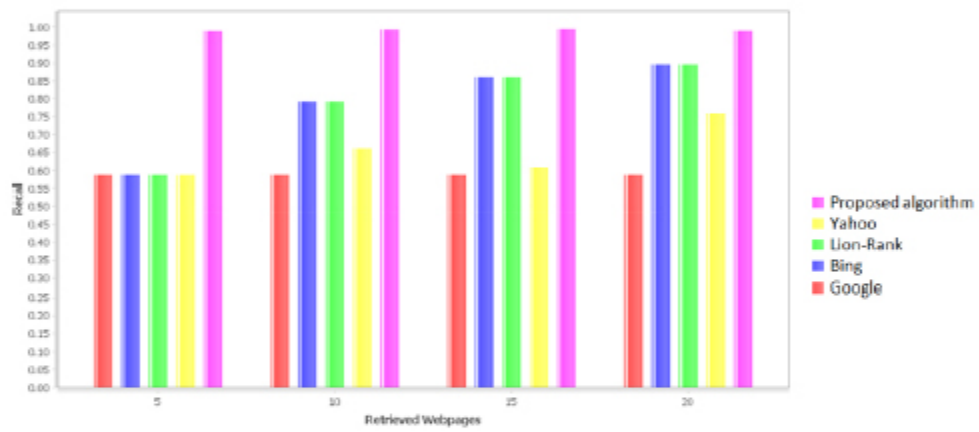
The comparative study of the suggested method performed using the query “information extraction management” by various websites is shown in Fig. (3a) shows the precision analysis in relation to the number of websites obtained using query-1. For five websites, Google calculated a precision of 0.50, Bing computed a precision of 0.50, Lion-Rank computed a precision of 0.50, and Yahoo computed a precision of 0.48, whereas the suggested method achieved a precision of 0.90. When the number of websites is raised to 10, 15, and 20, the suggested method outperforms others in terms of accuracy. With 20 websites, search engines like Google have a precision of 0.50, Bing has a precision of 0.833, Lion-Rank has a precision of 0.857, Yahoo has a precision of 0.889, while the suggested method has a precision of 0.889.

Fig. (3b) shows the studies of the recall of the number of websites that use Query-1. If Web Pages=5, the recall of search engines, such Google's 0.588, bing is 0.588, lion rank is 0.588 and the reminder of Yahoo is 0.550. The recall of search engines like Google is 0.587, Bing is 0.896, Lion rank is 0.896 and Yahoo is 0.750 and the recommended technique is a higher reminder of 0.990. The number of websites is 20.

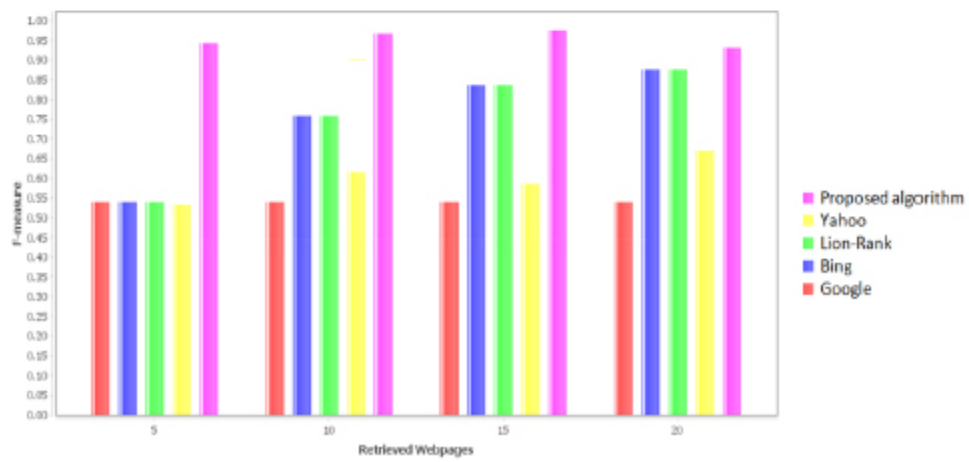
Fig.(8.5.1 c) depicts an examination of the F-measure in relation to the websites utilizing query-1. The F-measure of search engines is calculated by taking into account 5, 10, 15, and 20 websites. When the number of websites is set to 20, the F-measure of search engines such as Google is 0.541, Bing is 0.876, Lion-Rank is 0.876, and Yahoo is 0.710, while the suggested method achieved a superior F-measure of 0.932.



(a)



(b)

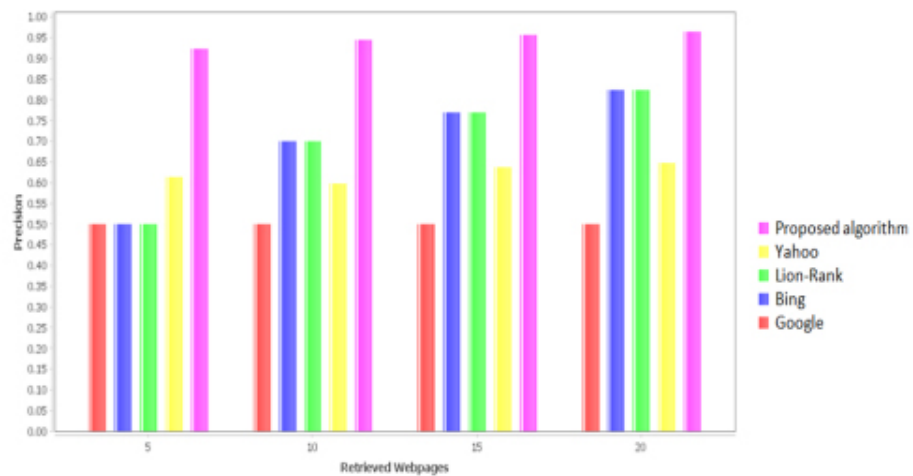


(c)

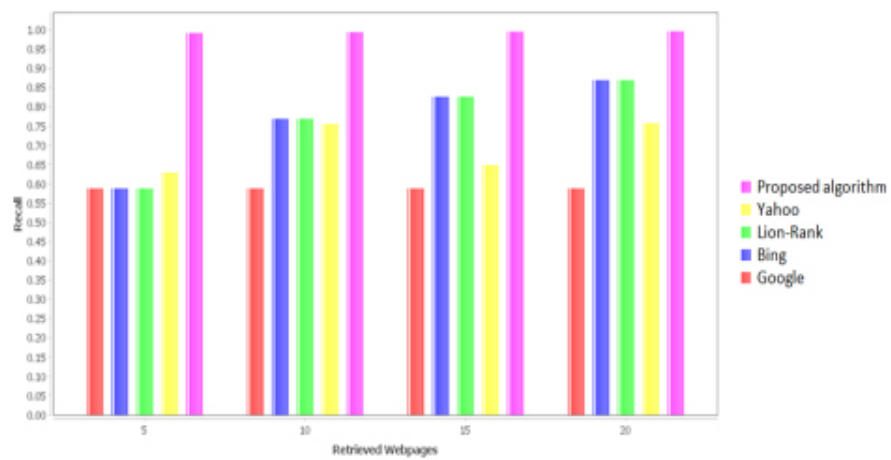
Fig. (3))

Comparative analysis on precision, recall and f-measure for Query 1.

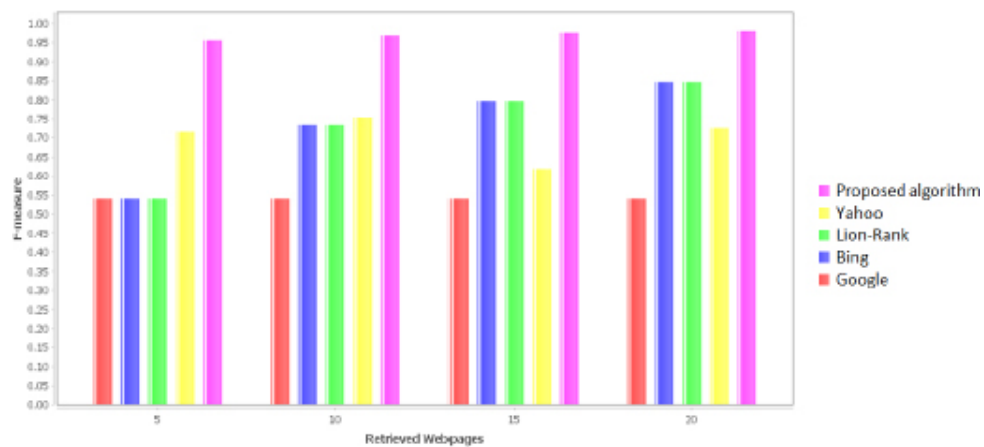
Fig. (4) shows a comparative analysis using the query “Web mining” of the proposed approach. The precision analysis of the number of web pages that are being accessed through query-2 is shown in Figure 8.5.2(a). The precision on 5,10,15 sites and 20 sites is correspondingly 0.50, 0.30 and 0.583, whereas the proposed approach achieves a greater precision of 0.923, for search engines such as Google’s, Bing, Lion Rank and Yahoo. By raising the page number by 20 search engines like Google's accuracy to 0.50, 0.824 for Bing, 0.824 for Lion and 0.653 for Yahoo and 0.964 for greater precision for the proposed approach.



(a)



(b)



(c)

Fig. (4)

Comparative analysis on precision, recall and f-measure for Query 2.

Fig. (4b) depicts the recall analysis in relation to the number of retrieved websites using query-2. The findings are evaluated by taking into account different web page widths. For 20 websites, the recall of search engines such as Google is 0.588, Bing is 0.870, Lion-Rank is 0.870, and Yahoo is 0.750, while the suggested method achieved a recall of 0.996.

The examination of F-measure query-2 is shown in Fig. (4c). The findings are evaluated by taking into account different web page widths. When the number of search engines, such as Google, has a F measurement of 0.541, 0.846 is Bing, 0.846 is Lions-Rank, and 0.740 is Yahoo, while the recommended approach has a F measure superior to 0.980. The time duration is computed in millisecond (ms), in lower order terms and in most cases.

Table. 1 **Time Analysis.**

Query	Proposed Method	GSA [24]	WOA [26]	GHO [25]
Data Mining	2256	2475	2462	2379
Image Processing	3145	3365	3238	3186
Software Engineering	2652	2976	2822	2956
Data Mining and Image Processing	2687	2759	2735	2718

The processing duration is comparable to other existing works [24-26]. A search input request is used to compute the processing time. The following search request gets 2145ms to complete a process in the data mining input search query. It takes a little time, to complete the job, such as 3034, 2541, and 2576ms, for Table 1 data mining, image processing and data engineering. As a result, it may get a minimal amount of time needed to finish a task. When compared to the suggested approach, the other methods utilized in the previous work [24-26] need greater processing time. Table 2, on the other

hand, examines the memory use of the proposed technique and other current methods. Table 2 analyses a memory measure.

Table. 2 **Memory Analysis.**

Query	Proposed Method	WOA [26]	GHO [25]
Data Mining	123451	1276686	121655
Image Processing	1002682	108564	1012656
Software Engineering	956798	969966	95696
Data Mining and Image Processing	1043423	1235963	116566

CONCLUSION

This study develops an optimum and efficient re-ranking method for the ranking of web pages based on the OGBEE algorithm. The suggested algorithm consists of four phases: web page creation, pre-processing, extraction of features and re-ordering. Finally, the reordered measurement of the web is performed through factor-based and semantic characteristics. But, as a metric such as accuracy, F-measurement and recall values of 0.964, 0.980, and 0.996, the suggested algorithm achieved superior performance. In the future, some other optimization algorithms may improve the efficiency of the re-ranking process.

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Text Analytics

Divanu Sameera^{1, *}, Niraj Sharma², R.V. Ramana Chary³

¹ CSE Department of SSSUTMS-Sehore, MP, India

² Department of CSE, SSSUTMS-Sehore, India

³ Department of IT, B V Raju Institute of Technology, Narsapur, Medak, Telangana, India

Abstract

This chapter covers text analytics definitions, how to get started with text analytics, examples and approaches, and a case study. The chapter gives examples of existing text analytics applications to show the wide range of real-world implications. Finally, as a guide to text analytics and the book, we give a process road map. Chapter 2 (How to Get Started with Text Analytics) briefly explains the Analyse Your Data, Use BI Tools to Understand Your Data and Final Words. Chapter 3 (Examples and Methods for Text Analytics) explains various Text Analytics Approaches 1: Word Spotting Text Analytics Approach 2. Manual Rules Text Analytics Approach 3. Text Categorization Approach 4: Topic Modelling Approach and 5. Thematic Analysis. Applications of Word Spotting Text Analytics Approach, Manual Rules, Text Categorization Approach, Topic Modelling Approach and Thematic Analysis are discussed with real-time examples. Chapter 4 discusses the case

study, the following real-time application, Word Cloud Explorer, to illustrate its analytic capabilities.

Keywords: Text Analytics, Word Spotting Text Analytics, Thematic Analysis, Text Categorization.

* **Corresponding author Divanu Sameera:** Department of SSSUTMS-Sehore, MP, India;
E-mail: ameerach24@gmail.com

1. INTRODUCTION

1.1. Text Analytics Basics

Text analytics is the act of automatically converting an enormous amount of indefinite text into numerical data in order to identify insights, trends and patterns. This technique, when combined with data visualisation tools, allows businesses to comprehend the story behind the numbers and make better decisions [1-3]. You may have come across comparable ideas like text mining and text analysis while

looking for a definition of text analytics. Therefore, before we go into the details, let's revise the key distinctions between these phrases [4].

Text mining, text analysis, and text analytics are the terms that refer to the process of examining disorganized text so as to extract insights. Text analytics, however, gathers these results and turns them into something that can be quantified and represented through charts and reports, whereas text mining (or text analysis) provides qualitative insights [5].

Text analysis and text analytics are frequently used in tandem to provide a comprehensive knowledge of all types of text, including emails, social media posts, surveys, and customer service tickets, among other things. For example, text analysis technologies have major use in determining how people feel about a company on social media (sentiment analysis) or in comprehending the major issues in product evaluations (topic detection) [6]. Text analytics, else ways, uses the results of text analysis to discover patterns, such as a spike in negative comments, and gives you actionable insights that you can use to enhance your product, such as addressing a defect that is causing your users to be frustrated [7].

1.2. Text Analytics Examples

Among the most important intriguing applications of text analytics, customer feedback analysis is primarily in business. This could entail assessing product and service reviews to determine how your consumers rate your business, processing the findings of open-ended customer survey replies, or monitoring what customers are saying about your brand on social media [8, 9].

Let's imagine you want to look at open-ended NPS replies to see which components of your business are being discussed by your customers.

The first step would be to perform a topic analysis, which is a text analysis approach that tags NPS replies based on predetermined categories like Feature Requests and Customer Service. Listed below are a few examples:

Feature Request: "I wish you had the option to export the data."

"Customer service is extremely helpful and friendly!" Customer Service

After you've categorised each NPS response, you can use text analytics to find patterns and insights throughout the entire dataset, and then display the results in charts or reports. Recently employed

text analytics to map common subjects in NPS replies, as shown in the graph below:

According to (Fig. (1)) , the majority of negative comments are about product features, while the majority of positive comments are about product usability and customer support. According to the negative feedback, adding popular feature requests could help improve customer happiness [10].



Fig. (1)
Response Tags Analysis.

Customer service is another fascinating application of text analytics in business, in addition to customer feedback analysis [11]. Text analytics, for example, found major use in examining the content of support requests to better understand your customers' needs, motivations, and expectations, as well as providing suggestions on how to rethink your customer experience approach [12, 13].

Text analytics can also be utilised to extract crucial data from various customer support channels like as email, chat, and social media, in addition, to support tickets shown in Figs. (2 & 3). For example, we looked at telecommunication business customer service conversations on Twitter and discovered that T-Mobile has the most favourable interactions [14, 15].

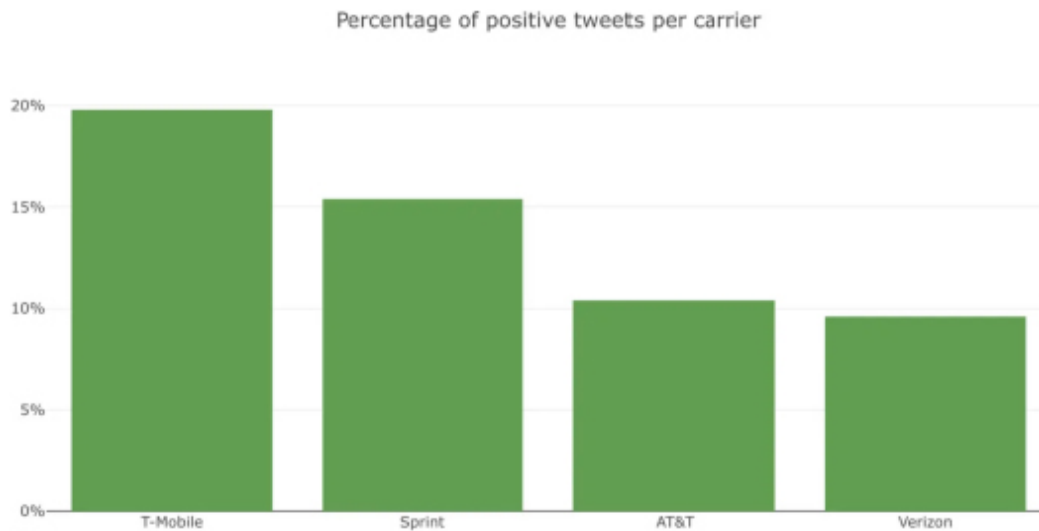


Fig. (2)
Percentage of Positive tweets per carrier.

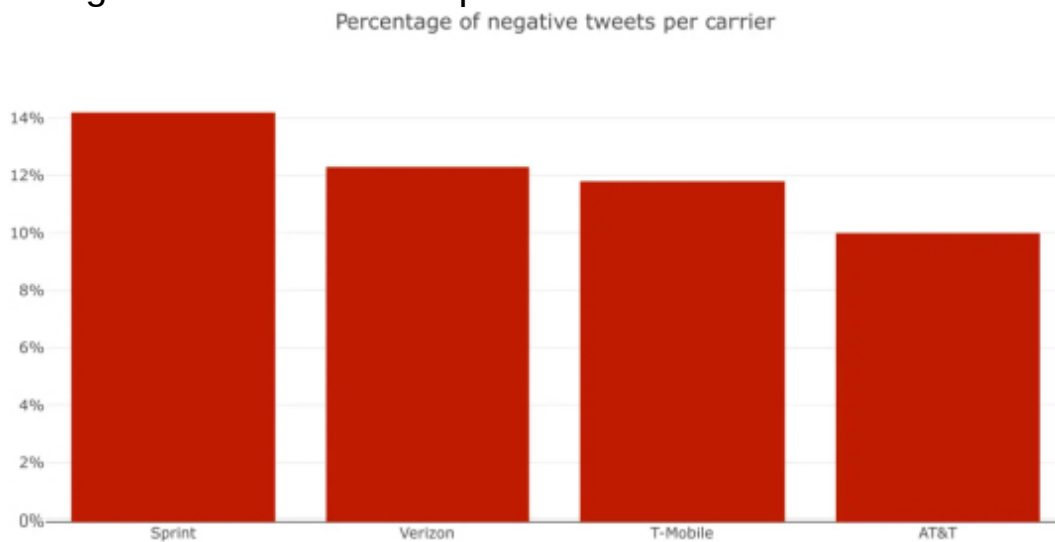


Fig. (3)
Percentage of Negative tweets per carrier.

Sprint, contrarily, has the most negative Twitter mentions: Detecting a spike in customer service complaints in a timely manner, for example, may indicate that you need to provide additional training to your agents, improve the routing process for support tickets, or make your help centre more user-friendly [\[16\]](#).

2. HOW TO GET STARTED WITH TEXT ANALYTICS

Now that you've learned about the various applications of text analytics, you're probably wondering how to get started. Fortunately, even if you don't have any programming skills, there are several online tools that allow you to take your first steps in a very intuitive manner [[17](#), [18](#)]. Let's take a closer look at each step:

2.1. Analyze Your Data

You'll need to analyse the data you wish to use text analytics on first. MonkeyLearn provides a variety of pre-trained machine learning models that you may use right immediately to perform text analysis. Observe this sentiment analysis model for an illustration of how they work [[19](#)].

You may also create your own customised machine-learning models for text classification and text extraction, which is a terrific alternative in the case of distinguishing specific subjects or keywords within a field [[20](#)].

Following these 5 steps, you can make a custom model:

1. Selecting a model type: Classifier or Extractor.
2. Importing your data: You can utilise a CSV or Excel file, as well as third-party connections, to import your data.
3. Choosing the tags that find use in the analysis.
4. Manually tag multiple instances to train your model. After a few examples, the model will begin to forecast on its own. If you're not happy with the findings, you may always train your model to enhance its accuracy.
5. Analyzing fresh data with your trained model: you can upload new data in batches, utilise one of the third-party app integrations to analyse data automatically, or use the MonkeyLearnAPI [[21](#)].

2.2. Use BI Tools to Understand Your Data

You can generate a data visualisation of the results once you've finished your text analysis. Business intelligence tools such as Google Data Studio, Looker, and Tableau, among others, can assist you in creating visually appealing and interactive reports and infographics to communicate your data's major insights. Here's an example of how Looker can help you visualise client data shown in Fig. (4) [22]:



Fig. (4)
Customer data visualization.

You're ready to share your findings with the rest of your team once they look like this.

2.3. Final Words

Text analytics enables businesses to extract useful information from a large assortment of data sources, including customer support requests and social media interactions [23].

Text analytics can reveal patterns, trends, and actionable insights that you can use to create data-driven decisions by combining the findings of text analysis and using business intelligence tools to put the numbers into easy-to-understand reports and images. You can use text analytics to help you detect opportunities for improvement and adapt your product or service to your clients' needs and expectations

after analysing customer feedback (like product reviews or NPS responses) or examining the content of customer support tickets with text analysis tools [\[24\]](#).

3. EXAMPLES AND METHODS FOR TEXT ANALYTICS

Here's my breakdown of these methods into five primary approaches that are now in use.

3.1. Text Analytics Approach 1: Word Spotting

Let's begin with some word spotting. First and foremost, it isn't a thing!

Such an approach is not well received in the academic world of Natural Language Processing, and for good reason. In reality, word spotting is a term used in academia to describe the recognition of handwriting (spotting which word a person, a doctor perhaps, has written) [\[25\]](#).

Keyword spotting is another technique that depends on speech processing.

But, as far as I'm aware, word spotting isn't employed for any kind of text analysis [\[26\]](#).

However, I've heard about it frequently enough in meetings to include it in this review. It's popular among DIY analysts, Excel wizards, and many consumer insights experts.

The basic premise underlying text word spotting is that if a word appears in text, we can infer that the content is “about” that term. When words like “price” or “cost” appear in a review, it indicates that the review is about “Price.” [\[27\]](#).

3.1.1. The Simplicity of the Word-spotting Approach is What Makes it so Appealing [28]

Below ten minutes, word spotting can be integrated in an Excel spreadsheet by you.

You could also write a Python or R script. Here's how to do it.

In about ten minutes, you can create a Text Analytics solution. To group comments into “Billing,” “Pricing,” and “Ease of Use,” you can use a formula like this one in Excel: And voilà! As shown Fig. (5), Column B of a Net Promoter Score survey comprises open-ended responses to the question “Why did you give us this score”:

	A	B	C
1	NPS Category	Comment	Category
2		3 Always cost	PRICE
3		1 Because I find my gas bills are crazy, even in summer.	BILLING
4		10 Very easy to understand web site	EASE OF USE
5		8 easy to pay and different payment options	EASE OF USE
6		10 Everything was easy to do and I had no problems!	EASE OF USE
7		8 it was easy	EASE OF USE
8		9 It's easy and fast	EASE OF USE

Fig. (5)

Word spotting approach 1.

I think it took me below ten minutes to make this, and the end product is fantastic! But hold on...

Simplicity appeals to everyone. However, in this case, simplicity isn't a good thing shown in Fig. (6).

This method can quickly lead to a slew of problems.

I've annotated them for you here.

	A	B	C	D
1	NPS Category	Comment	Category	Notes
2		3 Always cost	PRICE	
3		1 Because I find my gas bills are crazy, even in summer.	BILLING	Should be "Price"
4		10 Very easy to understand web site	EASE OF USE	Missed the Website aspect
5		8 easy to pay and different payment options	EASE OF USE	Missed the Ease of Payment aspect
6		10 Everything was easy to do and I had no problems!	EASE OF USE	
7		8 it was easy	EASE OF USE	
8		9 It's easy and fast	EASE OF USE	Missed "Fast"

Fig. (6))

Word spotting approach 2.

Only three of the seven comments were accurately classified. The term "billing" refers to "Price," and three other comments failed to mention further themes. Would you stake your customer data on something that is just 50 percent accurate?

3.1.2. When Word Spotting is Acceptable?

It's easy to see how the formula above could be improved. Indeed, I've spoken with businesses that have created enormous custom spreadsheets by hand and are very pleased with the outcomes [29]. This method can be used if you have a dataset with a few hundred replies that you only need to evaluate once or twice. On the condition that the dataset is tiny, you may rapidly evaluate the results and ensure that they are accurate.

When word spotting doesn't work

What about the disadvantages? Please refrain from using the term spotting:

- If you don't have time to examine and fix the accuracy of each piece of text
- If you want to visualise the results (Excel will hear you swear)
- If you wish to share the results with your colleagues
- If you want to keep the data constant over time

There is a slew of other drawbacks to doing your own word spotting, which we'll go into in the following post. I'll also discuss what truly works and what constitutes a good strategy. Check out our in-depth guide: How to develop your own feedback analysis solution if you want to build your own Text Analytics solution [\[30\]](#).

3.2. Text Analytics Approach 2. Manual Rules

Word spotting is closely related to the Manual Rules technique. Both methods use the same premise to create a match pattern, but these patterns can get fairly complicated.

A manual rule, for example, might need the usage of regular expressions, which are difficult to apply in Excel [\[31-33\]](#). From the popular corporate system Medallia, below is a rule for assigning the category “Staff Knowledge” in Fig. [\(7\)](#).

“Staff Knowledge”:

- i. knowledge
- ii. know NEAR where within 1 words
- iii. know CONNECTED TO products

Fig. (7))

Staff Knowledge.

The majority of Text Analytics providers, as well as many other smaller companies who sell Text Analytics as a supplement to their core product, provide an interface that enables creating and managing such rules simply. They may also provide professional assistance with the formulation of these rules [\[34\]](#).

The nicest part about Manual Rules is that they are simple to comprehend. They are explicable, they can be changed and adjusted as needed.

But the simple line is that creating these regulations requires a significant amount of time and work. You must also guarantee that they are accurate and that they are maintained throughout time [35].

Some organisations provide pre-packaged rules that are already grouped into taxonomy to help you get started. For example, they might have a category called “Price” with hundreds of words and phrases already pre-programmed, as well as sub-categories like “Cheap” and “Expensive.”

They may also build up specific categories for specific sectors, such as banks. If you're a bank, the first thing you need to execute now is to add your product names to the taxonomy, and you're ready to go [36].

This method has the advantage of allowing you to run millions of feedback pieces and receive a good overview of the main categories specified in the text once it is set up.

However, this strategy, like any manual rule or word-detecting tool, has a number of drawbacks:

3.2.1. Multiple-word Meanings Make it Hard to Create Rules

The most prevalent cause of rule failure is polysemy, which occurs when the same word can have multiple meanings shown in Fig. (8):

It's Hard to Find a Rule That Works Well

<i>friendly</i> OR <i>friendliness</i> -> Staff friendliness		
I was impressed by how friendly the person on the other end of the line was	Staff friendliness	✓
The lady who helped me was friendly	Staff friendliness	✓
Friendliness of staff	Staff friendliness	✓
Your website is very user friendly	Staff friendliness	✗
The young man on the phone was very pleasant	Other	✗

Fig. (8))

Comparison Table.

3.2.2. Mentioned Word! = Core Topic

Just because a term or phrase appears in a book, it does not necessarily imply that the text must describe the issue. When a consumer describes the circumstances that guide to an issue, for example, “My credit card was declined, and the clerk was incredibly helpful, patiently waiting while I searched my bag for cash.” This remark is not about credit cards or cash; it is about the staff's behaviour [37-40].

3.2.3. Rules

Knowing the broad category is insufficient. Are individuals pleased or unhappy when they think of the word “Price”? It's hard to capture sentiment using manually pre-set rules in Fig. (9). Many people are unaware of how unique and diversified our language is.

	Positive or Negative?
My coffee was great	positive
My coffee was awful	negative
My coffee was not great	negative
My coffee was not that great	neutral?
I did not think my coffee was great	negative
I did not expect my coffee to be this great	positive
I was disappointed with the quality of the coffee	negative
I was not disappointed with the quality of the coffee	positive

Fig. (9))

Rules.

As a result, modelling a sub-category like “expensive” is incredibly tough. Someone can say, “I didn't believe this product was so pricey.” To classify this comment into a category like “excellent price,” a

complicated algorithm to identify negation and its scope would be required. It's not going to be enough to use a basic regular expression [[41](#), [42](#)].

3.2.4. Taxonomies Don't Exist for Software Products and Many Other Businesses

Non-standard products and services will not have pre-defined taxonomies with rules. This is especially difficult in the software sector, because each product is distinct, and consumer feedback addresses extremely specific issues [[43](#)].

3.2.5. Not Everyone can Maintain Rules

Even if you possess a rule for working based on taxonomy in any business, someone with good linguistic understanding will need to regularly maintain the rules to ensure that all of the feedback is correctly classified. This person would have to be on the lookout for new expressions that people make on the fly, as well as any emergent themes that hadn't been examined before. It's a never-ending and extremely costly process [[44](#)].

Despite these drawbacks, this approach is the most commonly used commercial use of Text Analytics, dating back to the 1990s, and there is no clear path for resolving these concerns.

So, are Manual Rules sufficient?

My answer to this is No. The majority of people who use Manual Rules are unhappy with the amount of time it takes to establish a solution, the expenses of maintaining it, and how actionable the insights are.

3.3. Text Analytics Approach 3. Text Categorization

Let's clear up the murky subject of Advanced Text Analytics and the many vendors' and data scientists' pitches for it.

We'll look at Text Categorization first, as it's the first of the three ways that usage of algorithms are automated.

3.3.1. What is Text Categorization, and How Does it Work?

Machine learning is a key to this strategy. The basic concept is that a machine learning algorithm (of which there are many) analyses previously manually classified instances (training data) and determines the rules for categorising new examples. It's a method that's been overseen in Fig. (10) [45-48].

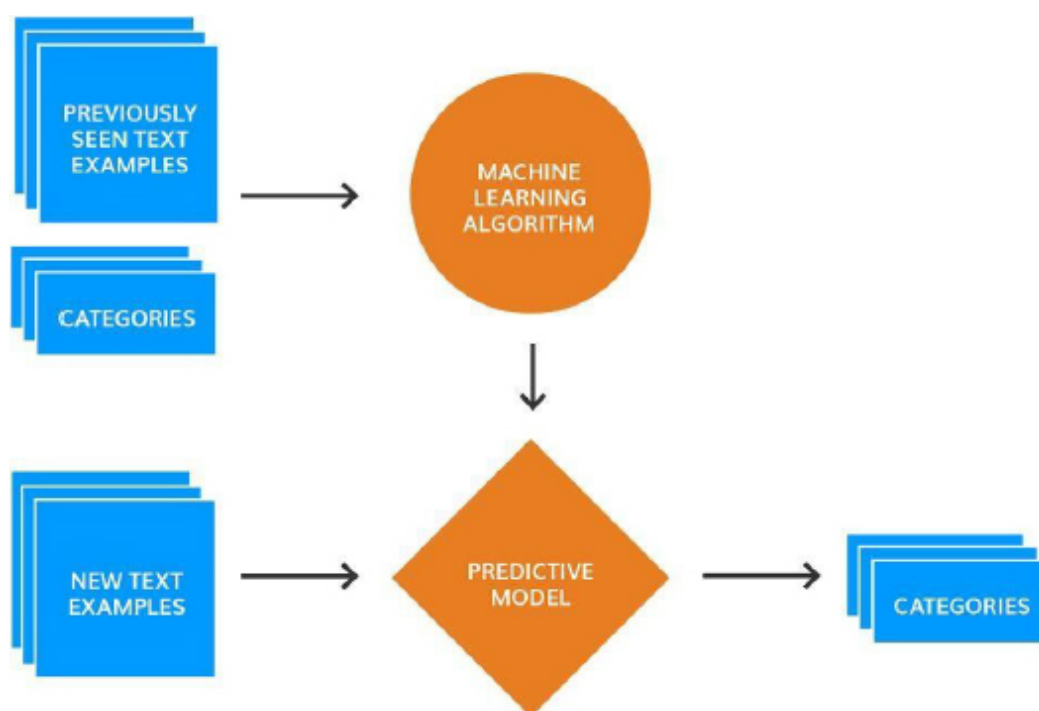


Fig. (10))
Text Categorization.

The strength of the classification of text is that, unlike the two previous approaches, you only need to give samples; no human construction of outlines or instructions is required.

Another benefit of text classification is that it should potentially be capable of expressing the relative importance of a word's appearance in the text. Let's look at an example from one of our previous posts. "My credit card was refused, and the clerk was incredibly supportive,

standing by persistently waiting patiently while I looked for cash in my bag,” a client may be clarifying the circumstance that prompts an issue. This remark is not about credit cards or cash; it is about the staff's behaviour. The comment's theme of “credit card” is unimportant, but “helpfulness” and “patience” are. With the correct training, a text categorization technique can capture it [\[49-51\]](#).

Seeing similar occurrences in the working out of data is the key. Near perfect accuracy... but only with the right training data. There are scholarly research articles which display that categorization of text can attain near flawless accurateness. Compared to old algorithms (Naïve Bayes), deep learning algorithms are more influential [\[52\]](#).

But all the researchers approve that compared to **the training data, the algorithm isn't as important.**

Both the quantity and quality of the training data is the deciding factor in how fruitful this methodology is for managing input. Hence, how much is sufficient? Indeed, it relies upon the quantity of classes and the algorithm implied to make a classification model.

The more the categories and they are more closely related, the additional training data is required to aid the algorithm to find out the differences between them [\[53\]](#).

The start-ups of a few up-to-date Text Analytics depend on the categorization of text, which offers the tools by making it simple for the people to get trained with the algorithms, so that they get better over time. Yet, do you have time to **wait for the algorithm to become improved, or do you want to act on the feedback of customers today?**

Aside from demanding to train the algorithm, there are four problems with categorization of text to analyse the feedbacks of people.

3.3.2. You Won't Notice Emerging Themes

You will only understand perceptions about the categories for which you have trained for and will omit the unknown unknowns. This is exactly similar to the disadvantages of word spotting and manual rules. It is important to monitor the incoming feedback continuously for miscategorised items and for developing themes [\[54-59\]](#).

3.3.3. Lack of Transparency

It is difficult to comprehend why it works in the way it is working while the algorithm improves over time and, in this way, effectively changes the outcomes [\[60\]](#). Subjective analysts have said that the major reason is the lack of transparency why text categorization did not take off in their world. For instance, if the accuracy to differentiate between two themes, “wait time to install fiber” and “wait time on the phone to set up fiber” is poor, suddenly, what amount of training data is needed to add, till the algorithm quits committing these errors?

3.3.4. Preparing and Managing Training Data is Hard

The real issue is the deficiency of training data. It's difficult to begin without any preparation, and most organizations need more or sufficient data to train the algorithms. Indeed, organizations consistently overestimate what amount of training data they have, which makes execution fall beneath assumptions. Lastly, in the event that you need to refine one explicit classification, you should re-mark the entirety of the information without any preparation.

3.3.5. Re-training for Each New Dataset

Transferability can be truly dangerous! Just think you own a working text categorization solution for one of your divisions, for example, support, and then you need to analyse the feedback that arises through the surveys of customers, such as CSAT or NPS. Once more, you would have to re-train the algorithm.

I just had a call with the person who is the subject matter expert on the analysis of survey who revealed this story to me: A group of data

scientists expended numerous months and made a solution that she eventually needed to excuse because of accuracy deficiency. The organization didn't have the opportunity to trust that the algorithm will improve after some time.

3.4. Approach 4: Topic-Modelling

Topic-modelling is additionally an ML method, however an unproven one, which implies that this methodology gains from the crude text. Sounds energizing, correct?

Infrequently, I receive perceptions experts allude to any ML approach as “topic-modelling”, yet data scientists generally mean a particular algorithm when they say “topic-modelling” in Fig. ([11](#)).

It is known as tongue-twisting Latent Dirichlet Allocation (LDA). It's an exquisite numerical model of language that catches subjects (arrangements of comparable words) and how they range across different texts [[61](#), [62](#)].

To applying topic modelling to beer reviews, for example,

1. The input is surveys of different lagers.
2. A topic is a group of parallel words like espresso, coffee, dark or black chocolate.
3. A catalogue of topics is composed in each survey. In the mentioned example, The Kernel Export stout London has four topics allotted to it.

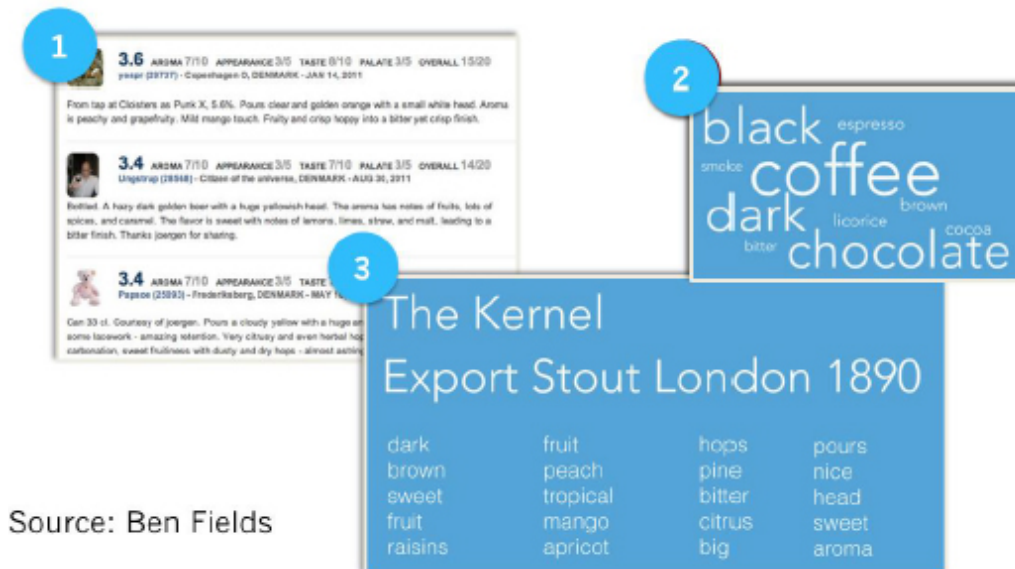


Fig. (11)
Topic modelling.

The topics can likewise be biased. For instance, a client remark like “your client care is dreadful, kindly get a telephone number”, could have loads and topics as follows:

- 40% support, service, staff
- 30% bad, underprivileged, terrible
- 28% number, phone, email, call

3.4.1. What’s Incredible Regarding Topic-modelling

The finest thing regarding topic modelling is that it needs no info other than crude client criticism. In contrast to the classification of text, it’s unsubstantiated. In basic words, the learning occurs by seeing which words show up close by different words in which surveys, and catching these data utilizing statistical probability. In case you are into maths, you will cherish the idea, clarified completely in the Wikipedia article, and if those equations are a bit excessive, I suggest Joyce Xu’s clarification [63].

There are new text analytics companies that utilize point demonstrating to give an investigation of criticism and other content

datasets. Different organizations, similar to Stitch Fix, for instance, are using topic-modelling to drive item suggestions. They broadened conventional topic-modelling with a deep-learning method called word embeddings. It permits us to catch semantics in a more exact manner (more on this in Part 5).

For what reason is topic-modelling a deficient strategy for analysing feedback?

When utilized for *analysing* feedback, *topic*-modelling has one fundamental inconvenience:

The importance of the topics is truly hard to decipher.

Every topic catches some part of language, however, in a non-straightforward algorithmic way, which is not quite the same as how individuals get language [\[64\]](#). For example, how might you decipher the second and the fourth themes for the bold lager in the above model.



Fig. (12))

Non-transparent algorithmic.

While the first and the subsequent topic can be, by one way or another, “named” as pleasantness and fruitiness, the other two points are only an assortment of words.

Any information researcher can assemble an answer utilizing public libraries that can rapidly let out a fairly significant yield. In any case, transforming this yield into outlines and charts that can support business choices is hard. Observing how a specific subject change over the long haul to build up whether the activities taken are working is significantly harder [\[65, 66\]](#).

To summarize from Fig. ([12](#)), in light of the fact that topic-modelling produces results that are difficult to decipher as it needs straightforwardness actually like text-categorization calculations do, I don't suggest this methodology for examining criticism. Notwithstanding, I remain by the calculation as one that can catch language properties genuinely well, and one that functions admirably in different undertakings that require Normal Language Understanding.

3.5. Approach 5. Thematic Analysis

The entirety of the previous methodologies referenced has inconveniences. In the best case, you'll get alright outcomes solely after spending numerous months setting things up. Furthermore, you might pass up obscure questions.

The expense of acting late or passing up significant experiences is immense! It can prompt the loss of clients and stale development. This is the reason, as per YCombinator (the startup accelerator that delivered more billion-dollar organizations than some other), “at whatever point you're not dealing with your item, you ought to be addressing your clients”.

After Topical partook in their program, we've been requested exhortation multiple times by means of an overview, once through an individual email, and furthermore, face to face. YCombinator additionally utilizes Topical to sort out all the input they gather [67].

With regard to client input, three things matter:

1. Accuracy, explicit and noteworthy investigation.
2. Capability to see arising subjects quick, without the need of setting things up.
3. Clearness in how results are made, to acquire area skill and good judgment information

In my exploration, I've discovered that the lone methodology that can accomplish every one of the three necessities is Thematic-Analysis, joined with an interface for effectively altering the outcomes.

3.5.1. Thematic Analysis: How it Works

Topical Examination approaches separate subjects from text, as opposed to order text. At the end of the day, it's a base-up examination. Given a piece of criticism, for example, "The airline steward was useful when I requested to set up a child bed", they would remove topics, for example, "airline steward", "airline steward was useful", "useful", "requested to set up a child bed", and "childbed" [68, 69].

These are all meaningful phrases that can potentially be insightful when analysing the entire dataset.

In any case, the most significant advance in a Thematic-Analysis approach, shown in Fig. (13), is consolidating phrases that are comparative into subjects and coordinating them in a manner that is simple for individuals to survey and alter. We accomplish this by utilizing our custom word embedding execution; however, there are diverse approaches to accomplish this [70].

THE FLIGHT ATTENDANT WAS HELPFUL WHEN I ASKED TO SETUP A BABY COT

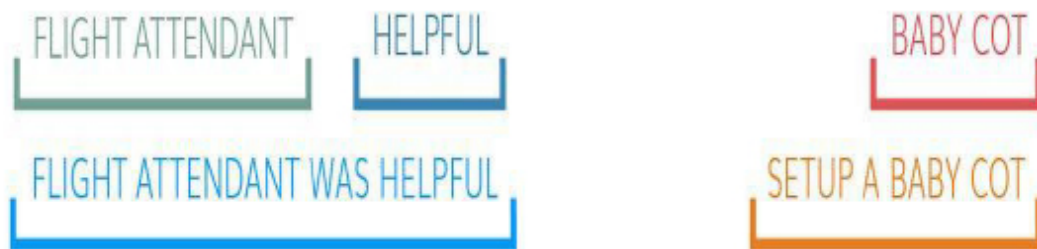


Fig. (13))
Thematic Analysis.

For instance, here is the manner by which three individuals talk about exactly the same thing, and how we at Thematic-gathering the outcomes into subjects and sub-subjects shown in Fig. (14).



Fig. (14))
Thematic Analysis Example 1.

3.5.2. Advantages and Disadvantages of Thematic Analysis

The benefit of Analysis is that this methodology is unaided, implying that you don't have to set up these classifications ahead of time, don't have to prepare the calculation, and accordingly can undoubtedly catch the obscure questions.

The disservices of this methodology are that it's hard to carry out accurately. An ideal methodology should have the option to blend and

sort out subjects in a significant manner, delivering a bunch of topics that are not very nonexclusive and not very enormous. In a perfect world, the subjects should catch basically 80% of verbatims (individuals' remarks). Furthermore, the topics extraction should deal with complex invalidation conditions, for example, "I didn't think this was a decent espresso" [\[71, 72\]](#).

A portion of the set-up greater players have carried out Thematic-Analysis to upgrade their Manual Principles approaches yet will, in general, deliver a clothing rundown of terms that are difficult to survey.

Traditional Text-Analytics APIs planned by NLP specialists additionally utilize this methodology. Be that as it may, they are once in a while planned considering client criticism and attempt to take care of this issue in a nonexclusive manner. For instance, when we tried Google and Microsoft's APIs, we found that they weren't gathering subjects out of the crate.

Accordingly, simply 20 to 40% of input is connected to the top 10 subjects: just when there are solid likenesses in how individuals talk about explicit things. By far, most of the criticism is uncategorized, implying that you can't cut the information for more profound bits of knowledge.

At Thematic in Fig. [\(15\)](#), we have fostered a Thematic-Analysis approach that can undoubtedly investigate input from clients of pizza conveyance administrations, music application makers, land agents and more. We accomplished this by zeroing in on a particular sort of text: client input, dissimilar to NLP APIs that are intended to deal with a book. We have carried out complex invalidation calculations that differentiate positive from negative topics, to give a better understanding.

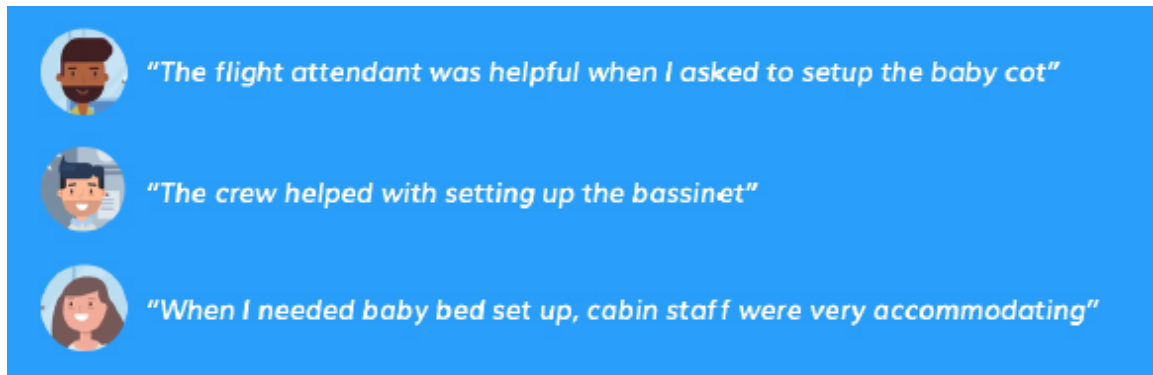


Fig. (15))

Thematic Analysis Example 2.

3.5.3. Human in The Loop

Each dataset shown Fig. (16), and now and then even each review question, gets its own arrangement of topics, and by utilizing our Themes Editor, experiences experts can refine the topics to suit their business. For instance, Thematic may discover subjects, for example, “quick conveyance”, “fast and simple”, “an hour stand by”, “moderate help”, “delays in conveyance” and gathering them under “speed of administration”. One understanding proficient may re-bunch these into “moderate” and “quick” under “speed of administration”, another into “quick assistance” > “fast and simple”, and “moderate help” - > “an hour stand by”, “delays in conveyance”. It's an abstract errand [73, 74].

Approach	How it works	Best fit	Data required	Advantages	Disadvantages	Effort to setup	Effort to maintain	Accuracy	Transparency
Manual Rules and Taxonomies	Manually crafted and maintained rules.	Companies in industries that don't change their offering.	Any amount	Easy to understand.	Requires a lot of time to create. Can't capture unknowns or sentiment.	Months	Professional services: 1 person 1 day a week.	Moderate to Good	Good
Topic Modeling	Topics (sequences of related words) are extracted from data.	Companies with expert NLP data science teams who do a one-off analysis for quick insights.	At least several thousand of feedback pieces.	No training required.	Difficult to interpret. Impossible to tweak.	Days to weeks	Depends on the data scientist's skill set.	Moderate	Poor
Text Categorization	Categories trained on pre-categorized data.	Companies who manually and consistently tag feedback and don't make major changes to their offering.	At least a few hundred pieces per category.	Can be highly accurate and capture context.	Requires training data. Can't capture unknowns. Difficult to tweak.	Months	Not much, if categories don't change.	Good to Very High	Poor
Thematic Analysis	Themes are extracted from text, similar ones merged.	Companies in any industries who have at least a few hundred pieces of feedback per month.	A few hundred pieces of feedback a month.	Does not require training data. Captures unknowns. Easy to understand. Can be accurate and capture context.	Difficult to implement correctly, e.g. negation must be handled accurately.	Days	Anyone can maintain. 1-2h per week for Thematic to review.	Good to Very High	Very Good

Fig. (16))
Text Analysis Approach.

I accept an ever-increasing number of organizations will find Thematic-Analysis, in light of the fact that dissimilar to any remaining methodologies, a straightforward and profound examination doesn't need preparing information or time for making manual principles.

We've created a cheat sheet which lists the text analytics approaches; check it out below

4. CASE STUDY

We'll show you an application that uses Word Cloud Explorer Fig. (17) to illustrate its analytic capabilities. Reuters Corpus Volume 1 (RCV1) is used as an example corpus. It is made up of a massive collection of news articles that have been manually classified and made available for research purposes by Reuters Ltd. From August 20 to August 26, 1996; we used the first week of articles from the corpus [75].

During that week, three significant worldwide sporting events dominated the news. First, there was the summer Olympic Games in Atlanta, Georgia, that year. Second, the Wimbledon Tennis Championships, which started a few days later earlier in London. The third event was the U.S. Open, which began that week in New York. As a result, we limit the Reuters corpus to sports news by choosing those items that have been classified as such [76]. We utilise a task from the qualitative user study described in the next section as an analysis example: “Who are the most commonly mentioned male and female Olympic champions during that week?”

We begin by selecting an alphabetically sorted word cloud in the Word Cloud Explorer, as shown in Fig. (4). The most often occurring terms in this cloud are sports-related phrases like won, played, match, and game. We can also make out the names of countries, towns, and sporting events, such as Wimbledon, the Olympic Games, and the U.S. Open. We ponder the term Olympic because we are interested in Olympic champions. We can notice that the word statistics panel appears in 87 sentences in the text corpus. The info box goes on to say that it appears 90 times in those 87 phrases, with seven of those occurrences being within multiword like Olympic Committee [77].

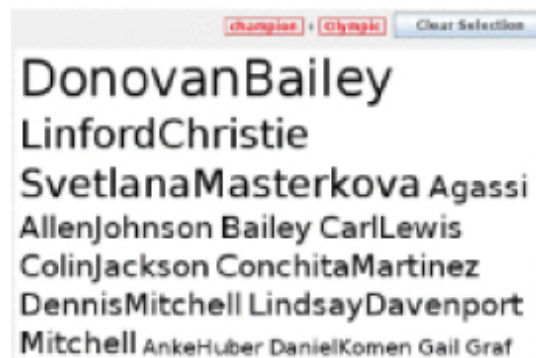
When we look at the Olympic co-occurrence cloud (Fig. (12)), we can observe that the phrase champion is used the most. Many sportsmen had their first and last names contracted by the multiword feature among the less common phrases in the cloud. We also see numbers that represent scores, years, and distances, among other things.

Following that, we add a champion to the term filter and select frequency-based ordering. The named entity filter is then used to show only people in the cloud (Fig. (12)). The most frequently referenced Olympic champion in the studied part of the corpus is Donovan Bailey, as we can observe right away. We open the text viewer to learn more about him and see a list of all phrases that contain his name. We find that he is a sprinter from Canada who set a world record at the 1996 Olympic Games.

Looking at the other names in Fig. (12) frequency sorted co-occurrence cloud, we can immediately identify the female Olympic winner who is referenced the most. Svetlana Masterkova is her name, which is the third phrase in the cloud. To produce the co-occurrence cloud displayed in Fig. (12), we choose this phrase and deselect the Person filter. This cloud offers more details on Svetlana Masterkova, such as the fact that she looks to be Russian and that the distances of 800, 1,000, and 1,500 metres appear to be linked.



(a) The alphabetically ordered co-occurrence cloud for the term *Olympic*.



(b) The frequency ordered co-occurrence cloud for the terms *Olympic* and *champion* filtered by person names.



(c) The frequency ordered co-occurrence cloud for the terms *Olympic*, *champion*, and *Svetlana Masterkova*.

Fig. (17))
 Word clouds of the application example.

Switching to the text viewer to confirm these assumptions and obtain additional information, we see that Svetlana Masterkova is a Russian sprinter and double Olympic winner in the 800 and 1,500-metre distances. We also learned that shortly after her double Olympic gold, she established two world records, one in the 1,000-metre track at a tournament in Brussels and the other in the mile distance at the track and field event Weltklasse Zürich. Those four victories in a brief timeframe drew a lot of attention in the news this week.

The application exemplifies some of the benefits of text analytics based on word clouds. Users can rapidly acquire a general understanding of a text's term space by glancing through the word cloud at the start of the study. The most commonly used terms and subjects are instantly transmitted, providing a suitable beginning point for the investigation. In the application example, we began our investigation with the term Olympic.

The many varieties of word clouds make organised exploration easier by allowing users to organise the term space according to their information requirements. In this case, the frequency-based layout assisted in identifying the people who were most frequently cited in conjunction with the terms Olympic and champion. Users can filter the term space and investigate parts of it in greater depth using co-occurrence clouds, as we did with the name Svetlana Masterkova. Co-occurrence highlighting, contrastingly, was excellent for exploring and discovering word relationships interactively. We were able to focus solely on human names thanks to the named entity functionality. Finally, we might utilise the text viewer to go back to important areas of the original text to double-check assumptions and obtain additional information.

CONCLUSION

This chapter begins with a quick overview of Text Analytics Fundamentals and Examples. The second chapter, How to Get Started with Text Analytics, covers the topics of Analyse Your Data, Use BI Tools to Understand Your Data, and Final Words. The third

chapter, Examples and Methods for Text Analytics, discusses numerous Text Analytics Approaches. 1: Text Analytics Approach based on Word Spotting 2: Manual Rules Text Analytics Methodology 3. Text Categorization Methodology 4: Topic Modelling Methodology and Thematic Analysis. With real-world examples, the applications of Word Spotting Text Analytics, Manual Rules, Text Categorization, Topic Modelling, and Thematic Analysis are addressed. The following real-time application Word Cloud Explorer is used in Chapter 4 to demonstrate the analytic capabilities presented.

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Human Activity Recognition System Using Smartphone

R. Usha Rani¹, M. Sunitha¹ *

¹ CVRCE, CSE, Hyderabad, India

Abstract

Recognition of human activity has a wide range of applications in medical research and human survey systems. We present a powerful activity recognition system based on a Smartphone in this paper. The system collects time series signals with a 3-dimensional Smartphone accelerometer as the only sensor, from which 31 features in the time domain and frequency domain are created. The quadratic classifier, k-nearest neighbor algorithm, support vector machine, and artificial neural networks are used to classify activities. Feature extraction and subset selection are used to reduce dimensionality. In addition to passive learning, we use active learning techniques to lower the cost of data tagging. The findings of the experiment demonstrate that the categorization rate of passive learning is 84.4 percent and that it is resistant to common cell phone postures and poses.

Keywords: Activity Recognition, Gyroscope, Segmentation.

* **Corresponding author M. Sunitha:** CVRCE, CSE, Hyderabad, India; E-mail: m.sunitha@cvr.ac.in

1. INTRODUCTION

The demand for understanding human activities has increased in the health-care domain, particularly in elder care support, rehabilitation assistance, diabetes, and cognitive disorders [1]. A significant amount of resources can be saved if sensors can assist caregivers in constantly recording and monitoring patients and reporting automatically when any abnormal behavior is detected. The study has benefited other applications, such as a human survey system and a location indicator. Many studies have successfully identified activities using wearable sensors with very low error rates, but the majority of previous work has been done in laboratories with very constrained settings [2, 3].

Although readings from many body-attached sensors produce a low error rate, the sophisticated configuration is not practical in reality. In our research, we employed low-cost, commercially accessible cell phones as sensors to detect

human activity. Smartphone's' increasing popularity and processing capacity make them a perfect choice for non-intrusive body-attached sensors. According to US mobile subscriber statistics, around 44 percent of mobile users in 2011 had Smartphones, and 96 percent of these smartphones had built-in inertial sensors such as an accelerometer or gyroscope [4, 5]. According to research, gyroscopes can

aid in activity identification, however, their contribution alone is not as effective as accelerometers [6, 7]. As gyroscopes are not as widely accessible in cell phones as accelerometers; our method exclusively employs 3-dimensional accelerometer measurements. Unlike many others, we removed the limitations of connecting a sensor to a set body position with a fixed device orientation, unlike much previous research. In our concept, the phone may be positioned anywhere around the waist, including the jacket, pocket, and jeans pocket, with arbitrary orientation. These are the most prevalent postures in which individuals use their cell phones [8].

When a new activity is added to the system, a training procedure is always necessary. Due to sensor variation, parameters of the same algorithm may need to be learned and changed when the algorithm runs on various devices. However, labeling time-series data takes time, and it is not always viable to ask users to label all of the training data. As a result, we advocate for the use of an active learning strategy to expedite the training process. Given a classifier, active learning intelligently searches the unlabeled samples and learns the parameters based on the right labels returned by the oracle, which is generally a human. In this manner, users label just the examples requested by the algorithm, reducing the total number of training samples necessary. To the best of our knowledge, no previous research has been conducted on applying active learning to the problem of human activity identification. In this paper, we created a lightweight and accurate system for recognizing human behaviors on a Smartphone. Furthermore, active learning algorithms are being developed to minimize labeling time and stress. We determine the optimal learning algorithm for our system in terms of efficiency and accuracy by testing and comparing several learning algorithms.

2. LITERATURE REVIEW

For years, experts have researched human activity recognition and presented several solutions to the problem. Existing techniques usually employ a vision sensor, an inertial sensor, or a combination of the two. The use of machine learning and threshold-based algorithms is common. Machine-learning algorithms are often more accurate and dependable, but threshold-based methods are quicker and simpler. Body posture has been captured and identified using one or more cameras [8, 9]. The most frequent methods are several accelerometers and gyroscopes coupled to different body locations [10-13]. Approaches combining visual and inertial sensors have also been considered [14]. Data processing is another critical component of all of these algorithms. According to some works, the performance is greatly influenced by the quality of the input characteristics concentrated on extracting the most valuable characteristics from a time series dataset [15]. The most typical method is to examine the signal in both the temporal and frequency domains. The active learning approach has been used in many machine-learning situations when identifying samples is time-consuming and costly. Speech recognition, information extraction, and unconstrained handwritten recognition are some examples of applications [16, 19, 20]. However, this method has yet to be applied to the human activity problem.

3. MAIN TECHNIQUES

K-Nearest Neighbor is a supervised learning technique that classifies the results of new instance queries using the majority of K-Nearest Neighbor categories. It is one of the most common activity detection algorithms. The KNN algorithm's goal is to store all existing objects and classify new ones using attributes and training samples. The KNN classifiers are memory-based and do not use any models to fit. As the prediction value of the new query instance, the KNN algorithm employed neighborhood classification. The KNN Algorithm is depicted in Fig. 1. The k-nearest neighbor algorithm (K-NN) is a non-parametric (or distribution-free) method for classifying objects in the feature space based on the closest training samples. KNN is a memory-based learning method, where the function is just approximated locally, and the entire calculation is postponed until after categorization. The KNN algorithm is one of the most basic machine learning algorithms available: An object is classified by a majority vote of its neighbors, and the object is assigned to a class among its k nearest neighbors (k is a small positive integer). If $k=1$, the item is simply assigned to the class of its nearest neighbor. Each of the training examples is a vector with a labeled class in a multi-dimensional space. The feature vectors and class labels of the training samples are stored throughout the training (data pre-processing) phase

of the algorithm. k is a user-defined constant in the classification phase of an activity recognition system, and an unlabeled vector is classed by it. Selecting the most common label from the k -training samples for continuous variables, Euclidean distance is typically applied as the distance metric

$$\text{Euclidean} \quad \sqrt{\sum_{i=1}^k (x_i - y_i)^2} \quad (1)$$

Hamming distance is used a distance metric for discrete variables, such as text classification.

$$D_H = \sum_{i=1}^k |X_i - Y_i| \quad (2)$$

If the distance metric is learned using specific methods, including Large Margin Nearest Neighbor or Neighborhood component analysis, the classification performance of KNN can be greatly enhanced in Fig. 1.

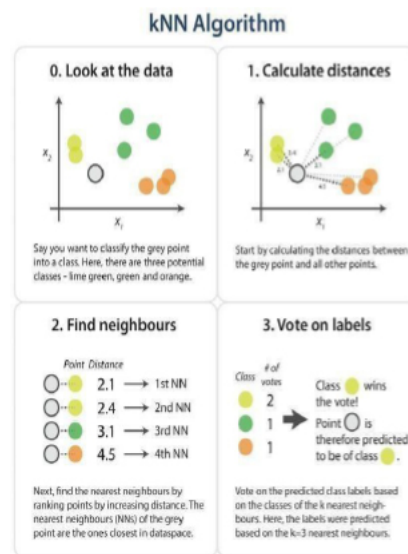


Fig. (1))
KNN Algorithm.

4. SMARTPHONE HAR COMMANDS

This section covers a smartphone-based human activity recognition system. On Android phones, the clustered KNN classifier is used to identify five major activities: walking, running, lying down, standing, and sitting. As illustrated in Fig. 2, the procedure is separated into two steps for this purpose: data pre-processing and classification.

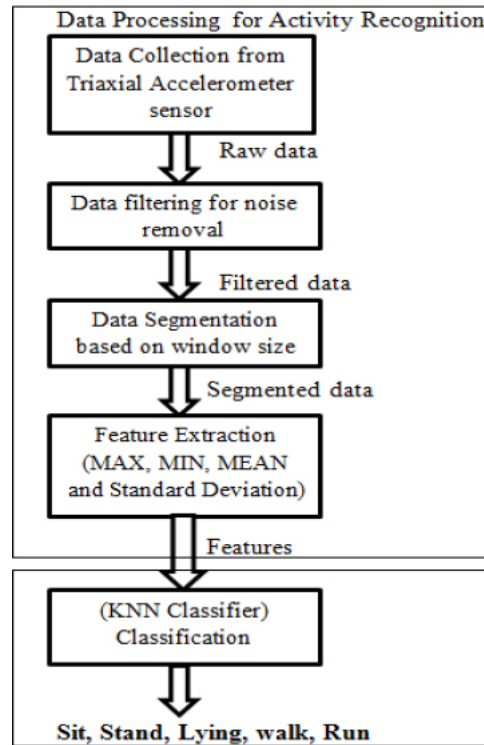


Fig. (2))
Human Activity Recognition System.

The ground truth data, *i.e.*, which behavior is genuinely performed by the user, is logged in order to track the performance of the classifier. To accomplish this, the application sends repeated voice commands to perform an action. The sequence of activity is predetermined in the system, however, the length of activity is supplied directly to the system as user input in the unit of seconds. During our tests, each activity lasts 60 seconds every cycle. Finally, utilizing these ground truth data, *i.e.*, activity tags, the effectiveness of activity identification, and other performance measures [16].

4.1. HAR Section 1: Data Cleaning Through Preprocessing

To Perform Cleaning of Data initially need to gather the data From Different Sources: Data is gathered using an application we developed called activity logger, and this element is in charge of gathering training data for each activity individually from sensors, *i.e.*, accelerometer data. In this application, the user will pick the activity to be done, place the phone in their pocket, and begin the relevant activity in Fig. 2. This programme is in charge of producing various training data files for each activity, in which raw data from the accelerometer's three axes is logged [17-19].

4.1.1. Data Filtering

The acquired data may contain noise, and the data is processed to eliminate the noise utilizing noise filters and the low-pass filter approach. To filter relevant information within acceleration and rotation signals, time-domain and frequency-domain characteristics have been extensively exploited. We employed four characteristics in this paper: MIN, MAX, MEAN, and standard deviation. These statistical procedures computed characteristics.

4.1.2. Data Segmentation

Data segmentation is an essential phase in the activity recognition process; generally, a sliding window method is used for segmentation, but there is no clear agreement on which window size should be

employed preferred. Intuitively, shrinking the window size provides for quicker action detection while also consuming less resources and energy [20]. Large windows, on the other hand, are often chosen for distinguishing complicated actions. Using windowing methods, the filtered sensor data is separated into discrete chunks for feature extraction.

As in any other data mining task, extracting the right features is critical to the final recognition performance. For activity recognition, we can extract features in both time and frequency domains [21, 22].

The extraction of features is a critical pre-processing step in activity recognition. It is frequently divided into two parts: feature creation and feature selection. When an algorithm's input data is too vast to process and is thought to be meaningless, the input data is turned into a reduced representation set of features. Extraction is the process of transforming raw data into a set of features. It is assumed that even if the extracted features are correctly chosen, the features set will extract the most relevant information from the input data in order to complete the specified task utilizing this smaller representation set of features rather than the full-size input. Prior to applying the KNN algorithm to the original data, feature extraction is performed [23].

4.1.3. Reduction

As illustrated in Fig. 2, Dimensionality Reduction is in charge of lowering computational complexity, which in turn would reduce reaction time for the recognition process while guaranteeing recognition accuracy. It is made up of two parts: the Feature Selection Component and the Instance Selection Component [24]. The Feature Selection function is in charge of detecting and deleting unnecessary, irrelevant, and redundant attributes from the dataset. Instance Selection is in charge of eliminating unnecessary, irrelevant, and redundant instances from the collection.

4.1.4. Selection of Feature

The classifier will cut processing costs by removing unnecessary and superfluous characteristics while maintaining recognition accuracy. Those features that are repetitive are those that do not add information to the recognition process. In other words, features that provide no/are irrelevant to the currently selected feature set should be removed. As a consequence, model interpretability will increase, training time will be reduced, and generalization will be improved by decreasing overfitting [25].

4.2. HAR Section II: Procedure to Perform Classification

When the window is full, the classification phase begins, and the average, minimum, maximum, and standard deviation values of the data in the window are obtained and evaluated one by one with the values in the compact training sets established during the pre-processing stage. The K-nearest sample to the test data is chosen from the training sets, and voting is done by looking at the final list of activities. The data in the associated window is labeled as the activity for which we have the most data in the final K set. The standard deviation feature identifies the one that is closest to the standard deviation of that particular window as the identified activity. We will receive four labels in the end and finalize with a high score

Table 1 **Confusion Matrix.**

Window Size(sec)		0.5			1			2		
-		10	50	100	10	50	100	10	50	100
K	10	87.9	87.8	87.7	88.4	90.3	89.5	88.6	87.8	89.3
	50	91.1	90.0	91.4	91.9	92.1	90.8	88.9	89.4	91.0

To measure the efficiency of clustered KNN classification for every activity. Table 1 displays the confusion matrix for clustering KNN. The confusion matrix is a type of visualization tool that is commonly used in supervised learning approaches. The confusion matrix has the advantage of making it easy to identify if the system is confusing two classes. The examples in a predicted class are represented by each column of the confusion matrix, whereas the occurrences in an actual class are represented by each row. A confusion matrix comprises data on both known and expected class labels. When compared to the performance of activities such as running, lying down, standing, and sitting, the KNN classifier performs somewhat worse for walking, which is commonly categorized as running or standing. Nonetheless, the overall performance got an accuracy of 92%.

We also evaluated the impact of the K value on the performance of clustered KNN classification for each activity. As we expected, increasing the K value will affect accuracy rates positively. Whenever we consider the overall effect of all system parameters, we observe best results in the case where K is selected as 50, the window size is selected as 1 second and sampling interval is selected as 50msec.

CONCLUSION

For the process of analyzing the quality of clustered KNN categorization for each activity, the clustered KNN confusion matrix presents its measures of accuracy. The confusion matrix is a type of visualization tool that is commonly used in supervised learning approaches. We also investigated the effect of the K value on clustered KNN classification performance for each activity. As predicted, raising the K value has a favorable effect on accuracy rates

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Smart Water Bottle with Smart Technology

Dumpala Shanthi^{1, *}

¹ Maturi Venkata Subba Rao Engineering College, India

Abstract

Working people have forgotten the importance of water intake. Smart Water Bottle will blow an alarm if it is being used by homemakers or people at home. This device can also be configured to their smartphones, which would send a message or alarm to their smartphones, making the user remember the intake of water by keeping some time interval. If configured to their mobiles, it would send a record of the amount of water taken by the user. The latest emerging technology like IoT, Android and many others will definitely be useful to the user.

Keywords: Smart Water Bottle, IoT, AI & ML, Arduino.

* **Corresponding author Dumpala Shanthi:** Maturi Venkata Subba Rao Engineering College, India;

E-mail: Dshanthi01@gmail.com

1. INTRODUCTION

The Internet of Things (IoT) is expected to grow exponentially to 50 billion devices by 2020 [1]. Smart homes, transportation, healthcare and industrial automation are new uses. Smart objects in our environment can provide context awareness that is often lacking in activity monitoring [2, 3]. Smart Stuff, or smart objects with physiological and activity sensors, used for new non-invasive physiological monitoring options. Smartphones and smart watches make it easy to connect SmartStuff to wearable sensor networks [4, 5].

New smartwatches like the Basis Peak and Apple Watch continuously measure physiological parameters like heart rate, GSR, and temperature. Many ubiquitous health monitoring applications require smartwatches to receive messages and notifications. Smart sensors and smart environments enable the quantified self (QS) [6, 7].

The QS community tracks physiological, behavioral, and environmental data. New sensors and systems enable easy record collection and database integration, allowing for data mining and new insights. Standard toolsets like Apple Health

Book, Google Fit, Samsung S.A.M.I, and Microsoft Health vault support this trend.

Big Data analytics can support personalized health monitoring and interventions [3, 7]. One of the most important factors for health and well-being is proper hydration.

60% of our body, 75% of our brain, and 83% of our blood contain water. An intelligent hydration monitoring and management platform can provide automatic, accurate, and reliable monitoring of fluid

consumption and provide configurable advice for optimum personalized hydration management.

A smart water bottle can help a variety of people, from athletes to dieters to the elderly, who suffer from dehydration in group homes or alone. While most users who seek to improve their health do not drink enough water, some medical applications do. Patients with kidney disease or congestive heart failure (CHF) should limit their liquid intake while still drinking regularly throughout the day.

Besides getting notifications and reminders at the optimal times to drink, Current and historical intake, as well as progress towards personal hydration targets, can all be monitored by means of a unified hydration management platform. The original smart water bottle was called the Hydra Coach. There is a hydration indicator on the bottle, but it isn't wireless. Bluetooth wireless connections and specialised smartphone and smartwatch applications are featured on a number of recently released bottle designs [8]. This study demonstrates the functionality of a smart water bottle equipped with sensors.

2. EMBEDDED SYSTEM

Embedded systems are not intended to be general-purpose computers. Others may have low or no performance requirements, allowing the system hardware to be simplified and costs reduced [9].

An embedded system is often physically integrated into the device it controls. Embedded software is stored in read-only memory or flash convector chips rather than a disc drive. It frequently runs on computers with a small keyboard, screen, and memory [10, 11].

Embedded systems are controlled by microcontrollers or DSPs (DSP). The main characteristic is being dedicated to a specific task, which may necessitate powerful processors. For example, air traffic control systems use mainframe computers and dedicated regional

and national networks to connect airports and radar sites. (Each radar probably has its own embedded system).

As embedded systems are task-specific, designers can optimize them to reduce product size and cost while increasing reliability and performance. Some embedded systems are mass-produced, saving money.

3. ARDUINO NANO

This breadboard is used to connect a few connectivities shown below in Table [1](#).

Table 1 [Arduino Nano Specification \[9\]](#).

Arduino Nano	Specifications
Flash Memory	32 KB, of which 2 KB used by the Bootloader
Input Voltage	(7-12) Volts
Microcontroller	ATmega328P
Operating Voltage	5 Volts
PCB Size	18 x 45 mm
Power Consumption	19 milliAmps
PWM Output	6
SRAM	2KB
Weight	7gms

4. PIN DIAGRAM

4.1. Serial Communication

It can communicate with a computer, another Arduino, or other microcontrollers. The ATmega328 supports UART TTL (5V) serial communication on digital pins 0 (RX) and 1 (TX) (TX). An ATmega16U2 on the board sends serial data over USB and appears as a virtual com port to computer software. There is no need for an

external driver with the 16U2 firmware. However, Windows requires an.inf file. The Arduino software includes a serial monitor for sending and receiving simple text data. The board's RX and TX LEDs will flash when data is sent over the USB-to-serial chip and to the computer (but not for serial communication on pins 0 and 1) in (Fig. [1](#)).

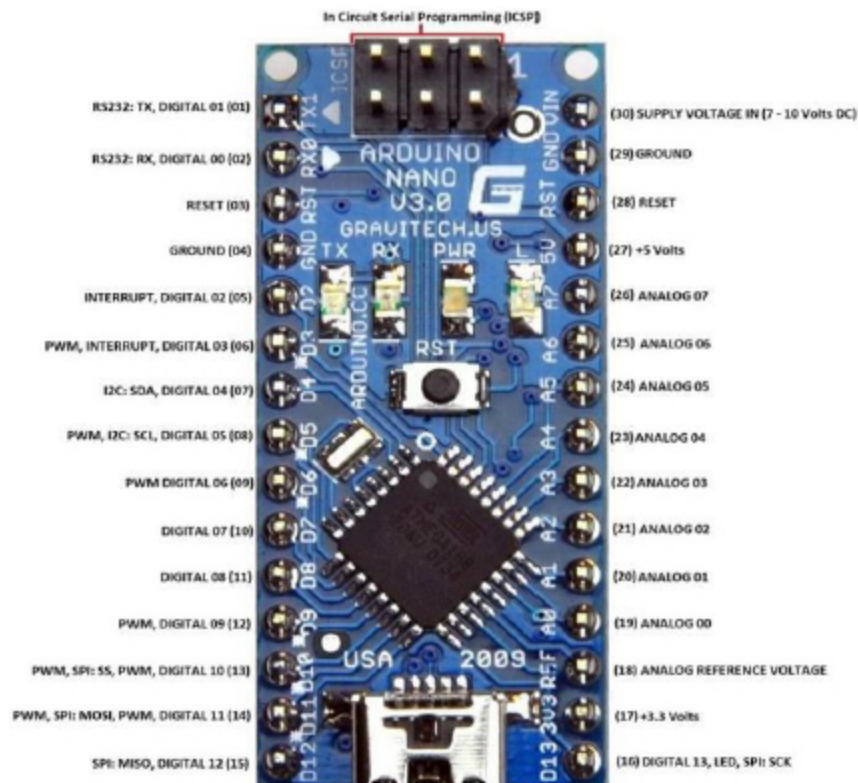


Fig. (1)
Pin Configuration of Atmega328.

It is Uno's digital pins that can be used for serial communication. The ATmega328 supports I2C (TWI) and SPI. The Arduino software includes an I2C library called Wire. Use the SPI library for SPI [\[12\]](#).

4.2. Water Level Sensor

A water level sensor will detect water in a tank, well, or soil. This sensor will go in the tank. It always checks for water. If the tank is dry, the comparator will be notified. This comparator will perform the action specified by us. After filling the tank, the sensor detects water

in the soil and well. The comparator will get the same data [\[13\]](#). Using the water level sensor information, it performs the action. Water level sensors and irrigation controllers are connected by 3,000 feet of wire. The water level sensor will then send data to the comparator [\[14, 15\]](#).

5. LDR: Working

As the intensity of the incident light increases, the resistance of a photoresistor decreases. It is also known as a photoconductor. It is a high-resistance semiconductor. Photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band. Less resistance is created by the free electron (and its hole partner) [\[16\]](#).

Extrinsic or intrinsic photoelectric devices exist. A non-efficient intrinsic semiconductor, like silicon, has its own charge carriers. Due to the limited availability of electrons in intrinsic devices, the photon must have sufficient energy to excite the electron across the band gap. Because electrons don't have to jump as far, lower energy photons (longer wavelengths and lower frequencies) are sufficient to trigger extrinsic devices. Extra electrons are available for conduction if some silicon atoms are replaced by phosphorus atoms (impurities). This is an extrinsic semiconductor.

It is a device whose resistance varies depending on the amount of light falling on its surface. Their resistance will be 1 MOhm in total darkness and 1 to 10 kOhm in bright light. Extrinsic or intrinsic photoelectric devices exist [\[17\]](#).

6. APPLICATIONS

(Fig. [2](#)), Use the backlight of a mobile phone to remotely turn on an LDR - call the phone from anywhere in the world, and it lights the LDR, and lighting can be turned on remotely [\[18\]](#).

There are two basic circuits using light-dependent resistors - the first is activated by darkness, and the second is activated by light.

In the circuit on the left, the led flashes when the LDR is dark. The 10K variable resistor adjusts the required level of darkness before the LED turns on. For best results, replace the 10K standard resistor with one of at least 1K to avoid damaging the transistor.

Using the 10K and 10K variable resistors instead of the LDR activates the circuit. The LED will turn on when enough light hits the LDR (manually adjusted with the 10K variable resistor), (Fig. 2).

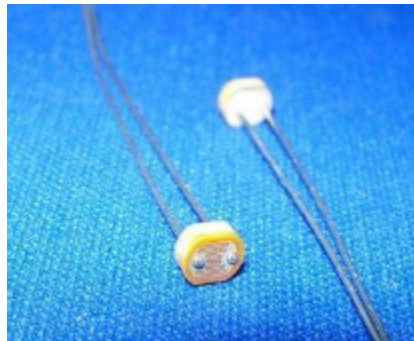


Fig. (2a))
LDR.

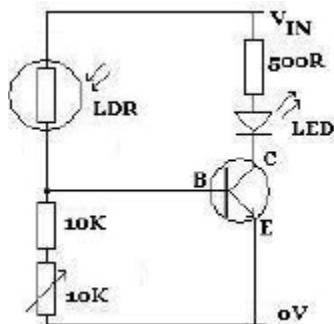


Fig. (2b))
LDR circuit.

The circuits shown above are not practically useful. In a real-world circuit, the LED (and resistor) between the positive voltage input (V_{IN}) and the collector (C) of the transistor would be replaced with the device to be powered in (Fig. 3).

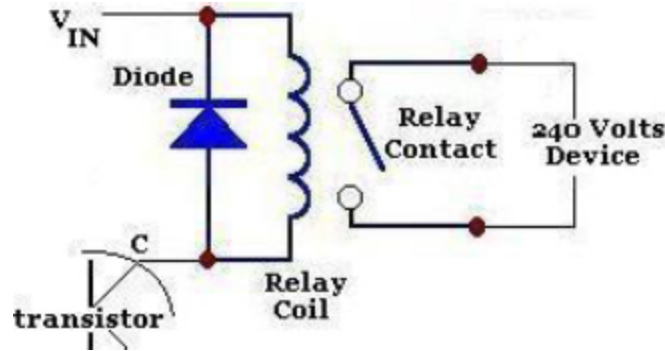


Fig. (3)

LDR circuit with the coil.

A relay is usually used, especially when the low voltage light detecting circuit is used to turn on/off a 240V device. Above is a circuit diagram of that part. When it gets dark (if the LDR circuit is set up that way), the relay triggers and the 240V device (like a security light) turns on.

6.1. Light Dependent Resistor (LDR) Measurement

The RL resistance (Fig. 4) and light intensity $RL = 500 / \text{Lux Kohm}$

The LDR's output voltage is $V_o = 5 \cdot RL / (RL + 3.3)$ when connected to 5V via a 3.3K resistor.

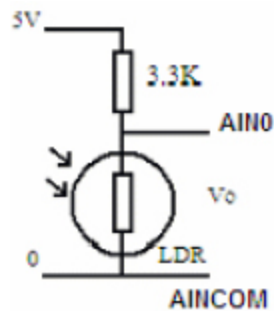


Fig. (4)

LDR resistance.

Reworking the equation, we obtain the light intensity $\text{Lux} = (2500/V_o - 500)/3$

6.2. Message Management General Description

Playback and record operations are managed by on-chip circuitry. There are several available On-chip circuitry controls for playback and recording. Depending on the operation, several messaging modes are available. Modes control message management, message length, and external parts. So, before designing, the designer must select the appropriate operating mode. Operating modes have no effect on voice quality; see Sampling Rate & Voice Quality for details. MSEL1, MSEL2 and M8 OPTION pins in Figs. [1](#) and [2](#) define five message management modes:

Random access with 2, 4, or 8 messages multiple variable-duration messages on tape provides two options:

- Autoplay
- Normal

Modes can't mix. Changes in mode after the first message are not advised. Unpredictable message fragments from the previous mode may remain present and audible on playback in the new mode if modes are switched after the initial recording. These snippets will vanish after a Record operation in the new mode. Table [1](#) defines the decoding required to select a mode. The APR9600's message management features include the ability to audibly alert the user to changes in the device's status *via* “beeps” superimposed on the device's output. To use this feature, set the BE pin to logic high.

6.3. RA Mode

Random access mode supports 2, 4, or 8 fixed message segments. Any of the selected messages can be recorded or played back at random. That's the total recording length (determined by the sampling rate) divided by the total number of segments enabled (as decoded in Table [1](#)). Random access mode indexes message segments easily.

6.4. Tape Mode

This mode manages messages like a cassette tape recorder. Tape mode has two options: auto rewind and normal. In auto-rewind mode, the device automatically rewinds to the start of the message after recording or playing it. Messages must be recorded or played back sequentially in tape mode, just like a cassette tape recorder.

This device's design helps simplify microprocessor-controlled message management. Message control To control messages, the microprocessor essentially toggles pins as described previously. The /BUSY, /STROBE, and /M7 END pins are included to simplify microprocessor-APR9600 handshaking.

It is low when the device is busy and cannot accept commands. That means it's ready to accept and execute commands from the host. When a memory segment is used, the /STROBE pin is low. The host processor can count pulses on this pin to determine how much recording time has been used and how much recording time remains. The APR9600 has 80 memory segments. The /M7 END pin indicates that the device has stopped recording or playing back. A low pulse indicates that all memory has been used. A low pulse indicates the last message has been played. Several APR9600 devices can be linked together using microprocessor control to increase total recording time. The speaker and microphone signals can be connected in parallel in this case. Using the /CE pins, the microprocessor controls which device drives the speaker. A continuous message cannot be recorded on multiple devices because the transition between devices causes a noticeable delay. As a result, message and device boundaries should always coincide.

The APR9600 samples incoming voice signals and stores them in non-volatile FLASH memory cells. Memory cells support voltages from 0 to 256. The 256 discrete voltage levels correspond to 8-bit ($2^8=256$) binary values. A continuous signal is then amplified before being fed to an external speaker during playback.

For aliasing errors to be eliminated, the highest possible frequency component introduced into a sampling system must be equal to or less than half the sampling frequency. To meet this requirement, the APR9600 automatically filters its input based on sampling frequency. Higher sampling rates improve voice quality but use more memory cells per second. Using less memory cells increases device duration but reduces incoming signal bandwidth. The APR9600 supports sampling rates up to 8 kHz and down to 4 kHz. Controlling the sampling frequency affects the quality/duration trade-off.

The APR9600 uses an internal oscillator to sample. The resistance from the OscR pin to GND can change the oscillator frequency.

6.5. Automatic Gain Control (AGC)

The APR9600 has an integrated AGC. The AGC affects the mic input but not the ANA IN. The AGC circuit ensures proper signal amplification. The AGC applies maximum gain to small signals and minimum gain to large signals. Assuring optimal signal levels for inputs of varying amplitudes. The AGC amplifier has a fast attack and slow decay.

The RC network connected to pin 19 controls this timing. The English language works well with 220K and 4.7uF. Languages, speakers from different countries, and music may all necessitate modifying the AGC RC network's recommended values.

6.6. Sampling Use

These reference schematics show how a recording system might be designed. It is shown in one of three modes: Random Access, Tape Mode – Normal, and Tape Mode – Auto Rewind. Several applications connect one or more of the /BUSY, /STROBE, or /M7 END pins to LEDs to show device status. Fortunately, these pins and signals were designed to work with both microprocessor and manual LED indication.

To power the electrets microphone's internal circuitry, a bias must be applied. This bias network's ground return connects to /Busy.

This setting saves power when recording. To disable the DC biasing voltage, pins 18 and 19 must be AC coupled to the microphone network. 3 show the device in random access mode. In this mode, the device uses the maximum eight Message segments. For modes with less than eight segments, unconnected message trigger pins can be left unconnected, except for pin /M8 OPTION, which should be pulled through a 100k resistor.

7. MODULE

We use LCD Modules to show interactive messages. We look at a two-line intelligent LCD with 16 characters per line connected to the controllers. The display protocol (handshaking) is shown. Auxiliary pins are +5V, -5V and GND. Data lines are D0 to D7th bit. Enable, Read Write, and Register Select are all pins.

In the display, there are two byte-wide registers: one for commands (RS=0), and the other for characters (RS=1). It also has a user-programmable RAM (character RAM) that can generate any dot matrix character. The hex command byte 80 will be used to indicate the display RAM address 00h will be chosen. Port 1 handles commands and data, while ports 3.2–3.4 handle register select and read/write levels.

The display's functions take varying amounts of time. For example, bit 7 of the LCD is monitored for busyness. LCD (Liquid Crystal Display) (Fig. [5](#)) is a very useful tool for both user interface and debugging. The most common LCD controller is the HITACHI 44780, which connects the controller to an LCD. These LCDs are inexpensive and easy to connect to the controller [[13](#)].



Fig. (5))

2x16 Line Alphanumeric LCD Display.

Some common ALPHANUMERIC displays are 1x16 (Single Line), 2x16 (Double Line), and 4x20 (four lines & Twenty characters per line).

8 (or 4) data lines and 3 control lines (RS, R/W, EN). Each mode has a different data line count. 8 data lines + 3 control lines = 11 lines required for 8-bit operation. In 4-bit mode, 4 data lines + 3 control lines equals 7 lines. How do we choose a mode? Simply put, if you have enough data lines, you can use 8 bit mode, and if you need a faster display, you must use 8 bit mode because 4 bit mode takes twice as long as 8 bit mode [\[14\]](#).

Table 2 **Symbol and Function.**

Pin	Symbol	Function
1	Vss	Ground
2	Vdd	Supply Voltage
3	Vo	Contrast Setting
4	RS	Register Select
5	R/W	Read/Write Select
6	En	Chip Enable Signal
7-14	DB0-DB7	Data Lines
15	A/Vee	Gnd for the backlight
16	K	Vcc for backlight

When RS is zero, the data is a command. When RS is high (1), the data sent should be displayed on the screen.

According to (Fig. 6), When R/W is low (0), the data bus writes to the LCD. It reads from the LCD when $RW = 1$. Since the LCD is rarely read, this line can be connected directly to Gnd, saving one controller line. The ENABLE pin latches the data from the data pins. To latch data, a HIGH-LOW signal is needed. When the EN line is lowered, the LCD interprets and executes our command. Enlowing EN prevents your command from being executed [12].

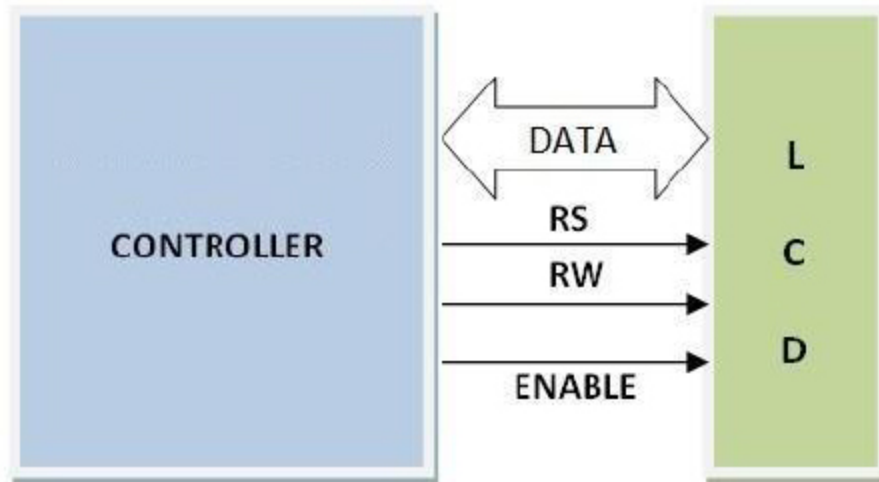


Fig. (6))
Data controller.

7.1. Types

WIFI technology supports two types, infrastructure and ad hoc. Ad hoc Ethernet networks can connect without a router or access point. Ad hoc networks are always preferred over infrastructure networks, but they have issues. Ad hoc Ethernet devices provide minimal network security against intruders. Unlike infrastructure mode, ad hoc Wi-Fi devices cannot disable SSID broadcast. Ad hoc Network attackers will not need much effort to win.

However, using alternative infrastructure will provide full-strength singles. 802.11g Ethernet networking standard requires ad hoc mode of communication with 11Mbps bandwidth. In infrastructure mode, Ethernet devices can transfer data at 54 Mbps, while in ad

hoc mode; only 11 Mbps can be achieved. For this reason, ad hoc mode is slower than infrastructure.

7.2. Security Concerns

Ethernet networks are easy to set up but difficult to secure because Ethernet access points do not use encryption. As the network is enabled, it is required. Hackers can easily breach a secure Ethernet network to steal personal data. Non-harmful guests can still use network resources and slow down the network.

7.3. Other Devices' Interference

Ethernet transmits data at 2.4 GHz, making it susceptible to interference from Bluetooth-enabled devices, mobile phones, cordless, microwaves, and other communication devices.

7.4. Streaming Media of Poor Quality

High-definition media pushes today's fastest Ethernet standards to their limits. Because of the slower transfer rate, high-definition video and audio cannot be viewed flawlessly.

Even today's fastest Ethernet standards are pushed to their limits by today's high-end media. High-definition audio and video files require fast delivery, and typical wireless networks lack both speed and consistency. This issue is exacerbated when multiple devices are connected to the same network because the bandwidth must be shared.

7.5. Specifications

IEEE 802.11 is the global standard for Wireless Local Area Networks (WLAN). In short, it is a standard for computer and wireless device communication. IEEE LAN/MAN Standard Committee set standards in 5GHz and 2.4GHz public spectrum bands.

A number of mechanisms and services in IEEE 802.11 relate to present station mobility through the network mass. IEEE 802.11 is a wireless LAN. The interchangeability of IEEE 802.11 and WiFi is based on market demand. For wireless networks from 1997 to 2008, it is the basic protocol. It has changed a lot, but the basic “high performance” remains the same. Its range is from 100 to 5000 meters, and its frequency is 2.4-5 GHz. These services include system authentication, de-authentication, data privacy, MSDU delivery, association with files and the system, disassociation and re-association. IEEE 802.11 allows for independent and infrastructure configuration of a station. IEEE 802.11 was born out of the need to wirelessly connect multiple computers in places where wires cannot go.

For top layers, IEEE 802.11 required accepting devices to connect peer-to-peer using access control and data transformation services.

Several physical layers signaling methods and interfaces exist. IEEE 802.11 transports user data in privacy and security over wireless media. Infrared and spread spectrum PHYs are part of the IEEE 802.11 protocol. IEEE 802.11 defines MAC and physical layers with distinct specifications. Infrared is one of two physical layers. Infrared PHY offers 1 Mbit/s with 16-PPM and 2 Mbit/s with –PPM, while Frequency Hopping Spread Spectrum Radio PHY offers 1 Mbit/s with two levels and 2 Mbit/s with four levels.

Now 802.11a, b, and g are mostly used to network homes, offices, and businesses. 802.11 uses 802.1X for verification, RSN for link maintenance and AES-based CCMP for privacy, veracity and basis. IEEE 802.11 is reliable due to modem and DSL service propagation. People always want to set up home networks to share their high-speed Internet connection. Many free networks allow anyone within a certain range, including passers-by, to connect to the Internet. Unpaid helper groups are also hard at work launching wireless community networks to provide public wireless connectivity. With 802.11, wireless networking has a bright future. IEEE 802.11 will become more dynamic in the near future.



Fig. (7))

Smart Water Bottle integrated into a personal body area network.

We built several smart water bottles to test user experience and performance shown in (Fig. [7](#)). a) Body area network integration; b) Wi-Fi cloud system integration.

The PPG sensor on the bottle and the smartwatch can also be used to identify the user, which is useful if the bottle is held by the opposite hand. Touch sensor can assess finger-to-PPG sensor contact. The peak-to-peak amplitude of the PPG for each heartbeat was investigated as a function of touch sensor capacitance (Fig. [8](#)) shows the results. For the 13-bit precision of the PPG, the light touch is shown as measured capacitance between 10 and 27 pF. The amplitude improves with pressure (28-33pF), but decreases with high pressure due to loss of finger pulsation. This study used National Instruments' Battery Simulator NI PXIe-4154. To communicate with the server, the Photon consumes between 3 and 4 Amps (320-450 mA). Power consumption is 4 mA in standby and 76.5 mA in deep sleep.

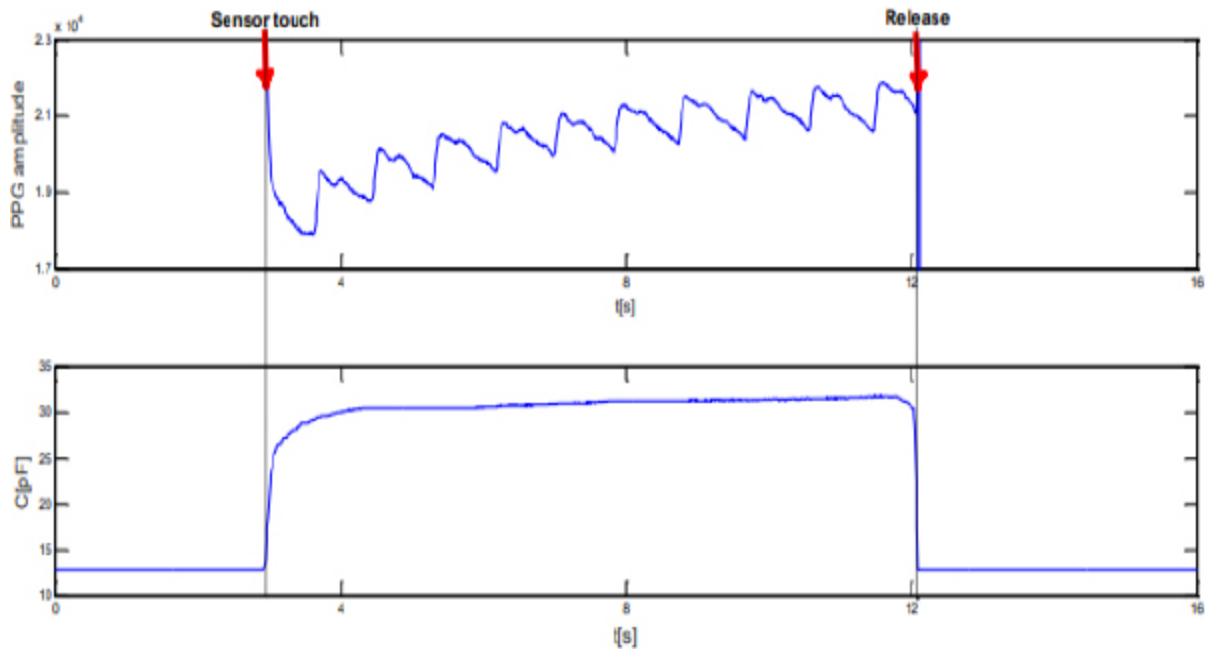


Fig. (8)

PPG Signal and Sensor Capacitance from the Pulse Sensor on the Smart Water Bottle.

Preliminary testing of the smart water bottle's physiological sensors shows excellent signal quality for non-invasive physiological monitoring. (Fig. 8) shows the PPG signal from the bottle's Pulse Sensor, while (Fig. 9) shows the capacitance change caused by touching the sensor.

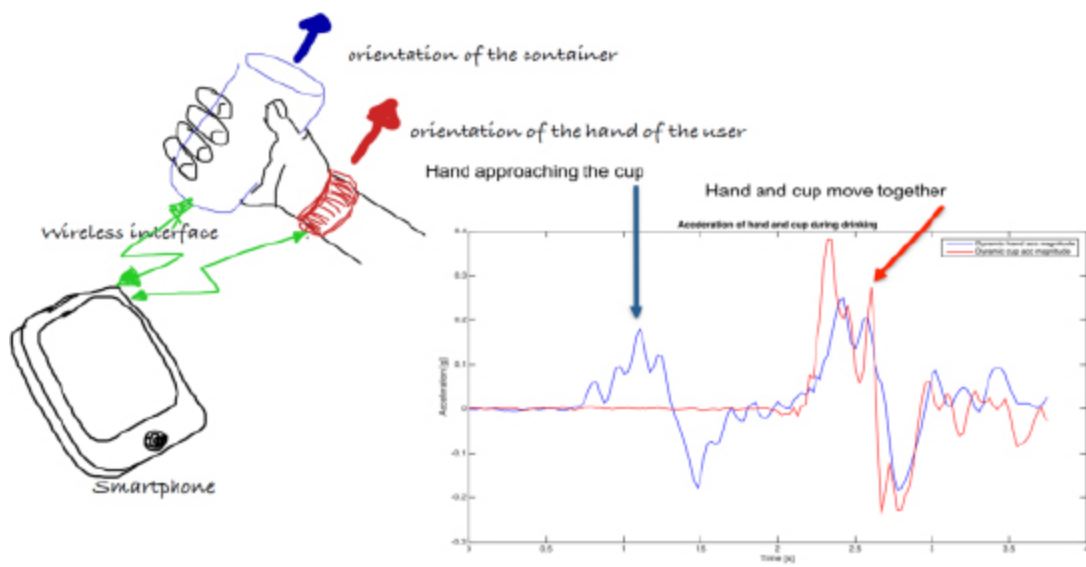


Fig. (9))

Alerting us to drinking water.

CONCLUSION

The proposed model of smart bottle architecture for hydration reminder to remind humans to intake sufficient water on time to keep the body stable is successfully implemented. Not only smartphones but also basic mobile phones can also be used for verification and getting execution successfully. Further, some possibilities for maintaining the record of daily usage of human sipping h₂o are kept as future works and can be used in other fields.

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Real World Applications of Machine Learning in Health Care

Kari Narmada^{1, *}, Sanjay Kumar Singh¹, Dumpala Shanthi²

¹ Amity University Rajasthan, Jaipur, Rajasthan, India

² Maturi Venkata Subba Rao Engineering College, Hyderabad, India

Abstract

Machine learning (ML), a subset of artificial intelligence, is used to construct algorithms for monitoring, diagnosing, forecasting, and predicting clinical results. Health is a major concern for human beings. The current success in ML is due to deep learning (DL), using huge artificial neural networks. In the past, machine learning has demonstrated its usefulness and skills in detecting cancer. It is one of the most feasible solutions for top healthcare pioneers to detect anomalies. When healthcare companies succeed in using predictive models, they face challenges in demonstrating their value and gaining trust across the company. Recently, established standards for reporting machine learning-based clinical research will aid in connecting the clinical and computer science communities and realizing the full potential of machine learning techniques. The

researchers have many objectives in the design of machine Learning Algorithms for different applications. Many papers discussed how machine learning algorithms are involved in health monitoring which will be updated so that patients, doctors, or any individuals can view the information. The main goal of this paper is to discuss basic types of Machine Learning and the challenges faced by Artificial intelligence (AI) in health care. The possible risks in clinical research give practical information on how to accurately and effectively analyze performance and avoid frequent pitfalls, particularly when dealing with applications for health and wellness contexts.

Keywords: Machine Learning , Health care , Applications , Challenges in AI .

* **Corresponding author Kari Narmada:** Amity University Rajasthan, Jaipur, Rajasthan, India;
E-mail: narmadakari@gmail.com

1. INTRODUCTION

Nowadays, technology provides more rapid analysis to offer a proper treatment plan for human lives and health. Artificial Intelligence and Machine Learning are currently ruling the world. We know how significant advancements in Machine Learning Algorithms can benefit the healthcare industry in the medical field. IoT

(Internet of Things) is the most common technology is Internet with things, providing excellent advantages in real-time applications.

In healthcare technology, machine learning algorithms are better at detecting malignant tumors, and it provides self-learning neural networks that improve treatment quality by analyzing data regarding the patient's health, such as numerous tests, X-rays, and screenings are all available. Deep learning algorithms are increasingly being utilized to detect cancer cells and improve clinical analyses. In general, Machine Learning can help with the treatment process in a variety of ways, such as automated messaging warnings [[1](#)].

A voice-controlled healthcare assistant functions as a computer-generated nurse for patients, providing information on a variety of ailments, health issues, and medications. Data engineers are developing solutions for overall health monitoring and assisting in treating or preventing disease.

The total number of cancer-diagnosed persons in the last two decades has almost doubled from a record 10 million in 2000 to 19.3 million in 2020. In today's world, one out of every five people will have cancer at any point in their lives. It is estimated that, in the coming years, there will be a much greater rise in the number of people living with cancer, and by 2040, it will be almost 50 percent over 2020. Cancer is the world's leading cause of mortality; almost 10 million deaths occurred in 2020. The WHO will hold the first consultations on World Cancer Day to develop a new global breast cancer plan starting in 2021 [[2](#), [3](#)].

Maintenance of good health with the present day-to-day life activities is not easy. So many humans does not have proper diet and exercise, leading to many health issues. Many human beings neglect their health and give priority to their daily lifestyle. This gives a clear chance for the diseases to attack easily. All over the world, around 8.9% of the population is suffering from diabetes in India and 9.3% worldwide. Similarly, the blood pressure patients are 13.8% in India and 19.3% worldwide. These values are exponentially increasing in the past decade, giving future generations bad results. Diagnosis done at the proper time will lead to control of these values. As an engineer, we can diagnose the patient, but if we create a channel

between the patient and doctor, it helps the patients control their problems. Diseases are generally classified into two types [4].

- Acute diseases
- Chronic diseases

Nowadays, technology has been improved drastically; wearable systems and gadgets for health monitoring are manufactured by various readily available companies. Modern simulated tools like IoT, machine learning, artificial intelligence, and data management are combined, and some superactive machines can be designed with these tools, which helps doctors to treat patients easily. A patient also easily interacts with the doctors without any disturbances [5]. The training of the data is done by using the data sets. Later testing is done. Testing always determines the non-regularities of the levels. Statistics is applied to the data set over the cloud, and analysis is done; finally, the accuracy is calculated. Acute conditions aren't too bad but last for life, while chronic conditions need surgery. Here are some examples of common acute disorders. Accuracy in the design of health monitoring systems is crucial to their success in curing many ailments, and in the modern world, everyone wants advanced technology; they want everything to be automated. Some businesses even use speech or voice recognition software to speed up processes [6].

2. LITERATURE REVIEW

The authors employed three machine learning algorithms to predict cancer cells. They extracted data from trials at the National Cancer Institute of Tehran using the ICBC dataset. On the ICBC dataset, they used three machine learning methods (ANN, D-Tree, and SVM).

Suji, R. J. *et al.* propose a system that combines CFRSFS, K-Mean clustering, and LS-SVM. CFRSFS is a feature selection method suggested by the author. To continue feature processing, CFRSFS selects just eight features and assigns them to K-Mean clustering. Their approach has a 99.54 percent success rate [7].

Rana, M *et al.*, the authors used various Machine Learning algorithms based on the dataset. They obtained experimental results using the WDBC dataset. They assessed the accuracy of algorithms to be 68 percent for SVM-linear, 68 percent for logic regression, 72 percent for KNN, and 67 percent for Naive Byes [8].

Singh, S. *et al.* employed Neural Networks to identify breast cancer. They used 1808 cancer cases. For testing, they used 387 instances. They assessed the accuracy of the Neural Network method to be 95.80 percent.

Shattuck *et al.* updated the EM algorithm, and they made sure that the advanced EM was used for all the images. Wells *et al.* proposed an important technique called Bayesian tissue classification. They also designed an EM algorithm [9].

Davatzikos *et al.* discussed how the boundaries are extracted if it's in snake curves. The moving curves are very difficult to operate. They used a predefined algorithm to obtain the accuracy [10]. Cohen *et al.* proposed an excellent technique that is used now also. They designed an algorithm for energy minimization. This algorithm mainly operated for 3D binary images with spherical topology [11]. Christensen *et al.* again updated their research and worked on the vector value flow field. They used partial differential equations in their design. It also uses continuum mechanics [12].

Imadail S *et al.* developed an advanced e-health service supporter that used an IPv6 protocol network for processing [13]. A record maintained named PHR (personal health record) is done by combining the devices in this network and storing all the health-related values through this internet protocol. After these developments, many devices are designed to work with this key protocol. There is proper security in this design, as given:

- Secure authentication and authorization
- Secure bootstrapping of objects and transmission of data
- Security of IoT data
- Secure access to data by authorized persons

Stefano *et al.* are hardware engineers who proved with a design of a detection system that again detects the movements of the patients as the previous paper, but their research uses different hardware with minimum cost and operation speed is high. They also proved the accuracy with 97.3%, which is an adequate result [14]. Even nowadays, many researchers use this technology as a reference and have designed many systems. But now, this technology is not used in the market; many new technologies have been developed with additional applications.

Mohammed *et al.* discussed the problems raised by the data which is captured and stored, and transferred. Security is the major problem for the data. To secure the previous data paper has given some methods. These authors have also given some algorithms to secure the data.

Franca *et al.* briefed and designed a new generation system. This system continuously screens the patients from time to time. The monitoring is perfect. The various parameters of the patients are noted and processed personally [15]. They focused on wireless technologies, which have added the advantage of low processing time.

Debiao *et al.* explained the required securities in Patient Health Monitoring System. They also focused on authentication schemes for RFID which is dependent on ECC (elliptic curve cryptography). Advanced encryption and decryption algorithms are used in this design.

Researchers embedded the IoT with networking and cryptography. Proved with the best technology and used for some years by multinational companies [16].

3. TYPES OF MACHINE LEARNING

Machine learning allows machines to learn on their own. It accomplishes this by creating basic models to address challenges.

Every time the machine learning system crawls through the data and discovers new patterns, it modifies the model. This method promotes learning and produces progressively accurate results. This learning process is carried out by an algorithm without the need for programming. There are four types of machine learning: supervised, unsupervised, semi-supervised, and reinforcement learning.

3.1. Supervised Learning

This learning procedure is based on a comparison of computed and expected outputs, *i.e.*, learning entails computing the error and modifying the error to get the desired result. For example, if a data set of homes of a given size with real prices is supplied, the supervised algorithm should produce more of these right responses, such as what the price of a new house would be.

3.2. Unsupervised Learning

Unsupervised learning is defined as when a machine learns on its own by discovering and adapting to a pattern of input. As the data is separated into multiple groups in this process, it is referred to as a clustering method. Google News (URL news.google.com) is an example of where clustering is applied. Google News collects fresh stories from throughout the web and organizes them into news articles.

Unsupervised learning is the challenge of discovering secret structures in machine learning unlabeled data. Since then, there have been no labels on the examples offered to the learner. To test a possible response, use an error or incentive signal [[17](#)].

3.3. Reinforcement Learning

Reinforcement learning is based on the output of how an agent should respond in a given environment to maximize a long-term reward. Correct output is rewarded, while incorrect production is penalized. In contrast to the supervised learning issue, proper

input/output pairings are never offered, and sub-optimal behaviors are never explicitly corrected in reinforcement learning.

3.4. Recommender Systems

Recommender systems are learning approaches that allow web users to tailor their sites to fit the preferences of their customers. For instance, an internet user can when he/she gets a rating of a product or/and related goods. Due to the existing recommender, you may search for things. As a result, the way individuals look for items has changed.

4. MACHINE LEARNING APPLICATIONS IN HEALTH CARE

As smart medical devices grow more common, technology-enabled healthcare is becoming a reality. The healthcare business encourages innovation, which is why AI in healthcare has a promising future. Google has already launched an algorithm that successfully detects cancer in mammograms was previously published, and Stanford University researchers used Deep Learning to detect skin cancer [[18](#), [19](#)].

- **Diagnosis and Identification of Diseases:**

This is a wonderful place to start because ML is excellent at diagnosing; in fact, it is one of the most effective regions since certain cases of cancer and hereditary illnesses remain difficult to detect, ML could help in the early stages of many of them. IBM's Watson Genomics is an excellent example of this. To help accelerate cancer detection, this effort incorporates cognitive computation with genome-based tumor sequencing [[20](#)].

- **Improvement of Health Records:**

Despite all of these technical advancements, maintaining health data remains a pain. Machine Learning-based OCR recognition approaches and Vector machines could be used to classify records. Google's Cloud Vision API and MathWorks' ML handwriting recognition technology are two of the most prominent examples.

- **Diabetes can be Projected:**

Diabetes is one of the most prevalent diseases and is likely to be deadly. It does not only damage the health of a person by itself but produces several other harmful conditions too. Diabetic renal disease, heart disease, and nerve damage are the most common complications. Machine Learning could aid in the early detection of diabetes, perhaps saving lives. KNN, Decision Tree, and Naive Bayes are examples of classification algorithms that could be used to develop a diabetes prediction system. In terms of performance and calculation time, Naive Bayes is the most efficient [\[21\]](#).

- **Finding the most Effective Treatment:**

Another great use of Machine Learning is in the early stages of drug research for patients. Microsoft's Project Hanover, which attempts to invent tailored medicine combinations to cure Acute Myeloid Leukemia, is now leveraging AI-based technology.

- **Image Analysis is used to Make a Diagnosis:**

With its Inner Eye project, Microsoft is transforming healthcare data analysis. To make a diagnosis, this start-up uses Computer Vision to process medical images. Inner Eye is becoming more well-known in the field of healthcare analytics software as technology advances. Machine Learning will become more efficient shortly, allowing for the analysis of even more data points to establish an automatic diagnosis [\[22\]](#).

- **Individualizing the Treatment:**

Medicine is making remarkable strides with machine learning. IBM Watson Oncology stands out as a pioneer in this field, offering a variety of therapy options based on a patient's medical history. Effects will get even recovering when it arises to establishing tailored treatment regimens as improved biosensors become more widely available, offering more data for algorithms.

- **Changing one's Conduct:**

This is a fascinating area to observe. Giving cancer prevention advice based on your regular activities-That's exactly what Matrix, a B2B2C company, is doing with its app.

- **Making use of Crowdsourced Medical Data:**

This will be the source of future progress in the field of medical machine education. What is the importance of healthcare data analysis? Medtronic-IBM partnership has already enabled the deciphering, accumulation, and delivery of insulin data in real time. Moreover, the procedure of diagnosis and issue of medicinal prescriptions would improve public data [[23](#)].

- **Widespread Management:**

Experts will have access to satellite data, social media trends, news websites, and video feeds as far as data analytics are concerned in 2020. All this may be handled using neural networks, as may be concluded from epidemics everywhere. Before they inflict widespread devastation, dangerous illnesses could be averted. In Third World nations, where modern medical systems are lacking, this is particularly crucial. Pro-MED-mail is perhaps the greatest example in this field, an Internet-based reporting tool that monitors global epidemic data. Food safety is also employed by artificial intelligence to aid epidemic disease prevention [[24](#)].

- **Surgery Using Artificial Intelligence:**

It perhaps the greatest important field for machine learning, and it is much more popular shortly. The following groups can be categorized as robotic surgery [\[25\]](#):

- Suturing is done automatically
- Surgical workflow simulation
- Material advancements for robotic surgery
- Evaluation of surgical skill

4.1. The Most Challenges For AI In Healthcare

- **Data management and administration:** Medical records stay confidential and are not publicly available. According to a study performed by the Wellcome Foundation in the United Kingdom, just 17% of people respondents are opposed to their medical evidence being shared with other parties [\[26\]](#).
- **Algorithm transparency:** Transparent algorithms are not only important to satisfy strict standards for drug development but also to help people understand how algorithms generally make conclusions.
- **Improving electronic records:** A lot of information is scattered across many databases that have to be organized better. When this circumstance improves, personal treatment solutions will progress.
- **Recognizing the value of data silos:** The healthcare business should reconsider the value of data and how it may be used to provide value in the long run.
- **Experts in data science:** It is critical for both the healthcare and pharmaceutical businesses to attract more Machine Learning expertise and data science specialists.

4.2. Possible Risks to Generalizability in Clinical Research and Machine Learning In Health Care

- Changing patterns of practice through time

When low-tidal-volume ventilation is used in the intensive care unit (ICU), patient outcomes improve. This will affect how well models work that were made when higher tidal volumes were the norm. In most countries, leucodepletion of blood for transfusion became the standard of treatment verification; blood transfusion and outcome models must be recalibrated before the practice changes.

- Variation in patient demographics

Why COVID-19 hospitalization risk prediction models based on data from Italy, where there are a large number of elderly people in the population will not perform well in countries with a different age distribution, such as low- and middle-income nations, which usually have a younger population.

- Variation in genotype and phenotype in patients

Model performance is related to the makeup of the training cohort in terms of disease genotypes, phenotypes, or both. These models are not going to transfer effectively to groups with a distinct genotypic or phenotypic makeup; a few Sepsis and acute respiratory distress syndrome traits, for example, might be over-represented or under-represented in many contexts.

- Data capture hardware and software variations

Bedside monitors with varying sample rates for capturing physiological signals that are monitored constantly will be more susceptible to artifacts and will have an impact on models that use time-series data as input. Computer-vision models for automatic CT scan interpretation are sensitive to the equipment used to generate the pictures.

CONCLUSION

The medical field is an important area for all humans to research. This paper shows the basic idea of how machine learning is used in

the medical field. The best mark goes to machine learning's extraordinary capabilities in classifying and categorising health data, as well as speeding up physicians' clinical decisions and any type of forecasting that might save lives or make surgery less difficult. In this paper, various applications of machine learning are explored. This improves the patient's health quality. Prevention, cure, and treatment are given to the patients on a priority basis. Without a doubt, human life is the most essential thing. Healthcare machine learning nowadays offers technologies that contribute directly to increased medical diagnosis and therapies. There are other options, such as artificial intelligence in nutrition, which we will examine in future postings

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Investigating and Identifying Fraudulent Behaviors of Medical Claims Data Using Machine Learning Algorithms

Jyothi P. Naga^{1, *}, K.V.S.N. Rama Rao², Rajya Lakshmi³, S. Suresh⁴

¹ Department of CSE, GITAM Institute of Technology, GITAM, Visakhapatnam, A.P., India

² Department of CSE, K L Educational Foundation, Hyderabad, A.P., India

³ Department of Computer Science and Engineering, JNTUK UCEV, Vizianagaram, A.P., India

⁴ Anil Neerukonda Institute of Technology & Sciences, Sangivalasa, Bheemunipatnam, Vishakapatnam, Andhra Pradesh, India

Abstract

Healthcare is essential in pandemic times, but it is crucial for the well-being of daily life. Many countries allocate substantial funds towards providing high-quality healthcare services. As healthcare expenses escalate, policymakers and funders are increasingly focused on investigating the underlying factors driving the high costs of medical resources. A comprehensive analysis carried

the required expenses towards identification, valuation, and measurement of resources utilized for the diagnosis process. The objective of the chapter is to provide how the data analysis is carried which helps to identify fraudulent behaviors. The generated model assists health management organizations in identifying suspicious behaviors toward claims. Healthcare fraud is a severe threat to global health results, and could lead to misuse, scarce resources, and negative impacts on healthcare access, infrastructures, and social determinants of health. Healthcare fraud is associated with increased healthcare costs in most of the leading countries. The proposed research work provides an estimation mechanism for utilizing health resources and their impacts on healthcare costs. This chapter proposes strategic ways of handling healthcare data to prevent future healthcare fraud, decrease healthcare expenditure, and adequately use resources to benefit the population. This chapter works on three primary datasets and a synthetic dataset aggregated from the primary datasets. The data preprocessing is carried out at different levels of the model, which truly enhances the data quality. The model is constructed at three levels; the first level analyzes datasets in which it extracts the primary features and provides constructive decisions and outcomes on the processing of data. Regressive analysis of the hierarchical grouping mechanism helps to know the detailed features that could affect healthcare and prevent resource misuse.

Keywords: Healthcare, Machine Learning, Fraud detection, Statistical, Supervised, Decision.

* **Corresponding author Jyothi P. Naga:** Department of CSE, GITAM Institute of Technology, GITAM, Visakhapatnam, A.P., India;

1. INTRODUCTION

Many factors affect the health insurance systems in most countries, like increased population or fraud/deficient patients, fraudulent physicians, and providers, which automatically reduce the allotted funds by the government or companies. There is a significant increase in fraud, abuse, and waste of services, consequently straining finance. The scam includes suspected diagnosis proof, low insured willingness, recurrent claims, late filing of claims, inconsistent documents, insufficient damage level information, high payments, and poor reliability on the certificate. Each claim submitted must be approved by the medical professional to justify the chapter's weakness. It incorporates age, sex, name, date of birth, type of claim, identification number, resting period solicited, identification of medical expert, identification of employer and past and present income records.

Experts usually detect fraud within healthcare through auditors and investigators by searching numerous records manually and digging out suspicious or fraudulent behaviors. As the volume of medical data increases, it is tedious for experts to detect fraud. It may lead to weakness when compared to more automated data mining and machine learning methods. The global economic crime of 2018 by PwC reports that 49 percent of organizations suffer from fraud. According to the NHCAA, about 9% (47.9 billion dollars) of the United States, the annual healthcare expenditure is increased due to fraud and corruption. In Europe and Australia, the losses due to fraud and corruption correspond to 10% of the total expenses, according to the respective Healthcare Anti-Fraud and corruption institutions, EHFCN and PHA, Private Healthcare Australia. Every year this count increases to billion-dollars in business. The estimation of Insurance Fraud was nearly \$80 to \$100 billion in 2014, equating to the cost per insured household between \$400 and \$1000 in the United States [1, 2]. According to the Federal Bureau of Investigation, in 2013 cost of

all insurance fraud was about \$40 billion, with about 20 percent being in the property-casualty sector [35]. European Insurance Anti-Fraud Guide has estimated the cost of fraud cannot be less than €8 billion or approximately 2 percent of the total annual premium income from all classes of the European insurance market in 1996. In Australia, 10 percent of the total amount is paid for fraudulent claims each year which is about AUS\$1.4 billion [6].

In contrast, in Latin America and the Caribbean, the fraud cost is estimated at 19 percent to 35 percent of the annual revenue of the insurance industry. Only due to claim fraud/policyholder fraud, South Africa had lost 100 million rands in 2010 [7]. In Kenya, Kuria, and Morongo, it is estimated that 40 percent are fraudulent claims, and it is between 10 percent to 30 percent in Nigeria [8]. Health Insurance Portability and Accountability Act (HIPAA) is a US regulatory system to punish the criminals involved in insurance fraud based on the extent of the fraud. The possibilities of fraud are co-evolving with technology. Fraud analytics is a probe towards digital technology across industries. Fraud prevention and detection have to chapter simultaneously to stop the occurrence of fraud and handle it in the best possible way. In practice, detecting fraud starts with the prevention of false alarms, like identifying the tradeoff between the costs of analyzing a fraudulent case and saving by investigating it by taking it into consideration. This chapter highlights opportunities and opportunities and challenges researchers by handling multiple attributes that assess the medical service cost for different procedures. Medical service costs will somehow decrease the high costs of resources, increase the scrutiny by payers, and reflect the payment to payers' actual negotiated price. Therefore, resource utilization will be more effective and may be reduced to low-end procedures.

To summarize, some factors are influencing the cost of International Health Insurance in a majority of countries like:

- There is an increased demand for the high-quality healthcare
- The increased cost of healthcare
- Increased regulations and regulatory systems

- Continuing challenges related to fraud

The issues of bribery and fraudulent claims remain the same as the most significant challenges to healthcare sectors. Hence there is a need for automated models to meet the above requirements.

2. ROLE OF MACHINE LEARNING ALGORITHMS

Manual Processing of health care data may lead to wrong diagnoses and analysis. The power of using electronic technology for recording patient information to improve the chapter flow is somehow remarkable. Using advanced analytics, it can provide better information on patient care to doctors. There are several successful cases that machine learning algorithms are able to identify cancer tumors, skin cancer and diabetic retinopathy. Machine learning helps for quicker decision-making. AI and ML can automate the routine task in the forthcoming years. This could open more number of possibilities for human to take over more sophisticated tasks. The various applications of ML in the medical domain like robot-assisted surgery, virtual nursing, administrative assistants, fraud detection, dosage error reduction, clinical services and automated image diagnosis. These outcomes can help to reduce the procedure cost, and the claim can be estimated approximately to treatment.

Most ML applications help to predict the disease and diagnosis procedure. This book chapter discusses the use of ML algorithms on health care claims data which provides a decision about whether the claim is justifiable or fraudulent.

3. VARIOUS KINDS OF HEALTHCARE DATA

Healthcare has a diversified nature of data which are as follows:

- Administrative enrollment data
- Billing records data(charges and payments)

- Medical records data
- Hospitals management data
- Physicians related data
- Health plans information(private/public)
- Procedures data(diagnosis)
- Patient-related data(in/out)
- Treatment-related data(clinical, symptoms, disease)
- Demographic details(patient/hospital)

Fraud in health insurance is a planned act of concealing, deceiving or misrepresenting facts that result in healthcare paybacks to a single or a group. The system operates on policies supported and managed by the private sector or administrators.

The fraud includes various characteristics like suspected diagnosis proof, low insured willingness, recurrent claims, late filing of claims, inconsistent documents, insufficient damage level information, high payments, and poor reliability on the certificate. Each claim submitted must be approved by the medical proficient for extenuating the chapter's weakness. It incorporates age, sex, name, date of birth, type of claim, identification number, resting period solicited, identification of medical expert, identification of employer and past and present income records, *etc.*

Fraud prevention and detection have to chapter simultaneously to stop the occurrence of fraud and handle it in the best possible way. In practice, detecting fraud starts from the prevention of false alarms, like identifying the tradeoff between the costs of analyzing a fraudulent case and saving by investigating it by taking into consideration. According to the European Anti-Fraud Organization classification, the different typologies of corruption are based on the characteristics in stand-ard and financial flow involved in the fraudulent processes.

4. EXISTED WORK OF DIFFERENT MODELS ON FRAUD DETECTION

The literature within healthcare is categorized into three, among which the first category focuses on the statistical methods that can apply to fraud detection. The second is applying unsupervised and supervised methods used to find the usefulness in detecting fraud. This type helps to choose a method for fraud detection and can compare with the other methods. The third type is to develop general approaches and models to improve the performance of the existing methods for the prevention and detection of fraud detection. This chapter uses both statistical and supervised models that focus on multiple healthcare datasets, building data views, data analysis, and applied machine learning techniques to improve the performance results.

4.1. General Model

Peter Travaille *et al.* give electronic fraud detection (EFD) techniques. The results of fifteen review papers include an overview of methods for fraud detection and various features like industry. An objective of the chapter is to find the kind of fraud and techniques used with results and findings. The literature study discusses the typology for the detection of fraud was four kinds based on the methods. Moreover, the lessons are helpful for supervised models. The drawback of the methods is that there is no novel fraud scheme for detecting the new fraud at the input. Similarly, unsupervised techniques to detect fraud patterns in telecommunication areas, but these were a high false alarm rate. Commercial enterprises do not want to lose profit. Therefore, they implement essential measures to remove fraud from the system. Businesses and enterprises give priority to on-time and fraud-free payments. The latest is fully automated models incorporating the business intelligence tools for anomaly detection by combining the geographically and socio-economically adjusting the models. The limitation of the research chapter is a theoretical approach, getting a view on fraud schemes and detection techniques. The further direction is to evaluate the current technologies and tools for the prevention of fraud [\[9\]](#).

4.2. Statistical Model

Aral *et al.* gave a conventional prescription auditing tool for fraud detection. The prototype applies a new distance based on the DM methods for estimating the prescription fraudulent risk. The risk is evaluated by the number of times the medicine is issued to the corresponding unordered categorical entry. Five two-dimensional spaces define the problem as sub-domains by creating the incidence matrices, risk matrices, and risk estimation. Further work addresses building such a system because the fraudulent behaviors related to a cross-feature are outliers when considering the total dataset [[10](#)].

4.3. Supervised Models

MelihKirlidog *et al.* discuss some cases for finding fraud by analyzing the previous claiming behavior in Turkish health insurance data using DM software. DM algorithms with various applications to recognize fraud claims. The method is done at different intervals at health centers on excessive claims. The investigation is based on high amount claims, and the general average number of claims to anomalous claims average on some criteria like freelance doctors, pharmacies, medical visualization centers, hospitals, polyclinics, diagnostics centers, and others at individual health centers analyzes them. The fraudsters are identified and further explored to reveal original and unknown patterns [[11](#)].

Yang Xie *et al.* developed approaches for using large-scale insurance claims to grow the expenditure. The technique employs the division of the whole general population into sub-populations and applies the decision tree procedure. It predicts the fixed number of days for every customer constructed on their pre-successive number of years. There are various components for Hospital information, like claim forms, date information, and hospitalization for several days, among which hospitalization is a component for health expenditure, and some perspectives estimating the hospitalized days will enable to reduce/increase the cost of payment. The model is built in three levels: customer-level information, hospital admission level, detailed information about the customer, provider, and claims. Hospital Procedure claim level describes item type, accommodation,

prosthetics, etc. Admission records may contain one or more procedure claims undergone by the customer in the clinic. The dataset includes hospital claims data from the Hospital Contribution Fund of Australia (HCF) [12].

The combination of bootstrap with ensemble RF method is done by Weiwei on insurance data using big data analytics. The data imbalance, and feature selection are handled at the pre-processing stage. Increased sales by identifying salesman strengths to increase new users by considering the experience of a sales executive, the outcome of sales of new products was calculated by users. The classification model had chosen RF, Bootstrap-based KNN, and SMOTE to analyze the insurance data. The future work intends to seek the help of Medicare experts to get more labeled data and apply some supervised models [13].

Nitesh V.Chawla discussed oversampling the minority class (abnormal) and under-sampling the majority (normal) class for various DM techniques with class performance metrics. The SMOTE is learned with under-sampling C4.5, Ripper, and Naïve Bayes classifiers on synthetic examples. The big data application domain has pointed to the attention concerning class imbalance problems [14].

Richard A Bauder *et al.* use four different metrics to identify fraudulent providers and improve performance by diminishing the class imbalance problem by applying the learning methods. Training and evaluation were done with BACC (balanced accuracy), MCC (Matthew's correlation coefficient), F-measure, and G-measure. For the experimentation, they used hybrid learners to increase their performance. MARS (Multivariate Adaptive Regression Splines model is a two-fold and non-parametric regressive approach to detect outliers with a single variable. The model is a generalized Bayesian probability model using standard language to find abnormal behavior (outliers). The two sampling methods are applied to handle problems of class imbalance. Further, identifying fraud providers using machine learning approaches is carried out on hyperparameter tuning [15].

This outlines providers' facilities and agrees with them to spot fraud and abuse in the Brazilian healthcare market by Jackson Cunha *et al.* using classification algorithms with various class distributions. The experiment is designed with four steps. Random Forest, C4.5, Ripper, Naïve Bayes, and SVM algorithms had defined metrics to measure the performance loss, performance prediction, and performance recovery for the given class distribution. Table shows the comparison of different treatment methods like Random oversampling, SMOTE, and MetaCost methods with each other. The limitation of the chapter is the usage of three datasets for claim authorization. As if the training data size increases, the process of computation will be impractical. The outcome suggested a direction to carry the next level, by selecting an appropriate dataset to not affect the negative learning and with an efficient learning algorithm [[16](#)].

Some researchers have tried using big data for fraud detection to handle multiple Medicare data sources. The proposed model by Matthew Herland *et al.* improved anti-fraud healthcare systems. The aggregated CMS dataset is used for processing the experimental study. The pre-processing is carried out for categorical data, with one hot encoder and reduced feature set created by join operation on NPI, provider type, and year. The essential learners' Logistic Regression (LR), RF, and GB (Gradient Boosted trees) are used to evaluate. Comparison of prototypes is made with F-score, AUC, mean, SD, maximum, and minimum, performed ANOVA testing the individual datasets, learner, dataset learner, and residuals. The mechanism of logistic regression yielded the best performance for the part b dataset [[17](#)]. The results generated based on the hypothesis and which part of Medicare data will commit cannot identify fraud. The chapter concluded by finding fraud behaviors in submitted payments. Further work will include employing data sampling techniques to combat the imbalanced nature of known fraud events in evaluating the different Medicare datasets.

Matthew Herland *et al.* (2019) extend to severe class imbalance effects while removing the fraud instances to determine the class rarity on fraud detection performance. Performed experiments using RF under-sampling, GBT, and LR on both training and testing and

assessed the potential improvements in fraud detection on Part B and Part D and combined dataset. The AUC, Std, r Min, and Max metrics were used to compare both train and test data [\[18\]](#).

Matthew Herland *et al.* (2020) extend their work towards identifying medical fraud in provider claims data using data mining strategies. The two baseline algorithms used are Multinomial Naïve Bayes, and Logistic Regression are used to investigate fraud on the U.S Medicare part B dataset. Compared to their previous work, the accuracy of prediction increased. The prediction of the physician's specialty is based on the type and number of procedures performed. By adding class grouping and class removal to the baseline algorithms, they might be able to predict normal or non-suspicious behaviour, which would lead to more complex results [\[19\]](#).

Most of the research addresses problems like processing claims, identifying the anomalies specific to treatment procedures like dental claims, heart disease, cell transplantation, analysis of doctor prescription, and estimating the number of days expenditure for hospitalization towards treatment, *etc.* There are certain gaps:

- Reducing the cost carefully, where they affect the other quality measures
- To collect more labeled data and apply some supervised models
- Building a model for identifying the fraudulent behaviors related to a cross-feature are outliers when considering the total data
- To employ sampling techniques to combat the imbalanced nature of known fraud events in evaluating the different Medicare datasets

The motivation of the chapter is to progress by collecting all the claims related to different treatments and procedures initiated at various locations to process data, understand the current scenario, impact various features, and investigate the claims for reducing fraud [\[20\]](#).

5. MECHANISM FOR INVESTIGATING AND IDENTIFYING THE FRAUDULENT BEHAVIORS OF MEDICAL CLAIMS DATA

Medical data analysis and information about the consistency of submitted data and providing decisions are typical tasks for most healthcare organizations. The fraud which is associated with Medicare is a threat to health outcomes globally. It is also negatively affecting the misuse of resources, infrastructure, and various social determinants of health. Healthcare fraud is associated with increased costs in most of the leading countries. The chapter provides an estimation mechanism for the utilization of health resources and their impacts on healthcare costs [21].

5.1. Dataset Description

In this context, the research chapter shall analyze healthcare claims in all dimensions using data mining techniques. This ideal helps analyze a wide variety of medical data gathered from multiple data sources from CMS, CHHS, NTRVSS and aggregated datasets, as shown in Table 1.

Table 1 **Datasets description.**

Name of the Dataset	Number of Attributes	Dataset Size
CMS(HHS) 2005-2006 Receipts of fiscal year data	Fiscal year, ALJ number, Received date, appeal category, request type, Medicare-type, requestor type, state, decision, disposition, appeal status, <i>etc.</i>	56,753
CHHS(FFS)	NPI (national provider identifier), provider owner number, service, location number, legal number, provider type code, provider specialty code, specialty description, out-of-state indicator, in/out state, county code,	173,424

Name of the Dataset	Number of Attributes	Dataset Size
	county name, address attention, addressline1, city, state, Zip, etc.	
NTR- Vaidya Seva Scheme (NTRVSS)	Cardnumber,Card-Midnumber, age,sex, caste,category-code,category-name,surgery-code,surgery-name,village,mandal,district,pre-authorized,pre-authorize-amount, claim-amount Hospital-name, etc.	1,60,949
Aggregated CMS data	Provider number, NPI, Specialty Description, Category_Name, Surgery_Name, Clm_id, Clm_from_dt, Clm_thru_dt, Card_No, Age, Sex, Claim_Amount, Preauth_Amount, Diff_amt, Claim_Date, etc.	173384

5.2. Preprocessing

Every country will have a different data collection format related to healthcare, and it has highly complex features and heterogeneous typed data. Analyzing healthcare data is a role undertaken by the government and commercial companies for accepting appeals and sanctioning the claims of those insured. As the data of healthcare is robust and made up of heterogeneously typed columns, traditional approaches consist of a laborious and time-consuming process. The feature selection is the most important, as the evaluation performance model highly depends. The supervised learning method models the parameters to identify the complex interactions in the medical claims domain and retrieves the falsified, suspicious, and abnormal records under different fraud categories. The limitation is likely to be overcome in this research by using mixed techniques for the individual and aggregated dataset, which implicitly detect the complex nonlinear relationship between features and the ability to detect all possible interactions between the predictor variables [22].

5.3. Model Construction

The model is constructed at three levels: In the first level, the analysis of datasets extracts the primary features and provides constructive decisions and outcomes on data processing. The methodology is depicted in Fig. [1](#). Develop a mathematical model to infer the behaviors of features in medical claims data to analyze and identify fraudulent behaviors. Understanding the current state of health care claims and expenses about the severity, frequency, and consequences of occurring fraud is a severe problem facing insurers, insurance consumers, and regulators. The existing systems will not only increase the cost but also affect the financial strength of insurers. Understanding and processing such data sets and finding the role of each attribute for data analysis are tricky tasks that this research will attempt to ease. The second level uses the SODAC approach to identify suspicious and abnormal claims. The third level introduces the finding the fraudulent behaviors by using different machine learning algorithms [\[23\]](#).

5.4. Mathematical Model for Experimental Analysis of Healthcare Claims

The level uses Hierarchical Grouping (HG) with an experimental model to handle the complex data and analysis of the categorical data consisting of heterogeneously typed columns. The procedure shapes a structure by quantitatively estimating the most excellent likenesses as for the predetermined features. It is an estimation technique employed in a panel data set that allows one to control time-invariant, unobserved individual characteristics correlated with the observed independent variables. Almost all the features are multiclass and of considerable size, and multi-collinearity occurs between the components. To compute aggregation on multiple columns at a time and pivot the results it is a sophisticated operation group by data frame. The metric Incorporates split, apply, and combine the metrics used to prepare the CMS dataset in Fig. [2](#). The traditional strategies for analysis take much cost and a long-time but deliver little because there is a lack of trust towards inconsistent data

quality and the high cost for multiple iterations. This experimental analysis helps make decisions for the different groups' data like claims, medical services, status of claims, etc. The motive is to provide a supportive chapter for outlier detection, prediction, decision-making, and prescriptive tasks for multi-dimensional data. It provides a new efficient approach to analyze Medicare datasets [24].

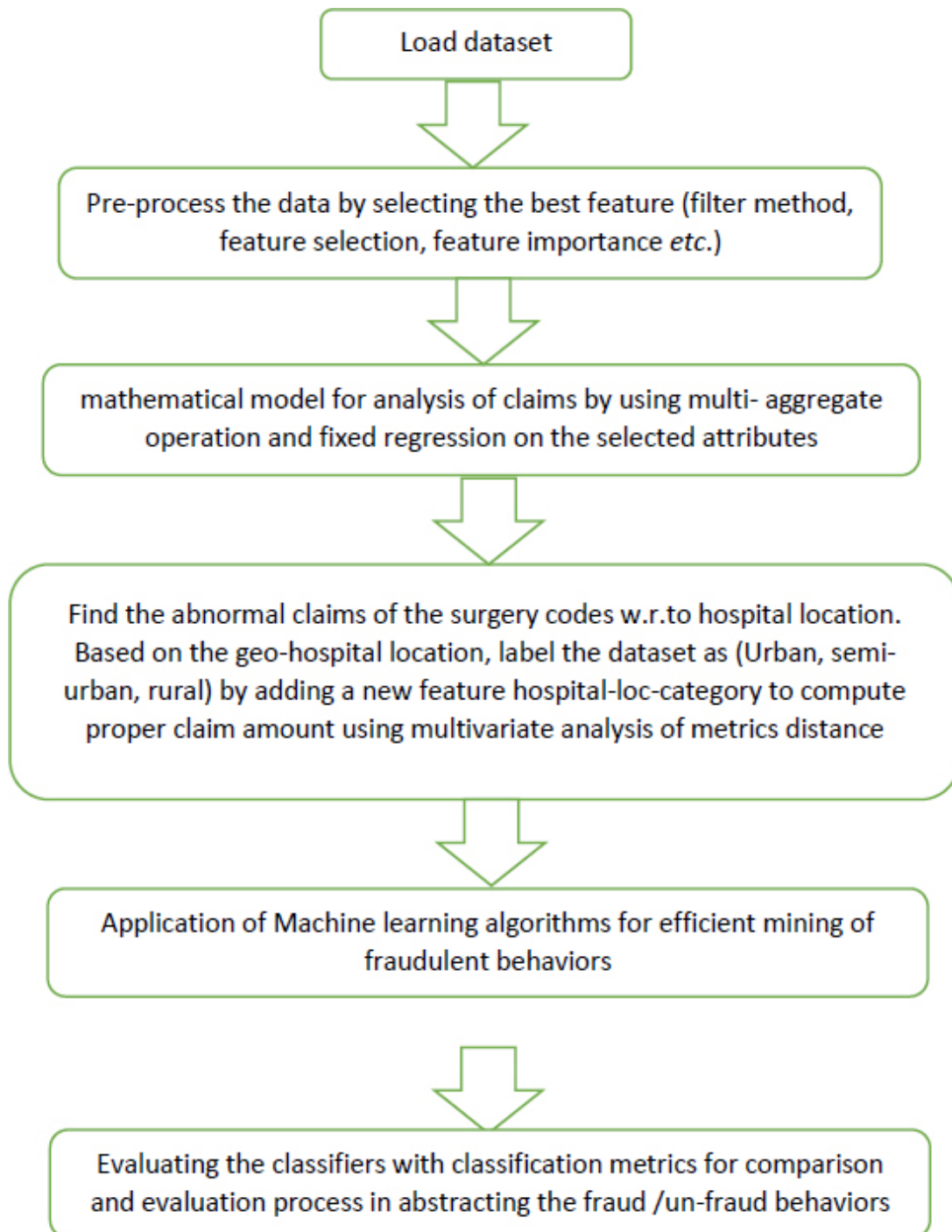


Fig. (1)

Stages of finding the fraudulent behaviors of medical claims data.

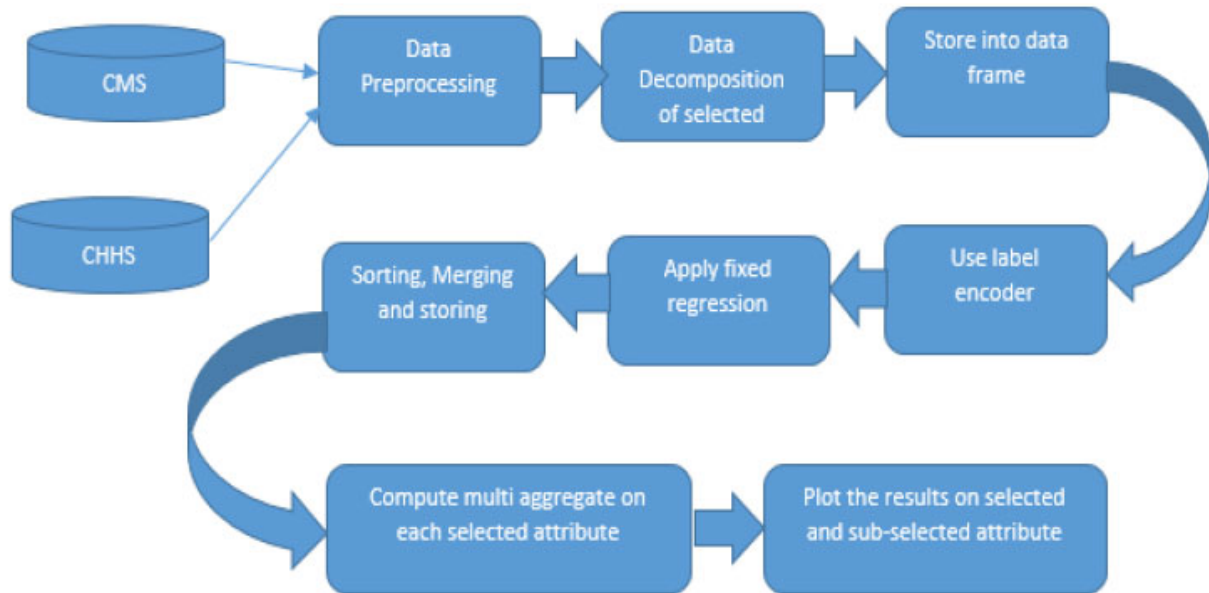


Fig. (2)

Block diagram for Mathematical model for experimental analysis.

5.4.1. Outcomes of the Experimental Analysis

The outcomes were discussed in light of various datasets while considering the HHS dataset the critical features as appeal category, hearing type, appeal status, request type, Medicare part, and disposition for both fiscal years 2005 and 2006. The scope of claiming is based on various features in the dataset for different organizations. The statistical analysis helps insurance organizations to estimate the utilization of services and payment associated with claims can be minimized and properly used. This difference and discriminant aggregate data analysis could determine the price differences between participants and insurance companies. It, in turn, assists the control groups in adjusting healthcare spending for the benefit of the population implicitly. The results suggest some care management or coordination programs demonstrate the savings for potential companies (high cost) and improve healthcare. In continuation, these cost savings will help the beneficiary and control group with the rapid spending of Medicare services over some time.

This model suggests ways to reduce expensive services and improve the potential benefit to providers. The research method resulted in an estimation approach and acknowledged the risk of overestimating savings, costs, and services. These automatically incur preventive efforts and can be adjustable for overall cost savings. This study has a limitation not to consider all the features for a calculation to produce results. The dataset is constrained based on the requirement.

Comparative analysis on appeal category w.r.t Fiscal year (Fig. [3](#)) discusses the claims under various sub-classifications and their most extreme and most minor recurrence of usage of appeal category over both the fiscal year of 2005 and 2006. The 2005 fiscal year has 23 appeal sub-categories, whereas 2006 has 32 appeal sub-categories. So, to normalize the data fields of appeal sub-categories, the common sub-categories have been considered. Otherwise, the appeal sub-categories for the fiscal year 2005 should be raised to 32 appeal sub-categories making their utilization zero. To reduce the over-fitting of data, we minimize appeal categories of the 2006 fiscal year, as discussed in Fig. [3](#). The outcomes for both budgetary years represent the annual spending and utilization of resources on Medicare data. Similarly, the analysis is carried out on various features for the fiscal year to minimize correlated features. It helps to sort the extent the services can be extended and whether data is a handful to analyze and process the claims shown in Figs. [4](#) and Fig. [5](#). The discriminate aggregate data analysis was shown in Fig. [3](#) from the above results in all categories have increased from 2005 to 2006.

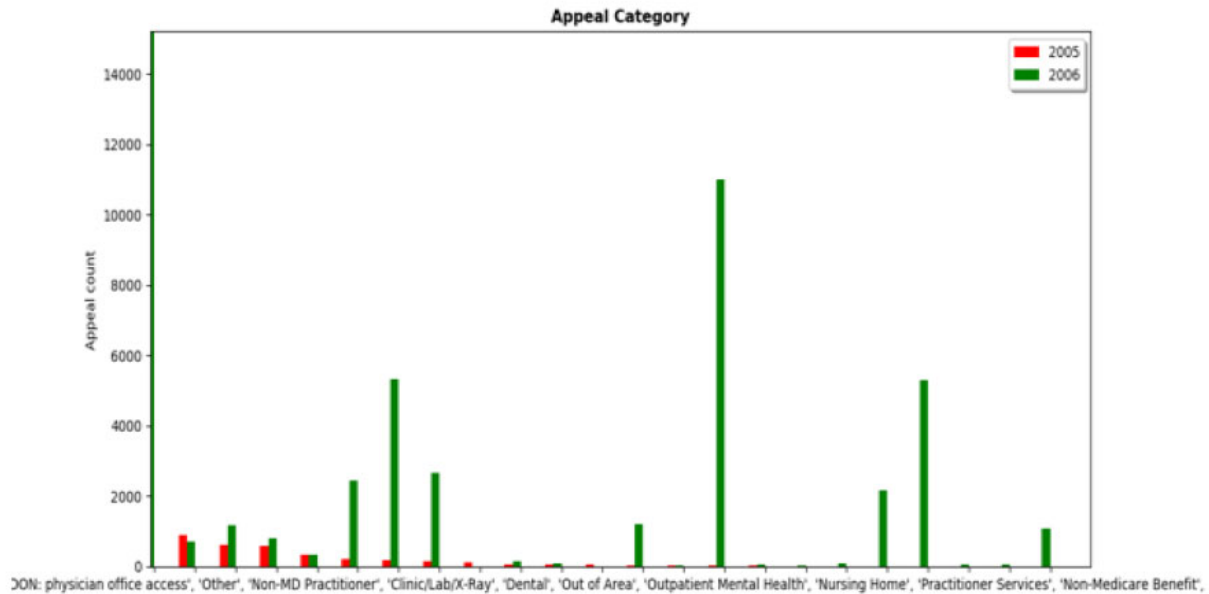


Fig. (3)
Comparative aggregate data analysis for 2005-2006 Fiscal year datasets based on appeal category on the x-axis. By considering the common 22 sub-categories of the appeal category, the analysis.

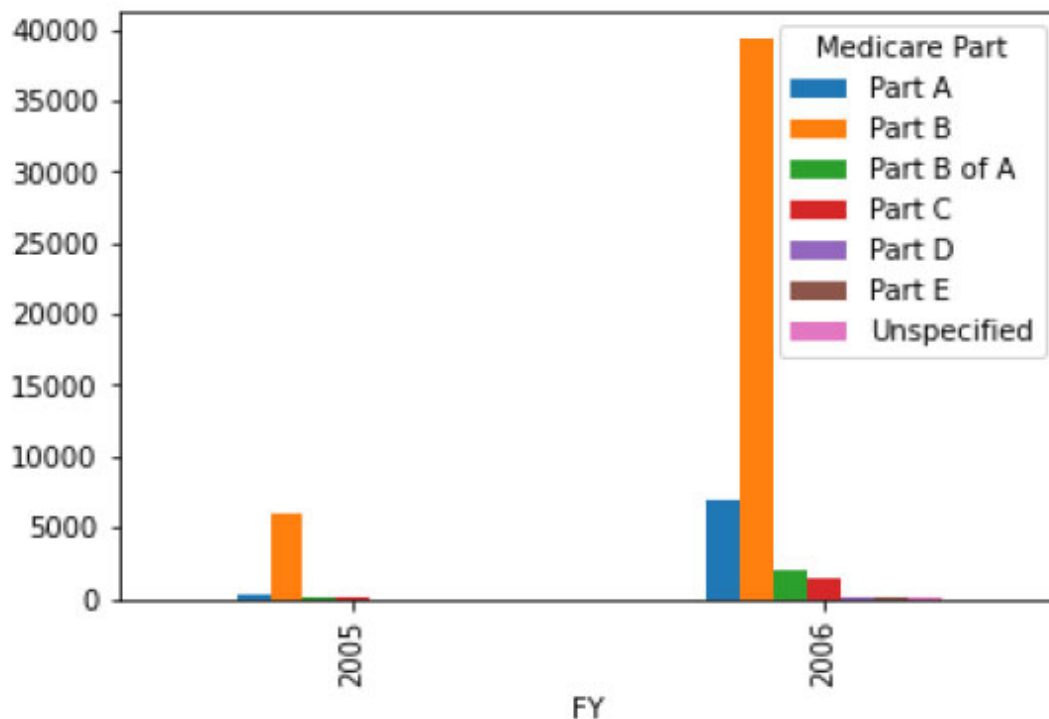


Fig. (4)
Discriminant aggregate data analysis between the F.Y. and Medicare part.

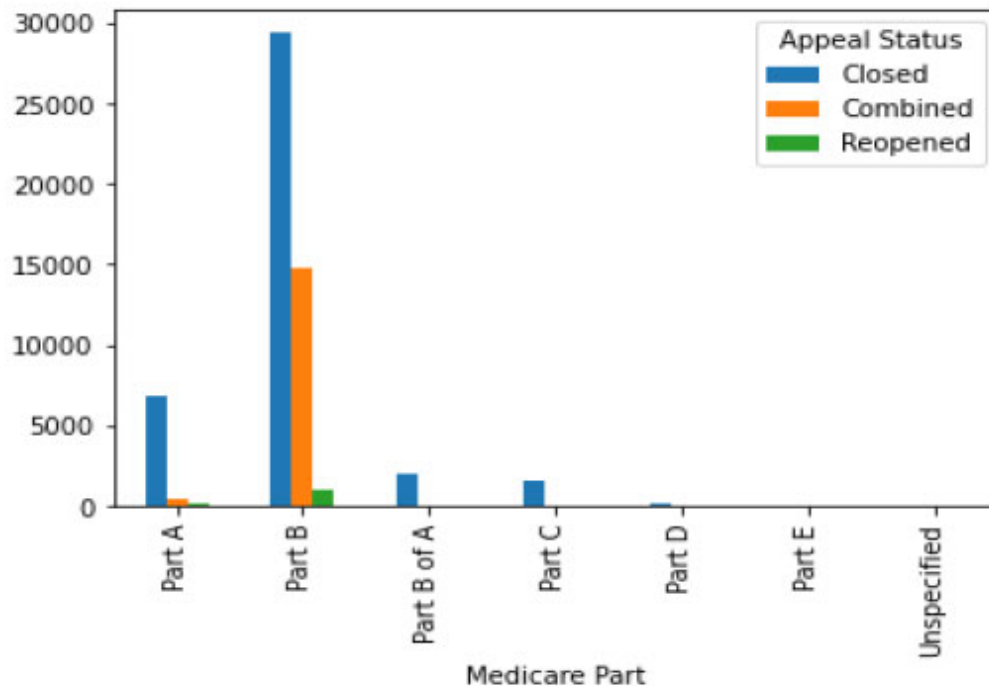


Fig. (5))

Aggregate data Medicare part and appeal status.

The provided claims have undergone the OMHA program with a decision by the jurisdiction. The dataset describes this decision by disposition characteristic. Most of the case's conclusion is done by the Inspector General after hearing or should they process the claim positively or negatively. The analysis shows that the claims that do not possess clear documentation of that appeal is rejected based on their claiming behavior from year to year. Similarly, the analysis is carried on various features for the fiscal year to minimize correlated features. It helps to sort and how can be the services be extended and whether data is a handful to analyze and process the claims. The fixed regression helps to know the impact of features for analysis, and it sorts out which factors interact with each other for making out decisions.

5.5. Methodology for Detection of Abnormal /Suspicious Claims in Healthcare Data

The recommended system identifies suspicious behavior in medical claims where it associates with patients or professionals submitted to the organization. The analysis is carried out on features and their cross-references with number of times of claiming at various locations for different surgeries. The claim amount is determined by its category code and surgery code w.r.to different hospital location codes. Preprocessing involves feature selection. It enables the algorithm to converge and train faster, reduces complexity, easier to interpret, and improves efficiency if the proper subset is chosen. Finally, it allows over-fitting. For experimentation purposes, the chapter has applied filters to select the subset of attributes. The selected set of features is prompt for the outlier codes detection of the claimed amount for surgeries for various categories undergone in a different hospital at various locations. The relatedness and dependencies of each feature were considered, and did the transformation of some features was done. In the next stage, generalization of the dataset is done by adding a new feature for the experimentation to measure the similarity of the data as all the records are unique. The supervised outlier code detection technique is based on the previous instances or records and detects suspicious/abnormal claims. The statistical mean is applied to measure the distance.

Definition: Let N_p is a set of all instances of D , d_1 is a subset of all instance values w.r.to to an object X_p . The value of X_p is determined by the average statistical mean measure from all sets of objects in instances values w.r.to A_p ,

let $avg\ dist\ (x, y) \geq 0$ be the measure of distance between the instances x and y

The Statistical mean measure for object X_p is defined as:

$$\overline{X_p} = \sum_{m=1}^p \frac{x_m}{n} \quad (1)$$

If X_p is calculated w.r.t to an object N_p , k and k_1 are the number of instance values of D and d_1 .

The formula calculates the average distance of N_p , of X_p

$$\bar{D}_{x_p} = \frac{1}{k} \sum_{x_i, x_i \in N_p} \frac{X_p}{k_1} \quad (2)$$

The inner distance of all instances of X_p for an object x_p is evaluated by the normal deviation, maximum and minimum values of X_p . The distance-based outlier is determined by the:

$$dod_{x_p} > \max(\bar{D}_{x_p}) \quad (3)$$

$$dod_{x_p} < \min(\bar{D}_{x_p}) \quad (4)$$

For example, in dataset D , the k are instances of category code $M1, M2, \dots, M12$ and $S1, S2, \dots, S17$. Let $M1$ be the category code where it retrieves nearest claim values of different surgery code ranges from $\{M1.1, M1.2, \dots, M1.7, M1.8\}$ as sub-instances k_1 . The claim of surgery code $M1.1$ compares with all instances of the whole dataset and takes the local claim amount. The claim amount $M1.1$ w.r.to each hospital-loc-category $\{\text{Rural, Semi-Urban, Urban}\}$ is evaluated. The outlier claim amount of surgery is calculated by comparing the claim amount w.r.to the previously stored values and identifies the fraud claiming behavior. According to the dod values, the data instances arranged of $dod_{M1.1} < dod_{M1.1}$. The highest values of dod for each surgery code are termed as the maximum claim amount. The outlier codes are determined by the claim amount, which exceeds the maximum or minimum of particular surgery codes [23].

5.5.1. Outcomes of the SODAC Approach

The two baseline methods for detecting the outliers are the Statistical mean measure and distance measure for identifying the suspicious behavior of a patient or professional. The analysis of Medical claim

Dataset (NTRVSS) requires domain knowledge as all are of domain-specific features that take much time to select and define.

To apply data transformation, data reduction, and normalization, requires explicitly defining new features for processing selected features for analysis. The dataset is of 1,60,949 records and 25 attributes such as card number, Card-Mid number, age, sex, caste, category-code, claim-mount, Hospital-name, Hospital-type, hospital location, hospital district, surgery-date and discharge date etc., having multiple labels for each data instance. The application of derived multi aggregates for the dataset on each surgery code under different categories and w.r.to hospital location shows in Table 4. The changes of claims amount from initial to computed values for each tuple, in Table 4, as we see the considerable change of claim amount is observed for specific surgery codes. The initial value for carrying the M10.2 surgery at semi-urban is 15000, but they claim they applied later as 32067 from this observation is suspected as abnormal. Similarly, specific surgery codes go beyond and below the initial computed values shown in Table 2. Out of 40000 tuples, the observed outlier surgery codes are 3500 at different hospital location categories for all surgeries. The comparison of claiming amounts at different hospital locations for some categories is shown in Table 3. It discusses the M5 category with M5.1.1 to M 5.6 surgery code w.r.to rural, semi-urban; urban; the variation of claim amounts may differ relatively small for large for some surgeries.

Table 2 **List of Outlier codes for sub-surgery code M10, M11 w.r.to hos- loc-cat of category which exceeds the amount of prior maximum Cost.**

M 10			
Surgery_code	Rural	Semi-Urban	Urban
M 10.2	--	15000	--
-----Exceeds Maximum amount-----			
Surgery_code	Rural	Semi-Urban	Urban
M 10.2	---	32067	---
M 11			
Surgery_code	Rural	Semi-Urban	Urban

M 10			
M 11.1.1	--	15946.97	--
M 11.1.2	---	16600	16333.33
M 11.2.4	--	14500	--
-----Exceeds Maximum amount-----			
Surgery_code	Rural	Semi-Urban	Urban
M 11.1.1	--	71250	--
M 11.1.2	---	30000	25000
M 11.2.4	--	32067	--

Table 3 **Comparative claim amounts of average statistical mean value and outlier cost for M5 sub-category surgery codes.**

Surgery Code	Rural	Semi-Urban	Urban
Initial Claim amounts of M 5 Surgery codes			
M5.1.1	19400	18486	20786
M5.1.2	23542	23855	24257
M5.1.3	6000	18949	18225
M5.6	-	15285	-
Outlier surgery codes claim amounts.			
M5.1.1	30000	25000	60000
M5.1.2	45000	36000	25000
M5.1.3	16000	30000	35200
M5.6	-	18000	-

Table 4 **Classifying the instances of combined CMS dataset with different fraudulent categories.**

Fraud Category	Type of Party	Type of Fraud Scheme
Fraud_by_patient	Patient, Provider	Unbundling
Fraud_by_NPI	Provider	Duplicate billing
Fraud_by_Disease	Patient, Provider	Excessive services
Fraud_by_patient_diagnosis	Patient, Provider, Physician	Duplicate billing, Excessive services

Fraud Category	Type of Party	Type of Fraud Scheme
Fraud_by_diff_amt	Patient	Upcoding, Excessive or unnecessary services
Fraud_phy_diagnosis	Patient, Physician	Phantom billing, Bill Padding
Fraud_diff_amt_60	Patient	Excessive services

5.6. Methodology for Fraudulent Behaviors of Healthcare

The designed model is carried out in various phases to obtain successful results. The stages include pre-processing, labeling the dataset with various ML techniques. The classification rules are defined for every category and assess the true and false records. The rules identify the frequent patterns and their associations, such as sequence of diagnosis (procedure) codes, provider/physician includes diagnosis codes, and frequency of patient undertakes the number of diagnosis codes. The following classification rules define fraud schemes shown in Table [5](#).

5.6.1. Outcomes Using Machine Learning Algorithms

This section presents the details of the existing world's problem of imbalanced data; the investigation is carried out and configured with parameters and then discusses mathematical tests used to contrast and relate to performance with different metrics of the approaches. The instances of the dataset after adding all classification rules are shown in Table [6](#).

Comparison with existing models Table [6](#) is compared with the present and proposed model, by assessing the dataset and learner performance of Medicare fraud detection systems. The favorable results have been obtained by combining RF and SMOTE, the better scores with the smallest variability between the classifiers. Therefore, based on the exploratory performance results, the paper

demonstrates when a physician, provider involves in Part B, Part D or a combined dataset, using RF with SMOTE, it indicates better overall fraud detection performance. Evaluation of the RFSC model after pre-processing the model has been updated with classification rules; the number of instances is characterized into true and false instances. The parameters help the model to be best for performance with a comparison with other models [24, 25]. The quality of classifiers is given by various metrics for evaluation of different models are F-score, accuracy, precision, AUC and recall. Evaluation is a factor considered for analysis, performance and guidance for the constructed classifier. The efficiency of the RFSC model and R, SVM and RF with performance metrics are shown in Fig. 6.

Table 5 **Classification Report 2 shows the comparison of different models.**

Parameter		LR	SVM	RF	RFSC
CV Score	Mean	0.50	0.50	0.500428	0.867336
	Stand.Dev	0.0064	0.0064	0.0067229	0.0028613
	Min	0.487	0.489	0.4896601	0.8625037
	Max	0.5188	0.513	0.5133004	0.872521
Duration in seconds		37.189	39.4	39.445	45.322
Precision	0	0.17	0.17	0.17	0.89
	1	0.83	0.83	0.83	0.82
Recall	0	0.04	0.00	0.04	0.74
	1	0.96	0.76	0.96	0.93
F1-score	0	0.06	0	0.06	0.81
	1	0.89	0.91	0.89	0.87

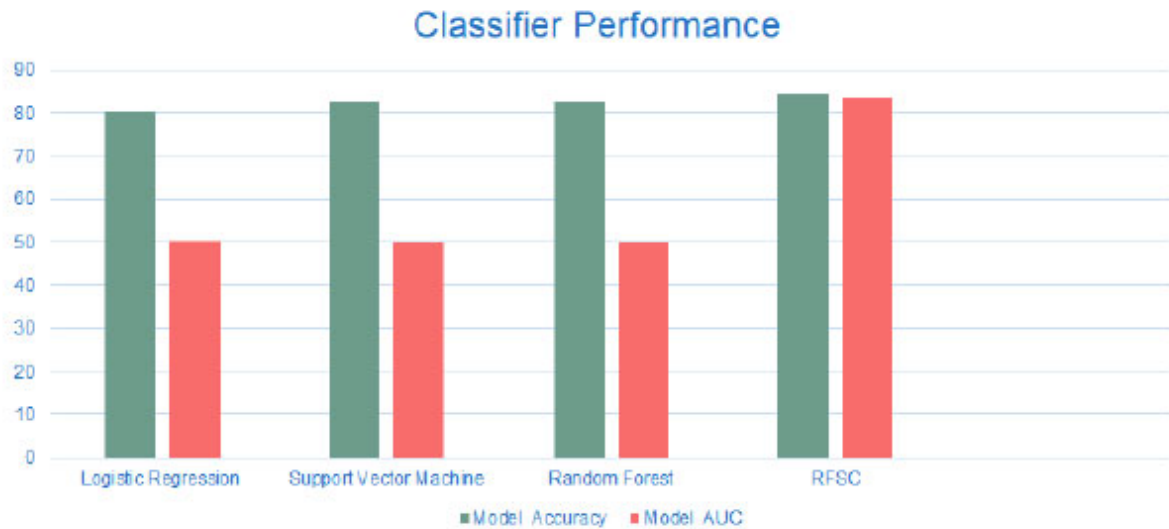


Fig. (6)

Comparison with classifier metric like Accuracy and AUC for the different classifiers.

Table 6 **Comparison of different classifier performance with accuracy and AUC metrics.**

Model	Classifier Performance	
	Accuracy	AUC
Logistic Regression	80.2	50.2
Support Vector Machine	82.6	50
Random Forest	82.6	50
Random Forest with SMOTE	84.5	83.5

CONCLUSION

The work that is outlined is perhaps creating an impact on the healthcare domain. Typically there is a large quantity of information and is primarily categorical apart from numerical data. The key objective of the chapter is to reduce Medicare fraud and keep fair and reasonable healthcare costs. Handling quality research to detect fraud is necessary for providers to maintain reasonable healthcare costs for medical services. However, other areas have used statistical

tools for the identification of fraud and suspicious activities. Irregularities are most common in finance, accounting, money laundering, insurance, the web, and many more. Whereas plagiarism can also be called fraud, where the manuscript needs to compare against the article database from the web with that statistical measure, and the originality is returned. The critical issue in all of this work is depicted incrementally, how effectively analyses are carried out, suspicious identification w.r.t claims, and identification of fraudulent claims by considering the different classification rules. The proposed approach leverages work to investigate the robust dataset, as it is a dubious job for any medical provider. This present accurate healthcare information is of multi-record and heterogeneous composing attributes.

While coming to the analysis, the Hierarchical Grouping strategy of the mathematical model is to disentangle the process of utilizing services to measure the quality of Medicare data. The technique depends on the selected attribute and subcategories, as they influence the number of different services offered by the medical administration. Quality metrics improve the cost-saving of medical providers. The outcomes are examined in the CMS (HHS) and CHHS (FFS) datasets by taking diverse selected attributes. The next stage of the incremental model is outlier detection in medical claims, which tends to be proactive for finding the suspicious (fraud) activities and identifying the medical fraud w.r.to claiming behavior [\[26\]](#).

The recommended methodology uses the distance-based and statistical mean measure to find how the outlier object deviates from the normal object from its neighborhood objects. The distance-based outlier codes analyzed the properties of the objects and identified the false distribution of instances. As far as the limitation, it underwent testing with a single dataset. The final stage of the work is applying RFSC as a unique approach that can be applied to large datasets. RF is a base, which generates decision tree ensembles, and it gives a good performance. It shall assist the government and private health providers to increase the economy and reduce fraud occurrences. The methodology tested for quality in terms of accuracy and f-

measure, and the design technique has evolved with fraud policies for the data.

The results show that the RFSC algorithm performs well for the imbalanced class problem. The overall performance of the aggregated dataset indicates a fraudulent system where it precisely targets the behaviors of medical procedures, physicians, services, procedures, claims, and providers. This work displays an underlying stage for examining Medicare information and accommodating for recognition of anomalies, and developing a predictive model for various applications. Preparing the healthcare claims data is a precarious errand. The work handles in an ideal way. The overall strategy is to build an approximate method for analysis, investigation, and identification of the suspicious events with large and small dramatic scores, but those will be closer to estimation. For different circumstances, there are many challenges with statistical tools and methods for finding fraud. The positive conclusion of the proposed approaches can be general and provides opportunities to identify fraud in healthcare claims. The scalability of the proposed methodology is high, and the computational cost is average. The real-world applications can be extended to high dimensional data also, but the efficiency in terms of time complexity (execution) goes low.

The limitation of the work is that more real-time labeled data is required to develop an automated supervised model. As such, it can work in real-time and for new incoming data. As there is a huge demand for healthcare data of its service, there is a requirement to continuously fine-tune data. The other challenge is a lack of trust arising from the inconsistent data quality and the high cost of multiple iterations. Further extension of work is to access data from any source, understanding clinical insight, and processing the claims with operational efficiency requires a deeper instigation of parameters for serving as AI-based decision support systems. As part of upcoming work, develop an integrated AI-based data model for concurrent data processing to access the data systems. The prototype can handle structured, unstructured, and semi-structured data with the help of AI algorithms for providing recent decisions on Medicare data. AI Model has a strong influence on claim adjudication, claim processing, claim

submission, and fraudulent claim detection. AI must be able to adjust the claim process based on the patient's characteristics and the severity of the case. As well it should be able to maintain the confidentiality of data. More research is also needed that addresses the current lack of understanding of optimal functionality and usability of the systems and how they can play a beneficial role in supporting healthcare systems.

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Security Threats and Detection Mechanisms in Machine Learning

K. Madhuri¹ *

¹ Anurag University, Hyderabad, India

Abstract

Machine Learning refers to computer programming or data science that can learn from data. For example, the performance of T is increased with experience E if a computer program is learned from E in some tasks T, and performance is measured as P. The learning of computers is allowed without explicit programming in Machine Learning, which is a field of research. Machine learning can be used in various applications, such as banking, travel and tourism, healthcare, marketing, insurance, and human resources. Machine learning is a powerful tool for implementing security applications.

Keywords: Machine learning, AI, Security, threats, Classification, Regression, Fraud Detection, Spam Detection.

* **Corresponding author K. Madhuri:** Anurag University, Hyderabad, India; E-mail: madhuris97@gmail.com

1. INTRODUCTION

ML is a subcategory of AI that uses algorithms to achieve automatic performance without prior programming. For predicting new output values, historical data will be used in machine learning algorithms.

Machine learning includes different categories, such as:

1. Reinforcement machine learning
2. Semi-supervised machine learning
3. Unsupervised machine learning
4. Supervised machine learning

For training the algorithms, the labelled data sets are used in supervised learning for data classification or determining the correct outcomes. Supervised learning is one of the types of machine learning that uses the 'labelled' training data for training, and the results are predicted using this data. The marked data describes marking some input data with the correct output. The training data provided to the machine considers a supervisor in supervised learning, and the machine learns about how to predict the output accurately. It also applies the concept of students learning under the guidance of a teacher. The process of providing inputs and outputs correctly for machine learning models is supervised learning. It has the objective of determining a mapping function to correlate the input variable (x) and the output variable (y). Supervised learning involves real-time applications, such as spam filtering, fraud detection, image classification and risk assessment. Based on the labelled data sets, the training of the model is carried out in supervised learning, and each type of data is learned by the model. The model will test using

the test data after completing the training process, and the results will be predicted [1].

1.1. Supervised Learning Example

For example (Fig. 1), if we have a data set that includes different types of shapes like polygons, triangles, rectangles, and squares. For each shape, the model is trained in the first step. The shape will mark as a square when a given shape includes four sides, and they all are equal.

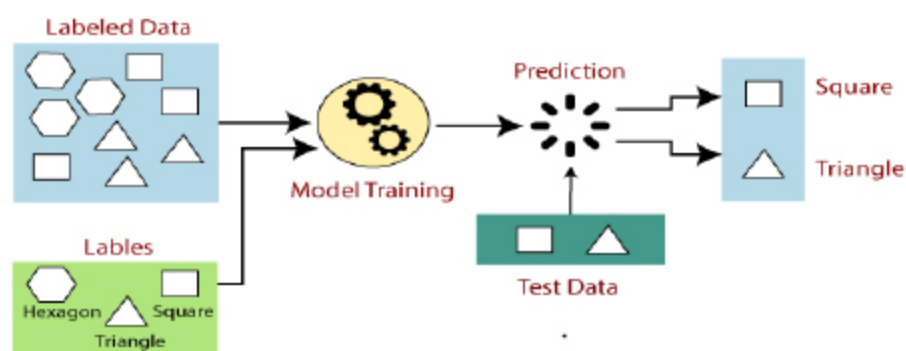


Fig. (1)

Supervised Learning Example.

If a shape includes three sides, it will mark as a triangle. Hexagon will be marked when a shape includes six equal sides. For testing the model, the test set is used after training. The model has the task of recognizing the shape. The training of a machine has been performed on different shapes. Based on the edges, the shapes will be sorted out if a new shape is found and the result is estimated.

1.1.1. The Steps Involved in Supervised Learning

Primarily, the training data set type needs to determine and collect the labelled training data, which is categorized into a verification data set, test data set, and training data set. Sufficient knowledge should have been included for computing the input characteristics of the training data set that leads to the accurate prediction of output. The appropriate algorithms estimate for the model, such as decision

trees, support vector machines, *etc.* The algorithm runs on the training data set. The validation set is required as a control parameter, which is a subset of the training data set. A test set is provided to evaluate the model's accuracy. The model is accurate when the model predicts the correct output [2].

Similar to the learning of things by students in the presence of a teacher, supervision is required for training the model in supervised learning, which can be utilized for two different problems, such as regression and classification, as shown in Fig. (2).

1.1.2. Supervised ML Algorithm Types

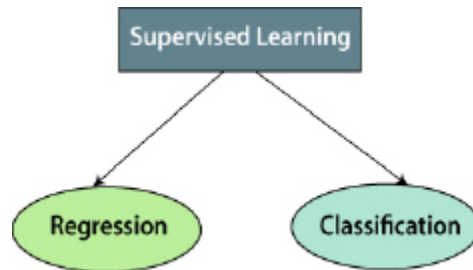


Fig. (2))

Supervised ML algorithm types.

1.1.3. Regression

The regression algorithm is used when a relationship exists between input and output variables. It is used to predict continuous variables like market trends, weather forecasts, *etc.* Some popular regression algorithms are included in supervised learning:

- o Linear regression
- o Regression tree
- o Nonlinear regression
- o Bayesian linear regression

o Polynomial regression

1.1.4. Classification

When the output variable is classified, the classification algorithm is used, which means there are two classes, such as Yes/No, Male/Female, True false, *etc.*

1.1.5. Supervised Learning Includes the Benefits such as

1. The outcome can be predicted by the model using previous experience in the supervised learning. In supervised learning, we can accurately understand the category of the object.
2. Supervised learning models help us to resolve different problems of real-time applications like spam filtering, fraud detection, *etc.*

Supervised learning disadvantages are:

1. The supervised learning model is inadequate for dealing with complicated tasks.
2. The correct output cannot be predicted by supervised learning when the test data is distinct from the training data set.
3. A lot of computing time is required for training.
4. Sufficient knowledge about object classes is needed in supervised learning.

1.2. Unsupervised Learning

A set of training data is not monitored in the model in unsupervised learning, which is a machine learning technique. From the provided data, the hidden insights and patterns will determine by the model itself. It has been compared to the human brain learning that occurs while learning new things. It can define as the machine learning type,

where a set of unlabelled data is used for training a model, and the actions to be taken on the data are allowed without supervision. Unlike supervised learning, it cannot be implemented to classification or regression problems directly. Input data is included but the corresponding output data is not there [3].

Unsupervised learning includes the objective of determining the data set's underlying structure, grouping of data using similarity, and the data set represented in a compressed way. For example, an unsupervised learning algorithm includes the input data set that has images of different types of dogs and cats. For a given data set, the algorithm is never trained, which means the data set characteristics are not known. The image characteristics identification is the main task of unsupervised learning algorithms. It performs the task based on the grouping of image data sets into the groups based on the similarity between images. For two different types of issues like Association and Clustering, unsupervised learning can be exploited.

1.2.1. Differences between Supervised Learning and Unsupervised Learning

- For training, tagged data is used by supervised learning algorithms. Whereas in unsupervised learning algorithms, unlabelled data is used for training.
- To validate whether the correct outcome is predicted, direct feedback is used by the supervised learning model. The feedback doesn't accept by the unsupervised learning model.
- The outcome is predicted using the supervised learning model and the hidden patterns of data are discovered using the unsupervised learning model.
- Input and output data are provided for the model in supervised learning. The input data is only provided for the model in unsupervised learning.
- If new data is provided, the prediction of outcome based on the trained model is the objective of supervised learning. Additionally, it requires estimating the hidden patterns and useful data from unknown data sets.

- Supervision is required for model training in supervised learning. Supervision doesn't need in unsupervised learning for training the model.
- Supervised learning can categorize into regression and classification problems. The association and grouping problems are included in unsupervised learning.
- If the corresponding input and output is known, supervised learning can be utilized. If input data is there only, but the output data is not corresponded, unsupervised learning can be used.
- The accurate results generate by the supervised learning model. Less accurate results can provide by unsupervised learning model than the supervised learning.
- Due to the training of a model for each data piece and predicting the correct outcome later, supervised learning doesn't provide accurate performance relevant to true artificial intelligence. As unsupervised learning is similar to student learning every day through experience, it is closer to true artificial intelligence [\[4\]](#).

2. REGRESSION ANALYSIS IN MACHINE LEARNING

One or more independent variables are used for regression analysis, which is a statistical technique for modelling the relationship between the independent variable (predictor) and the dependent variable (target). The understanding of the dependent variable value in regard to the changes of the independent variable is involved in the regression analysis if other independent variables are fixed. The actual or continuous values are determined, including prices, wages, age, temperature, *etc.*

The correlation between variables is determined and the continuous output variables are predicted using one or more variables of the predictor in the regression, which is a supervised learning method. The causal relationship between variables is determined using time series modelling. For a given data point, a graph is drawn between the most suitable variables in the regression. The data predictions

can be performed using the machine learning model based on this graph. Simply put, “Regression displays a straight line or curve, which passes through all the data points on the target prediction graph, so that the vertical distance between the data points and the regression line is the smallest.” The distance between the line and the data point indicates whether the model captures the strong relation. Some examples of regression included temperature and other factors utilization for predicting the traffic accidents that caused by hasty driving, market trends, and rainfall. In regression analysis, the dependent variable is used as the major factor to understand and predict the dependent variable, which is also known as the target variable.

2.1. Outliers

In comparison to other observations, outliers are included very high or very lower values based on the observations. They can hamper results so that avoid these outliers.

2.2. Multi-collinearity

The situation is known as multi-collinearity when independent variables correlate with each other highly than the other variables. In the dataset, it shouldn't display due to the problems caused while ranking the most influential variables.

2.3. Underfitting and Overfitting

It's possible for the overfitting problem to occur when our algorithm suits for training, not for the test dataset. A mismatch problem results if the algorithm doesn't provide better performance even on the training dataset. The continuous variables are predicted by the regression analysis, as mentioned earlier [\[5\]](#).

In real-time cases, different scenarios exist for making some estimation in the future, such as marketing trends, weather conditions, forecasting of sales, etc. Some technologies are needed

for this case to make accurate predictions. The regression analysis is needed in this case and it is used as a statistical technique for data science and machine learning. Some other reasons are used for regression analysis: The relationship estimates between the independent variables and the objective using the regression.

For data trends, it is used to forecast the continuous or actual values. The least important factors, the most important factors, and how each factor impacts the others can estimate using the regression.

3. TYPES OF REGRESSION

In machine learning and data science, different types of regression are used. The independent variables impacting on dependent variables is analysed by all regression technique, and each type has its own importance in different settings.

Some important regression types are given below:

- o Linear regression
- o Logistic regression
- o Polynomial regression
- o Support vector regression
- o Decision tree regression
- o Random forest regression
- o Ridge regression
- o Cyclic regression
 - Support Vector Machine for classification problems
 - Random forest for regression and classification issues

- Linear regression for regression issues.

3.1. Linear Regression

For determining the relationships, regression is used as a set of techniques. A set of data items is taken as input in the linear regression and returns the value as output.

$Y = f(x_1, x_2, x_3 \dots)$, where Y refers to the dependent variable.

Logistic regression is used as a supervised learning classification algorithm for estimating the target variable's probability. It is used for different classification issues like cancer detection, diabetes prediction, spam detection, etc. and is considered as one of the simplest ML algorithms. Based on a given set of independent variables, continuous dependent variables are predicted using linear regression, which uses for forecasting the categorical dependent variables for a given set of independent variables. The categorical variable value is predicted in logistic regression.

To model the binary variables that consider two values only 0 and 1, logistic regression is used and it is one of the widely used ML algorithms. A mathematical equation development is the objective of logistic regression that can provide a range of scores from 0 to 1. The probability of variable considers the value of 1 for this score [\[6\]](#).

4. APPLICATIONS OF LOGISTIC-REGRESSION

4.1. Spam Detection

A binary classification problem is the spam detection. We should have to classify whether it's a spam or not for a given email. We mark the email as 1 if it is spam. If not, it marks as 0. The below-mentioned features extract from emails for implementing the logistic regression to the problem of spam detection. In email, the number of

typographical errors like 'gift', 'prize', 'offer', *etc.* and the sender of email are included.

For training a logical classifier, the resulting feature vector is used that provides a score output within the range of 0 to 1. We consider the email as spam if score value is more than 5. We will not mark as spam in other cases [\[7\]](#).

4.2. Credit Card Fraud

As the banks spending millions or hundreds of dollars on fraud for every year, it's essential to detect the fraud problem of a credit card to the banking industry. The bank will have to be attentive towards various factors if a credit card transaction has occurred.

For example, type of purchase, location, amount, date of transaction, *etc.* A logistic regression model is developed using these factors to know whether the transaction was fraudulent. It marks as a fraud when the amount is higher and a high purchase hasn't made by the person.

4.3. Tumor Prediction

To identify the type of a tumor whether it's a benign or malignant, the logistic regression classifier is used. For extracting various features of tumors, different techniques of medical imaging are utilized. For example, the affected area, tumor size, *etc.* For detecting this, these features enter into a logistic regression classifier [\[8\]](#).

4.4. Marketing

The powerful algorithms are used for predicting whether you are interested in certain content when you browse for news feeds of Facebook every day. These algorithms can be seen as complex variants of logistic regression algorithms, where the question to answer is simple: will the user like this particular ad on their news feed?

5. NEAREST NEIGHBOUR ALGORITHM (KNN)

To determine the solutions for regression and classification problems, a simple and supervised machine learning algorithm can be used. It's easier to understand and implement the KNN algorithm. Due to the used data is increased; it has a drawback of reduced speed significantly. The working principle of KNN is included the determining of distance between query and the data. The particular number of examples (K) has chosen nearer to the query and vote for the average label (for regression) or the frequent tag (for sorting). For prediction of classification and regression issues, K-nearest neighbour algorithm (KNN) is used as a supervised ML algorithm.

Specifically, it uses for prediction of classification problems. KNN-Lazy Learning Algorithm can define using two attributes as it doesn't include a phase of dedicated training and all data use for training in the classification. KNN is also considered as a non-parametric learning algorithm due to the making of assumptions regarding the underlying data [\[9\]](#).

5.1. Working of KNN Algorithm

To forecast the new data point's value, K-nearest Neighbour Algorithm (KNN) is used as 'feature similarity'. Here, new data points will assign as a value according to the matching of training set points.

It works with the help of the following steps:

Step 1: A data set is needed for implementing any algorithm. The training and test data is required to load in the primary step of KNN.

Step 2: K value is required to choose and it is the closest data point. Here, K refers to any whole number.

Step 3: The following operations can perform for each point in the test data:

3.1-Use any of the following methods to calculate the distance between the test data and each row of training data: Euclidean distance, Manhattan distance or distance of Hamming. The most used method to calculate the distance is Euclid.

3.2-Now, arrange them in ascending order according to the distance value.

3.3-Next, you will select the first K rows of the ordered matrix.

3.4-Now, you will assign a class to the test points based on the most common class in these rows.

Step 4: Finish

For example (Fig. [3](#)), the concept of K is needed to understand – A data set can plot as follows:

For classification of data points (at 60, 60), the black points are used and enter into the red or blue category. $K = 3$ is assumed to find three closest data points. A black dot includes for the three nearest neighbours of data point as shown in below figure. Among these three, two represent the red classes and the black spots will assign to red classes.

Using the feature similarity, we classify new data items into a particular class based on the majority matching the above diagram, black spot is matching (nearer) with two red dots, so we classify black dot to be belonging to red dot's class.

6. MACHINE LEARNING AND SECURITY TASKS

6.1. Spam Detection

Spam can take many forms. A lot of it is just bothersome news that people are circulating to divert attention from their bogus information. Some of them are phishing emails designed to trick users into installing malware or visiting harmful websites. In most cases, they simply do nothing to address the recipient's concerns. So that users may read legitimate emails in their inboxes, we must limit the flagging so that the spam emails can be discovered using the spam detection algorithm. It needs to be done so that it adapts to the latest fads, whether they be related to the upcoming election, a global pandemic, a sudden interest in cryptocurrency, *etc* [10].

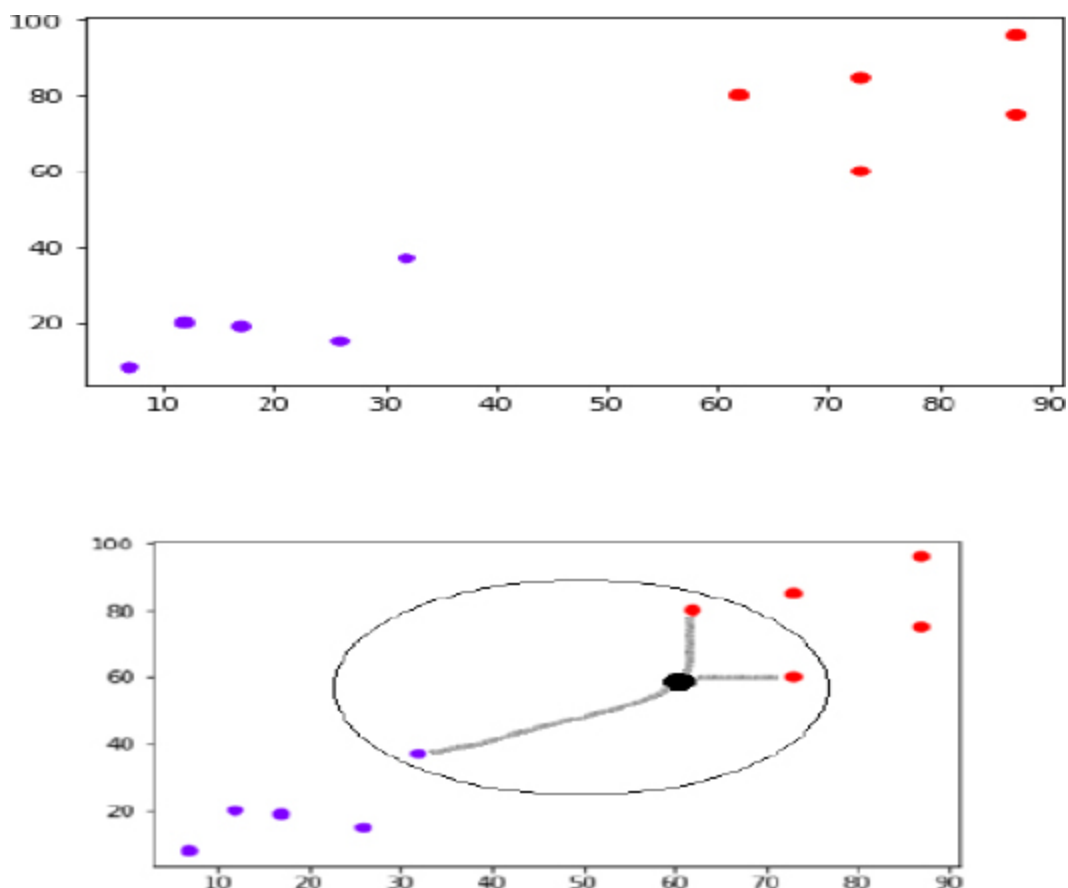


Fig. (3))

Classification of attributes using K-Nearest Neighbour.

A spam filter that separates spam from other messages can be used as an example. The spam filters maybe the first method of dealing with network security tasks.

Recently, unpopular emails known as spam that becomes a major issue on the Internet. People who are sending the spam are termed as spammers, who collect the email addresses from different chat rooms, websites, and virus. The benefits of bandwidth, storage capacity, and network time will be prevented to gain by spam. A devastating impact will result on the user time, CPU power, communication bandwidth, and memory space by a large number of spam emails over the computer network. The spam threats have increased every year that represents over 77% of global email traffic.

Spam recipients did not inquire about their discomfort. This has also made many users vulnerable to spammer schemes and other fraud. They send emails pretending to be from trustworthy firms to get bank verification numbers, passwords, etc. It causes massive credit card and economic loss.

In 12 years, the spam volume has fallen to the lowest point based on the report from Kaspersky Lab, 2015. Since 2003, the amount of spam has fallen below 50% based on the statement passed by the anti-virus software developer. The spam amount has reduced to 49.7% in June 2015 and it was further decreased to 46.4% in July 2015. This decline can be attributed to the billions of fewer top botnets responsible for spam. According to reports, the number of malicious spams in 2015 remained unchanged. Kaspersky Lab detected between 3 and 6 million spams in 2015. In contrast, as the year draws to a close, the number of spam emails has increased. Based on the Kaspersky Lab report, Dec 2015, spam emails were started to increase with malicious attachments, including JavaScript, malicious macros, ransom ware, and malware. The number of spam emails had quadrupled by March 2016 compared to 2015 [\[11\]](#). Kaspersky Lab discovered the number of spam emails as 22,890,956 in March 2016. In the first quarter of 2016, the spam volume would score to 56.92% on an average. The spam volume was accounted for 56.87% of everyone's mail traffic based on the latest statistics. The dating and health spam emails are the most familiar types. The resources are not used effectively on the Simple Mail Transfer Protocol (SMTP) due to the spam: Unsolicited email.

The major email providers like Outlook, Yahoo Mail, and Gmail integrate with different ML technologies such as email filters and neural networks for effective management of the threat that is posed by spam. Based on the assessment of various emails on the large number of computers, the spam and phishing emails can learn and detect by these technologies. The pre-existing rules do not utilize by spam filters of Yahoo and Gmail for verifying the spam due to the adaption of machine learning to different conditions. By using the knowledge that they learnt and continuing the spam filtering operations, new rules are generated on their own. Google has developed the machine learning model that can identify and filter the spam or phishing emails with an accuracy rate of about 99.9%. This means that one in a thousand emails successfully by passed your spam filter. Google statistics show that between 50 and 70% of emails received by Gmail are unsolicited. For detection of websites that have malicious URLs, the detection model of Google has combined with a tool known as the Google Safe Browsing. By introducing a system, the performance of Google's phishing detection has improved to review the phishing messages by delaying certain Gmail messages delivery for a certain time period. Analysing it centrally will help to find them. Sending suspect emails has delayed deeper inspections while allowing more messages to arrive on time and changing the algorithm in real-time. Emails are delayed by 0.05%. To limit the spam problem, different spam filtering techniques have been used extensively [12].

6.1.1. Content-based Filtering Technology

To create automated filtering rules and machine learning techniques, content-based filtering is used usually for classification of emails. ML methods include neural networks, K-nearest neighbours, support vector machines, and naïve Bayes classification. The distribution and occurrence of phrases and words in the email content have analysed using this method and the generated rules are used for filtering incoming spam emails.

6.1.2. Case-based Spam Filtering Methods

One of the spam filtering techniques is sample or case filtering. The collection model is used for extracting all non-spam emails from emails of each user. For conversion of emails based on the client interface, the steps of pre-processing are performed. It includes the email data grouping, the selection and extraction of functions, and the processes evaluation. Two sets of vectors are classified from the data. To know whether the incoming email is spam or not, ML algorithms have exploited for training and testing the data sets.

6.1.3. Rules-based or Heuristic Spam Filtering Technology

For evaluating the large number of patterns, the rules or heuristics are used in this method. Usually, it includes regular expressions for selecting the messages. The message score will increase by several similar patterns.

In contrary, any pattern will deduct from the score when it doesn't correspond. A certain threshold is exceeded by the email score that will be filtered out as spam. Otherwise, it will consider as valid. Even though some rules of classification won't change over time, remaining rules should be constantly updated to effectively deal with the threat from spammers. Spammers continue to introduce new spam e-mails, which are easy to escape, but not detected by the email filter.

Spam Assassin is a good example for a rule-based spam filter.

6.1.4. Spam Filtering Technology Based on Priority Similarity

The instance-based or memory-based machine learning techniques used in this technique for classifying incoming emails according to the similarity to stored samples. For creating a multi-dimensional space vector, the email attribute is used. This vector utilizes for drawing the new instance as a point. Among the most recent K-training instances, the new instance is assigned to the most popular class. The favourite neighbour (KNN) is used in this technique for filtering the spams.

6.1.5. Adaptive Spam Filtering Technology

In this technique, the spam detects and filters with the grouping of spam into various categories. Several groups are categorized by an email corpus and a flagship text includes in each group. For computing the possible group to which the email belongs to, each received email is compared with each group and a similarity percentage generated.

Different spam classification methods have been proposed by many scholars and researchers and they are exploited for data classification. The artificial immune system, decision tree, and probability are included in these techniques.

Content-based filtering techniques can be used to identify certain characteristics (usually keywords frequently used in non-spam emails), so that these classification methods can be used for spam filtering. Each feature's probability estimates that appear in the email and measures based on the threshold. The emails that exceed the threshold classify as spam.

The biological neural network function imitates the artificial neural network, which is a non-linear model. Simple processing components known as neurons and the computational operations performed based on the processing of data. By implementing the content-based filtering, neural networks are used for classification of spam. The attributes are determined using this method based on the computation of keywords or patterns occurrence rate in email messages. A medium ranking achieves by the neural network algorithm for email filtering that shows in the literature.

Static rules can be useful. For an instance, a very short body of text includes in BCC with too many recipients and some signs of spam include an upper-case subject. In a similar manner, email addresses and sender domains can associate with spam. Spam detection is based on the email content analyzation in most cases.

The statistical models are used in machine learning algorithms to make the data classification. The trained ML model should have to estimate whether the words sequence found in the email are closer to the words' sequence in the security email or spam in spam detection. Spam can be detected by using various ML algorithms, but the 'naïve Bayes' algorithm is the most attractive type. Based on the 'Bayes Theorem', Naïve Bayes is implemented that demonstrates the events' probability using prior knowledge. The independent characteristics are assumed so that the 'Bayes theorem' is known as 'naïve'. For example, you can use Naïve Bayes machine learning for determining whether it will rain or not. Here, the characteristics would be humidity and temperature and the expected event would be rain. In various fields, Naïve Bayes become popular due to its efficiency and fast machine learning algorithm.

For spam detection, our target variable is to find out whether an email is 'spam' or 'not spam' (also termed as 'ham'). In the email body, features like words or word combinations are found out. In other words, the spam probability of an email is to calculate based on the text. But the issue is that the features are not independent necessarily.

Consider the terms like 'sandwich', 'cheese', and 'grilled'. Different meanings can include for these terms based on their appearance one after the other or in different parts of message. Another example is considered as the words like 'interesting' and 'not'. In this case also, the appearance of message decides the meaning that can be different completely. In natural language processing tasks, the naïve Bayes classifier is effective if configured correctly and if the functions are independent in text data complicated way. The detection of data spam is a supervised machine-learning issue. It means that you need to mention the machine learning model with a set of examples for irregular messages and spam. The relevant patterns can find out to separate two different categories. The large tagged email data set includes most of the email providers. You can provide the training data of Google for machine learning algorithms when you mark the email as spam in your Gmail.

A weight assigns to each term according to the number of times the spam or irregular emails have appeared while training a machine learning model on the trained dataset.

When 'hitting the jackpot' is one of the features and which appeared in spam only, it's considered as spam. If the email mentions "important meetings," it's likely not spam. The spam can filter out by the machine learning model after data processing and assigning weights to the functions.

By using the formula of Bayesian, the tokenization of text and execution are performed when a new email is received. In the message body, each item multiplies by its weight, and the weights summation estimates the email probability as spam. For various text classification tasks, like spam detection, the naïve Bayes ML algorithm has proven to be effective.

6.2. Limitations of Naive Bayes

The language context doesn't understand by naïve Bayes like other ML algorithms that are dependent on statistical relations between words for estimating whether the piece of text is relevant to a certain class. If the sender adds some non-spam words at the message end or replaces the spammy terms with other relevant words, a naïve Bayes spam detector can enter into the overlooking of a spam email.

In the processing of sequential data, including text and email messages, the efficient algorithms for spam detection are transformers and recurrent neural networks (RNN).

6.2.1. Regression

It can be implemented to detect the frauds. The probability of fraud can be determined by the characteristics.

6.2.2. Detecting Credit Card Fraud

When detecting the fraud of credit cards, the main challenges involved are:

- Every day, the data with large amounts are processed and the model should be established in faster way to respond to the scams in a timely manner. Unbalanced data, that is, most transactions (99.8%) are not fraudulent, so it is difficult to detect fraudulent transactions.
- Data availability, because most data is private.
- Misclassified data can be another major problem, as not all fraudulent transactions are detected and reported.
- The adaptive technology used by the scammer against the model.

Machine learning models designed to detect such fraudulent transactions must be simple and fast to detect anomalies and classify them as fraudulent transactions. The incidence of fraudulent transactions including the credit cards, has increased by the use of credit cards for financial transactions. The financial losses could be avoided by the fraud detection when issuing the credit cards to customers by the banks. Credit card fraud occurs when a cardholder's personal information is used without their knowledge. Two different methods of fraud detection are included as anomaly detection and abuse detection.

6.2.3. Abuse Detection

Abuse detection is the task of identifying abusive behaviours, such as hate speech, offensive language, sexism and racism, in utterances from social media platforms. Based on false and normal transactions, the system training will be performed while detecting the abuse and will detect the fraudulent behaviours. For training purpose, the normal transactions are utilized in the anomaly detection system so that the detection of new frauds is possible.

On behalf of the financial industry, the most digitalized part is the payments that make it more vulnerable towards the digital fraud. The

reduction of number of verification stages is allowed by the rise of competition towards the best consumer experience, online or mobile payments. The less effective rule-based approach has resulted. The payment and bank companies turn to the development of artificial intelligence-based techniques, machine learning, and data analysis.

Several methods exist for detecting payment fraud during the checkout process, all of which analyse the user's actions. For example, a particular supermarket visits by a customer at 9 o'clock at every night that is located near the customer's home. The amount of payment is from \$10 to \$40. Every two days, the customers drive to the gas station.

The activity will be considered suspicious in the algorithm and a higher level of fraud probability is assigned after transaction takes place in different parts of the town and in bars and the total amount is \$40. A verification request will send to the cardholder by the system for validating the transaction.

To analyse the behaviour, it's useful to consider descriptive statistics, such as high/low values, standard deviation, and mean. The individual transactions comparison is allowed in these indicators with a group or individual internal benchmarks. It's suspicious when payments include larger standard deviations. In such deviations, a verification request is required to be sent to the account owner.

6.2.4. Anomaly Detection

In data science, it is one of the common anti-fraud techniques to categorize all available data objects into two groups, such as outliers and normal distribution. Outliers consider as potential fraud and are objects deviating from normal conditions. For fraud detection, various data variables exist.

The below questions can be answered by the anomaly detection algorithm with the analysis of these parameters: Does any inconsistencies exist in the user-provided data? Is the client

accessing the expected services? Does the behaviour of a user normal? Are they typical transactions?

As the anomaly detection technique offers a simple binary response, it is the simplest method. Several verification phases need to be performed by the user in the system when the transaction is similar to suspicious. The frauds will not identify by conventional anomaly detection [12].

6.2.5. Advanced Fraud Detection System

The anomalies and existing patterns can be determined by the advanced system that targets specific fraud cases. In anti-fraud systems, two widely used ML techniques are included such as unsupervised machine learning and supervised machine learning. For creating more complex anomaly detection algorithms, these algorithms can be utilized alone or in the form of integration. Based on the labelled historical data, the training algorithms are involved in supervised learning. The target variables are marked by the existing dataset in this case, and the training has an objective of a system that predicts the variables in future data.

Unlabelled data is processed by the unsupervised learning model and is classified into different groups. The hidden relationships between data items' variables are detected, as shown in Fig. (4).

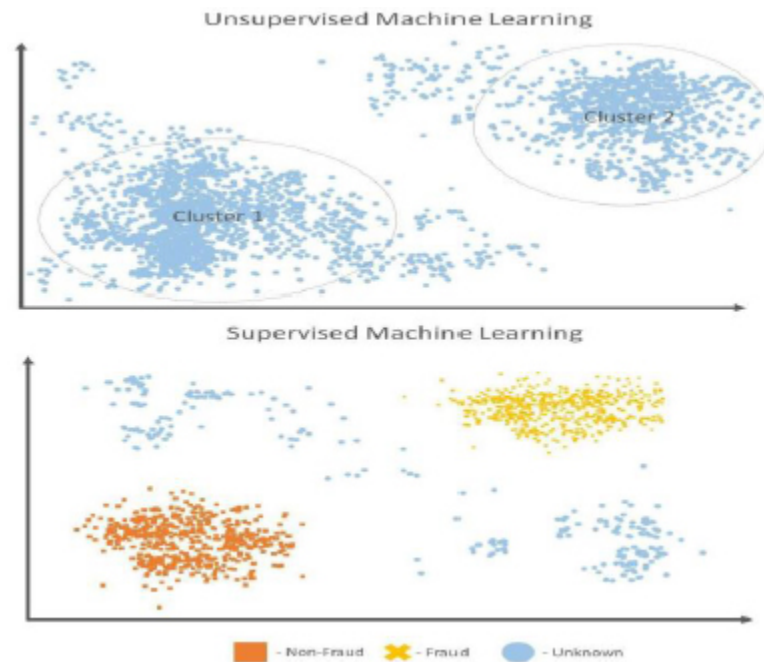


Fig. (4))

The hidden relationships between data items' variables are detected.

7. SOURCE: ALTEXSOFT

As scams are very domain-specific, they are primarily based on private data sets owned by the organization. For example, many banks with cases of fraud in their data use supervised fraud detection technology to prevent potentially fraudulent monetary transactions, taking into account variables such as time, location and the amount of the transaction [13, 14].

Some of the security mechanisms provided by machine learning for fraud detection are:

- Safeguarding Electronic Payments
- Security against fraud activities and risk prediction
- Diverse and massive set of partners or merchants
- Assess, classify, and verify the e-commerce websites for compliance
- Exploit the text mining based on NLP for categorizing with high sentiment accuracy automatically

- The widest data sets mash-up – usage logs, customer profiles, transaction history, service usage historical data, and known cases of fraudulent behaviour
- Identify misuse, anomalies, and fraud *via* operationalized classification model
- Analyse high-risk sites accurately and automatically –92% classified correctly
- Eliminate the false positive – normal sites are not classified as high risk
- Cost and time for resolving the fraud case radically

CONCLUSION

The most recent tool to be utilised in the process of detecting security vulnerabilities and proposing models to circumvent these dangers is machine learning. These methods provide assistance to the user so that they may put into practise the algorithms that enable them to design and model their systems in a manner that will detect and get rid of the dangers.

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